

Blockchain Organizing for Managers

First edition, May 2018
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Both authors work at Weconet Blockchain Technologies (www.weconet.org). Weconet is a spin-off of the Weconomics Foundation. Weconomics is primarily a way of thinking, a way of organizing projects with three pillars: organizational model, infrastructure and transition program. This book is part of the Weconomics program. Most of the Weconomics books are in Dutch at the moment. When you are member of the Blockchain Organizing Center of Knowledge (see www.weconet.org/block), **we will translate** the references in this book from Dutch to English for you.

The following books are included in the Weconomics program:

- Book1: 'Every day when the sun rises: the history of the community economy' (2010)
- Book2: 'Weconomics: how will you survive the 21st century as information worker?' (2013)
- Book3: 'Weconomics analysis: why our welfare machines need to be replaced' (2013)
- Book4: 'Weconomics theory: organizational science for Weconomics' (2013)
- Book5: 'Weconomics practice: practical advice for setting up communities' (2013)
- Book6: 'Blockchain Organizing: foundations for a new social-economic order' (2017)
- Book7: 'Blockchain Organizing for Managers: The Reinvention of Management' (2017)

Book 1, 2, book 3, 4 and 5 (also called the Weconomics Trilogy) and book 6 are only available in Dutch at the moment. More information about Weconomics program: www.weconomics.org (Dutch) or www.weconet.org (English)

Published via: Mijnmanagementboek
ISBN: 9789463678223

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For more background information with this book, including a literature list and blockchain knowledge center, see also the website: weconomics.org (Dutch) and weconet.org (English).

Blockchain Organizing for Managers

- The Reinvention of Management -

Paul Bessems
Walter Bril

“The most important, and indeed the truly unique, contribution of management in the 20th century was the fifty-fold increase in the productivity of the manual worker in manufacturing. The most important contribution management needs to make in the 21st century is similarly to increase the productivity of knowledge work and knowledge workers. The most valuable assets of a 20th-century company was its production equipment. The most valuable asset of a 21st-century institution (whether business or non-business) will be its knowledge workers and their productivity.” (Peter Drucker, 1999)

Content

Preface

1.	Introduction.....	9
1.1	What do you need to do business?	12
1.2	Is it time for a paradigm shift?	18
1.3	Blockchain by Design	22
1.4	Some concepts	28
1.5	What is Blockchain Organizing?	32
1.6	Blockchain within your organization?	34
2.	Blockchain as organizational concept.....	39
2.1	Context: digitalization	41
2.2	Basis: trust, values and transactions	44
2.3	Old organizing	50
2.4	New Organizing.....	51
2.5	Five basic principles of organizing	52
2.6	When hierarchies become markets.....	62
3.	Blockchain as technology	69
3.1	History.....	71
3.2	The Bitcoin blockchain	78
3.3	Different types of blockchains.....	85
3.4	Blockchain concepts and tools.....	95
3.5	Blockchain for communities.....	114
3.6	Blockchain standardization.....	121
3.7	Blockchain is industry agnostic	129
4.	What does it mean for your organization?	133
4.1	Which problems can you solve with Blockchain Organizing?	134
4.2	Blockchain for the organization of money, trade and economy	141
4.3	Organizing work.....	144
4.4	Organizing supply chains	151
4.5	Organizing government tasks.....	155
5.	How to start with Blockchain Organizing?	159
5.1	Blockchain lead	160
5.2	How do you start and how do you move.....	169
5.3	Project-related approach	178
5.4	Dealing with resistances	184
6.	The future of Blockchain Organizing.....	191
6.1	Blockchain power.....	191
6.2	Blockchain stages.....	193
6.3	Where blockchain should not lead to	196
6.4	Innovation Strategy: towards a decentralized cooperation perspective.....	200
6.5	Blockchain in the near future	202
6.6	The future of Blockchain Organizing.....	208

About the authors

Preface

What if selling your house would be as easy as sending an email? Big chance you would not believe that this will become reality in the near future? But when internet started nobody could imagine, that it once would be possible to operate Spotify on internet. As Halfdan Mahler, former director World Health Organisation once said: *“What sounds idealistic today becomes realistic tomorrow”*. But from ‘today’ to ‘tomorrow’ is a long and difficult road where leadership and management can play an important role. One of the key stakeholders, in the digital transformation and the transition to a mainstream usage of blockchain, will be leadership and management. Will they act pro- or reactive? Will they be ambassadors or resisters of change? Will they give up some autonomy and believe in decentral decision making? One thing is clear: leadership and management must innovate more fundamentally then they did till now. According to fintech expert Chris Skinner the real problem is legacy leadership, not legacy technology.

When you ask someone what the most important invention has been during the last few centuries, you will probably hear: steam engine, electricity or, more recently, the internet. Not many people will say: management. In general, management is not regarded as an invention or technology. Charles Perrow, an American emeritus professor of Yale and Stanford once said: *“Organizations are systems for getting work done”*. In our digital era you could say: Organizations are digital technology to share information, make decisions and process transactions.

In our opinion: processing ideas into tools to combine humans and resources in order to perform activities in the most productive way is also a form of technology. So we could agree on the idea that management is also a technology. But when we compare management with other technologies we can conclude that it hardly innovates, at least not in a fundamental way. We have hardly seen any fundamental innovations in management since its first invention at the beginning of the twentieth century: Scientific Management (1911) and the first assembly line of Ford Motor Company (1913). With management technology we don’t see an innovation pace that is comparable with other technologies such as information technology. But we believe that this will change in the next generation.

Harvard Business School professor Clayton Christensen coined the term disruptive technology. In his 1997 book, ‘The Innovator’s Dilemma’, Christensen separates new technology into two categories: sustaining and disruptive. Sustaining technology relies on incremental improvements to an already established technology. But the light bulb was not invented by the continuous improvement of the candle. Therefore you need a quantum leap, you need a disruptive innovation. We strongly believe that Blockchain Organizing, as a management concept or technology, is an example of the latter. But disruptive technology often has performance problems because it is new, appeals to a limited audience and may not yet have a proven practical application. A lot of people find it abstract but they use ‘abstract’ not in the proper way.

Abstract means: let go of non-functional parts or back to basic. And Blockchain Organizing is back to basic, back to the fundamentals of organizing. A lot of people find this hard to understand because it deviates from what they are used to and the complex world they live in. It seems that they avoid simplicities because they don't know what to do with the surplus that technology can create.

Innovations are often associated with matter, but you can also innovate ideas and concepts ('products of the mind'). Management is above all a type of social technology because we believe that management tasks such as decision making must be performed by people and cannot be done by protocols. Our claim as authors is that management, as an invention, as a technology, has not changed fundamentally since the 'first' Ford Motor Company assembly line started working at the Highland Park Assembly Plant in October, 1913 in Michigan. Management was invented to align capricious people: farmers, housewives and craft-men, into the systematic way of working on an assembly line, between nine and five, in a building away from home. 1913 is also the year when the Chartered Institute of Personnel and Development (CIPD) was founded in England, the first, what we now would call, HR community. Since then, management is not reinvented in a fundamental way.

Yes traditional management did its job: it doubled our standard of living on average every thirty years. It brought us prosperity, but the context changed from factory work to knowledge work at the end of the twentieth century. And for that matter, management didn't adapt in a way our productivity growth could keep up with the rising costs of healthcare, education and safety. Management did a good job, improved over time but is still not replaced by something better as the enemy of good as a result of a changing context. The candle has continuously been improved, but it has not been replaced by the light bulb as mentioned before.

Some adjustments have been made to management, such as the recently introduced agile and self-steering teams, or before that Lean Six Sigma and the implementation of performance indicators. But these concepts have never challenged 'the firm' as a superior way to organize work. They have, especially in the last decades, made insufficient contributions to the required growth in productivity and improvement of trust. They are relatively small adjustments within an existing invention that mainly involves the use of buildings, hierarchies, planning, informing, assessing, deciding, awarding and punishing. The way we connect people and resources to tasks, projects and targets in order to increase the productivity has not fundamentally changed since Taylor and Ford.

Because many people do not consider management as an invention, there are many different views regarding management innovation. The term innovation is mainly associated with process and product innovation or ICT; not management innovation. Most managers will say that the current work method is fine because everything has been going well for quite some time. However, there are more and more undercurrents that indicate that things are not going so well after all. These include, among

other things, privacy concerns, regulation, bureaucracy and complexity, power concentration, inequality and lagging productivity. Management is and was meant to organize people and tools in such a way that the productivity and therefore the 'standard of living' improves. But traditional leadership and management are no longer capable to tackle the social problems we are facing.

There is however a rapidly growing management technology that can perform many of the existing (often mechanical) management tasks decentral, autonomously and much more efficiently: it is called Blockchain Organizing. This technology will create surplus time. We can move from the third (after agriculture and factories) work sphere (offices) into the fourth 'work sphere'.

Management (as technology) develops itself at a much slower speed than other organization technologies, so management, in its current shape, could quickly become less relevant. Many traditional managers are still trapped in an existing paradigm of what management is and the best way to handle it. On the other hand, more and more organizations, such as Svenska Handelsbanken or the distributed company Automatic, adopt new organization methods and make maximum use of technology to connect people and tools to activities. This means for example that the span of control within these organizations is more than 1:10, while this is 1:5 within traditional organizations. So there are management concepts that operate fundamentally different from the way most organizations and managers are used to. One of these concepts is, as we as authors call it: Blockchain Organizing. Blockchain Organizing is the combination of a disruptive technology (blockchain) with a fundamental new model of organizing supply and demand (Weconomics). This book addresses the questions: What is Blockchain Organizing, why is it important and what does it mean for managers and management innovation?

Since the seventies people started to do other work (from factory work to knowledge-based and service work), but the methods with which we organize work have hardly changed. The question that has been keeping us authors occupied for years is: what is the reason that we, despite increasingly more and 'better' information technology, share information increasingly worse, cooperate increasingly worse and trust each other increasingly less? Why are there so many intermediaries required between supply and demand to reduce uncertainty? Why is the capacity utilization not improved, why can't we safeguard our privacy better, trust each other better and share knowledge better? Why has internet not made us more productive? Why have markets not become more transparent? And why have we not become less dependent on large companies and central governments? Further on in this book we'll see that the number of managers and staff officials continues to increase despite the present technology. You expect that technology would make us sufficiently productive but that is not the case. And now a new technology is emerging: blockchain. So, will this organization technology indeed provide an improvement? Why would this technology succeed where others failed?

To keep our organizations, economy and society going, we use tools such as management, money, contracts, brands and laws. These are ideas created by human minds. These tools are organized by companies, banks and governments and are meant to reduce insecurity and to install trust within our economy and society. In the organization theory we call these tools agents or institutions. Without these institutions we would be unable to organize work, to trade, to make economies work and to enable our society to function. However, these institutions have the disadvantage that they make our society unnecessarily complex and bureaucratic. And as a result we need more specialists to understand this complex world. These specialists must be attuned to each other and therefore we need even more institutions, etcetera. In short: we are all very busy, without making enough progress.

Many of the institutions that we use date from the time of Columbus and the Industrial Revolution. Nowadays we live more and more in a digital society. Trust can be programmed as it were. That is in any case an important promise that is inherent to blockchain and one that is proven by the Bitcoin blockchain until now.

The purpose of this book is to describe how to use blockchain technology for solving social, organizational and management problems. We do not want to stimulate and sell a new technology. In this book we do not extensively discuss Bitcoin as money, as we might not need any money (physical or digital) in the future as a way to exchange values. The hype surrounding the price of a bitcoin, cryptocurrencies, futures or the Initial Coin Offerings, illustrates again that when changing a technology, the behavior of people does not automatically change with it. That will require more, and most managers will be aware of this. Human behavior is mainly focused on satisfying our conviction and needs. Too often useful new technologies are used in old organization methods with the purpose to strengthen existing positions or organizations and therefore weakening new business models.

More and more banks buy fintech start-ups. Is that because they are interested in solving social problems or to strengthen their existing position? Banks will start to compete with each other on 'who has the best blockchain'? But what if money and banks become irrelevant? Therefore, it is important to realize that Bitcoin can take many forms. Blockchain Organizing mainly concerns the exchange of values between seller and buyer. And Blockchain Organizing does this with minimum friction, and at minimum transaction costs. Blockchain Organizing primarily concerns solving management problems, secondly it looks as the organization model that can be used best and tertiary it looks at the tools.

We believe that technology can actually add value to our society. That was also the promise of the internet. But there was no clear problem to solve and we have used the internet within an old management model. With the internet we did not fundamentally organize supply and demand differently, mainly the medium has changed: the brochure was replaced by the website and the letter by e-mail. The relationship

between producer and consumer, government and citizen and employer and employee has not actually changed.

We see the shortcomings of the internet more and more, it is a beautiful technology that is still used within the boundaries of a traditional management style and an old socio-economic system. The internet did not increase our productivity sufficiently, privacy is increasingly more difficult to safeguard, American internet companies are becoming increasingly more powerful, elections are being manipulated and we find it increasingly more difficult to distinguish between lies and the truth. More and more people are aware that internet alone does not really help us to progress as we would like: it gives us more and more insecurity and we trust each other less and less. We can conclude that the internet is far from being completed; it is actually only in its infancy. With the introduction of blockchain we get another chance to do it better. Blockchain can be considered as the next stage in the development of the internet. Not only does it enable the sharing of information, but values can be moved from seller to buyer, without having to trust each other and, in principle, also without money as exchange tool.

So, 'the blockchain' can be regarded as an additional layer on the internet: a layer of trust. With the internet we can share information without any significant friction. Sending e-mails (bits & bytes) costs almost nothing. With blockchain technology you can transfer values (also bits & bytes) from supply to demand without irrelevant third parties, without any significant transaction costs. This does not just apply to buying a training course or consultancy service, but also doing the daily shopping up to selling your house on a blockchain. To bring supply and demand together against the lowest possible costs is not just the essence of organizing work, but also the essence of organizing our economy.

The essence of organizing is: look for a tool that can transfer values from supply to demand against minimum friction (or transaction costs). Blockchain has a lot less friction than the existing coordination mechanisms such as hierarchies, building, management development programs and permanent contracts. We as authors of this book support the potential of blockchain technology, but we are also worried about the 'hype' around 'the blockchain'. A hype can frame or slow down a fundamentally good idea. A lot of sense, but also a lot of nonsense is written about blockchain. We believe that a major breakthrough of using blockchain technology in several areas, might take a decade unless we quickly become collectively smarter and also innovate management technology in addition to blockchain technology. But we are slightly less optimistic about this. Institutions that have been around for centuries cannot be changed 'overnight'. We believe that the core of the success of blockchain will not be the technology, which normally finds its way, but leadership and the organizational ability. This is a broad perspective that also includes: laws, politics, power, adaptability and management.

So, we must not overestimate the power of technology alone. Technology is ‘only’ a lever: an ability to realize something. Without being firmly grounded, without foundations, a lever does not work. When we start with new digital leadership, connect a new technology to a fundamentally new management model the effect on the existing organization methods can be huge. As a result, enormous resistances can arise. However now may be the best time for renewal. For us Blockchain Organizing is a fundamentally new way of organizing that can provide a large contribution to social challenges, such as the crumbling welfare state, aging population, growing inequality, fraud, tax evasion, propaganda, cyber security and climate change. It is definitely not a panacea for all problems but it can contribute in resolving many social problems. It has fundamentally different answers to the question how physical and data flows should be organized. Although the impact can be huge in the long term, the embedding of Blockchain Organizing in our daily lives will be rather evolutionary than revolutionary. Changes will be made step by step comparable to the development of the assembly line or internet for example.

In this book we mainly focus on the fundamental management innovation by operating organization technology, such as the internet and more recently blockchain, in a smarter way. Tools will not have the desired effect if we, with the technology not also change the methods and the institution with which we organize trust, work and economics. If we stay trapped in the old leadership and the management innovation of Taylor and Ford, blockchain will add little or even make it worse. Traditional companies, banks and governments have had their days. They become irrelevant. They are no longer the superior coordinating mechanism with the least friction. They keep us unnecessarily trapped and at work. More and more people will become aware that we create fake work to keep the illusion of full employment alive and avoid real work at home, in local communities and to organize a more sustainable prosperity.

By operating Blockchain Organizing we become a lot more productive and we save more time. We can use this surplus for the next step in our evolution: the sustainability of the prosperity we enjoy and to secure this for the future of our children. That is also growth and progress. Growth is not just more in the ‘here and now’, but also better and in the ‘there and later’. The first step to use Blockchain Organizing is the transition from hierarchy to market. To differentiate between traditional markets (markets between two hierarchies), we call these new markets also: network organizations. Management and leadership can play an important role in this transition, both in a positive as well as in a negative way. When you see technology as a danger, you will resist the introduction as much as possible, whether or not anonymously. When you see technology as a basis of our evolution and progress, you will embrace it and discover what your new added value will be. This book will also help you with this.

Apart from a positive contribution to our productivity growth, Blockchain Organizing also ensures that personal data can be better protected. In the coming years more and more organizations will face problems due to the new and much stricter

European privacy law (GDPR). The problems that we expect can no longer be solved by more IT, lawyers or 'Data Protection Officers'. The anticipated problems can only be solved with another mind-set, new thinking, a new IT-architecture and a fundamentally new way of organizing supply and demand in a digital context. This will increasingly happen with humans as the unity of design, as the starting point of organizing supply and demand instead of firms, banks and governments. People will become owner of their personal data. Blockchain Organizing and concepts like Personal Data Service, Bring Your Own Data and Data Self-Determination will help with this. The future of our digital society can be seen as an American barbeque party whereby you bring your own food and drinks. When you buy a service from a company or government authority you yourself provide the data that this organization requires to perform the business process, such as the printing of an address label by the company Amazon to send you this book.

In addition to increase our productivity and securing privacy we also want to use Blockchain Organizing to become less dependent on large tech companies. Blockchain technology can provide an important contribution to decentralization and the construction of a general data common whereby we are less dependent on companies like Facebook and Google, for using data, sharing files, messages or promoting products. This way, we the people will also be able to, jointly, curb propaganda in a much better way.

A final incentive for companies to introduce Blockchain Organizing is to make processes more efficient. This can be useful in the short term, but in the long term more and more processes will have to be transferred to shared information- and transaction networks. This means that processes must be 'prepared' to be transferred from company to network. Currently for some organizations blockchain technology is only considered as a technology to make processes within the existing system more efficient while the system is the problem.

Our intentions with this book

How can you start with Blockchain Organizing in your own organization, supply chain or ecosystem? Which innovations do you require for this and how do you organize these? With this book, we want to help start-ups, enterprising people and organizations to operate the management concept of Blockchain Organizing. This book can help you to set up and implement a blockchain project within your organization or network. We, ourselves, have helped to set up many of these projects as consultant, project leader and sometimes as an entrepreneur. These projects have led to the design, development and setup of shared information and transaction networks. Through this book we like to share our acquired knowledge, experience and methodology. For example, this book offers some kind of organizing capacity, a process and proven method for your own project. Purpose is to implement your idea in terms of Blockchain Organizing within your organization or maybe within a new foundation, limited company or cooperative to be established. You could regard our method of Blockchain by Design and our way of project-related working as some

Kickstarter with the difference that you not only could bring money together, but also other building blocks, like a network, methods and techniques, a Blockchain Organizing Center of Knowledge (BLOCK), infrastructure, transition program et-cetera. Our intention with this book is to further strengthen the movement towards a more sustainable prosperity with the help of organization technology.

What to do next?

If you would like to proceed with Blockchain Organizing after reading this book, we advise you to appoint a 'blockchain lead' within your organization or network. This lead is the first appointed person who keeps track of developments, advises management and, for example, sets up and supervises projects. A blockchain lead prepares the organization to use Blockchain Organizing and is therefore some kind of fore-runner or quartermaster. Some examples of the tasks:

- recording developments, visiting conferences, establishing network
- designing and setting up Proof of Concept projects and experiments
- creating conditions for a successful implementation
- coordinating different internal and external roles

Finally

This book is full of ideas about the use of blockchain technology, but not all ideas have to be executed as described. A good book doesn't do the thinking for you but makes you think. Play with the idea of Blockchain Organizing and try to imagine what it can do for you and your organization, without immediately having to decide what you actually will do with it. To realize a fundamental change there will have to be a good balance between the existing and new world. This means that we, with this book, focus on both 'traditional' managers, consultant and policy makers as well as dissenters, evangelists and pioneers and everything in between. The fact that you have picked up this book probably means that you are fascinated by the idea of Blockchain Organizing and that you in any case would like to know more about it. We are aware that we sometimes might have chosen an explanation or example that technically is not 100% correct or definitely has to be. On the one hand, this is due to the fact that blockchain technology is still developing feverishly. In a number of areas, no consensus has yet been found. On the other hand, we have chosen to give a simplified (technical) explanation as this book does not primarily focus on blockchain technology or blockchain developers, but on management problems that we want to solve and on organizational aspects of blockchain. If you want a further deepening (both organizational as technical), we advise you to read our book 'Blockchain Organizing: foundations for a new social-economic order' (called book 6, see also colophon, only in Dutch at the moment).

We hope you'll enjoy reading this book.
Paul Bessems and Walter Bril
Eindhoven, the Netherlands, May 2018

1. Introduction

Imagine you lived a long time ago and looked at the sky, see the moon and stars....You must have been thinking that you were the center of the universe and that everything else was revolving around you. That was the prevailing view until somebody called Nicolaus Copernicus came up with a completely different model. He developed a different world view and stated: *“Those things which I am saying now may be obscure, yet they will be made clearer in their proper place.”*

The majority of the people we as authors engage with nowadays don't immediately understand the new world view we are promoting. They 'think' in a similar way as people did before Copernicus: that their job, department, company or industry is the most important dot in the universe and that everything else revolves around it. Most companies see themselves positioned between purchase and sales. When a new technology comes to their attention, they will probably integrate this new technology from the perspective of their existing job or organization. The way they will organize trust, work and economics will not be fundamentally changed as result of that new technology. Their reasoning will be: 'Let's put this new technology in our organization and strengthen our business model'. They do not want to cannibalize their existing revenues. They assume or hope that it is just a new technology that will not impact their business.

Well, Blockchain Organizing is not business as usual disguised as innovation. Blockchain makes value exchanges as easy as sending an e-mail. Blockchain has the potential to organize transactions and economics without the traditional institutions such as money, contracts, brands and hierarchies. It even has the potential to disrupt modern institutions such as peer-to-peer business platforms and non-transparent algorithms. Uber, Facebook and other platforms are examples of new technology dressed up in an old 'business model'. You can put lipstick on a pig, but it's still a pig. Blockchain Organizing has the potential to reinvent management and turn hierarchies into markets and eventually into shared information and transaction networks. Not for a few industries as internet did, but for most of the industries we know of.

Digitalization is not the same as automation. Automation started roughly in the mid-twentieth century in a context of expanding industrially based organizations. It automated relative simple and routine processes such as bookkeeping and document management and it stored data in databases instead of filing cabinets. All of this work was done with 'the firm' as unity of thinking, analyzing and design. If you automated processes at that time, it was within a department or a company, not within a business ecosystem. As a result of this 'firm based IT development' process, we expect that the vast majority of IT service companies still uses the 'firm' or a department as starting point for their automation, which they unfairly call digitalization.

You could say that digital transformation is automation plus the change of the organization model. Digitalization is processing transactions between the smallest organization units (humans and resources on the supply side and activities on the demand side) in an autonomous and decentralized way. This means actor-oriented design with the smallest building blocks in mind: humans and resources, supply and demand. And when both technology and organization model fundamentally change, most of the organizations that are unwilling or unable to adapt, will become irrelevant with respect to organizing trust, work or economics. If they are no longer the most efficient way to coordinate supply and demand, they will disappear eventually. In that context, we expect that more and more IT-companies in Europe will become utility companies that use and share a data common, similar to other utilities such as electricity grid, roads- and water infrastructures. It is possible that Europe, the USA and China will develop different data infrastructures: Europe more private/public, the USA more private and China more state controlled infrastructures.

Blockchain Organizing is a new management concept and concerns embedding a disruptive organization technology (blockchain) in a fundamentally new organizational model (Weconomics). It promises to reorganize trust, work and economy. It claims to largely replace money, contracts, administrations and hierarchies by programmable trust. Blockchain Organizing does not just improve productivity; it also ensures more privacy and limits the power of large companies, banks and central governments. The influence of Blockchain Organizing will be substantial in the long term but its use in our daily lives will go step by step. Important is leadership and the manager's contribution to this process. In any case, you as a manager can no longer ignore this new management concept. In the coming years it will not just affect your position but probably also your organization and perhaps change it fundamentally or even make it redundant. We can see 'the blockchain'¹ as the next stage of 'the internet'², but also as an additional layer to the internet.

Internet makes it possible to send a message in a secure way from sender to receiver. But basically every time you send a message, you make a copy. The email you send is in your 'out-box' and in the 'in-box' of the receiver. We find it okay to copy information when we send hundred carbon copies of an email. But when I, as a sender, have to pay you and your neighbor each twenty euro and make a photo of a twenty-euro bill and email it to you and your neighbor with the remark: 'it is paid for', you probably will say: 'that is not okay'. We find it not okay to copy money, or more

¹ A 'single blockchain' does not exist. There are several chains of blocks. There is not one blockchain. There are private blockchains (similar to intranet), public blockchains (similar to public websites) and more hybride types (similar to login websites). They use different principles, techniques, governance models, protocols etc. When we use 'the blockchain' or 'blockchain' in this book, we mainly mean this as a theme, as a (organizational technology) concept. Concepts like blockchain technology, blockchain organisation, blockchain community and blockchain infrastructure are part of this.

² A 'single internet' does not exist either. It is a collection of concepts, techniques, media, agreements, protocols, websites etcetera.

general: values. And the receiver probably also wants to know, whether or not, the sender is the rightful owner of the twenty euro bill. So basically, blockchain adds two import functions to internet that already has secure transport from sender to receiver in its protocol. These two functions are: are you the owner of the value you want to send and do you send it only once? A blockchain secures these three important value transaction characteristics.

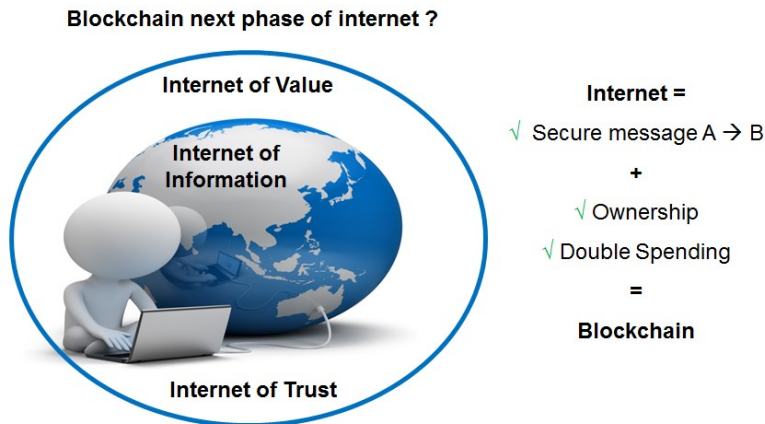


Figure 1.1: Blockchain: additional layer to internet: internet of value or trust

Internet makes it possible to share information without almost any friction. With a blockchain we can do business with each other, without having to trust each other. The ‘only thing’ we must trust are the underlying mathematical proof, the organizational principles, the protocol and the people who design, develop and maintain it. Blockchain Organizing ensures the proper organizational embedding of blockchain technology. This allows us to coordinate, validate, process and store transactions. It functions automatically and autonomously without the involvement of third parties, such as notaries and accountants, and without any significant transaction costs³.

Suppose you have a community for demand and supply of management services like: conferences, training courses, consultancy services and interim managers. With Blockchain Organizing, and more specifically a smart contract, you can facilitate the following processes: profiling, searching, selecting, approving, contract preparation, ordering, confirming, planning, invoicing, paying, evaluating, registering, administrating and archiving. You will not do this on your own, but with and in your business ecosystem. The core of Blockchain Organizing is transferring values from seller to buyer. The ‘challenge’ is to do this at the lowest possible costs. The processes, required for these transfers, are mainly protocols with if-this-then-that-statements⁴.

³ See also the transaction cost theory of Ronald Coase, for example in his article: ‘The Nature of The Firm’ (1937).

⁴ Also called IFTTT.

Another example: suppose you want to sell your house. You need a notary for this. This reliable third party normally performs four checks:

- Are seller and buyer the person they claim to be (*authenticity*)?
- Is the seller the owner of the house (*ownership*)?
- Does the seller sell the house only once (*double spending*)?
- Is the transfer from seller to buyer done safely (*security*)?

In principle, these steps can be programmed, as has been done with Bitcoin. This means that we can, to a large extent, replace the organizing of trust by means of the use of social institutions (such as traditional management), by the organizing of trust through the use of blockchain technology. Social institutions have the disadvantage that the complexity increases. Blockchain technology, provided they are used correctly, has the advantage that productivity will increase, privacy can be better assured and the power of mainly American tech-companies will be reduced.

Blockchain technology is a type of organizational technology that makes it possible to work much more efficiently, especially within offices where many knowledge workers are engaged. Blockchain Organizing makes it possible to autonomously and decentral perform almost all validation, administration and archiving processes without third parties and many existing manual management processes.

The best definition of a blockchain is a chain of blocks; a block is a collection of transactions that (logically) have taken place at the same time and about which there is an agreement that they have occurred. A transaction is transferring the value from the seller to the buyer. The offering, searching, booking and paying of management services can also be performed with blockchain technology. So, hardly any people are required anymore to do this. Whether the implementation of blockchain technology will succeed, does not depend on the technology but on the used management model and the guts to let go of existing institutions and striving for full employment (making full use of the production capacity).

1.1 What do you need to do business?

Why does the number of managers in healthcare grow with an average twelve percent per year?⁵ Why does bureaucracy cost us 1.3% in productivity growth? And why does a ten-kilometer ambulance ride costs seven hundred euros? All questions that are related to the core of organizing: uncertainty reduction and the organizing of trust. As we can see in figure 1.3 organizing in its core includes: specializing, coordinating and finalizing. Coordinating capacity brings specialists together to complete activities. In the example of the ambulance (finalization), these seven hundred euros do not just cover the costs of the direct people and resources (specialization). A large

⁵ See for example: The number of NHS managers has grown by almost 18% in the four years since the government introduced a 'bureaucracy-busting' shakeup of the health service, according to the latest official data, see article in 'The Guardian: 'NHS gains 4,000 more managers but loses GPs' (15-04-2017).

part of the costs is spent on coordination between demand and supply, actually on organizing trust and reducing insecurity. For example, we find it important that the ambulance arrives on time, that the staff is well trained and that the equipment complies with the latest safety standards. We must be able to trust that the ride with the ambulance does not make us sicker. Organizing this trust, reducing insecurity, costs money. The question is not whether coordinating capacity costs money or not. The question is which coordinating capacity costs the least money with the condition that that way trust is organized must not lead to a more complex supply chain.

Complexity is growing

According to research by Gary Hamel and Michele Zanini⁶, increasingly more people are employed in the trust industry. This creates bureaucratic organizations. While the number of people in primary processes has been reduced, the number of managers and supporting staff has almost doubled between 1983 and 2014. Despite, or maybe thanks to, all kinds of investments in lean and mean programs, ICT technology and management development programs, the number of managers is increasing. Managers have an important role: they reduce the insecurity in transactions⁷: they organize, coordinate and provide trust so the primary process goes as expected. But they also have an impact on the transaction costs. And when the transaction costs increase and the primary turnover stays the same, productivity reduces.

Because the complexity is growing, the transaction costs increase. As a result, our welfare becomes far too expensive. The productivity growth cannot keep up with the increase in welfare costs. Healthcare, safety and education become too expensive and the welfare state is crumbling, which partly explains the rise of populism. The core of our welfare problem is that the costs of our welfare are rising much faster than our productivity. It is okay to earn more or spend more on welfare as long as the system becomes more productive. But despite investments and innovations like internet, IT, cloud services, social media, management training courses and so on, productivity has insufficiently risen to keep up with the rising welfare costs. This phenomenon is also known as the productivity paradox.

We can no longer justify our welfare state with sufficient productivity growth. This is not due to the absence of appropriate technology but because we strive for full employment and use the wrong resources to organize trust and work. To break this pattern, we must reinvent management and go back to the fundamentals of organizing work.

Economy is driven by complexity

⁶ Gary Hamel and Michele Zanini: 'More of Us Are Working in Big Bureaucratic Organizations than Ever Before' (Harvard Business Review, 5-7-2016).

⁷ The 'cooperation' between two colleagues or between a colleague and a flex worker etcetera can also be seen as a transaction with transaction costs.

At the moment we create, often unconsciously, a complex world because we strive for full employment in addition to innovation. For some reason we refuse to accept a simple world because we are afraid to be left without any work. We create fake work at the office to avoid real work at home. Plus, in accordance to the system we 'must' grow two percent per year. Almost every entrepreneur has growth DNA. But the result is that we do not become more prosperous, but instead make our world more and more complex and erode the earth. The only way to correctly deal with complexity is to go back to the basics; to the smallest building blocks and the fundamentals of organizing work.

This is an important role for managers and the consultants who support them. There is an important role for professionals who no longer earn their money with complexity but who want to contribute to provide more welfare for more people. The strength of Blockchain Organizing is the simplicity of the underlying organizational model. It is based on a limited number of basic principles of organizing work and trust. All that Blockchain Organizing requires are the small building blocks: man and resources and the connections between them. That is all! It is the perfect way to cooperate in a smart way within and mainly between organizations. The beauty of this disruptive management concept lies in the simplicity. It makes it possible to coordinate demand and supply without expensive institutions and with a good balance between scalability and flexibility, competing and cooperating and the individual and the community.

Trust paradox

When executing transactions, we are prepared to incur costs to reduce insecurity. But if the various resources to reduce insecurity are not correctly aligned, the complexity increases and therefore also the insecurity. In case of bad alignment, the insecurity reduction creates a new problem: complexity. And an increasing complexity again leads to more insecurity and distrust. This way a vicious circle of cause and solution is created: in economic theory also known as reflexivity. This also leads to active inertia: we do a lot, but we do not move. So, a paradox is born when organizing demand and supply: by reducing insecurity, the insecurity increases. This paradox is also called the trust paradox: *In order to improve trust, distrust grows.*

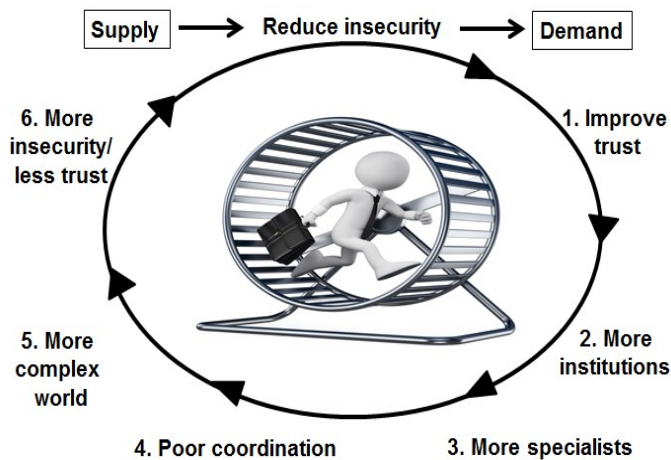


Figure 1.2: The trust paradox: in order to organize trust, distrust grows

What is trust?

Stephen R. Covey⁸ formulated it as follows: *“Trust is the glue of life. It’s the most essential ingredient in effective communication. It’s the foundational principle that’s holds all relationships”*. Trust is a broad and difficult to grasp concept that is widely represented in our lives. All four living atmospheres (personal, public, private and politics), learning atmospheres (art, sports, education and religion)⁹ and working atmospheres (agriculture, industry, offices and welfare) require trust to make and maintain a connection. Managers wonder if they can trust their employees: for example, do you trust, as a manager, that your employee will perform the task correctly? Or do you trust that the intentions are good and the employees stick to their agreements?

You could divide ‘trust’ in human aspects (human relationships) and resource aspects such as craftsmanship. Emeritus professor Bart Nooteboom¹⁰ puts trust in a time dimension: *“Trust is of all times. It is pervasive and indispensable.”* In his book *‘Trust: Forms, Foundations, Functions, Failures and Figures’*, he discusses organizing trust within and between organizations. He also makes a link with the transaction cost theory of Coase and Williamson. In his book he answers questions like: can trust serve as a tool to manage relationships? Is trust a substitute, a condition or for example a result of a contract? According to Nooteboom trust is essential in order to be able to make and maintain connections.

⁸ Stephen R. Covey (1932- 2012), was an American bestseller author. He wrote, among others, the bestseller: *‘The Seven Habits of Highly Effective People’*.

⁹ See also the Agora model (the world is much simpler than you think) develop by René Gude (1957-2015), a Dutch philosopher.

¹⁰ Professor ‘Innovation Policy’ at the universities of Tilburg, Rotterdam and Groningen.

Organizing trust is a ‘core activity’ for manager. However, trust is complex and versatile. It would require a separate book to discuss this further. For the purpose of this book we can only share a few insights related to organizing trust¹¹. Blockchain Organizing, and the organizing of trust within it, only focuses on a limited part; trust that you require to initiate and execute a transaction and to check and register the execution. The correct execution of these processes subsequently has an impact again on the next transactions and other aspects of trust. We mainly need trust when we do not know the other party, for example a supplier, or do not know how this supplier will behave under certain circumstances. We need trust because purchasing and sales do not take place simultaneously and at the same physical location.

It is important to know who to trust and what to trust. There are major differences in how and to what extent we organize trust. A system that you must trust is more abstract than a contract you can use to trust someone. There are categories of trust whereby, for example, risk of damage, delivery time, distance, price, personal contact or not, are characteristics that influence the need or expectation of trust. It is also often difficult to establish afterwards what has betrayed the trust. Was it opportunistic behavior, a lack of research or the wrong competences? All these aspects results in the innovation of a large number of instruments to organize trust and there is Blockchain Organizing added to the long list of instruments to organize trust.

With blockchain technology we can organize trust much easier because we use a technical script based on mathematical proof. We start from organizational principles, the smallest building blocks and the connections between them. In principle, trust mainly concerns the connection between two people. Whether or not they will use any resources for this is secondary. They can also use their experience, intuition and hormones. But organizing trust is still difficult to make ‘feasible’, even with Blockchain Organizing, human interpretations remain possible. Blockchain opens a new road: trust between people starts where machine control stops.

Companies to organize trust

Many companies have turned organizing trust or reducing insecurity into a product or service. For example, banks, governments, auditors, accountants, software companies, lawyers and notaries. Hierarchies, positions, permanent jobs and money: they are all resources to reduce insecurity and to organize trust. But when these tools are not coordinated in a proper way, they create the opposite effect. In organization theory, we call these tools also: institutions or agents. When you merge a number of institutions, an organization is created with a building, employees, an organogram, a structure, a culture and an own ICT system. The most common order we use to coordinate different tools are: the firm as a structure with the market as coordinating capacity between firms. The firm itself is also a form to coordinate (internally) and to reduce insecurity.

¹¹ More about organizing trust in the book: ‘Blockchain Organizing: foundations for a new social-economic order’ (Bessemers & Bril, 2017), also called book6 (see colophon, only in Dutch).

When organizing work, we choose by default to do the work ourselves (via the firm) or to outsource it (via the market). An advantage of a firm is scalability a disadvantage is its inflexibility. For a market it is the other way around. In our fast-changing world we mainly need a hybrid form between doing it yourself and outsourcing, between the firm or hierarchy and the market. Internet and blockchain are very scalable (with six connections you know the whole world) and humans are very flexible. New organizational forms are decentralized platforms that bring specialists together and carry out transactions when needed. For this to work, we need a good balance between scalability and flexibility. Structured flexibility is one of the five organizational principles of Blockchain Organizing. More about this in paragraph 2.5.

Depending on third parties

When organizing trust, we depend more and more on third parties (agents or intermediaries), such as a notary and a bank when selling a house. We depend on institutions that we use to organize trust. Relatively little attention is given to the function IT could have when organizing and actually automating trust. Blockchain technology will change this. For this we mainly have to investigate the relationship between trust and control. Trust and control are both a supplement as well as a substitute. Blockchain technology in particular can make a difference in organizing control (input references, double spending, encryption, keys and digital signature) and therefore be a supplement to trust. You could see trust as 'giving more freedom', and control as limiting this freedom when you have transactions executed via a protocol.

You limit the freedom but provide more control and therefore probably more trust. When you have transactions executed with human interpretation, you provide less control and more freedom. In some cases man is better at organizing trust and in other cases (often routine-like common transactions) a protocol is better. But to realize this we need a paradigm shift in the way we organize trust, work and economy. Blockchain Organizing is a new way to organize management in relation to the organization of trust, work and economy. To understand the paradigm shift and the strength of Blockchain Organizing, in the next paragraph we will go deeper into the theory of New Institutional Economics (NIE) that explains, why, in our modern times, we need so many intermediaries, between demand and supply.

Who produces trust?

You could ask to what extent the blockchain software development process is sensitive to the perspective of the people who design and develop the protocol? We think that certainly on the design side, the designer is driven not only by a certain perspective on humans, society and societal problems, but also by the way communication is organized or management is done, or how our socio-economic order is working (or not working). It is probably no coincidence that the whitepaper of Satoshi Nakamoto was uploaded a few weeks after the fall of Lehman Brothers in 2008. A designer designs from certain values, perspectives, principles and goals, all part of the ten core connections of Blockchain Organizing (see also figure 2.6).

Software development plays an important role within digital transformation. We should not underestimate this role, nor overestimate it. Developers are an important part of the trust machine but not the whole trust machine. Before a programmer gets down to work, an IT architect has to develop a framework, a construction plan. And before an IT-architect can start, an organizational designer has to develop an organization model and before that people, groups and societies have to find a common ground to start from. First you have to identify the problem you want to solve, then you can choose the organization model and then the tool.

You could ask the question here: To what extent does the person who designs and programs blockchain software believe in the best aligned way to organize trusts him- or her selves? Only the tool doesn't change our behavior, a fool with a tool is still a fool you could say. It is also important to have a dialogue with designers and programmers. To start and develop a blockchain community you also need to think and talk about the ten core connections (see also figure 2.6). Besides this, an advantage of Blockchain Organizing is that most blockchain software is open-source, transparent and verifiable and is often scrutinized by dozens, hundreds or sometimes thousands of independent programmers for potential back doors, bugs and other imperfections. It is therefore difficult to build something that others cannot check and control. This characteristic of a blockchain community is also an important basis of trust.

1.2 Is it time for a paradigm shift?

One of the key questions in the philosophy is: Why is there something instead of nothing? Why is there management and why is there no management? Management is there because more than hundred years ago farmers, craftsmen and housewives started to work in the factory straight from home. These 'fickle' people had to fit within the rhythm of the assembly line. By inventing management, time measurements, labor analysis, rewarding and punishing, this became possible. Management got a function in the whole process. It ensured that people could be connected to tasks in a way that increased productivity and therefore welfare. Since then the world has changed but the way that we manage people and resources has not really changed during that time.

Connecting humans and resources to a task entails, with the current way of managing, the deployment of a large number of intermediaries and therefore transaction costs. In addition, the complexity and bureaucracy are increased unnecessarily. An important advantage of Blockchain Organizing is that this concept makes it possible to program trust and the execution of transactions in such a way that you do not have to trust anyone, 'only' the entire network and the protocol. Therefore, more and more institutions, which are now involved in reducing insecurity, will become redundant. And this also reduces the complexity. Because most institutions in the core are a set of instructions (technical script), we can program these institutions

rules as it were. Management will be divided into a social and technical script and the technical script can be programmed and executed autonomously. To better understand the influence of institutions on reducing uncertainty, we go briefly back in time and we look at a new trend emerging within economics.

New institutional economics

For a long time, economists assume that markets are perfect and that man is a rational acting being, who is fully informed and has absolute knowledge of the future. Within this context an ‘invisible hand’ ensures the establishment of a price. Based on these assumptions and under perfect competition, the coordinating capacity or market mechanism creates an optimal allocation of the production resources. Few institutions are required to realize this: established and enforceable property rights and a perfectly operating market mechanism, for example to prevent fraud, will suffice¹². We know that these assumptions, especially in the present time, are not correct.

However, many classical economists have followed the aforementioned theory for a long time. This theory is not complete and does not correspond with reality. For these economists the functioning of markets based on the argument of Adam Smith that economic actors who pursue their own interests and thus more or less automatically serve the general interest, is leading. The above classic image of reality is not correct. Directors, who award themselves bonuses, do not do this to serve the general interest. We know that actors in a market are not fully informed and sometimes even misinformed. Think of usury insurance policies, the influence of fake news and propaganda on elections. With elections there is also a market between supply (voting right) and demand (who will you vote for). This market, and perhaps especially this market, is also influenced by information differences.

In a market for example, a seller, knows more than a buyer or vice versa. Sometimes people make irrational decisions and markets need rules to function¹³. There are certain power and dependency relationships between market actors. To reduce insecurities in non-perfect markets, we innovate all kinds of tools that help us execute a transaction. These tools, as indicated before, are also called agents or institutions in the organization and economic theory. An important perspective, you could also call it a field of expertise within economics, that deals with these tools and the questions that we can solve with these, is called ‘New Institutional Economics (NIE)’.

Institutional economists expand existing economic insights with social and legal standards and rules. In general we call this expansion of tools: institutions. These are required to execute (market) transactions. A market is a form of institution (organized capacity), and stands in relation to seller, buyer and other institutions. Many of

¹² These elements can also be seen in ‘the blockchain protocol’; proving that you are the owner and ensuring that you can only spend a value once.

¹³ For example a competition regulator: a competition agency as a government agency. In the European Economic Area these authorities are coordinated by the ECA: European Competition Regulator.

these standards and rules are institutionalized and organized through institutes (organizations). In the Netherlands, there is for example an Authority Consumer and Market (ACM) that promotes or indeed limits opportunities and choices for buyers (including consumers) and sellers (including companies). The authority creates opportunities by fighting unfair competition and making it easier for sellers to enter an existing market.

The origin of NIE lies within two articles of Noble prize winner Ronald Coase: ‘The Nature of The Firm’ (1937), and ‘The Problem of Social Costs’ (1960). You can read more about Coase¹⁴ and his transaction cost theory in book6. In addition to Coase, the following people and Noble prize winners have contributed to this field of expertise: Oliver Williamson, Elinor Ostrom, Douglass North¹⁵, and Herbert Simon¹⁶. You can read more about this in the Weconomics theory book.

Blockchain Organizing and NIE

As seen before, high transaction costs are a main reason for insufficient productivity growth. After all, you have to share the same end result with more parties. People believe that we become more productive because specialists have started working more efficiently. But these people have ‘forgotten’ that the number of intermediaries, the coordination or the managing of specialists, involves more costs. As seen in paragraph 1.1, the number of managers and staff positions only increases, despite all kinds of improvement programs. This is because these programs, and therefore many tools, regard the company as analysis unit and not organizing capacity. You can make a company more productive but that does not mean that the entire chain also becomes more productive. More specialization, without appropriate coordination, makes the problem worse. That is the paradox of the insecurity reduction (see also figure 1.2).

The problem must be tackled at the core with a broader view, especially when you start designing blockchain organizations. Institutional economists adopt a broader view, towards more organizational aspects than economists. These aspects¹⁷ are also discussed in the Weconomics analysis book and can be summarized as follows:

- Psychological (people's perspective)
- Economic (broad perspective)
- Social (including demographic aspect)
- Technical (including innovation aspect)
- Ecological (including externalities)

¹⁴ Coase, Williamson and North are also at the forefront of International Society for the New Institutional Economics (ISNIE). The first chairman was Ronald Coase (1996). Now it is called: Society for Institutional & Organizational Economics (SIOE). See also the website: sioe.org.

¹⁵ See for example: ‘Institutions, Institutional Change and Economic Performance’, Cambridge University Press (1990).

¹⁶ See for example: ‘Organisations and Markets’, Journal of Economic Perspectives, vol. 5, no. 2 (Simon, Herbert A. 1991).

¹⁷ Also called the ‘PESTEP’-perspective.

▪ Politics (including democratization)

When innovating and applying a new management concept such as Blockchain Organizing, it is important that the natural inertia of people is taken into account. People cannot be forced to change, however people do change. People have a certain adaptability, especially when it comes to a new technology that threatens their job. Due to this inertia, organizations change slowly and an economy and society will require time to adjust itself to the new and often better ways of organizing. It requires time and attention before people dare to let go of existing self-evidences, conditions and institutions, even when they already know that the existing order does not work correctly. The new world that is being proposed is initially still abstract for many. They cannot imagine enough advantages and practical applications yet to dare to let go of the existing ones. This definitely also applies for Blockchain Organizing, which initially comes across as abstract, technical and complex.

Then the question quickly arises: how can a complex technique reduce the complexity? Initially, many will mainly feel the negative side (way of fear) of the innovation and not yet the positive one (towards happiness). It is easier to see what we can lose to robots than what we can gain for ourselves.

During the transition it is definitely important to clearly communicate that organizations can also use organizational technology to finish work quicker. At this moment organizations will probably respond to insecurity and complexity in two ways:

1. Reducing the insecurity and complexity by using more existing resources; for example, more management, staff, IT, structure and procedures.
2. Innovating in new technology and management concepts such as Blockchain Organizing. It is important that organizations are aware that innovation does not happen automatically and that it is necessary to invest time and money. Organizations will have to learn before they know what works and what does not. When you are too early, this investment can have negative results, but also when you are too late. The right timing is just as important as the innovation itself.

These two ways are reflected in our approach, in our consultancy practice and blockchain projects support: both solving short time problems as well as tackling long term challenges. You will find more information about setting up your own blockchain projects in chapter 5. In any case, the insights of NIE will help with the design and further development of blockchain organizations. But we are not there yet with just a theoretical basis. We will also have to change our way of thinking and designing. Concepts such as design thinking help to apply Blockchain by Design.

1.3 Blockchain by Design

Apple founder Steve Jobs once said: *“That's not what we think design is. It's not just what it looks like and feels like. Design is how it works.”*¹⁸ According to Jobs, creativity and imagination are important conditions for design thinking. When you ask creative people how they did something, they often feel guilty. They have done nothing, they only saw something new. But you can only see something ‘new’ if you first can let go of the ‘old thinking and acting’. Creative thinkers and dissenters must compete with the old establishment. According to the existing power, and the experts they consult, old is probably better, because it has proven itself. New is often abstract and insecure. Therefore, dissenters are often amateurs in the eyes of experts: they do not understand. According to the famous expert, the world is much more complex than the newcomer imagines. However, they forget to realize that the world is not complex but that it is made complex, that is a totally different thing.

Complexity, cynicism and work pressure are increasing

A recent study¹⁹ of the HIVA²⁰ shows that one in three surveyed employees ‘always’ or ‘often’ suffers from stress at work. This number is rising and is mainly due to the turbulent environment and complexity of their work. In addition, the number of care tasks at home is also increasing. Especially in combination with aging, this can become a fundamental problem for employment organizations. The number of sick reports and burnouts will probably increase further, and people become less productive. Many burnouts are caused by high work pressure in combination with cynicism: *We ‘have’ to work hard but we no longer know why.*

People often have to work overtime and on top of this we have to do tasks, such as filling in forms and doing the administration, which seem useless to us and are often about organizing trust. The Belgian minister of employment wants to counter this problem by offering employees more autonomy. However, autonomy might slightly reduce the stress but the complexity of and the cynicism about the work remains. After all, this is usually an identity issue and design problem.

The cynicism and the complexity of work will remain, unless we fundamentally innovate our way of organizing. If we do nothing, the complexity, as a result of the trust paradox, will continue to increase and we still do not give another meaning to work. To reduce the insecurity within the existing establishment, existing institutions are expanded or similar institutions are added, which makes the situation even more complex. The HIVA research shows that the complexity of work has increased by twelve percent between 1995 and 2015. The rationality, with which we try to reduce insecurity, starts to reach a level of irrationality and sometimes absurdity and stupidity. We solve complexity with even more complexity. The privacy problem is solved

¹⁸ See article in The New York Times Magazine by Rob Walker: ‘The Guts of a New Machine’ (30-11-2003).

¹⁹ See <https://lirias.kuleuven.be/handle/123456789/585820>

²⁰ HIVA: Research institute for Employment and society of the University of Leuven Belgium.

with even more IT, laws and procedures, which makes the organizations even more complex and increases the risk of data leaks. We can no longer deal with this with the current way of thinking and the current organizational design. We will have to go back to the drawing board and think about a new organizational design. Thereby we must see the role of designer much broader. It is not about product design but about organizational design. We need architects of the future. According to the inventor Richard Buckminster Fuller, a design is a cross between an artist, an inventor, an engineer, and objective economist and evolutionary strategist. For us, a designer must mainly have a wide (see also the aforementioned PESTEP perspective) and deep (historical) perspective and knowledge of organizational principles and theories.

So, design is not just about how something looks like, but mainly about how it works. Many management concepts no longer work. And if it does not work in the present, we must really worry about the future. Many managers and directors still think that progress can be made by implementing small adjustments in the existing system. For example: sending people on a training course, another IT system, strengthening the balance sheet of banks or replacing the management. But all these small adjustments have become a part, whether or not consciously, of a policy of active inertia: we move a lot, but we make little progress and it seems that we are okay with this. People become blasé and cynical. Many traditional managers and consultants still genuinely believe that they are doing it right. They think and act from the belief system that what they have learned is correct and that experience will show the way.

They do not know another way because they think and decide within the framework of their nurture, education and experiences. But the days of adjustments to the existing system are over. Usually we try to solve the problems with the same system and perspective of mind with which they occurred. The existing world is mainly engaged with solving short term and practical problems and not with applying fundamentally new management principles. We improve from K to L, but the total (A up to L) becomes worse. If you buy a painting, you probably like it otherwise you would not have bought it. But when you subsequently hang this new painting in an overcrowded living room, it does not necessarily make the room any prettier. This is also the case when applying management concepts within existing systems. The management concept itself is not the problem, but the system in which it is applied.

We must prepare ourselves for a new reality, a new system and that starts with awareness and a new way of thinking. Most innovations focus on even more jobs, the improvement of internal processes or the improvement of the competitiveness. But this will fundamentally change in the coming years, in the coming decades. They are outdated concepts that insufficiently help us solve the social organizational problems. We will start to cooperate more and better, not out of weakness, but out of conviction that it is necessary. This gives hope and according to the author Rebecca Solnit, hope can be found in the assumption that we do not know what will happen and that this insecurity offers us the space to do something.

Back to the core of organizing

Many people believe that we live in a complex world, but the world is not complex, people make it complex, that is a different matter. Plants and animals do not suffer from complexity. A forest simply exists because leaves seek sunlight and roots seek water. People and organizations unnecessarily make the organization of work complex with positions, departments and companies. Subsequently these intermediaries are not correctly coordinated and as a result the world becomes even more complex. In a complex world distrust increases. So, organizing trust creates more distrust. The only way to escape this *trust paradox* is to go back to the core of organizing. When there is one seller and one buyer, we do not need to coordinate a lot. For a one-on-one coaching talk we mainly need a diary. But we all know that more specialists are often required to achieve a goal. These must be aligned, these must be managed. Organizing work in all cases where there is no 1:1 relationship, contains the following three components: specializing, coordinating and finalizing, see the schedule below:

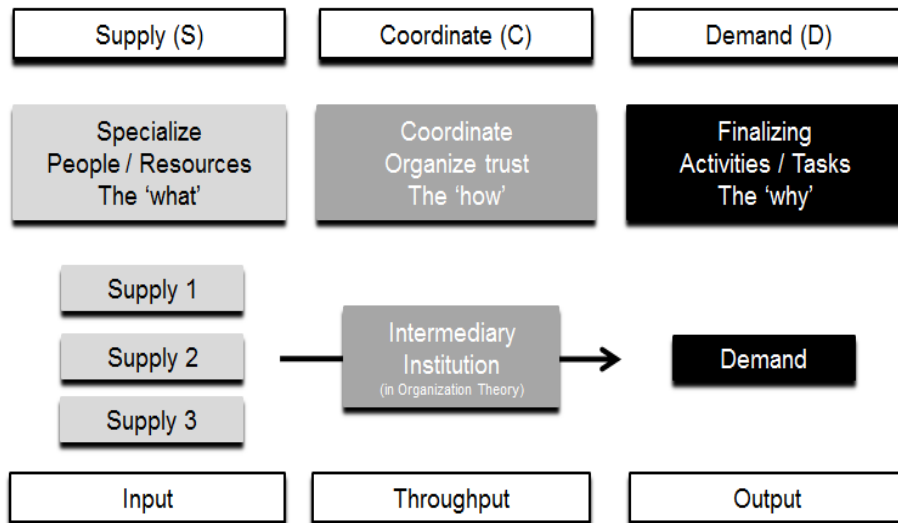


Figure 1.3: Core of organizing: supply, demand and coordination between them ($S \rightarrow C \rightarrow D$)

To understand the core of organizing and therefore the core of management, we have to go back to the question why we organize work. From the first moment that humans started to organize work, we focused on improving our standard of living. The predecessor of the Homo Sapiens, the Homo Heidelbergensis, 'discovered' about three hundred thousand years ago that giving up a piece of autonomy results in an increasing revenue per person: smart cooperation provides more food. Even animals that are much bigger or faster can be caught with the help of specialization, coordination and finalization. These three components have always been a part of organizing work. You can coordinate with different tools, such as: management, hierarchies and organizational culture programs. These will now be joined by a new tool: Blockchain Organizing. With the use of Blockchain Organizing the

coordination is programmed, so there is very little resistance and therefore the transaction costs will be lower.

From company to network organization

In order to apply Blockchain Organizing correctly, it is important to first *transform to a network organization*. It is important to note that your position or organization is NOT the center of attention, with the stakeholders revolving around it. Neither is it the center of all the business being carried out. To apply Blockchain Organizing correctly you first have to give up some autonomy, you will have to treat your organization as a hub in a network, a part of a business ecosystem together with your stakeholders such as: customers, suppliers, employees and governments. The transformation takes place step by step, comparable to the development of the assembly-line or internet for example.

This book mainly focuses on the organizational embedding of blockchain technology. In the future leadership will increasingly have to deal with the question: What is the best way for my organization to start with Blockchain Organizing? In the nineties, they were asked the question: Should we switch to the internet or not? Now managers are faced with the question: Should we switch to a blockchain or not? There will be supporters and opponents. According to John Hagel²¹, co-chairman of Deloitte Centre for the Edge, large-scale interventions, including the switch to blockchain technology, will involve two groups: *ambassadors* and *resistors of change*. The trick is to find out who belongs to which group and subsequently, before actually implementing the *large-scale intervention*, strengthen the ambassadors and weaken the resistors.

Practice above principles

The word ‘innovation’ is often associated with solving practical problems, creating product innovations or increasing the turnover. The relation between the system with which we organize supply and demand, and the management principles on which a system is based, are often not considered. But choosing practical solutions over principles for too long will ensure that you will have to go further and further back to take the right path for the future. In the movie ‘Doubt’, Meryl Streep’s character said it as follows: *“Every easy choice today will have its consequence tomorrow.”* Changing the DNA of your organization takes a lot of time and the longer you wait to adapt to new technologies and management concepts, the harder it will be to survive in the longer term.

Examples whereby practice was preferred over principles are IT debacles on government level or the choice of Eastman Kodak to postpone the digital camera for ‘a while’. Usually debacles are not caused by technology, but by the way of organizing and ignoring the question: which problem do we actually solve with the offered technology? IT debacles are mainly caused by:

- the idea that you can manage complexity (automating);

²¹ American author and consultant, who works for the Deloitte ‘Centre of Edge’.

- digitizing analogue processes;
- letting your organization drive/lead through technology (supplier);
- changing requirements (legislation);
- preferring practice over principles;
- implementing changing needs of employees and customers on a random basis.

Changing needs are also often caused by the automation paradox. Due to automation, people have less to do while they still have to be present till five o'clock. In the surplus time they create new needs, they start sending e-mails or writing memos with which they 'bother' other people. So more automation creates not less, but more work. Many traditional innovations focus on improving the internal processes, reducing costs or differentiation in the market. Usually innovation takes place per company and within a company per department or position. For example, IT and legal affairs will receive separate assignments to ensure that the organization complies with the new European privacy law. But the problems that have led to the new privacy law are much more fundamental. It is mainly the combination of a new reality (digital society) with an old management model (the firm as superior form to organize supply and demand with its own websites and databases), that leads to privacy problems.

IT, legal affairs and other positions will have to cooperate smartly to realize a new design, based on the principle that existing firms or positions within firms don't have to continue to exist. We have looked into this real case and used it to describe how you could setup a blockchain project (see Chapter 5). With most innovations the company as an organizational format or form is not quickly questioned. The form is not innovated, there is only innovation within the form. Because there are often many similarities between organizational problems, a number of organizational principles can be distilled and are further explained in paragraph 2.5. Principles do not change as often as practical situations. Therefore, when designing an organization, it is preferred to take principles as a starting point rather than practical problems. Placing current practices over principles as a guideline for change, will make it harder to adapt to a new fast-changing environment.

Blockchain innovation for existing organizations

Many policy makers, strategists and existing organizations will struggle to make a paradigm shift and to understand Blockchain Organizing in its purest form. They will soon find the new organizational technology and organizational form too abstract. This is not because new forms are complex and hard to understand but because policy makers have been conditioned. Many people confuse abstract with being different from what they are used to or learned. But abstracting literally means: omit what is not relevant. In addition, politics, also within business, plays an important role in the resistance of the old establishment. The power of the establishment depends a lot on the survival of old systems in which they have invested. They have established a network and have entered long term commitments with employees, customers and suppliers. This offers economies of scale but also has the disadvantage that traditional organizations cannot adapt as quickly to new technologies as start-

ups can. Start-ups have the disadvantage that they hardly have a network yet and that they must establish trust. Most start-ups also have limited financial resources en limited learning experience in innovation processes. Shell, for example, will have difficulty switching to renewable energy, but start-ups have little resources to invest in a wind farm.

Some companies want to start a new business within the existing company (structure). This idea of creating a 'new' company within an existing one has dramatically failed within the airline industry during the past thirty years when the low-cost carriers were created. This was also a major paradigm shift and large scale intervention. The real 'innovators' were companies like EasyJet, Ryanair and JetBlue. They are not a spin-off from the traditional airlines like Air France/KLM or Delta Airlines. It turned out that separating the 'new' from the 'old' could not be managed. To break this dilemma, we have to switch to a new form of innovation. Herein there must be room for a better cooperation between existing organizations and start-ups.

The government can act as a catalyst and launching customer. Many start-ups, also in the software sector, have been started by people who wanted to improve the world. But rather quickly they needed more money to grow and the perspective changed from 'don't be evil' to 'serve the shareholder'. Just as what happened when internet developed, many 'dissenters' spearheading the development of blockchain, are quickly becoming 'corporate executives in suits'. A hybrid form would be ideal whereby companies, in addition to their existing organization, start a new network organization (Greenfield Organizing). This new organization starts from scratch, with design thinking and without the ballast of old systems, interests or obligations. After the new organization is operational, they can transfer processes from the old organization step by step. This way, existing companies can renew. They can learn against acceptable failure costs.

With regard to blockchain innovations, there is still a lot to learn. Innovating in Blockchain Organizing means that you must discover several times what does not work before you know what does. The iPhone, as we know it now, was not sent from the drawing board to the factory in one go. We can no longer solve management problems insisting on the old frame of mind. We will have to switch to a more fundamental innovation that can no longer be realized by continuously improving the existing ones. The light bulb, for example, was not 'created' by continuously improving the candle. The light bulb is the result of design thinking: restarting from scratch.

Innovation by Design

Blockchain by Design is a form of Innovation by Design, with design thinking as a basis. When you are going to change, you usually do not start designing from scratch, but you adjust an existing design. When you are going to renew, you usually start from the foundations and drawing table with a new design. Blockchain Organizing is building something new. We design new network organizations or business ecosystems and transfer existing processes to this new construction. Improving existing

processes is mainly useful when, besides efficiency improvement, the focus is on ‘preparing’ processes so they can later be transferred to a new blockchain organization. With the transition from company to network organization you must take into account the old interest and forces or, as formulated by Thomas Friedman: *“If horses could have voted, there never would have been cars”*. You can read more about the transition from existing organizations to new blockchain organizations in Chapter 5.

In the next paragraphs we will further discuss what Blockchain Organizing actually is and what it can or cannot do for your organization. The organizational aspect of Blockchain Organizing is further explained in Chapter 2, the technical aspect and Chapter 3. The problems that can be solved with Blockchain Organizing will be further explained in paragraph 4.1.

1.4 Some concepts

Blockchain Organizing could be placed in a logical order between Weconomics (as a form of New Organizing) and blockchain technology. For a quicker insight it is useful to explain a few concepts in more detail. Because the topicality of blockchain technology can overtake us quickly, we will limit ourselves to a more general description of a concept. For an up to date overview we refer you to our Knowledge Centre Blockchain Organizing²².

See Figure 1.4, The Blockchain iceberg. There is a logical order between the concepts shown. Only the tip of the iceberg is visible. What we see is the daily application and specification or operation of the technology (like Bitcoin). The further we go below the surface, the more abstract and general it becomes. Bitcoin is part of cryptocurrency, cryptocurrency is part of the blockchain etcetera.



Figure 1.4: ‘The blockchain iceberg’

Bitcoin

²² See also the websites: weconet.org/block

Bitcoin is a digital currency and can be seen as the first, long-term, workable application of blockchain technology. The visible development of Bitcoin started with the publication of a whitepaper²³ by Satoshi Nakamoto²⁴. A blockchain is a distributed network of nodes (participants). The challenge with distributed system is the sequence of transactions, because there is not a central unit that determines this. Within the Bitcoin protocol, a block is a collection of transactions that, logically, took place at the same moment. A block is added to the chain of existing blocks (blockchain) by miners²⁵ every ten minutes.

Unconfirmed/unordered transactions can be place in a block by a node

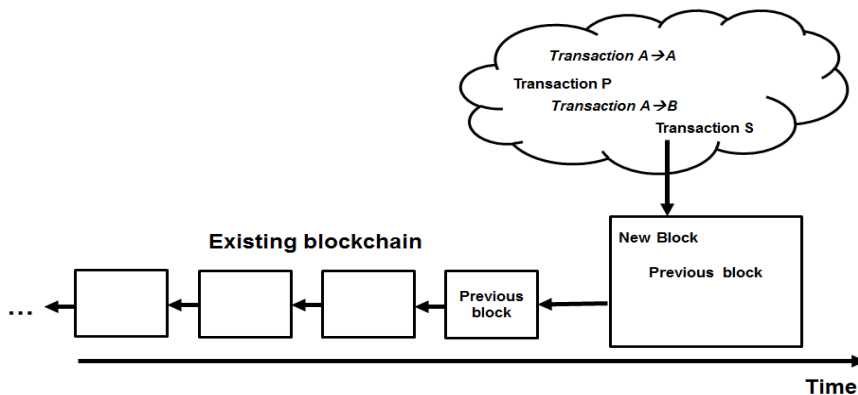


Figure 1.5: A blockchain is a collection of blocks and a block is a collection of transactions with the same timestamp. It is impossible that transaction $A \rightarrow A$ and $A \rightarrow B$ will be sequenced in same block because they use the same input-reference (no double spending).

Within the Bitcoin protocol, a transaction included in a block can no longer be changed²⁶. A transaction is transferring a value (bitcoin) from seller to buyer. Hereby the protocol ensures that the seller actually has this value and can only spend it once. Transactions are recorded in a decentralized accounting system and are distributed across the participating nodes²⁷ of the network.

Cryptocurrency

Currency literally means medium of exchange. Cryptography is a field of expertise that has been conducting research into the encryption of values and messages with the use of processors for over forty years, in order to prevent unauthorized third

²³ Satoshi Nakamoto: 'Bitcoin: A Peer-to-Peer Electronic Cash System'.

²⁴ A pseudonym for a hitherto unknown person or group of persons.

²⁵ A miner can add a block to the blockchain by being 'the first person' to solve a difficult puzzle. When a miner adds a block, he receives bitcoins.

²⁶ In 2016 two components related to Bitcoin and Ethereum were hacked. After this, a discussion arose in both communities on how to solve this. Ethereum (The DAO), has chosen a hard-fork (a split), so the (future) history can be changed.

²⁷ A node within a (computer) network is a computer or other device that is connected to a certain network. The node can have an own identity (for instance an IP address). The node can run software and therefore contribute to the operation of the network.

parties from reading a message. For over twenty years cryptocurrency has been using cryptography when moving money from seller to buyer. Cryptocurrency can be regarded as programmable money. Hereby the seller, for instance a healthcare insurer, can characterize a cash payment in such a way that this money can only be spend on a healthcare provider.

Blockchain

A blockchain is a distributed database that records the continuously growing and non-mutable 'administration system' of transactions, that are arranged in blocks and whereby the network will find consensus on a shared reality. This creates a way to organize trust without the complexity having to increase. A blockchain facilitates the secure execution of digital transactions. The underlying technology to make this possible can be called blockchain technology. The most famous and longest running application is Bitcoin. When we use blockchain technology with a fundamentally new organizational model, a disruptive management concept is created. For the first time in a hundred years a management technology is actually reinvented.

A blockchain is created because a new block is constantly added to an existing chain. Hereby a reference is always made to the previous block. A blockchain is safe and unchangeable 'by design'. A block is a collection of transactions that have taken place 'at the same time'. Because transactions distribute themselves across all nodes of the network, not every node receives a certain transaction at the same time. To solve that problem, blocks are mined and added to the chain. Once a transaction is included in the chain (with a previous and next block) it can no longer leave the chain. You cannot mutate a blockchain²⁸. The integrity of a blockchain is only guaranteed when all users check the transactions on this chain and at least half of the users are honest (not trying to spend a value, an input, more than once).

Blockchain technology

Blockchain technology is the underlying technology of Bitcoin. Blockchain technology is used within Distributed Ledger Technology (DLT)²⁹. This is a technology to maintain a distributed administration system. DLT is again part of a wider field of expertise: organizational technology. This is a technology to organize demand and supply in a smarter way by reducing insecurity without increasing the complexity. Insecurity is reduced by validating transactions, by recording these in a non-mutable way and using this information for a next transaction and to create a shared reality. Blockchain technology is a technology to check, validate and add transactions to the administration system that is on every full node of the network. Each computer that is connected to the Bitcoin network is a node. A full node is a node that fully executes all the rules of the protocol. Full nodes download all transactions made, in other

²⁸ That is not completely true. A hard-fork is an adjustment in the Bitcoin protocol that makes it possible to validate previous invalid blocks or transactions. Therefore, each node must be upgraded. A second chain is created whereby nodes that have not downloaded an upgrade are not included in the new chain.

²⁹ An example of DLT that is not a blockchain is Hashgraph.

words, the complete administration system. An administration system is literally keeping track of:

- how much value you have now
- when sending: what is deducted and what are you left with afterwards
- when receiving: what is added and how much have you got afterwards

Blockchain Organizing

Blockchain Organizing is a management concept that primarily takes an organization problem and design as the starting point and not the technology. It answers the question how you can organize demand and supply as efficiently as possible. Blockchain Organizing is a new management concept whereby your design is not based on existing ‘buildings’, for example companies, traditional markets and central governments, but on the organizational capacity within the smallest building blocks: man and resource. Smallest building blocks have values. A value must exist and you must be able to record its characteristics (facts, attributes). A value can transfer (flow) from seller to buyer. As soon as a transaction has taken place and is validated by the network, it is irreversibly recorded in a chain (the blockchain). This creates a single source of reality whereby you do not have to trust anyone, only the entire network including the protocol.

As this book is based on organizing work, we mainly look at what Blockchain Organizing means for the domain of HR, management and employment organizations. Blockchain Organizing used in employment organizations is based on task performance whereby people and resources are connected to a demand via coordinating capacity. For example, currently it is widely assumed that managers, development programs and hierarchies as coordinating capacities are essential, but that does not have to be the case. Traditional organizing is especially suitable for the production and consumption of physical goods via traditional markets and companies in a stable environment. Blockchain Organizing on the other hand is especially suitable for the production, storage, transport, repacking and consumption of digital values³⁰. It makes information and knowledge workers more productive. Or as Peter Drucker describes it in his book *de ‘Post-Capitalist Society’* (1993): *“It is time to develop a model that supports the growth of productivity of knowledge workers.”*

Weconomics

Weconomics is a socio-economic mental legacy, a body of thoughts, accommodated in the Weconomics Foundation. Weconomics facilitates the transition to a more sustainable welfare. It is a network of pioneers who independently and jointly contribute to the sustainability of our welfare. The core of this is the development and deployment of new management concepts based on organizational technology such as blockchain. The Foundation is a commonality and helps start-ups, entrepreneurs,

³⁰ Digital values are digital products and services such as an e-book or paid newspaper article via internet, but also information about products and services (for instance price, delivery time, suppliers).

intrapreneurs and organizations, from idea to implementation. Weconomics is a fundamentally new organization model, an infrastructure and a validated transition program.

The above are the most important concepts needed for the purpose of this book. Next we are going to answer the question: What is Blockchain Organizing?

1.5 What is Blockchain Organizing?

Blockchain Organizing is a new management concept. It is a combination of blockchain technology and the Weconomics organizational model³¹. Blockchain Organizing makes it possible to organize demand and supply against the lowest transaction costs. It is a concept whereby a design is not based on the company or traditional market as coordinating capacity, but on the smallest building blocks (human and resource, demand and supply), and the basics organizing principles. A blockchain protocol is the coordinating capacity between supply and demand. It is based on task performance whereby humans and resources (Supply) are connected to a finalization (Demand) via coordinating capacity (Coordination). In this book we also present a value flow (see also figure 1.3) as:

$$\text{Supply (S)} \rightarrow \text{Coordinate (C)} \rightarrow \text{Demand (D)}$$

Conventional organizing is especially suitable for the production and consumption of physical goods via traditional firms in a stable environment. As we saw earlier: Blockchain Organizing is especially suitable for the production, storage, transport, repacking and consumption of digital values. From an organizational point of view, blockchain can be seen as an additional layer to the internet. Internet makes it possible to share information without almost any friction. Blockchain makes it possible to execute transactions without any friction. Therefore, we also call this a value, trust- or transaction-network.

We talk about THE blockchain and THE internet while these actually do not exist. Internet is a network of connected nodes with an IP address that can exchange data (values) with each other via a protocol (TCP/IP) and connections. Blockchain technology uses internet and is therefore also a distributed network of nodes. Non-fraudulent nodes benefit if the 'accounting' system is correct. This means that the nodes (whether or not against remuneration) want to cooperate when checking and validating transactions.

The principle of the Bitcoin protocol relies on the fact that more than half of the network computer power is required in order to commit fraud. Within Bitcoin a node can include both wallet software as well as miner software. Special mathematical

³¹ For a detailed description of the Weconomics organizational model, see chapter 5 of the Weconomics theory book.

functions protect every part of the system including sending and receiving bitcoins. Within Bitcoin a blockchain is actually and literally a chain of blocks, whereby a block is a collection of bitcoin payments that logically take place at 'the same moment'. Meanwhile there is not one blockchain or chain, but there are several, whether or not private, also called permissioned blockchains, public, also called permissionless, or more hybrid forms.

We know the blockchain concept especially in relation to Bitcoin. Blockchain technology became best known after the publication of the whitepaper by Satoshi Nakamoto in 2008. Just as internet and the World Wide Web did not just fall from the sky³², Bitcoin and blockchain technology were created by further development of existing insights. Nakamoto's work continues to develop based on the knowledge of others. Bitcoin is mainly a form of cryptocurrency³³. A format of value that no longer includes cash or giro money but software code. For instance, it allows recording of rules that housing benefits may only be spent on rent.

So, it is possible to program money. In addition, Bitcoin does not have any central authority, like large national banks, but it is a form of distributed authority with a single source or reality. The essence of Bitcoin, blockchain technology as well as Blockchain Organizing is the recording (profiling) of a value with the seller, the selection of the value through the buyer and the transaction between seller and buyer. Value flows from sender to receiver and the protocol is the coordinating capacity that reduces insecurity and organizes trust. Specific algorithms and encryptions are used to make the connection meaningful, secure and reliable.

Blockchain Organizing integrates ideas and concepts like community, platform and sharing economy, peering, content sharing, digitization, democratization of internet, crown funding, crowd sourcing, cloud computing, social media and open innovation. Blockchain Organizing provides a new perspective on positions and concepts such as freedom, shared reality, fact and fiction, non-commitment, avoidance and evasion, decentralization, accountability, trust, value, trade and economy. It ensures that you can smartly cooperate with both your colleagues as well as your environment (portfolio workers³⁴, customers, suppliers and talents) in the same way. It provides an effective policy with regards to the organization of fixed, flex and more hybrid positions or projects. This would be no problem within Blockchain Organizing because it works with type connections (loose, fixed, definite period, indefinite period), smart contracts and prior consensus. Blockchain Organizing is especially suitable for digital

³² See also the Weconomics practice book, paragraph 3.4: 'Weconomics and the World Wide Web'.

³³ The word 'currency' is derived from the Middle English 'curraunt' which means 'in circulation'. Currency is a medium of exchange and therefore not just money. In general money has the function of medium of exchange, calculation unit and is a way to store value.

³⁴ Portfolio worker: collective name for freelancers, flex workers, self-employed people, hybrid workers, etcetera.

production and consumption and will make information- and knowledge workers ³⁵ much more productive.

For managers it is important to understand a number of relevant organizational principles of blockchain technology, to further explore these and consider what you can do with it. What are the principles of blockchain technology and how can you implement those? The question is how blockchain technology can contribute to organizing trust and solving paradoxes like the trust-, automation- and productivity paradox. Some key organizational principles of Blockchain Organizing are:

1. It is based on values (smallest building block) and transactions (connections between blocks).
2. *Panta Rei*: value flows and you cannot make the same transaction twice.
3. Seller must prove that he is the owner of a value.
4. It is impossible to spend a value more than once.
5. It is a neutral, distributed network with peer-to-peer connections.
6. It works with a community governance model.
7. Open source as much as possible, without exclusion and without (country or company) boundaries.
8. There are less traditional trusted third parties in order to reduce insecurity.
9. Transactions can be performed through an appropriate encryption, with an acceptable security level, the transport from supply to demand is secure.
10. The person who contributes to the community also benefits, for example with of tokens. The value that is registered (for example number of bitcoins), must be the same as, or without fiat/non-transparent medium of exchange, be exchangeable with, the value that the miners receive for their validation work. If that is not the case, a central position is needed that issues these values. This party can exercise control by rewarding selectively. The reward mechanism must be included in the blockchain protocol.

The next question to be addressed is: how can you apply Blockchain Organizing effectively within your organization?

1.6 Blockchain within your organization?

More and more organizations ask for useful applications of Blockchain Organizing within their own organization. But to put it bluntly: these are actually not available, unless you see internal transaction (interactions) as transactions, your organization as a collection of smallest building stones or unless you find consensus with your stakeholders. You can start Blockchain Organizing from your own organization, but you

³⁵ Information workers: collective name for people who are fully or partially involved in or associated with the processing of data into information, from information to knowledge and from knowledge to wisdom. Collective name for office workers, data processors, information processors, knowledge workers, service providers or service workers whereby secondary processes include the processing of data or information (planning, coordinating, communicating, sharing information, etcetera).

have to find other parties to cooperate with because normally within organizations we don't 'see' transactions. You don't bill a colleague for attending a meeting for example.

A blockchain within a company is like an assembly line within a craft. It is not just about technology, but mainly about the problem that you want to solve and the organizational model in which we use the technology. If you start with blockchain within your own organization there is a risk that it will lead a life of its own. Compare it with setting up an internal communication systems based on the internet protocol (TCP/IP) at the beginning of the nineties. After years of investments, organizations no longer fundamentally adjust these systems. The result is that we now share data, information and knowledge in an outdated way, by sending e-mails to each other and by coping data in our own systems.

Blockchain checklist

In general, there are no transactions within organizations in the traditional meaning of the word. Normally, no invoice is sent when you call a colleague for advice. This means that you have to look at possible applications within your business ecosystem. For instance with your employees (like employment contracts on a blockchain) and your suppliers (purchase orders and invoices on a blockchain). To test if Blockchain Organizing is useful for your organization you can ask a number of questions. These questions should primarily focus on the situation where you want to apply Blockchain Organizing within your own organization in relation to some suppliers.

Your 'own' blockchain could, for instance, be used for transactions with some of your suppliers. We advise not to start with core-suppliers or core-processes but HR-transaction for example. HR-instruments such as: courses or assessment products. We also advise to start with your suppliers and not your customers. Usually your customers will choose a solution themselves unless you consult with them about smart cooperating within ecosystems.

Let's assume that you want to set up a blockchain with some of your suppliers. You (always) will have to execute the following four basic processes that you require with value transactions:

1. profiling (what do your suppliers offer)
2. selecting (how can your organization select the offer of your suppliers)
3. purchasing (value flows from supplier to your organization)
4. paying (money flows from your organization to supplier)

To assess if it is useful to set up your own blockchain with a number of suppliers, there are a few questions to ask yourself. The more questions are answered with 'yes', the greater the chance that a blockchain can add something to your business:

1. We want to improve the productivity in the supply chain (aggregated productivity). First of all, we look at processes whereby we can cooperate with our suppliers. We look for the commonalities in our processes and solve the differences locally.

2. We do not know all our suppliers and we cannot trust them completely. Therefore, we need purchase conditions and contracts.
3. We must find consensus if and when a transaction has taken place.
4. We require a digital accounting system where both suppliers and our own employees must add, check and record transactions.
5. We need our employees and suppliers to check and validate the transactions of others and we are also prepared to 'pay' for this.
6. We do not need (full) control over the execution of transactions, the protocol with which we execute them or the accounting system in which we record and check them. If a participant in the ecosystem wants to adjust the software, other participants can mutually decide, for example by consensus or by majority.
7. If the software is not running correctly, an error emerges, the system fails or is hacked, we do not need to have full control over this and participants can decide how they will handle this.
8. We want to make certain transaction information publicly available to those with the right access rights.
9. There is sufficient knowledge and experience with employees/suppliers in terms of blockchain technology and its components such as: cryptography, database technology, network complexity, game theory, distributed networks and digital currencies. We might hire people for this, we know the blockchain market.

We think that it is unlikely that you will be able to answer 'yes' to most questions and this example is only about your connection with a number of your suppliers. To apply Blockchain Organizing in a useful way you have to give up some autonomy and to look at cases where you could apply blockchain technology with your stakeholders: your customers, suppliers, staff, governments and non-governmental organizations. Together they are a business ecosystem or shared information- and transaction network. Blockchain Organizing is much more useful for this. You start your ecosystem with 'growing' awareness, your starting point should be that you want to apply Blockchain Organizing on an equal basis and that you want to improve productivity not only for your own organization, but also for the total supply chain. Don't think that the world turns around you. In a blockchain community a buyer for example does not have more power than a supplier when stipulating how transactions must be executed.

A major problem in developing insights and answering questions related to blockchain is not so much the technology but the conditioning (actually framed idea) that we would require companies and governments to organize trust, work, commonalities and economies. As formulated by Noble prize winner Elinor Ostrom: *"There is no reason to believe that bureaucrats and politicians, no matter how well meaning, are better at solving problems than the people on the spot, who have the strongest incentive to get the solution right."*

Many organizations managers find it hard to imagine that 'their' superior management model (the firm with a hierarchy) can become redundant. They will have to let

go of the idea that they are the center of doing business. Just as the sun and stars do not revolve around the earth, neither does the world revolve around your organization or position. Many organizations and officials suffer from position centrism, in other words: making you position or organization the reference point for everyone else. Those days are over. It makes little sense to set up your own blockchain. A central blockchain is a ‘*contradictio in terminis*’. You are not going to innovate cars to subsequently have them pulled by horses.

Blockchain will only come alive if you let go of the current unit of analysis and organizational design and you go back to the smallest building blocks and organizational principles. If we are smart, we will change the organizing capacity at the same time. ‘The firm’ is only a form of organizing capacity. It is a form that we still have to deal with for a while, but that eventually will largely disappear because it is becoming the most expensive form of Organizing. We see the average ‘life expectancy’ of companies drop from 60 years in the 1950s to 20 years now. In addition, net jobs are almost only created in companies that are less than five years old. Organizational forms are already changing. “*The organization of the future might be ten million people who get together (online) for ten minutes, solve a problem and then go separate ways*”, as formulated by the management consultant Esko Kilpi

Transaction costs within companies?

Many managers still believe that the company is the superior form to organize supply and demand of work in a digital era. This image is probably based on the wrong assumption that there are no transaction costs within companies. But these costs are definitely there if we no longer take the company but man as the smallest unit of analysis and design. With an automation project you can take a department or company as design unit. Or you can take the smallest organizational building block: human and resources, as building block and build you ecosystem by connecting humans to resources and supply to demand. Within Blockchain Organizing man is the smallest building block and the world is the largest building. Everything in between moves by switching connections on and off.

To apply Blockchain Organizing in a useful way, we must look at the application of blockchain technology within business ecosystems. Blockchain Organizing makes more sense for shared information- and transaction networks than for individual organizations. The implementation of blockchain technology starts with companies that want to transform from a traditional company to a network organization. Managers can provide an important contribution to this transition, both in a positive and negative sense. Blockchain will bring little change if we do not change the management model simultaneously and focus ourselves on real social organizational problems.

Blockchain Organizing, and the principles on which it is based, still has a long way to go before it becomes self-evident for organizations. Many people still find it too abstract and too complicated, but that has more to do with the degree to which they

are conditioned than with the complexity of the concept. You can only oversee the whole picture by going back to the core. Something new starts with letting go of the old. Blockchain Organizing is actually very simple, because it is based on a limited number of organizational principles. These principles can be applied almost everywhere. The best chefs prepare food with principles and use recipes. Organizational science does not have a recipe for the solution of every local management issue, but it does have principles. With the knowledge and experience of the right organizational experts most organizational problems, also the major ones, can be solved.

This was an introduction in Blockchain Organizing. We now continue by taking a closer look at the organizational aspect of Blockchain Organizing. In chapter 3 the technical aspect is explained in more detail.

2. Blockchain as organizational concept

You might not realize it but you are constantly organizing. Even when you are asleep, your brain is trying to recover. It is organizing itself so it can perform again the following day. The word organizing is derived from the Greek word 'organon' that means tool or instrument. Historically, these were mainly physical tools, such as the plough. Organizing is applying or using tools to achieve a target. Apart from physical tools, 'organizing' nowadays mainly refers to applying knowledge and for example ideas, concepts and methods. The word 'organon' also means 'organ'. An organ is a subordinate group in a greater total. An organization is a tool that must have a position in the total picture, otherwise it should not be there.

A synonym for the word 'organizing' is 'completing' or 'managing'. This has to do with the purpose of organizing (finalizing). Organizing is a targeted activity that must lead somewhere. This meaning is more on the demand side. Another synonym for organizing is: 'providing'. This meaning is more on the supply side of organizing. This means that organizing is aimed at converting input into output, or as indicated before: aligning demand and supply. The capacity that does this with the least friction, or the lowest transaction costs, has the best chance to survive. Organizing is the total of specializing (supply), coordinating and finalizing (demand). Blockchain Organizing is a coordinating capacity with (in theory) lower transaction costs than traditional forms such as the company or the hierarchy.

The word 'organization' derived from organizing has two meanings: a 'fiscal-legal entity' and 'organizing capacity'. IBM for example is an organization and IBM has organizing capacity. Especially the second meaning is important for the purpose of this book. Your brain has a certain organizing capacity to repair cells. In this book we look at the emerging blockchain technology in the context of its organizing capacity. First at the demand side: how can this technology help us solve a number of organizational problems? Secondly at its 'coordinating capacity' between supply and demand and finally at the supply side: which techniques, specializations and concepts will help to solve problems. We mainly see Blockchain Organizing as an organizing capacity to help solve management problems and therefore social issues.

In these dynamic times, with major changes and social challenges, putting more focus on organizing capacity, and therefore organizational science as a field of expertise, is not an unnecessary luxury. During these stormy times, we can use some higher organizational science and 'organizational vitamins' in education, healthcare, safety, business and politics. Blockchain Organizing best finds its way in a new context, where old principles and traditional management concepts no longer work. We live in a time where data, information and knowledge have become the most important production factors. We now have a technology at our disposal to put people back at the center and to realize and share a better reality together. But still, despite the presence of better technology, we find it more and more difficult to organize trust and

for example make a distinction between facts and fiction. Plus, the power of large companies and central governments is increasing and it becomes more and more difficult to assure privacy.

We believe that a renewed vision of how we organize our information society is in order. We are mainly talking about organizing trust, work and economy as basis for our welfare. When we look more fundamentally at the underlying principles with which we organize our digital society, we see that most organizational forms still go back to the beginning of the twentieth century, for instance ‘the factory’. They are especially suitable for physical production in large quantities and a relatively stable local market. Traditionally we organize supply and demand with ‘the firm’ (or hierarchy), ‘the market’ and ‘the government’³⁶ as forms of coordinating capacity. Hereby the firm or government can decide to outsource a certain process via the market, but after sourcing out, the process is again performed via a firm. Often the firm is better scalable and more transparent, but less flexible while outsourcing via the market is flexible, but often less scalable and transparent. Firms and markets often work more efficiently than the government, but more often lead to inequality.

In the context of the twenty-first century we need more hybrid organizational forms that are scalable, flexible and transparent at the same time. We need to find forms that can better organize trust and that find a greater balance between competing and cooperating, between part and total. Forms with a new social economic order that reason from man needs and not from the supply side or existing business models. Forms that do not take permanent jobs, structures and business cultures as principle, but human demand and the pursuit of happiness and welfare.

Blockchain Organizing makes cooperation between firm, market and government much simpler because it reasons from the organizational problem that must be solved, from the smallest building blocks and organizational principles. Blockchain Organizing is an organizational model with a new unit of thinking. We work from a stratification (taxonomy) whereby changes at a higher level do not affect the lower levels. Blockchain Organizing reasons from the smallest building block: human and resource (supply, ontology) and connect this to a question or problem that you cannot solve by yourself. The unit of analysis and design is: coordinating capacity. This is the meaningful connection between supply and demand. Together they constitute an organizing capacity.

In this chapter we further elaborate on the basis of Blockchain Organizing: organizing trust and allowing values to flow as efficient as possible, based on five organizational principles or foundations that are applicable to any organization. We also

³⁶ The firm, the market, the bank or the government do not exist. When this book states the firm (or the market, the bank, the government) one should actually read ‘the firm’, whereby ‘the firm’ should be understood as an institution or a form, a way of organizing or as some kind of organizing capacity.

further explain the question what we mean by ‘Old’ and ‘New’ Organizing. But we start with the basis: trust and the organization of trust.

2.1 Context: digitalization

We believe that everything that can be digitized will be digitized eventually. And the pace at which the real physical world will be digitized will accelerate, or as the Canadian prime-minister Justin Trudeau formulated it: *“The pace of change has never been this fast, and yet it will never be this slow again.”* Your physical house is ‘no longer’ relevant in a transaction between house seller and buyer. Instead of the physical good, the digital ownership claim, registered within the land register is important. And digital ownership claims are just bits and bytes that can be transported in a ‘blockchain way’. That means: authentication of seller and buyer, secure transport between seller and buyer, the seller has to prove ownership and the seller can only sell it once. And when you need more independent actors to validate transactions, a distributed system can be the solution to reach consensus. Within the digital transformation, data, and especially the way we organize supply and demand of data, will be important. The first cars that were made were laughed at by people who used horses and wagons. The roads were made for horses, not for cars. Cars got stuck in the muddy roads that were made for horses. This metaphor illustrates that with the technology change (from horse to car), the infrastructure to organize supply and demand, should also change, otherwise it doesn’t make any sense.

The European Commission³⁷ expects that the number of people working in the data market will grow from 6.16 million in 2016 up to 10.43 million by 2020 and the number of data organizations from 255,000 to 359,050. That’s almost a doubling in five to six years. Revenues will more than double from €300 billion to €739 billion. They define data workers as: workers, who collect, store, manage and/or analyze, interpret and visualize data. The pace of the digital transformation will continue to accelerate and so do the problems that will originate when applying new technologies in old organization models without attempting to change behavior. Most digital strategies in traditional organizations don’t reflect how digital is changing our social economic order and management as an innovation. We see organization costs (management and staff) increase and transaction cost decrease. When transaction costs go to zero, interactions (internal transaction) within companies become transactions and can be executed more efficiently with Blockchain Organizing. In that case, hierarchies will become markets and traditional organizations will become irrelevant and eventually obsolete.

Most suppliers of IT- and consultancy services have old ‘firm-DNA’. They have customer complexity at the core of their revenue model. They benefit more from

³⁷ See also ‘European data market study’ on <https://ec.europa.eu/digital-single-market/en/news/final-results-european-data-market-study-measuring-size-and-trends-eu-data-economy>

complex than from simple situations. They mainly see the differences but forget that there are more similarities at the core of digitalization. Every value (products and services) can be translated into bits and bytes and can be transported from supply to demand in the ‘same’ way. If you take a closer look at the development of cloud computing for example you see a slow adaption. Cloudcomputing is not mainstream, most people still use e-mails with attachments to share information. You could see cloud computing as a predecessor of blockchain (see also Leslie Lamport and the concept of ‘Distributed systems’ in paragraph 3.1).

Because of the ‘firm-DNA’, there are so many applications that don’t have ‘the cloud’ as unity of thinking and design. This makes it almost impossible to migrate processes into the cloud. Despite the many advantages such as: lower costs, efficiencies, speed, privacy protection and transparency, the adaption is at a very slow pace, compared to the technological opportunities. For real digitalization you need a transition program and a new organization model that is industry agnostic with one basis in an ontology and a taxonomy, comparable with one electricity grid. For a modern IT-landscape you don’t need a database per company, you only need access to attributes to execute processes. Just as an electronic device in your house ‘translates’ electricity into motion, our sound, an IT-application translates data into information, or better gives answers to questions.



Figure 2.1: From power grid to ontology, feeding different applications

Leslie Lamport already developed the fundamentals for cloud computing in 1976. So the first step in your digital transformation is not a new technology, the first step is finding consensus about the problem you want to solve. The second step is to design and develop a new organization model that can change the way you process interactions and transactions. Start your project with identifying your stakeholders: market parties, facilitators, providers, governments, etcetera. The development of shared transaction centers is a big challenge, not that much from a technologically perspective, but from a social cultural one. Again we suffer from leadership legacy instead of technology legacy.

To put digital transformation in place you should start thinking about the ten core connections for communities we will explain later in this book (see also figure 2.6). It is not a proof of wisdom when you put Blockchain Organizing in bilateral

relationships between customers and suppliers. You need to think of and develop collective value chains with no single point of failure but with a consensus mechanism between the different actors which are equal in the way they process transactions. Giving up some autonomy is much more difficult than integrating new technologies into your existing organization model.

When management - as we know it and still use it - was invented in the beginning of the twentieth century, organizations were almost entirely managing physical flows and trying to adjust capricious people to the assembly line. In recent decades, knowledge work, self-organizing and the information and therefore data component has become more and more important. This was due to a changing context and available database technologies, internet, and in the last few years due to blockchain technology. All physical and human values can be converted into bits and bytes. As a result of this digital transformation, physical flows can be performed much more efficiently and we will become less depending on the separated parts of supply chains.

We can imagine that the siloization of knowledge stops and becomes wisdom. Old siloized organizations become part of an internet assembly line. But most digital strategies from organizations we researched, trained or advised do not reflect how a digital transformation will change information systems, leadership, management and our social economic order. In particular, analog processes are digitized. We call that automation and not digitalization. Most organizations are still hunters and collectors if it comes to data. Most of the energy we put in digitalization still didn't create an assembly line of data processors. The big question is: will blockchain change this?

Digitalization is the process of converting analog information, into a digital format with a new organization model. For example: a physical book that can be read with an e-reader and distributed online directly between writer and reader. Or the characteristics and transactions of an employee, that can be stored in a Personal Data Service, connected to a Blockchain As A Service for example. Digitalization also converts analog processes into digital processes. Digital processes transport, store and retrieve data in a much faster, cheaper and trustworthy way than old models do. This opens the possibility to change processes not only within organizations, but especially between them. Organization costs of traditional firms keep rising while transaction costs will go to zero. This means that the generally considered superior form to organize trust, work and economy will not remain 'the company' and 'the government' and thus 'the traditional market' between them, but will be replaced by shared information and transaction networks.

Digital transformation requires new and above all authentic leadership. A digital leader is not someone who knows everything about automation, about soft- and hardware, but knows what technology means for humans and how it can help us to pursue prosperity, welfare and happiness. In addition to automation, we need new organizational models such as Weconomics and a change of behavior. We don't see that behavior is changed by technology in crypto trade markets and their derivatives,

for example. It looks like we are heading towards a new ‘2008-crisis’. An important advantage of Blockchain Organizing is that the model is industry agnostic. This makes the world much less complex. In order to achieve this, at least two conditions must be met: authentic digital leadership and letting go of existing contexts (industries, sectors, organizations and jobs).

Western countries are developing from an industrial economy to a data economy. This digital transformation has far-reaching consequences for existing organizations, forms and leadership. We are in a transition period where physical buildings are replaced with ‘data warehouses’ and physical flows become relatively less important compared to data flows. If we organize this new context well, we will become more productive and solve privacy and security issues. We will become less dependent on big data companies and we can deal with fakenews and fakework in a better way. With Blockchain Organizing it is possible that (IT) technology finally makes us more productive and produces surpluses to organize a more sustainable prosperity.

We can conclude that we are in a transition from a physical to a digital society, or as we sometimes put it: *“Our world is undergoing a global digital transformation where digital and the physical reality are blurring into a single integrated modern way of living. This new world does not only need new organization technology such as blockchain, but especially needs new organization models to organize supply and demand of data”*. But we can also learn from the physical world. Although data and data processes have other characteristics as physical products or processes, some characteristics are the same. We can learn from the ‘old world’ how we should organize (or should not organize) supply and demand. For example we could learn from risk & compliance processes in order to organize privacy. Organization should take care of data and privacy in a similar way as they take care of money in an organization. A lot of effort is put in processes to assure that money is only spend when there is a budget, or the proper function in an organization has access to budgets or can transfer money etcetera.

2.2 Basis: trust, values and transactions

The core of Blockchain Organizing includes a number of concepts that are essential in understanding and using this new management concept:

- **Trust:** in order to do business with each other, you must trust each other. When you do not know each other personally and you cannot seal transactions with a handshake, you need institutions such as money and contracts. These institutions have been doing their job correctly for a long time. However now they become too expensive because they make the world unnecessarily complex. We can also program trust. A blockchain protocol ensures that you do not have to trust anyone only the entire network, or as formulated by Pavel Kravchenko³⁸: *“Blockchain a mechanism for reaching consensus regarding the state of a shared database between multiple*

³⁸ Founder of Distributed Labs, blockchain expert, teacher and speaker.

parties who don't trust each other". In fact, it is a mechanism for finding consensus on a shared reality as the basis for trust.

- **Value:** a resource has value when a person assigns it a certain value and when it can be recognized and distinguished from other values via attributes (such as color, price or supplier). A tree in the desert is for example 'nature' and has no economic value for a person, till someone decides to charge ten euros per hour to sit in the shade of that tree. Then you start to assign attributes to the tree (who is the owner, how much does it cost to sit in the shade for an hour, etcetera). When a value is initially placed in a blockchain we call this an 'Oracle'. In addition to the value of a physical product or a service itself, the information (collection of attributes) of a product or service also has a certain value and can be exchanged for money. The data that you generate at home (CO₂ emission, fuel consumption, number of hours that the light is on, etcetera), or from your body (Fitbit), your car or DNA-profile can have a certain value. Data from your house and can be sold to a utility supplier for example.
- **Transaction (Flow):** takes place when value transfers from seller to buyer. Another word for transaction is value flows. Usually there is a purchase (an order, for example the delivery of a book from Ben → Ann) and a payment (the delivery of 10 bitcoin, for example, from Ann → Ben)³⁹.

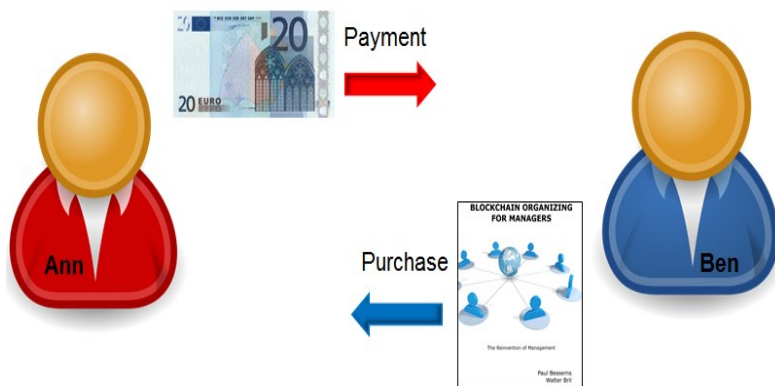


Figure 2.2: Example of a transaction: two value flows (purchase and payment)

³⁹ When we explain a transaction in this book we often use as payment: from Ann → Ben (abbreviated to A → B) and as purchase (for example delivery book): from Ben → Ann (abbreviated to B → A). It can differ per situation whether first the purchase is made and then the payment, vice versa or simultaneously. Usually we only use the word transaction in this book if we mean a purchase and an associated payment. In addition, 10 Bitcoin is a lot for a book, but it is just an example.

Organizing is about allowing value to flow as frictionless as possible from someone who has it (the seller) to someone who wants it (the buyer)⁴⁰. The most ideal form for this, is the one that provides the least friction. The same applies to⁴¹ the flow of water from high to low, whereby water seeks the path with the least resistance. Electricity goes from plus to minus and gas flows from high pressure to low pressure.

This book is about allowing values to flow as frictionless as possible from supply to demand. This is the core of organizing! We also call this flow of values 'value flows'. What many people seem to forget is that 'asking' a questions and receiving an answer between two systems can also be seen as a value flow. This enables us to organize information systems fundamentally different with concepts such as Bring Your Own Data.

Four essential processes

As we saw before in the example of the three in the desert, an object gets an economic value as soon as it has value for a person, or can be connected to a person. It must be possible to further describe a value in characteristics (also called 'tagging'). Within organizational design and IT architecture we usually call these attributes. A car is a value and has a number of characteristics, such as color, weight and make. Based on these characteristics a value can be offered, distinguished, found, delivered and paid for. This transaction can also be registered in an accounting system based on attributes, or for example with the tax office. The four main processes with Blockchain Organizing are:

1. Profiling (supply)
2. Selecting (demand)
3. Delivering (purchase, supply → demand)
4. Paying (payment, demand → supply)

In graphical format:

⁴⁰ When we explain the organization of supply and demand in this book, we often use: Supply → Demand (abbreviated to S→D), and vice versa Demand → Supply (D→S). The cooperation between two sellers (S→S) or two buyers (D→D), can also be brought back to seller and buyer. There are different types of buyers and sellers such as: employers and employees, producer and consumer, official and citizen.

⁴¹ Can also be brought back to the (philosophical) notion of 'Panta Rhei' by the Greek philosopher Heraclitus: everything flows and you can only stand in the same river once (you can only make the same transaction once).

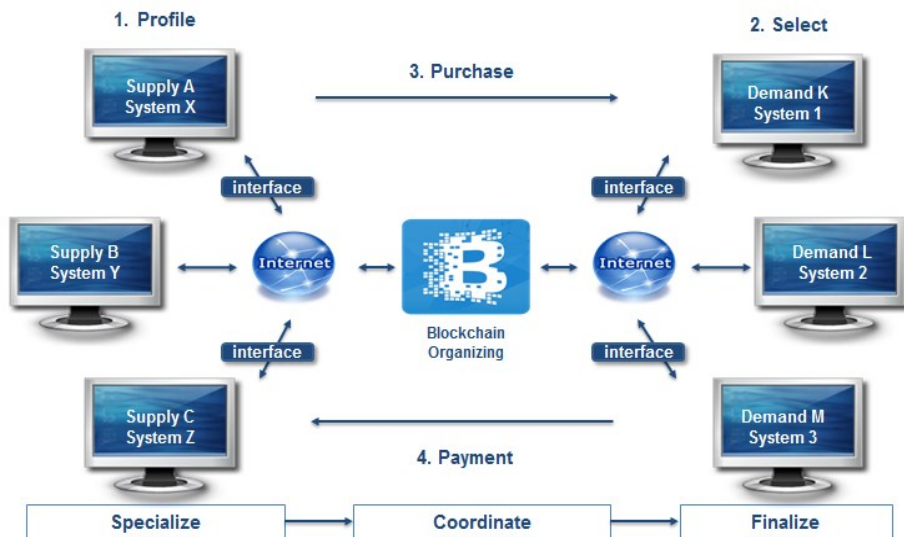


Figure 2.3: The organization of supply and demand and the role of coordinating capacity

Within Blockchain Organizing, process 4 can be sub-divided into:

1. Ann creates a transaction, including digital signature and public key for Ben.
2. This transaction is sent to the network and picked up by miners to be verified.
3. After miners have performed a number of checks, the transaction is recorded in a block.
4. This block is transmitted to the network. Nodes accept this block when the transactions in the block are valid and the amounts have not been spent previously. The chain (branch) with the most blocks continues to exist.
5. Ben receives the payment (recorded in blockchain).

Process 3 and 4 can also be exchanged. For example, you first have to pay for certain services before these are provided. Blockchain technology can especially be significant for process 4: paying. In addition, blockchain technology ensures that a transaction is logged non-mutable in a transparent shared accounting system. Therefore, information from this transaction can be used (process 1) with a next transaction in order to make a better choice (process 2).

The most common and used form of value is money. Money can go from a person who owns it (for example Ann) to a person who wants it (Ben). We call this a payment (process 4). A product or service is delivered in exchange for money (process 3). Often this money is used again later to buy another product or service. Money is a claim on a future purchase, money is a 'medium of exchange'. It is a catalyst that facilitates a process, but eventually it is no longer a part of it because it is used again. In addition to money, other exchange values or calculation units can be used for the transaction. For example: tokens, hours, credits, rights, energy, waste, other products

or services. Two hours of gardening by your neighbor can be exchanged for one hour of tax advice from you to your neighbor.

In addition to the value that is exchanged (purchase) for the counter-value (payment), values are also created within a blockchain community, like Bitcoin, for the persons who contribute to the verification of transactions (the miners: the persons who contribute to solving a mathematical puzzle, also called Proof of Work, see also paragraph 3.2). Apart from receiving Bitcoins for providing inspection work, you can also think of maintaining a Wikipedia page. You receive (service) tokens⁴² for this that you can exchange with sponsors of Wikipedia. So, within the Bitcoin community miners receive bitcoins for the inspection work they perform (similar to paying your accountant for checking the accounting system). Tokens are exchanged for making a contribution to the community. A token can then be used internally or externally (externally via exchange markets). Tokens are a resource and a derivative of the effort being made to maintain and/or further develop the community.

Apart from these four processes, three aspects are important with regard to values: ownership, (value) transaction (the flow from seller to buyer) and the recording of this transaction (which changes the ownership). In principle, blockchain is reducing the assets of the seller and increasing the assets of the buyer with the same amount. After all, the accounting system must also remain in balance with Blockchain Organizing.

Ownership

Before you can transfer a value, you must first prove that you are the owner of this value. It might also be required that you must prove that you are the person entitled to access, a will for example. You can program a message in such a way that at least two people must use their key to access a value. This can, for example, be useful when closing an escrow agreement. In paragraph 3.2 we will see how this is regulated within the Bitcoin protocol. Core of the solution to prove that you are the owner of a value, is to use input references. As owner, you have to prove where the value, that you claim to own, comes from. This is not required/not possible at the first transaction (also called Genesis block). Value is created from scratch in the Genesis block. In the example of the desert tree, you are the first person who claims the tree and the tree changes from nature to product.

(Value) transaction

With value we actually mean added value: it must add to what you don't own yet or what you cannot develop independently. When we use the term added value within the organization of work we see that this flows from employee, portfolio worker or supplier, to buyer (customer). If there is only one seller and buyer, not much coordination is required. But usually you have several sellers, for example with a project

⁴² In addition to service or utility tokens there are also virtual currency tokens and tokens to finance an organization (for example: equity, debt and participation rights tokens).

(see also figure 5.1). Blockchain technology can play an important role in this process like finding and connecting people and resources to a project target.

There is no such thing as a good or bad transaction, a valuable or non-valuable transaction, an authorized or unauthorized transaction. There are no spam transactions, illegal or immoral transaction. With Blockchain Organizing there are only valid and invalid transactions. An invalid transaction does not have any input reference and is not recorded in a block. A valid transaction complies with the consensus rules and it does not matter who the sender or receiver is or what the value of the transaction is. It is purely about the logistics of the transport between seller and buyer and if this has happened. This logistics is separated from the filter (interface) that you, for example, can install with the selection of values (see also figure 3.5).

For the execution of transactions we currently use a variety of agents, also called institutions in the previously discussed NIE theory. There are many variants conceivable to reduce the insecurity with the flow of value from seller to buyer. You can organize trust in several ways, via a social 'script' and a technical 'script'. The latter provides the advantage that it is less susceptible to the subjective interpretation of man. As we saw before, blockchain technology is a form of coordinating capacity which can be programmed.

Besides the function of blockchain technology in an existing transaction at time $T=0$, this technology can also have an effect on future transactions: time $T>0$. By logging transaction data in a non-mutable way, this registration can be used with the next transaction. Data can be used to formulate the demand better, to profile the supply better and to coordinate demand and supply better. This more or less 'automatically' further reduces the insecurity in a system. It creates a system that can, step by step, make a better and more efficient choice, and therefore makes the friction disappear through organizing. The 'machine learns automatically'. The allocation of humans and resources to finalization is increasingly improving. A reputation management system is automatically set up, with a shared reality. This can also be used for countering fake news or fake work for example.

Recording

The recording of a transaction (in a block and from a block in a blockchain) is further discussed in paragraph 3.2. An important technique that is used for this, is the hash function (a form of encryption), with which you record characteristics and the sequence of transactions (T) in a string of characters (H). If you would only change one character, for example a period or a comma, into T , then H will also change. In formula $T \leftrightarrow H$, there is in fact a 1:1 relationship and dependence between the two. Regardless of the size of T , the image to H (in the case of Bitcoin) has a size of 256 bits. Blockchain uses cryptographic hash functions, with the special characteristic that the hash is a unique fingerprint of T . Furthermore, a characteristic of the previous block is also recorded in a hash so that once a block is in a chain, you can no

longer adjust this block and therefore the underlying transactions, without adjusting the entire blockchain.

These were the three important aspects with value transactions. Before we look at New Organizing, of which blockchain is an example, first a bit more about what we mean by 'Old Organizing'.

2.3 Old organizing

For quite some time organizations have been struggling with issues that they will not be able to solve with the current perspective, their way of thinking and old management concepts. Government and business are continuously reorganizing without knowing where they want to go. Companies are addicted to forms and changes: from brainstorming to ISO certifications, from coaching to key performance indicators. These are all tools to organize trust and to reduce insecurity, but as a consequence distrust and complexity increase and productivity falls. This leads to nothing. We innovate to become more productive, but we are getting increasingly busier.

Fortunately, we meet more and more people in our consultancy practice, who look at it fundamentally different and also want things to change. They want to get rid of a system that is focused on continuous economic growth managed with factory DNA. We still 'suffer' under the system that has brought us much welfare, but now needs to be replaced. We have abolished slavery, but since then we have become more and more a slave of debts and the (capitalist) system that we have created. Even students are already being made dependent on 'the system' by student loan programs. We must constantly grow economically while we actually do not want it anymore and certainly cannot do that anymore. The Dutch government for example is struggling with administrative authorities like 'UWV' (administrative authority, commissioned by the Ministry of Social Affairs and Employment to implement employee insurances and provide labor market and data services), 'Sociale Verzekeringsbank' (organization that implements national insurance schemes), 'Belastingdienst' (the tax authorities) and 'DUO' (student finance agency). The tax authorities must juggle six hundred ICT systems while the secretary of education pulls the rug from under the ICT project 'Basic Administration', worth ninety million euros.

The election programs for the House of Representatives elections in 2017 and the subsequent formation of a new government didn't structurally address these problems. We as authors dare to say that, with regard to organizing our economy and welfare, it does not really matter much for which party you vote. Almost all of the parties are looking for solutions with the wrong perspective, the wrong frame of mind, the wrong assumptions, experts and organizational forms.

This book is therefore not only about a fundamentally new management technology that can bring us solutions. This book is mainly about a new context in which we can use this technology. This book is about a new perspective, and new way of thinking

and organizing. We will connect a promising new technology to a new management model. We, as authors of this book and as advisors in our consultancy practice, feel like pioneers. Pioneers of hope and trust. In addition to an explanation about Blockchain Organizing, this book also contains a new perspective on the question:

How can we organize trust without increasing complexity?

Blockchain Organizing is ‘only’ a tool to realize this. However, an important tool, with solid principles, foundations and building blocks. But it remains a tool that only works correctly when we start to think and behave differently. Principles of New Organizing are the foundation, the basis on which we can stand. Blockchain technology is the organizing capacity or the lever to renew the world. Or as formulated by Archimedes: *“Give me a place to stand and a lever long enough and I can move the world.”*

2.4 New Organizing

A large part of our life consists of sharing information and executing transactions. Trust, or actually organizing trust or reducing uncertainty plays a crucial role in this. From shopping in the supermarket to buying a house. These are all transactions. We already have internet to share information. Now there will be an additional layer on top: blockchain. Blockchain is also called ‘the internet of trust’ or ‘trust network’. Blockchain Organizing makes it possible to execute transactions almost frictionless, so without organizational costs, without having to trust each other. You only have to trust the mathematical proof, the organizational principles, the network and the person who has created and maintains the protocol. The starting principle of Blockchain Organizing is that the majority of the network benefits when there is no fraud committed while, at the same time, making the effort to check this. The current systems in use still require expensive intermediaries to organize supply and demand, to prevent fraud and to reduce insecurity.

Blockchain Organizing is a new way of organizing trust without increasing the complexity. It is a very efficient way to perform government tasks, to organize work⁴³, to do business with each other and to register transactions and assets in a shared and distributed ledger or log file. This creates a ‘shared reality of assets and transactions’, which is shared by the majority. And that is an important condition to trust each other. So, the buyer and seller no longer have to work with local ledgers, complicated contracts and conditions, they only have to trust the entire network and the executive protocol. They do not have to check after the fact because they create consensus in advance. They set up conditions under which transactions are executed and have a consensus on consequences if this is not done conform the agreement. For example, you will only get access to a taxi when you have sufficient credit on the application with which you want to pay the taxi. You can imagine that this option does not only have an impact on our work and the way we shop but also on our total economy

⁴³ In this book we mainly see work as providing added value (from supply to demand), whether or not economized, whether or not via an institute or market.

New Organizing and therein Blockchain Organizing can fundamentally change the economy and society over time. It is expected that Blockchain Organizing will have a similar impact on our life as internet did, but hopefully in a more positive way. Blockchain Organizing is a way to align supply and demand, that is so powerful that it will make the existing organizations, management concepts and positions irrelevant and therefore more and more redundant because they are no longer efficient enough and because they make the world unnecessarily complex and non-transparent.

New Organizing is about organizing supply and demand and mainly about the question what is the most efficient way to reduce insecurity associated with a transaction. Efficiency means: with the least possible resistance. The question that concerns us in this book is whether Blockchain Organizing is a more efficient way than the rigid company, the non-transparent market or the inefficient government. Can Blockchain Organizing give more meaning to our work, make us more productive, better protect our privacy, make us less dependent on large (tech)companies and central governments, protect us against fake work, fake news, propaganda and destructive populism and perform processes more efficiently?

In the next paragraph we will focus more on the five important organizational principles for New Organizing. The purpose of organizing is finding a good balance between supply and demand, finding a way to connect humans and resources to a tasks in a more productive way. The best coordinating capacity to reduce insecurity is therefore the one with the lowest transaction costs. To realize this, organizational principles or foundations play an important role. We will further discuss the five main principles for New Organizing:

1. 'specialize, coordinate, finalize'
2. 'programmable institutions'
3. 'structured flexibility'
4. 'apart together'
5. 'back to basics'

2.5 Five basic principles of organizing

This book is about a fundamentally new way of organizing supply and demand. Which coordinating capacity provides the least friction between suppliers, buyers and between supply and demand? Blockchain Organizing comes close. There are still many more innovations to come but the potential is definitely there and the principles are good, fundamental and sustainable. Blockchain Organizing is based on a limited set of principles. These will allow you to actually organize all work and it makes the world less complex. Many managers experience the world as being complex. The only way to understand a complex world is to go back to principles, smallest units and the interconnections between them. In this paragraph we further explain the five main organizational and management principles of Blockchain Organizing.

1. Specialize, coordinate, finalize

The main reason to organize work is: increasing the revenue per person (*standard of living*). Through evolution (variation, selection and inheritance) the hunter ‘finds’ out that you can catch bigger and faster animals through task division (see also figure 1.3). Giving up a piece of autonomy and more specialization leads to a better finalization. But when you specialize, you must also coordinate, otherwise explorers, chasers and spear-throwers will each go their own way and that does not lead to a better catch.

Apart from organizing work, it is important that you can trust each other when executing transactions. The chaser must be able to trust the spear-thrower that he will throw his spear on time. When demand and supply are further apart, both in time as well as distance, the need for reducing insecurity or organizing trust increases. But as we saw before with the trust paradox, when using traditional institutions (money, contracts etcetera) to reduce trust, the complexity can increase. This not only increases distrust but productivity and privacy will come under pressure.

Since the era of the hunter/collector we have seen many different ways of coordinating. An often used organizing form in the last hundred years is for example ‘the hierarchy’ or ‘management’, within companies. Now a new form of coordinating capacity is emerging. Blockchain Organizing is still in its childhood, a conceptual model for many people that in practice will have to be translated further into more common concepts, instruments and applications. There will be applications, based on blockchain technology, that bring the best suited specialist together to solve a problem or to achieve a finalization. Recruitment for example can more and more frequently be organized on a blockchain.

Back to the hunter: even animals that are much bigger, stronger and faster can be caught by organizing the hunt: that means by coordinating specialists into an activity. To realize a better task division and cooperation, coordination is an important condition. Good communication is an important condition for coordination and information is an important condition for communication, while data is an important condition for information. Within Blockchain Organizing we mainly talk about organizing supply and demand of data. The underlying data of a transaction is mainly data about the two smallest building blocks: human and resource and about the transaction. What are characteristics such as skills, price and availability? The demand should also be clear. Only when supply and demand are clear, a transaction can be made. Then a workflow (collection of conditioned processes) is important. This workflow can be programmed in a smart contract. Data also plays an important role in this.

Through communication the first person is quickly able to divide tasks, to specialize and to cooperate. So, the first person finds out that a specific work division and giving up a piece of autonomy can lead to a higher revenue per person. Different

tasks must be assigned. If everyone in a company is a seller and nobody produces a product, not much will happen. As soon as tasks are divided, the need for management, control systems, driver systems, assessment systems and rewards and punishment systems occur. Organizing comes before managing and includes: focus, shape and finally perform. When performing, management is especially used for planning, motivating, coordinating and controlling employees. A good organizational design reduces management.

Managers and staff positions currently play an important coordinating role within the existing design: the firm. Organizing is more than just managing and also includes an organizational design, setting targets, designing tasks and then assigning, organizing people and resources, coordinating, implementing feedback mechanisms to learn and to adjust. Machine Learning and Artificial Intelligence will play an increasingly important role in this. When your reasoning focusses on the basic activities, as is the case with Blockchain Organizing, there are many ways to coordinate and to reduce the insecurity in transactions, but only a few basics. In addition to deploying managers, think using an ICT system, the telephone, self-organizing capacity which is also a form of coordinating capacity. There are many ways to coordinate and the one with the lowest transaction cost will survive.

Throughout the centuries there have been various ways to organize our work. Blockchain Organizing is the next stage in the way we can coordinate supply and demand therefore can organize trust, work, trade and economy. Blockchain Organizing mainly focuses on renewing the organization of the work of service- and knowledge workers: no longer within the restricting walls of firms but within partnerships and shared information and transaction networks. Almost all professions and positions are as a matter of fact related to processing information and transactions (profiling, sharing information, saving, searching, communicating, consulting, having meetings, making transactions, etcetera).

It is estimated that more than half of the workers in western economics spend more than half of their time on processing information. This includes also people who provide services such as doctors and teachers. Because information (actually data) has become the main production factor⁴⁴, Blockchain Organizing mainly focuses on the improvement of the productivity of information workers by making transactions against the lowest possible transaction costs. Therefore, the company is no longer the only and often also not the best form of organizing capacity. We are ready for the next step in organizing work and welfare. We see the development⁴⁵ in 'forms', with which we have been and will be organizing our work⁴⁶ as follows:

⁴⁴ See for example Accenture Technology report 2013

⁴⁵ This development does not mean that there are no farming communities or crafts anymore but that the number of people, who organize work with this form, is relatively and absolutely reduced.

⁴⁶ Markets (a space between at least two entities) have always existed for modern man, however in the past hundred years markets have been given a different character, for example market

1. Hunting community (physical power coordinates⁴⁷)
2. Farming community (season and weather coordinates)
3. Craft/guilds (market coordinates)
4. Company/institute (hierarchy coordinates)
5. Network organization/blockchain community (network protocol coordinates)

2. Programmable institutions

Within the principle of specializing, coordinating and finalizing, programming of the coordinating position plays an important role in an increasingly digitizing society. When organizing supply and demand it is important that parties can trust each other. To realize this we have innovated all kinds of institutions to reduce insecurity. A small part of the population is still busy with the direct production of food and products (primary processes). The vast majority of the workforce is employed by institutions that act as a lubricant (secondary processes) for an efficient operation of primary processes and therefore our economy. But more and more of the existing institutions become less efficient when you compare them with the options offered by Blockchain Organizing.

Institutions to reduce the insecurity in a transaction can also be digitized and programmed. We already have programmable money. Money, a contract, a workflow and the accounting system, they are all institutions that can be programmed. And everything that can be programmed and contributes to improving the efficiency of the system, will eventually also be programmed. This means that designers, architects and software developers will probably become the new trust industry, also called fiduciaries. In other words: banks, accounting and law firms will increasingly become IT companies. The idea that you do not have to trust anyone on a blockchain is not completely correct. For example, when you buy bitcoins via a market place, you will have to trust a software developer. Because with examples of Mt. Cox⁴⁸ hack and 'The DAO-hack'⁴⁹ it is clear that we must mainly trust organizational designers and software developers and maintainers when we use blockchain technology⁵⁰.

3. Structured flexibility

Apart from programmable trust, which has less transaction costs, another organizational principle plays an important role when organizing. We usually organize work with the use of a (work) organization. We choose between doing it ourselves or outsourcing. A cross between both is called hybrid organizing and that is one of the

information and money are playing an important role now and a market is seen as a place between at least two hierarchies.

⁴⁷ Communication also coordinates in all examples.

⁴⁸ On 19 June 2011 Mt. Gox, the biggest Bitcoin exchange at that moment was hacked, whereby, among other things, 60,000 login details were captured. The rumour is that 850,000 Bitcoins were stolen.

⁴⁹ See also paragraph 6.9 in book6.

⁵⁰ See for example also article in American Banker by Angela Walch: 'Call Blockchain Developers What They Are: Fiduciaries' (09-08-2016).

pillars used by Blockchain Organizing. It provides a better balance between doing it yourself and outsourcing, between firm and market, between private and public. It improves the balance between scalability and flexibility. A new organizational concept will always have to find a balance between scalability and flexibility. We also call this 'structured flexibility'.

The relationship between scalability and flexibility is therefore an important relationship. Scalability and flexibility are important motives for our evolution. Both natural organisms (living organizations like mammals), living organizations (cities), as well as work organizations (firms) are continuously looking for a better balance between scalability and flexibility. *This balance is more and more lost within large-scale, bureaucratic, ego-centric and closed firms and non-transparent or poorly functioning markets.* Therefore, more and more people are looking for alternatives and find these in fundamentally new organizational forms. You can cooperate with a customer, a supplier or even a competitor. This can be with one relationship (1:1 relationship, bilateral), but also with several relationships (1:n relationships, market). However, more and more processes will be transferred to shared information- and transaction networks or modern partnerships (n:n relationship, community). For example searching and selecting flex workers or training courses. A community, cooperation or shared service center, which is connected to a data common, provides sufficient scalability on the one hand and sufficient flexibility on the other.

So, two main motives when organizing work and welfare are scalability and flexibility. Large companies often cannot grow any further without losing flexibility and markets or small organizations are often flexible but lack the scalability of a large organization. Therefore, we are looking for a better balance between scalability and flexibility and more and more often this balance is found in new concepts like Blockchain Organizing. The 'old' company can thereby act as permanent core or hub of a new network organization or 'n:n community'. A network organization is scalable and flexible at the same time. Information workers become much more productive, privacy can be assured better, the meaning of work increases and we become less dependent on large undemocratic companies and institutions. In addition, it is now possible to innovate more effectively and companies can provide an extra social contribution to the organization of a sustainable welfare. Internet might be the best example of a network organization. Internet is very scalable (in six connections you know the whole world⁵¹), and man is very flexible. By combining these, structured flexibility is created. Blockchain Organizing uses internet. Core of Blockchain Organizing is connecting flexible people (and resources) to a task via 'the' scalable internet and blockchain protocol.

4. Apart together

⁵¹ According to the idea of the 'six degrees of separation', everyone on this world is connected to each other in not more than six steps. See also the Weconomics practice book, paragraph 3.6.3: 'Profile -> connect -> collaborate'.

A fourth organization principle that we want to explain is the balance between cooperating and competing. Humans cannot exist without humanity and humanity cannot exist without humans. Ubuntu is an African philosophy and means, among others: 'I am because we are'. People are not just connected to each other, but also depend on each other to survive. An important basis of Blockchain Organizing is 'collective self-interest'⁵². Our society requires a good balance between: individual and group and between competing and cooperating. In the current spirit of time it appears that cooperating is becoming more important than competing. To solve the potentially disruptive problems such as climate change, debts, destructive populism and inequality we will have to cooperate more while competing appears to worsen the problems. The protectionist measures that many countries want to take in 'the aftermath of the 2008-crisis', will offer short-term solutions but bring less welfare in the long-term and cannot make this welfare sustainable. You could say that we must design a system whereby cooperating becomes the new competing. With Blockchain Organizing a new balance is found based on more cooperating in the agreements (infrastructure) and competing in the differences (social cultural). Blockchain Organizing makes it possible that we can temporarily connect many people and resources to the solve a problem.

5. Back to basics

There are both economists as sociologists who look at the origin and existence of organization from the organizational theory point of view to organize work and economy. 'The economist' usually looks through market glasses and uses the market transaction as principle. The 'sociologist' uses the company as the starting point. The organizational expert does not use the total or the building (firm, market or government) as principle, but the smallest building blocks and the connections between them. To build a house you basically only need bricks and cement. To build an organization you basically only need the smallest building blocks (humans and their resource) and the connection (coordinating capacity) between them to complete a task. In the following figures we see that the building blocks remain the same, only the connections change in time.

⁵² 'Collective self-interest' is actually a contradiction in itself, but important for solving the Tragedy of Commons, for example.

Organizing is (dis)connecting

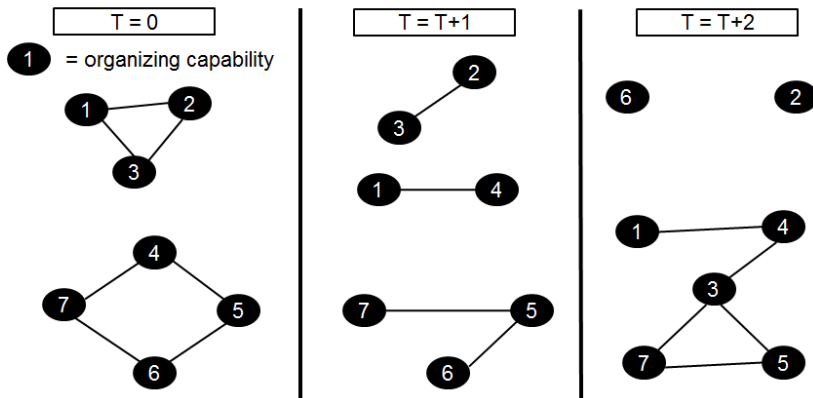


Figure 2.4: Organizing is switching connection on and off

You can always bring back a hierarchy from the smallest building block:

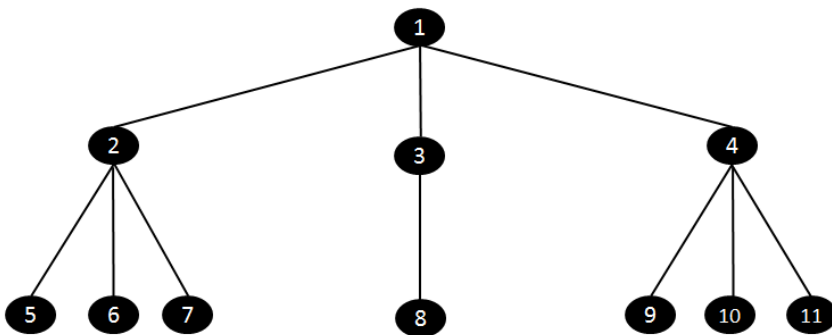


Figure 2.5: You can always reconstruct a hierarchy from the smallest building block

A problem of a large part of the organizational theory is that it is often based on the firm as ideal form to organize work, the non-transparent market to organize transactions and the central government to organize commonalities. Most theories assume that there would be a firm organization, a market organization and a government organization. This is right, there are indeed differences but there are much more similarities. When we reason from the smallest building blocks and connections, there are much more similarities. By reasoning from the similarities, the world suddenly becomes a lot less complex. Within Blockchain Organizing, the company is a collection of humans with the connection: 'is employee of' and a city a collection of citizens with the connection: 'is citizen of'. So a HR-systems that keeps records of employees is just a collection smallest building stones. And when one leaves for

another company, only the connection changes if that person has a Personal Data Service. The smallest building blocks remain the same, the connections change.

Most theories also suggest that there are no transactions made within firms. But transactions are also made within a firm, although these are not always paid for with money. And also a market can have a certain hierarchy, or there can be a concentration of power concentration. You can see a staff department within a company as some kind of commonality, just as the fire station and primary school in a city. Eventually everything can be flattened to the smallest building block and the connection between them. The type of connection may be different but it remains a connection. The civil servant has another type of connection than an employee. As a civil servant you have a professional confidentiality and you work under an administrative law. But a doctor, for example, also has to deal with the medical confidentiality and disciplinary law.

So far, we saw five organizational principles of Blockchain Organizing: ‘specialize-coordinate-finalize’, ‘programmable institutions’, ‘structured flexibility’, ‘apart together’ and ‘back to basics’. They are foundations for a better balance of cooperation between supply and demand, scalability and flexibility and competing and cooperating. To conclude this chapter about the organizational component of Blockchain Organizing, we mainly like to point out the importance of organizational science and of cooperating smartly within business ecosystems, or in blockchain term: shared transactions networks.

Cooperating is sharing information and executing transactions

A large part of cooperating consists of sharing of information and executing transactions. As we saw before, a transaction is very abstract: the flow of added value from the supply to demand, from specialist to finalization. More concretely, a transaction is purchasing a course, a computer or this book. We already have internet for sharing information. An additional layer is added to this. Blockchain Organizing makes it possible to share information, execute transactions and cooperate in a very cheap way. Currently we often require expensive intermediaries to organize trust or coordinate between supply and demand.

Blockchain Organizing is a new way to very efficiently perform government tasks, to organize our work, to do business with each other and to register transactions in a shared accounting system. The buyer and the seller no longer have to trust each other. They only have to trust the entire network and the protocol. You can imagine that this option will not only have an impact on your way of managing and cooperating, but also on our economy and society. Blockchain Organizing will probably have a similar if not bigger impact on our lives as internet. Blockchain Organizing is a way of organizing that is so powerful, that it will give another meaning to many existing management concepts, positions, organization and methods to organize trust. Existing forms will not all disappear but they will get a different interpretation. Introduction of the car has not lead to the total elimination of the horse-drawn

carriage. In cities like Bruges, Vienna and New York you can still use a horse and carriage.

The role of organization science

Is the market or the firm with a hierarchy and managers the most efficient way to cooperate? Or are there more efficient forms? Organizing is about solving management problems, is about solving other people's problems. If you could solve the problem by yourselves, you wouldn't need an organization. Organizational science plays an important role here. Organizational science is mainly engaged in studying, designing and developing organizations; management mainly with executing and managing existing firms. The design stage is especially important for Blockchain Organizing and in this book we mainly talk about Blockchain by Design. This means that we do not primarily focus on changing the existing firms or even partnerships, but on a fundamental renewal. We build new partnership organizations from the foundations and smallest building blocks where existing assignments, processes or projects can be transferred to the new organizations.

An important organizational issue that has been keeping politics, business and citizen busy for a while now is the organization of European cooperation. Currently this cooperation is mainly designed via the European Union. Especially after Brexit, more and more people are wondering how you can best organize this cooperation and whether existing perspectives and models are still working. At national level we wonder for example whether we still need provinces and how councils can best be organized to handle their new tasks as part of the participation society in the Netherlands for example. The possible separation of Catalonia and Scotland for example are also organizing challenges. And at organizational level more and more people have the feeling that cooperation within and between organizations is not really working. However, most of these partnerships between firms do have the ingredients for a good cooperation. There is a reason to cooperate, there is an ambition, participants in projects follow training courses and they use the best project management tools, but still... there is that nagging feeling that there is too much friction, double work, people bailing out or agreements are not met. How could this be possible?

Why is it that we hardly become more productive while we invest a lot in technology and smart cooperation tools? Are the tools not good enough or are these tools used in the wrong organizational forms? Is this not the time to thoroughly revise the formats that until now were regarded as ideal: firm, government and market? In this book we do not primarily focus on new tools or the application of tools in their existing forms. We mainly focus, with a new perspective, organizational model, knowledge platform and transition program, on fundamentally improving cooperation by renewing the organizational forms. We use the existing organizational forms less, which have become too expensive to keep up with the increasing welfare costs. There is a need for a fundamentally different perspective on cooperation and more knowledge of organizational science. Currently, mainly economists, politicians and directors of government, banks and large companies are 'controlling' our social-

economic system and join programs on national television. In an article of the Dutch Financial newspaper, only parties from the old establishment with certain interests, share an opinion on the bubble that Bitcoin is about to become⁵³. Maybe it is time to replace next to old ‘insights and experiences’, also people on television, radio or at the newspapers? Maybe it is time to invite more philosophers, anthropologist and organizational expert on television and less economist, CEO’s and politicians. We need people with a broader view who understand the key connections of communities and society.

Ten key connections of Blockchain Organizing

Within Blockchain Organizing we distinguish a number of different types of connections. We need these to design, set up and eventually manage a blockchain community and to allow added value to flow in your community:

1. Value: The first and most important connection that must be there is a shared value: what do you find important and what not and how would you act in certain circumstances?
2. Purpose: What is the purpose of your community? What is the function in the bigger picture? Why do you exist as community?
3. Principles: What organizational principles do you use?
4. Perspective (analysis, context, history and theory): What is your historical and organizational analysis of the problem? What is the context of the twenty-first century? Which assumptions and theories do you use?
5. Vision (where to?): What do you believe in, where are you going, what is your direction, what is the dot on the horizon? What is the space between here/now and there/later?
6. Ambition (how high do you set the bar and what is the time frame?): what is the SMART⁵⁴ objective of the next step and what is the target group?
7. Coordination (strategy): what is the connection between demand (connection 1-6) and supply (connection 8-10) ? How do you realize the ambition? Which capacity or technology do you use to connect demand and supply? How do you organize trust?
8. Governance: how do you make decisions, how do you find consensus, who controls what? How do you organize decision making? What is the political coordinating capacity?
9. Structure: which infrastructure will help you? How will you connect people and resources to the objective? What is the technical coordinating capacity? How do you connect your community to a data common?
10. Culture: What are the social cultural values? What is the coordinating capacity to create human interaction, to exchange ideas, to learn and to organize trust?

You could see 1 up to 6 as the demand side of organizing (D), 7 the pivot point: coordinating capacity (C) and 8 up to 10 the supply side (S). As we saw before value

⁵³ See article by Joost Dobber in Financieel Dagblad: ‘Are we witnessing a crackling bitcoin bubble?’ (16-09-2017, Dutch only).

⁵⁴ SMART: Specific Measurable Acceptable Reasonable and Time-bound.

flows from supply to demand, using coordinating capacity, or in formula: $S \rightarrow C \rightarrow D$.

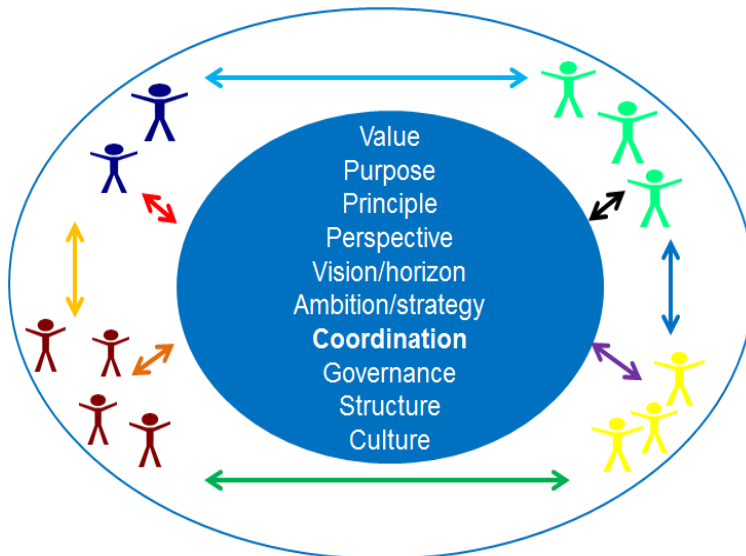


Figure 2.6: Ten key connection of a community

2.6 When hierarchies become markets

As mentioned before, people who heard of Copernicus 'new world view', did not believe him at first. But many decades later, Galileo Galilei proved in practice that Copernicus's theory and new world view was right. Even now, traditional organizations can hardly understand that they are no longer the center of doing business and therefore find it difficult to understand the impact of Blockchain Organizing. This is not so much because they cannot understand blockchain technology but rather it is more that they cannot imagine a world without the firm as superior form to organize work with, or without the government to organize commons. Private organizations see themselves between purchase and sales, public organizations between politics en civilians. And between these two traditional worldviews, we have traditional markets, where data and transactions are not processed in a way that aligns with technologies that are available.

The last disruptive view of how to organize work came from Ronald Coase in his important and famous work: 'The Nature of the Firm' (1937). In this work he basically explained why hierarchies emerge, exist and grow. Firms exist and grow because it is cheaper to do it yourself then to contract it out. When another organization has to do the work, you have to make transaction costs such as finding a supplier (who incurs marketing cost to be found), contract cost, and payment and security costs. So if the transaction cost are higher than your own organization costs, it is better to do it yourself. Between organizations (on markets) we have transaction costs, within organization we have interactions costs, also called internal transaction costs. These

are all examples of connection costs to connect humans and resources to activities when humans can't do it themselves.

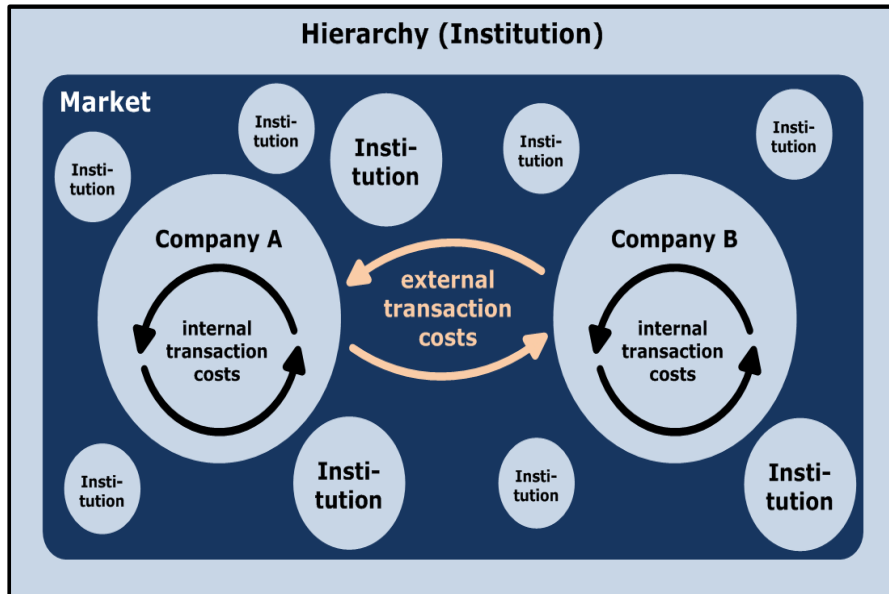


Figure 2.7: Transaction cost theory Ronald Coase

When we start seeing a company as a market and interactions as transactions, it will soon become clear that 'the firm' is not an efficient coordinating mechanism or capacity. If you, as an organization, want to apply digital transformation, it is important to let go of your own old world view and become (part of) a network organization. It is time to let go of the last management innovation of Taylor and Ford because the transaction cost (internal and external) will go to zero in a digital era. This transformation takes place step by step, similar to the development of, for example, the assembly line. The implication of the organizational aspect in this transformation will be far more difficult and will take far more time than the technical aspect. It is known that technical developments have a higher pace than social cultural changes. In this book we mostly focus on the organizational, the leadership and management aspect of embedding disruptive organization technologies into new organization forms that align better with new technologies. The first step within Blockchain Organizing is probably the development and application of blockchain technology within 1:1 and 1:n connections on a market, and after that within n:n markets. After that, hierarchies and markets will blur into one modern integrated single way of doing business, the sharing of information and execution of transactions will be as easy as purchasing this book via Amazon.

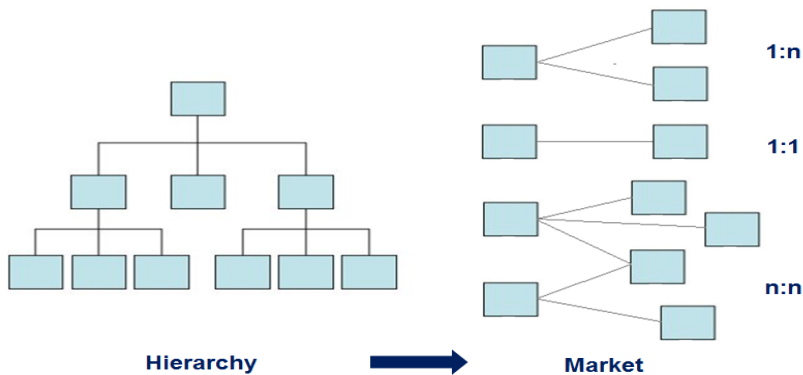


Figure 2.8: When interactions become transactions, hierarchies become markets

Hierarchy: no freedom within a so called free world?

You could see a hierarchy or firm as a none-free system within a free social economic order (capitalism). When you take a closer look at the way work is organized within hierarchies, you can conclude that a lot of activities are fixed. Not being able to choose how to do your work can lead to stress and burnout. And in a dynamic environment, with accelerating technological developments, a more flexible way of organizing should be considered, designed, developed and applied. The 'fixed' situation of the firm is not that strange, because that was the intention of designers such as Taylor and Ford. They had to align capricious humans, farmers, housewives and craft-men, into the very tight and systematic way of working on an assembly line, between nine and five, in a factory away from home. If you take a closer look at hierarchies you could see some characteristics that don't fit the free world most people think we are living in:

- Fixed wages: earn the same as your colleague, independent from the output.
- Other departments influence or even determine how you do your work.
- Workers can't choose because their boss will think for them.
- Employees 'know' their career because the only expected way is up.
- Most information workers, work between nine and five in the 'same' offices.

Transaction cost theory

As we will see in this book, Blockchain Organizing is based on the insight that we need a better balance between fixed and free. Once you start to see humans as the most important part of organizing work, instead of departments or companies, Blockchain Organizing will gain interest of academia, leadership and management because it provides a better balance between hierarchy and market, competition and cooperation, scalability and flexibility. It sees humans and resources and activities as the smallest building blocks and organizes work by connecting these building blocks via transactions at minimal costs.

When we start seeing a company as a market and interactions as transactions, it will soon become clear that 'the firm' is becoming the less efficient and therefore most

expensive coordinating mechanism. As mentioned before, probably the most important article on this issue came from Ronald Coase, and later from other economists such as Oliver Williamson who continued on the work of Coase. Coase was asking the question: Why do firms emerge, exist and grow? He tried to give an economic explanation of why individual farmers, craft-men and housewives, choose to cooperate under a nexus of contracts, to form partnerships, coops, companies and other business entities rather than trading bilaterally through contracts on a market. As we saw before, he came up with the transaction cost as antagonist of organization cost. Organization costs are cost for hierarchies, management, staff, inefficiencies and over-capacity. Organization cost will increase as a result of distrust and bureaucracy. In combination with increasing organization costs, transaction costs will go to zero as a result of digitalization. From the point where organization costs are higher than transaction costs: interactions become transactions and hierarchies become markets.

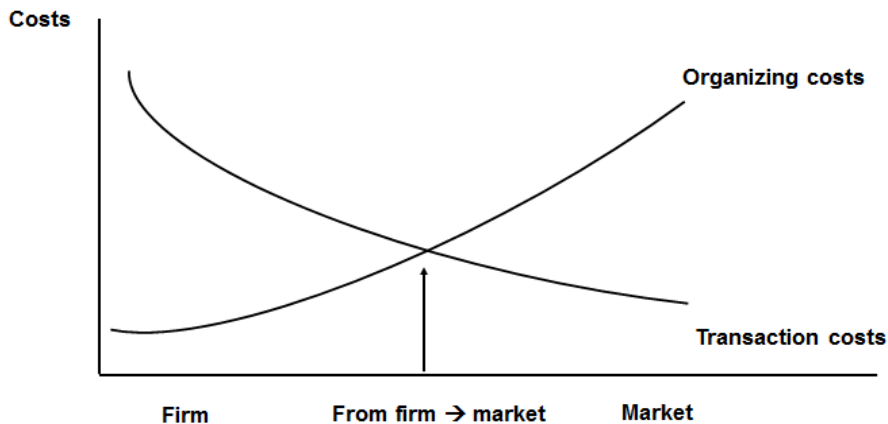


Figure 2.9: When the organization costs become higher than the transaction costs, hierarchies can become markets.

As mentioned before, one of the key questions in philosophy is: 'Why is there something instead of nothing?' Why is there management instead of no management? Management is there because, as we saw before, more than hundred years ago farmers, crafts-men and housewives started to work in factories. These 'fickle' people had to fit within the rhythm of the assembly line. By inventing management, with time measurements, labor analysis, permanent jobs, hierarchies, rewarding and punishing, this became possible. After more than hundred years this management invention evolved into a system with more disadvantages than benefits as the organization costs are rising because we make super specialists more efficient but we 'forget' to coordinate them in a modern and efficient way.

Because of super specialization and the natural movement towards more local efficiency instead of aggregated productivity improvement, more and more super specialists have to be coordinated, on a global scale, towards aggregated efficiency and finalization. No longer is traditional management the most productive way to

coordinate these specialists to activities, to connect supply and demand, but shared information and transaction networks. A research report⁵⁵ of the US based company ServiceNow shows that a lot of work that is done by humans is administrative, compartmentalized and routine work. ServiceNow commissioned the company Lawless Research to conduct a study about the state of work in companies with 500 or more employees. Lawless Research asked nearly thousand managers whether today's work processes are efficient or sap their productivity.

ServiceNow needed to quantify the productivity drain as a result of using email, phone calls and spreadsheets to perform routine administrative tasks within a company. The findings would prove the value for their products and services. So, this is not an independent and scientific research. ServiceNow is a market party with interests in the outcome. We as authors however believe that the results reflect the problem of productivity drain as the result of too many and poor coordinated and by technology facilitated intermediaries in the organization of work. Some highlights from the survey:

1. Managers say that they spend two days a week, on average, on administrative tasks as a result of internal transactions.
2. Nine in ten managers say their productivity depends on the efficiency of routine work processes provided by other departments, such as delivering marketing services, issuing purchase orders, and onboarding employees.
3. The majority of managers say these work processes are time consuming, and about half say they are frustrating.
4. Eight in ten companies, both large and small, still use unstructured manual tools such as email, spreadsheets and personal visits.
5. Four in five managers say that routine work processes cause significant delays.
6. Three in four managers want simple, self-service support processes, easy to use.

In our opinion, these results explain the rise of organization costs and therefore the interest of more organizations in a model that will boost productivity.

What if companies become markets (again)?

When you want people to work together, basically you can choose between two organizational structures: one based on hierarchy or one based on the market (a transaction sphere between at least two hierarchies). A firm is hierarchical power order: information is mostly bottom up and decision-making top down. The advantages are speed, clarity (one chain of command) and scalability. In this hierarchy, not everyone needs to know everything or have access to all information to find consensus or make decisions. Decisions are often made at the top and then scattered into the organization. A disadvantage of this structure is a *single point of failure*. This is the

⁵⁵ ServiceNow (www.servicenow.com) see report: 'Today's State of Work: the Productivity Drain' (April 2015).

opposite situation of fault tolerance systems that are at the basis of decentralized systems such as internet and distributed ledger systems.

In a market, decision-making takes place on a more decentralized basis and consensus is necessary between seller and buyer. All market participants need information to make decisions. The more suppliers and customers, the more data is potentially available in a market. Especially in markets with many actors, also called fat markets, who can share (if necessary on an anonymous basis) data, work will be much more efficient and better decisions can be made. We call these rich markets: many suppliers and customers with transparent data and algorithms. In this case, markets can perform better than hierarchies. The presence of a lot of data in a hierarchy does not necessarily mean that the company will make better decisions or function better because the data doesn't always reach every decision maker and people suffer from a limited cognitive capacity. When someone makes a bad decision in a market, the market as a whole will not suffer. But if the leader of a company makes a few stupid mistakes, the company will have to face the consequences.

Hierarchies and markets have both advantages and disadvantages in organizing data to decide on connecting capacities to activities. What we are looking for is a more hybrid form that reinforces the advantages and weakens the disadvantages. When we can model interactions within hierarchies as transactions and coordinate work as if it was a market, both will integrate into a network where supply and demand will connect in the most efficient way. Internet en blockchains are very scalable and humans very flexible. So, the basis of a new organization structure is structured flexibility, a system that is both scalable and flexible in itself. In addition to the mentioned article of Ronald Coase, 'The Nature of the Firm' and his key question, why do firms exist, we could ask ourselves the question now: *Why do firms still exist?* Why do we still use hierarchies and markets between them to organize trust, work and economies in an environment that requires something else? Just by asking these questions, you will become aware of the fact that most organizations still use an old-fashioned organization paradigm that is not suitable for the (digital) future.

As will be clear to many, and mentioned before, we are undergoing a digital transformation in which the real physical and digital world are increasingly converging into a single integrated modern way of living and working. More and more companies will become data companies. Not only data organization, data economy and data logistics play an important role, but also and especially digital leadership. We are developing from an industrial economy to a data economy. This digital transition has far-reaching consequences for existing organizations that are insufficiently (quickly) adjusting. When selling or insuring a house for example, the physical home is becoming less important and the safe storage and access to data about the house is becoming more important. These data must be complete, honest, reliable and transparent, and must take into account pragmatic, semantic and syntactic aspects. To insure a home, complete and reliable data is needed about home, environment and occupant. Selling a house means changing the digital connection 'A is owner of house

X', to 'B is owner of house X'. It is important that A proves that he/she is the owner of house X, house X can only be sold once (one digital transfer) and the digital ownership transport from A to B must be done safely. This can technically be done without land registry, notary and broker. Thanks to the rapidly developing of blockchain technology, more and more people can imagine that insuring or selling a house in the future is as simple as sending an email.

With regard to the organization of the supply and demand of data, many organizations suffer from certain choices from the past that were correct at that time, but have far-reaching consequences in the present. At the end of the last century, not the answer to a question was leading in automation, but the automation of internal processes. The known and so-called spaghetti IT-landscapes are the result. Companies and (especially) government (agencies) have to amortize hundreds of millions euro as a result of failed IT projects. Through an interweaving of large amounts of functions and internal interfaces, a large number of applications have arisen that also have to be accessed more and more frequently via the internet and external applications. Data interoperability is not realized by a joint ontology, taxonomy and semantics, but by API's, webservices and in the worst and in most cases by transferring data into their own systems. These are often Microsoft Office applications such as Word and Excel. IT management costs and therefore organizational costs increase while transaction costs decrease due to digitization. As we saw in figure 2.9: more and more organizations are coming to the point that hierarchies become markets because the organizational costs are higher than the transaction costs. If organizational costs continue to rise due to complexity, bureaucracy, distrust and fakework and the transaction costs continue to fall as a result of the digital transformation, hierarchies will become (again) markets.

As a result of the new European privacy law or general regulation, increasing pressure on the prevention of data breaches, fakenews and new technologies such as blockchain, supply and demand of data can be organized fundamentally differently. Not owning data becomes important, but access to it. The 'questioning' of another IT-system can be seen as a transaction, as well as receiving an answer. These transactions can then be validated on three points (are you the owner of the data, did you only send it once and has it been done securely). These validated transactions can then be stored on a non-mutually blockchain. Due to these developments, it is likely that data will increasingly be organized with people as a unit of analysis, design and development and no longer companies and governments. This also gives concepts such as Big Data, Machine Learning, Artificial Intelligence a different meaning because they assume that (a lot of) data is still present at central companies and governments, while this can also be decentralized to the people. And if people will not give access to their data, these concepts also take on a different meaning.

These developments will change the nature of most economies from an economy with factory DNA, firms and non-transparent markets to a rich data economy.

3. Blockchain as technology

For the purpose of this book we will clarify a number of important technical concepts within Blockchain Organizing. The intention of this is not to fully understand or fathom the technology but to give you a better idea what the possible organizational consequences of blockchain technology for management, your own organization or business could be.

Many people judge Bitcoin or blockchain technology on what it currently is. But if we had left internet for what it was⁵⁶, we would now only use it to send e-mails. We would not have explored the almost infinite possibilities that internet now offers us, in more detail. Blockchain technology and Blockchain Organizing are also far from being finished, we are just at the beginning. At the end of the second decade of the twenty-first century, blockchain technology, you could say is just in its early childhood. If we would limit blockchain technology to Bitcoin, then you could say it is finished, because Bitcoin has been in production since 2009 and the protocol is still running. It is not running with its initial code because we have witnessed several forks such as the one in August 2017. This fork has led to a split of the coin into Bitcoin (Core) and Bitcoin Cash. There are always new challenges and problems to solve, for example: hacks at market places, transaction speed, scalability, size of the database, centralization of mining pools, regulations etcetera. The technology, legislation as well as the development of ideas about what can and cannot be done, are far from finalized. In other words: blockchain refuses to be what it is. It is not a concept that does the thinking for you, not even a way of thinking. It is rather a concept that makes you think.

Bitcoin is a revival and further tightening of the cryptocurrency concept that was first described in 1998 by Wei Dai on a cypherpunk mailing list⁵⁷. The idea of Wei Dai, called B-money, was to develop an anonymous, peer-to-peer electronic payment system without intervention of a central bank. Organizing trust, that is now provided by banks and other intermediaries could be replaced by cryptography and software. This idea was later picked up and further developed into a working product: Bitcoin. The creator of Bitcoin, Satoshi Nakamoto, refers in his work to B-Money and he was in e-mail contact with Wei Dai⁵⁸.

⁵⁶ From 1969: An efficient and reliable way to enable universities that work with different computer systems for Defence/ARPA, to communicate with each other. It was chosen to cut up the little parcels that found the best way to the receiver via a distributed network and which are then assembled again into a message. This way the network is less dependent on the failure of nodes (single point of failure and fault tolerance systems. See also ARPA project in Weconomics practice book, paragraph 5.3.5: 'Free and open source'.

⁵⁷ A cypherpunk is an activist who uses cryptography and privacy improving technology for social and political changes.

⁵⁸ See for example the book by Paul Vigna and Michael J. Casey: 'The Age of Cryptocurrency, How Bitcoin and Digital Money Are Challenging the Global Economic Order' (2015).

Bitcoin is mainly a form of cryptocurrency, a form of value that no longer just includes money but also software code or programmable money. For example, the government can stipulate in advance that housing benefits can only be spent on rent. Bitcoin does not have any central authority over this, such as central banks have over fiat currencies. However, Bitcoin has the disadvantage that it is volatile at the moment and (exchange) transaction and savings (assets) are not guaranteed. The technology behind Bitcoin belongs to nobody, similar to technology to send an e-mail or the format language to develop a website. They do not belong to a certain company or a certain government. The software that makes it possible to pay in bitcoins is open source and is further developed and managed by Bitcoin communities, of which Bitcoin Core currently is the most important one⁵⁹.

Bitcoin (and its forks) can mainly be seen as proof that blockchain technology works long-term and stable in the practice of digital money. It is the longest running and probably most used application of blockchain technology as we know it today. But Bitcoin is definitely not the only application of blockchain technology and Bitcoin and blockchain are not the same. Blockchain technology is much broader and is used for instance by Ethereum for the development of smart contracts, IOTA for the communication between machines, Stellar for cross border payments or Cardano for a more scientific and research driven approach. Bitcoin is mainly a blockchain application for digital money. And money, whether digital or not, also has specific disadvantages such as speculation, calculating interest and saving.

Money is a tool ('intermediary' or kind of 'catalyst') to exchange two values. You can sell your house to someone and exchange this value for a lifetime of daily grocery shopping. The new owner of your house ensures that you no longer have to do any grocery shopping for the rest of your life. You probably will not go for this option but demand money for the sale of your house. Then you have more choices against which value you will sell your house. With the proceeds, you can buy more than just groceries. Money is then a claim on buying other values in the future. It is the intention of a central bank to keep this claim as stable as possible. However, you do trust that you can buy a similar value with the money you receive. Traditionally, money as currency, as medium of exchange, has a number of functions such as calculation unit, means of payment and savings/investments. We see these functions with digital money, as well as the disadvantages such as speculation, money laundering and decoupling from real values. An advantage of crypto currency is that people can spend it themselves (for example within own communities) and exchange it again for fiat currencies such as dollars and euros on exchange markets⁶⁰. Furthermore, you can also program digital money and allocate it in advance.

⁵⁹ See also website bitcoincore.org

⁶⁰ It is still debated if this is legally allowed, but how do you define a value?

3.1 History

You may know the metaphor of the dwarf sitting on the shoulders of a giant. Isaac Newton formulated it as follows: *"If I have seen further it is by standing on the shoulders of giants."* Each new invention builds on the strength of previous inventions. An invention does not just drop from the sky. We see more and we look further than our predecessors, because we have more history and knowledge to build on. The 'invention' of Bitcoin and the blockchain technology did not fall from the sky either. Satoshi Nakamoto continued to build on the knowledge and expertise of others. You can see blockchain technology as a puzzle that has been put together by Nakamoto and was 'finished' by adding the missing 'pieces of the puzzle'. As an example: Nakamoto has solved the long outstanding mathematic Byzantine Generals problem, partly due to inexpensive computing capacity, storage capacity and internet connections. In this paragraph, we have a closer look at the 'invention' of Bitcoin. For the purpose of this book we will provide a simplified and non-technical overview of the origin of blockchain technology, with special attention to the following 'pieces of the puzzle' (with in brackets the period/from time in which you can place these pieces of the puzzle):

- Cryptography (3000 B.C.)
- Database technology (1962)
- TCP/IP Internet Network Protocol (1974)
- Cipher Block Chaining (1976)
- Distributed systems (1978)
- Byzantine Generals Problem (1982)
- Commercial internet (1995)
- Digital currency and cryptocurrency (1998)

Cryptography

The use of cryptography, hidden or secret language, goes back to ancient Egypt. It is a technology to encrypt messages by replacing letters with numbers for example. With each transport of value (so also of messages) we are dealing with a sender, receiver and medium. When a message is not encrypted, a third party, not being the sender or receiver, who has access to the sender, receiver or medium, could read the message with reasonable efforts. By encrypting, the chance this will happen is reduced. Cryptography especially took a huge flight during the Second World War, partly thanks to the efforts of Alan Turing⁶¹. Nowadays most e-mails are encrypted by default to prevent abuse. Currently quantum cryptography is the latest development in terms of cryptography. Quantum cryptography is used, amongst others, to exchange keys which can later be used to encrypt and decrypt messages. It can be detected if a connection is being drained. But quantum cryptography is also sensitive to hackers, albeit with a lot of effort⁶². Cryptography is an area whereby, after a certain period, a new method is undermined by an action, that makes it possible to

⁶¹ See also Weconomics analysis book, paragraph 6.4: 'Internet in historical perspective'.

⁶² See for example article by Stephen McGann 'Technology Reporter van BBC News: 'Tricking the perfect code machine' (13-08-2011).

intercept and decrypt messages. It is an action-reaction ‘game’. Absolute security will never be achieved. But what really matters is security that cannot be cracked at an acceptable cost. In addition, it is interesting to investigate if you can remove the reason for a hack for instance. If a thief does not have a customer to sell to, he probably will not steal it in the first place.

Database technology

The database principles that we use today, also for decentralized databases, largely stem from the 1960s. The Oxford English Dictionary reports that the term ‘database’, was probably first used in 1962 in a report of the company: ‘System Development Corporation’ in California. A database can be seen as a storage tool for data (attributes and their values). It can be compared to a warehouse where goods are stored and can be retrieved from. The database that is most used by managers is probably ‘the excel sheet’. To create a database, you must develop a database model from an ontology and taxonomy and then a database. Within the blockchain infrastructure, these form the core for carrying out meaningful, reliable and secure transactions, checking and recording of data.

Blockchain technology will not change much to the (im) possibilities of a database. The organizing, storing and retrieving data from storage is also possible via a distributed database, whereby the participating nodes in the network store attributes in their own database via consensus rules. It does not matter for the storage whether a database is organized centrally or locally. The basis and technical principles remain comparable, the governance model is slightly different.

So, you can compare the basis of a database with a warehouse to store products that are in a box, in a (horizontal) rack. You can compare the box with a cell in a spreadsheet and the product with the content/value of a cell. An empty rack in a warehouse can be compared with an empty row in a spreadsheet. By placing boxes on a rack, the cells in a row in a spreadsheet are filled. So, the rack contains products that belong for instance to one seller, buyer, employee, consumer or citizen. By organizing horizontal racks this way, you can put the same products (attributes) in the same column (comparable with a column in a spreadsheet). There are three main processes in a database/warehouse. You can: bring an attribute/product into storage, keep it in storage and retrieve it from storage. There can be an operation between ‘retrieving from storage’ and ‘bringing into storage’ (again) that changes the value of an attribute, sets it to zero or empties it. You can take a box from the warehouse and attach a label with the sell-by date on all products and subsequently put them back again. In terms of a database, there are four main functions, you can:

- create and store data for the first time
- retrieve data
- delete data
- maintain data: adjusting the value of an attribute (update = retrieve + delete + create)

In addition to performing operations on an existing database, the design and maintenance of the database structure is also important. This means adding an attribute (like adding a column to a spreadsheet), cancelling an attribute (deleting a column in a spreadsheet) or replacing an attribute by another attribute (deleting column and adding a new one). You can compare this with adding or removing racks in a warehouse.

The important requirements for a data model and database are:

- data may not be lost;
- data may only be accessible to those for whom it is intended (correct rights and range for a certain role);
- it must be easy to write (store), change (mutate), retrieve (read) and remove (delete) data, without affecting the operation of the entire database;
- the integrity (accuracy, timeliness and completeness) of the data must be assured (via reference keys or an unique identification reference must be made to the correct position in a table);
- the consistency (coherent and free of contradiction) between attribute, entity, record and table must be assured;
- data may not be corrupted (by implementing adjustments, contamination of the database, double data).

These requirements do not have to be related to the way that records are stored. These requirements must be met with both central as well as local storage. But the perspective we have when designing databases will change. When the first databases were designed in the early 1960s our society and way of organizing processes was still in the 'Industrial Revolution phase'. The firm, the government and the bank were the thinking units for the organization of supply and demand and with that the organization of databases, trust and work. These entities had to keep records of employees, customers and citizens. Because this perspective is changing, the functional and technical need for data models and physical databases will also change. An important development for instance is that people themselves, and no longer companies, keep personal data to perform a process. The individual keeps a file and provides data (Bring Your Own Data) when a third party requires this for the performance of a process that the individual has requested.

With blockchain we move to a kind of network of databases that keep each other informed, controlled and updated. A network that (technically) can be shared with everyone and where everyone can connect to, at least where it concerns a public or permissionless blockchain. But locally organized databases are relatively expensive compared to centrally organized databases. So, there must be an advantage to switch. A local architecture offers mainly advantages when we no longer design with the firm as thinking unit, but consider the process and separate actors in a business-ecosystem. Processes will less and less take place within organizations and transform to processes between organizations (bilaterally, 1:1 and via markets 1:n) and eventually processes within shared transaction networks (n:n). With the use of more local but indeed (partly) synchronized databases, actors in the network can perform better and

check each other to accomplish a shared reality as basis for trust. Checks on whether the sender has correctly claimed the value or if this value is not spent twice and the transaction took place in a secure way. Being able to perform checks yourself means that trust can be organized in a different way. Therefore, we require fewer intermediaries. The advantage of fewer intermediaries is that the complexity does not increase so the transaction costs remain low and the productivity increases.

So, when you create a blockchain design you must consider the basic elements, principles and (im) possibilities of a database. Processing a blockchain transaction can in its core be reduced to retrieving, processing and replacing (the values of) attributes from/in a database. The important difference with the (desired) functionality of not blockchain databases in relation to a blockchain database is that once a value is in the database it cannot be mutated or deleted. Once a transaction took place and the network agreed on this event, this event cannot be altered or delete. On a blockchain you can't unscramble scrambled eggs.

TCP/IP Internet Network Protocol

In 1974⁶³ Vint Cerf and Robert Kahn designed something that later turned out to be revolutionary: the TCP/IP protocol. When someone asks you: How are you doing? You usually answer with: Yes great, and how are you? It is not written anywhere that you must give this answer, but it is some kind of communication protocol, a kind of automatism. If someone asks A, you reply with B. You can do this much more precisely in an instruction for a computer, so it does exactly what you want. The same holds true for sending and receiving messages via a 'telephone line'.

Within the TCP/IP protocol the IP address acts as the unique identification of the sender and receiver and the TCP protocol provides the connection, the secure transport between sender and receiver. The core of the methodology is 'package-switching': messages are cut up and sent from A to B in parts and via different routes (multiple simultaneous sessions). This method improves the efficiency and robustness (reliability) of the network. The 'invention' of the TCP/IP protocol might be the most important building block of what we now call internet. TCP offers the possibility to send reliable, structured and faultless bytes from A to B on an IP network. Widely used applications such as e-mail, internet browsers, communication platforms, administration packages in the cloud and file sharing applications rely on this protocol.

Cipher Block Chaining

The name blockchain probably first appeared with the block encoding technology also called: Block Cipher. It is a method to convert a piece of text (block) into a block

⁶³ See also publication of The Institute of Electrical and Electronic Engineers (IEEE): 'A Protocol for Packet Network Intercommunication' (May 1974).

of numbers via encryption and decryption. The easy and quick way to do this is via Electronic Code Book (ECB). A disadvantage of this is that a third person can easily add blocks and remove a message or blocks. When you connect blocks via a link with a previous block (like input reference with blockchain technology) this becomes more difficult. A new block that has to be added then becomes dependent on the previous blocks from the chain. We call this method: Cipher Block Chaining (CBC).

Distributed systems

Since the 1960s the network architecture has gone through a number of phases; mainframe, client-server, web/cloud and now decentralized and distributed computing. Distributed computing is more and more possible due to falling prices for hardware and connections, increasing computing power and virtualization. Distributed computing is a technique that is the foundation for applications such as Dropbox and Google Drive whereby you can work on a shared document simultaneously, with several people, from different computers. It is a technique whereby an operation is not performed by one computer but by a group of computers that are connected to each other and find consensus about the last change in a document (also a sequencing challenge just as blockchain).

This technique was already developed by Leslie Lamport⁶⁴ in the 1970s. He developed a technique to describe the behavior of systems. This technique is also called the Lamport clock. It is a logical clock that ensures that the clocks of the participating computers logically coincide. Therefore, it is possible to formalize 'previous actions' in a system and determine the sequence of events. Initially the purpose of distributed computing was primarily the sharing of computing capacity. When sharing local computing capacity each computer has much more computing capacity and relatively little data has to be sent via the connection. Nowadays this technique can be used to synchronize distributed accounting systems. A number of advantages of this technique are:

- less dependency on a central server that can fail;
- because several users produce and consume data together a more democratic governance structure can be implemented;
- nodes that have surplus computing or storage capacity can make this available for nodes that have a shortage or for commonality purposes.

Apart from the logical clock, Lamport also made a great contribution to finding consensus within distributed systems. How do you determine the sequence in which events take place within all participating computers? How can you make a decision if not all computers are switched on and participate in the decision-making process at the same time? Lamport has described this mathematical problem based on an example of a parliament with part-time participants. This example is known as 'The

⁶⁴ Leslie Lamport (1942 -) American computer scientist and winner of the prestigious Turing Award in 2013. He worked for DEC/Compaq and now Microsoft Research.

Part-Time Parliament problem⁶⁵, or ‘The Paxos Parliament protocol’. Currently distributed computing is used for projects that require a large amount of computing capacity and whereby a lot of data must be processed. Examples are projects in the field of climate change, genome and space research.

Byzantine Generals Problem

The Byzantine Generals Problem refers to a mathematical problem. This problem is described in a paper with among others the aforementioned Leslie Lamport⁶⁶. The authors make a comparison between the way that decisions are made in the army and the decision making process used in the control of computer (programs) for the performance of a joint task. Computers (or networks) depend on malfunctioning parts when processing data (also called fault-tolerant systems). Parts that do not function as intended, for example due to a power failure. When computers in a network asynchronously cooperate, the result depends on the functioning of, and the relationship between, the different nodes. The different nodes can give conflicting messages to each other. A value can be issued twice till another node discovers that there is no input reference for the issue of a value. The Byzantine Generals Problem is an important challenge for blockchain technology to solve. In order to apply blockchain technology, nodes must find consensus about the transaction sequence in a distributed network.

The Byzantine Generals Problem in brief means that armies must be able to trust each other before they attack a fortified city or not, in the simplest variant. Hereby it is important that there is consensus about the decisions to be made or about the plan of attack. After all, you will not attack a city if the enemy is too strong, or your fellow army is too weak or consists of traitors who want to confuse others. Coordination between the attackers (‘supply side’) and between attackers and city (‘supply’ and ‘demand side’) is important. Only a coordinated attack can be successful. The assumption for a solution to this problem is that all loyal generals must come to the same plan and the not loyal generals or traitors count for a minority. In Byzantine times there was no secure internet, telephone or telegraph. Generals had to communicate via messengers. These moved between the armies. But a messenger can be intercepted and transfer an unwanted message. Another attacker or the defender can intercept a messenger and change the message (defraud). Then the question is: how can you trust each other when you cannot trust the message on the basis of which you have to make a decision?

Let’s assume that the messengers used written messages. A message has a unique number and the message shows whether or not it has been opened (for example with the use of a seal). With written and non-mutable messages, the Byzantine Generals problem can be solved with any number of generals (nodes) and a minority of

⁶⁵ See also paper by Leslie Lamport: ‘The Part-Time Parliament’. This article appeared in the magazine ‘ACM Transactions on Computer Systems’(May 1998).

⁶⁶ Together with Robert Shostak and Marshall Pease he wrote the paper: ‘The Byzantine Generals Problem’ (1982).

possible traitors (fraudulent nodes). This situation is comparable with the Bitcoin network. The Byzantine Generals problem can be solved with the blockchain algorithm or protocol. The basis of this protocol is to check if the sender has the value: did a general send a certain message or not and was he the right owner of that message? In addition, it is checked if a general has sent a message more than once, for example by giving a message a unique number. The challenge is to find an algorithm to make sure that the generals can come to a consensus, to shared decision (based on a shared reality). Satoshi Nakamoto took on this challenge and solved it with the use of the concept of Proof of Work. If you want to defraud a message, for example by spending a value more than once, you must have more computing capacity (Proof of Work) than half of the (miner) network. This is almost impossible without centralization.

Free and public internet

An important condition for blockchain technology is the existence of the free and public internet. An important condition is not only the use of standards such as TCP/IP, but also a certain governance structure that guarantees free access. In general Tim Berners-Lee and Robert Cailliau are seen as the founders of the World Wide Web. It is important to mention that internet, as technology (ARPANET), already existed much longer than the commercial use of internet and the World Wide Web. In 1995 the National Science Foundation Network (NSFnet) was dismantled and internet was fully released for commercial purposes. Till 1995 NSFnet was the backbone of internet and was mainly used by researchers and scientists.

Digital currency

Digital currency is also called digital money, but it is not the same⁶⁷. A currency is a medium of exchange and money is a form thereof. Digital money distinguishes itself from physical money because it is programmable. Digital money can be transported from sender to receiver without the intervention of intermediaries. You can compare this with the sending of an e-mail, from sender to receiver. Digital money can be managed both centrally as well as locally, regulated and unregulated. Decentralized, unregulated digital money is often seen as virtual money. In 2012 the European Central Bank defined virtual money as follows⁶⁸: *“A type of unregulated, digital money, which is issued and usually controlled by its developers, and used and accepted among the members of a specific virtual community.”*

The ECB presents the following schedule to clarify the distinction between the different currencies.

The Money Matrix	Money format	
	Physical	Digital
		None crypto Crypto

⁶⁷ Money is a form of currency.

⁶⁸ See also page 5 from a paper of the ECB: 'Virtual Currency Schemes', (October 2012).

Legal status	Unregulated	Centralized			
		Decentralized			Bitcoin
	Regulated				

Table 3.1: The Money Matrix of the ECB

Besides the aforementioned Wei Dai, the American computer scientist Nick Szabo was one of the first people to be involved with digital contracts and digital currency. In 1998 he already designed a mechanism for decentralized digital currency what he called ‘Bit Gold’. Bit Gold was never implemented as Bitcoin was at the beginning of 2009. However, Bit Gold is seen as the direct predecessor of Bitcoin and there is also speculation that Nick Szabo Satoshi is Nakamoto, but he denies that⁶⁹. However, he has indicated that there are many similarities between Bit Gold and Bitcoin.

So far, a more detailed explanation of the creation of digital money and Bitcoin and the building blocks that serves as basis for blockchain technology. Now we continue with a more detailed explanation of the Bitcoin blockchain. As stated previously, there are several (types of) blockchains. The Bitcoin Core (BTC) blockchain is one of them.

3.2 The Bitcoin blockchain

As explained before, the story of Bitcoin started when Satoshi Nakamoto sent a message to the Cryptography Mailing List. The message was sent on 31 October 2008, with as subject: ‘Bitcoin P2P e-cash paper’. This message included a link⁷⁰ to a pdf (white paper). Figure 3.1 shows a copy of this message⁷¹.

⁶⁹ See for example article by Martin O’Leary in The Huffington Post: ‘The Mysterious Disappearance of Satoshi Nakamoto, Founder & Creator of Bitcoin’ (5-5-2016).

⁷⁰ See also website: bitcoin.org/bitcoin.pdf

⁷¹ For a list with e-mails and postings, see also the: Satoshi Nakamoto Institute.

From: Satoshi Nakamoto <satoshi@vistomail.com>
 Subject: **Bitcoin P2P e-cash paper**
 Newsgroups: **gmane.comp.encryption.general**
 Date: 2008-10-31 18:10:00 GMT (4 years, 52 weeks, 1 day, 3 hours and 23 minutes ago)

I've been working on a new electronic cash system that's fully peer-to-peer, with no trusted third party.

The paper is available at:
<http://www.bitcoin.org/bitcoin.pdf>

The main properties:
 Double-spending is prevented with a peer-to-peer network.
 No mint or other trusted parties.
 Participants can be anonymous.
 New coins are made from Hashcash style proof-of-work.
 The proof-of-work for new coin generation also powers the network to prevent double-spending.

Bitcoin: A Peer-to-Peer Electronic Cash System

Abstract. A purely peer-to-peer version of electronic cash would allow online payments to be sent directly from one party to another without the burdens of going through a financial institution. Digital signatures provide part of the solution, but the main benefits are lost if a trusted party is still required to prevent double-spending. We propose a solution to the double-spending problem using a peer-to-peer network. The network timestamps transactions by hashing them into an ongoing chain of hash-based proof-of-work, forming a record that cannot be changed without redoing the proof-of-work. The longest chain not only serves as proof of the sequence of events witnessed, but proof that it came from the largest pool of CPU power. As long as honest nodes control the most CPU power on the network, they can generate the longest chain and outpace any attackers. The network itself requires minimal structure. Messages are broadcasted on a best effort basis, and nodes can leave and rejoin the network at will, accepting the longest proof-of-work chain as proof of what happened while they were gone.

Figure 3.1: Message from Satoshi Nakamoto to a cypherpunk forum

On 9 January 2009 Nakamoto sent the message that the first bitcoin was released and invited others to participate in the network:

"If you can keep a node running that accepts incoming connections, you'll really be helping the network a lot. Port 8333 on your firewall needs to be open to receive incoming connections."

When we as authors are invited for a keynote for instance, we are often asked to name specific examples of blockchain applications for a certain industry, for a certain context, for example the recruitment industry, financial sector or government domain. But blockchain basically doesn't see any context. Blockchain is industry agnostic you could say. For a blockchain it doesn't matter if the transaction concerns a bitcoin, hundred bitcoin, this book, a house or a 13.7 billion aircraft carrier. However, as soon as we start to use Blockchain Organizing from the paradigm of a certain domain, we immediately miss the essence of Nakamoto's intention. In one of his blogs Nakamoto already wrote that he wants to be able to accept every form of value and every transaction (not just bitcoin): "...*The nature of Bitcoin is such that once version 0.1 was released, the core design was set in stone for the rest of its lifetime. Because of that, I wanted to design it to support **every possible transaction** type I could think of. The problem was, each thing required special support code and data fields whether it was used or not, and only covered one special case at a time. It would have been an explosion of special cases. The solution was script, which generalizes the problem so transacting parties can describe their transaction as a predicate that the node network evaluates. The nodes only need to understand the transaction to the extent of evaluating whether the sender's⁷² conditions are met.*"

⁷² Please note, just like with the Weconomics model, Nakamoto puts the sender in the centre (in the sense of: initiator of a transaction). So, the individual who makes the transaction, for example ordering a book at Bol.com, is at the centre and not the order procedure of Bol.com.

Bitcoin as money

Bitcoin is a number in an accounting system (ledger) what at some point represents a certain value (has a number) a certain trust. This value, this trust, is only beneficial to you if other participants in your community will do the same as you do. The numbers get practical value if someone wants to exchange bitcoins for bread, a car or a dollar and believes that others will act accordingly. Cryptographic digital money, like bitcoin, will only become a generally accepted medium of exchange (currency) if others trust that they as well can use it as a medium of exchange, as a claim on future value exchanges. Bitcoins and numbers in a ledger only have value, like the euro and the dollar, because you and many others believe that they have value. When the trust in a coin or a number in an accounting system disappears, you cannot exchange the value that is recorded in the books. When the trust is reduced, the medium of exchange will lose its value. Bitcoin is currently not supported by a government or a central bank, but by a protocol that ensures that a certain amount of bitcoins can come into circulation. It is supported by a community with members who believe that the value of bitcoins will remain constant or increase. This of course attracts speculators and in this respect, it does not differ from regulated currency. We also speculate with the dollar and euro.

However fiduciary currencies (fiat currencies; see also table 3.1: The Money Matrix of the ECB), such as the dollar or euro, are indeed supported by central banks. There are regulations and instruments such as monetary policy, including quantitative easing for example, to maintain or change inflation and the value of, or confidence in, the euro or dollar. In general you can say that money, whether it is digital or not, has the purpose to reduce the insecurity when transactions take place. The higher the trust level is, the lower the insecurity. But besides the advantage that money offers as a medium of exchange, money also has other functions that often involve disadvantages such as accumulation, charging interest, derivatives and speculation. Fiduciary currencies, like the euro, are no longer supported by real values such as gold or the gross domestic product. For more information about money and the role of money in our economy and society we refer to the Weconomics analysis book, paragraph 5.4: 'Economy and the role of money'.

In order to send and receive bitcoins you install a software program on your computer. Bitcoin Core is the original program but now there are also variants at the client side. The software program contains a wallet, what is actually an asymmetrical key pair. The client software is some kind of key holder with which you can 'sign' your transactions from your wallet. The private key (call it a pin code) provides access to the wallet, the public key (call it your account number) is the address of the wallet. In between there is also a digital signature (call it digital fingerprint). A digital signature works by using these two different keys that are connected to each other: the private key for creating the signature and a public key for having the transaction checked by others. You can see the private key as a real password, the signature as proof that you have the password without having to reveal the identity of the sender.

Wallet software

Not everyone with a wallet needs to assist in the maintenance of the network. A (Bitcoin) wallet has two asymmetric keys: a public key and a private key. A wallet is a piece of software that you can download, say on your smartphone. Using this software you can make payments and check your balance. Wallet software sends a transaction from the sender to the receiver. Subsequently this transaction is (if it complies with the rules) included in the blockchain during the next acceptance moment. Acceptance is done by miners. You can compare wallet software with an App for internet banking. In the core it is client software to send and receive payments and to check the balance. There are several providers of wallet software. After you have downloaded the wallet software, you can also create Bitcoin addresses of your friends. You can compare this with the address book used in internet banking or your e-mail address book. The sending and receiving of bitcoins is similar to sending and receiving e-mails. Paying and receiving Bitcoins is even possible by holding two phones together with NFC technology⁷³. The 'sent to' address can be scanned via a QR code. In figure 3.2 you can see an example of Bitcoin wallet:

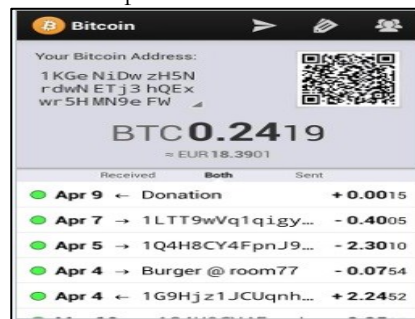


Figure 3.2: Example of a Bitcoin wallet

A distinction can be made between offline wallets (for example on your PC), that can be used to download the complete database, and online wallet software (also called eWallet). With the latter, you do not have to download any software but you are dependent on the supplier. You do not have the wallet in your hands and when the supplier goes bankrupt or has bad intentions, you can lose your bitcoins.

Mining

Mining is finding consensus on the state of a distributed ledger. Nodes verify transactions (keys, signatures, input references and double spending) and find either consensus on it or not. Mining is also a way to arrange transactions in a certain chronological order in a distributed system (sequencing). A central database (of a bank for instance) has no problem with sequencing: first come first served. Sequencing on a distributed network is necessary because not all nodes receive a transaction at the same chronological time. To accept a transaction, you must wrap it in a block so to

⁷³ NFC, Near Field Communication: contactless communication which uses a NFC chip and has a range of about ten centimetres. This technology can be seen as an extension of the RFID technology (Radio Frequency Identification).

say, whereby, with a hash, a reference to the previous block is made, which creates a 'fixed' chain (blockchain). Strict and rigorous rules are applicable to this, based on mathematical proof. Mining is some kind of competitive lottery whereby you, as individual, almost never can guess the winning ticket. Not even when you buy a lot of tickets. Therefore, an individual node can never gain control over the network and for instance, issue an input reference twice. When a block is included in the longest (chain) branch a miner earns bitcoins (Proof of Work). Therefore, other miners check the offered block if everything is correct. In addition, they always check if they have the longest chain. If this is not the case, they automatically switch to the longest chain.

The protocol that ensures that only a limited number of bitcoins can be mined, mainly applies to rewarding the miners. It does not say a lot about the value of the bitcoin. After all, a bitcoin is programmable money that you can divide and exchange via exchange markets. A total of 21 million bitcoins can be made. It is estimated that the final bitcoin will be brought into circulation somewhere in the year 2140. You can divide each bitcoin again in 100,000,000 units (also called Satoshi's). Bitcoins can be divided up to 8 decimal places (0.000 000 01) and possibly even smaller units if this might be required in the future when the average transaction size reduces. It is not known what the value is or will be of this smallest part. This value is usually determined on exchange markets. You can compare these with (border) exchange offices.

Mining is like searching for bitcoins by solving a puzzle. The person who solves the puzzle first, receives a reward that gets smaller and smaller (in numbers). The reward is halved after every 210,000 mined blocks. Mining bitcoins, solving the puzzle, was initially done with a processor (CPU). The level of difficulty of the puzzles to be solved was much lower then. Nowadays mining is more and more done using the graphic card of your computer. Special cards have been developed for mining such as the ASIC (Application Specific Integrated Circuit). During mining not only a puzzle is solved but the transactions are also verified.

If you install bitcoin software to verify transactions, all transactions up to and including the first transaction, are downloaded on your computer. They are not just downloaded but also verified. Please remember that you execute transactions on a blockchain with strangers that you cannot trust. You must be able to check every transaction yourself in the copy of the ledger on your computer. Downloading and verifying the software can take up to twenty-four hours. But fortunately, you only have to do this once. On January 1st 2016, the entire file was: 53.7 GB, on January 1st 2017: 96.3 GB and on January 1st 2018: 149.3 GB.

Bitcoin network

As soon as a new node is added to the network, it is provided with some kind of address list to make contact with the other nodes⁷⁴. These become the 'nearest neighbors'. Later on, new addresses are 'discovered' from this list and added to the address list. You could compare this with LinkedIn. You start with the people that you know and have met personally (tier 1) and from tier 1 you start to build a tier 2 of people that you do not know personally (yet), etcetera. The address list of the Bitcoin network is expanded to create an unstructured peer-to-peer network, in which each node 'communicates' with a large number of other nodes. Communication is done in such a way (with mathematical substantiation and proof) that this makes the network sufficiently secure.

In addition to the protocol that is used to create transaction messages and to verify transactions and blocks, there must also be a 'system' to develop, offer, accept and process software for example. There is discussion about the size of the block for example. Not all Bitcoin nodes follow the same rules.

It is important to which layer (stack) a rule applies and to what extent this rule influences other layers. So, following different rules does not have to be a problem, as long as the entire system functions as intended. Not all developers will follow the same rules when implementing the rules. However, developments within Bitcoin Core⁷⁵ require community-consensus before adjustments to the rules can be implemented. This is important because other application layers depend on this. An example of such a rule is the size of the block. Within the Bitcoin network we can distinguish four layers:

1. Core with consensus rules about the level of difficulty of the puzzle that must be solved with the mining.
2. Peer-to-peer: these rules determine how full nodes share data and which data they share. It contains rules related to sending and receiving transactions, blocks and special data packages.
3. Application Programmable Interface (API) and Remote Procedure Call (RPC): this is a communication layer between the peer-to-peer infrastructure and the application. For example, Bitcoin wallets and block explorers communicate with the network via this layer.
4. Application (wallet): contains rules about formats for addresses, generating private key and making backups of your wallet.

The consensus rules of the core are the most important rules of the Bitcoin network. They organize a number of important matters such as the block size and block

⁷⁴ See also article in Bitcoin Magazine by Aaron van Wirdum: 'Why Some Changes to Bitcoin Require Consensus: Bitcoin's 4 Layers' (26-2-2016).

⁷⁵ Bitcoin Core (formally Bitcoin-Qt) is currently the third Bitcoin client and co-developed by Wladimir van der Laan (from Eindhoven, The Netherlands) and based on the original work and the code of Satoshi Nakamoto. Wladimir is the successor of Gavin Andresen who resigned in 2014 as core maintainer. Andresen was the successor of Nakamoto.

reward, the level of difficulty for solving the puzzle (hash rate) and the type of Proof of Work that is required. When all nodes follow the same consensus rules, it guarantees that the same copy of a blockchain is maintained. When nodes accept other consensus rules, they could also accept other blocks that have been rejected by other nodes. This could create different versions of the same accounting system. As a consequence it does not provide a shared reality and opens the way to fraud.

From Bitcoin to other blockchains

So, we now understand the blockchain concept specifically in relation to Bitcoin and its underlying blockchain technology. Bitcoin is a form of crypto currency, a form of value that no longer includes physical or digital money but software code or programmable money. The Bitcoin Core community has no central authority like the European Central Bank (ECB) but it is a form of distributed authority with a shared reality in assets and transactions. Core of Bitcoin, the underlying blockchain technology, as well as Blockchain Organizing, is the transaction between sender and receiver, using coordinating capacity ($S \rightarrow C \rightarrow D$). Hereby value flows, via a transaction, from supply to demand. Algorithms and encryption is used to make the connection meaningful, secure and reliable. In fact, the coordinating capacity between demand and supply is programmed.

Bitcoin is regularly in the news, in a positive sense but often also in a negative sense. Just as it would not have been smart to assess the value of internet solely based on e-mail as a replacement of the letter, it will not be smart to judge the power of blockchain technology based on Bitcoin only. But in our opinion, it would be a good development when more people invest time in creating blockchain communities instead of spending time on the speculation what the Bitcoin price will do. Instead of creating jobs in trading, or worse speculating on something that has to be developed, it's better to invest time in the design and development itself.

Bitcoin is one of the many manifestations of blockchain technology and certainly of Blockchain Organizing. Bitcoin is proof that blockchain technology can work long-term, is stable and useful. Bitcoin shows that blockchain technology technically works well and cannot be easily adjusted involuntarily (at acceptable costs). To evaluate if blockchain can do something for your organization it is important not only to appraise Bitcoin but also to study the underlying blockchain technology. To assess Blockchain Organizing it is important to look at the underlying organizational model of the Weconomics Foundation.

So far, a more detailed explanation of the Bitcoin (BTC) blockchain. Now we continue with an explanation of the different types of blockchains that are currently available.

3.3 Different types of blockchains

Many innovations and their practical applications are similar to their predecessors. The first television broadcasts were mainly a copy of the radio broadcasts (for example: radio plays or classical concerts with images). The underlying technology was of course completely different. To embrace new technology it is smart that it is similar to what we are used to. For example, the first websites were mainly copies of paper brochures. A skeuomorph is something that seems functional because it looks like something that we already know. Most use cases or Proof of Concept projects we see in our consultancy practice come from existing organizations and contain existing processes. And therein lies a great danger. Blockchain is *not business as usual, disguised as innovation*. Blockchain is not meant to digitize existing processes. It is not a change within a system. It is mainly a change of the system. The system that especially will change is the firm as superior form to organize work and transactions. But when you invest too long and too much in a new technology within an existing system it will lead a life of its own. This happened with the development of internal communication systems based on the TCP/IP protocol in the nineties.

Because old thinking models and interests prove to be persistent, this currently leads to the development of all kinds of flavors of blockchain. We see an explosion of use cases and everybody wants to be first and unique and this is where it is already going wrong. Digital and blockchain are industry agnostic. A blockchain doesn't care which value is in 'the box' that it transport from A tot B.

Basically, there are two extremes: a variant that is closest to the current way of organizing (permissioned blockchain) and a variant that is the furthest away (permissionless). We mainly believe in the form in between: blockchain for communities or business-ecosystems. Although blockchain has a lot of standardization in 'itself', there is no standard yet, like the TCP/IP protocol for internet. Time will obviously tell which flavors will remain, whether or not there will be a good (good for the people) standard and how this will look like. Till then it is important to understand the thoughts and drivers behind these various flavors and developments.

We, as authors of this book, are not neutral when it comes to what flavor a blockchain should have or get. We combine blockchain technology with a, although for many new, but existing and proven organizational model of the Weconomics Foundation. We believe that the principles of Weconomics correspond to the principles as described in the white paper of Nakamoto. But Weconomics is not value-free. It is a community that wants to use organizational technology to create a surplus to make our welfare more sustainable. You the reader, might not want sustainable welfare but only more welfare? The Weconomics Foundation uses certain values, intents, perspectives, visions and ambitions. So, we also supply a certain blockchain flavor, but we try to 'cook' with widely supported organizational principles as much as possible and to stay as close as possible to the original organizational principles that we believe are meant by Nakamoto and his/her predecessors.

According to us, Blockchain Organizing can be best applied within partnerships, ecosystems, smart networks, shared transaction networks, also called communities. It makes little sense when a bank, insurance company, government or webshop starts its own private blockchain. Then we must exchange values between blockchains. In essence Blockchain Organizing is meant for peer-to-peer transactions within value chains and communities. Based on this insight you can distinguish the following blockchains:

- B4B: Blockchain For Business (permissioned)
- B4P: Blockchain For People (permissionless)
- B4C: Blockchain for Communities (hybrid)

An important difference in flavor is the difference between private (permissioned) and public (permissionless) blockchains. We believe that combinations of these two extremes are also possible. Usually we call these combinations: hybrid blockchains. We see these three flavors on the internet as well: open internet, login internet and intranet. According to us, there are essentially two main flavors: a decentral and a central flavor. The central flavor (top-down) is mainly fed by existing organizations with old DNA, old conditionings and the idea that we require certain institutions to organize trust. The decentral flavor (bottom-up) is expressed in the original idea from Nakamoto. Central or old thinking is not necessarily wrong, but when you continue too long with old thinking in a new (digital) context, things could go wrong. We have seen this with companies like Eastman Kodak or Blockbuster. They probably went bankrupt because they continued too long with old thinking in an environment that digitizes more and more. The tragedy with Eastman Kodak is that they invented the change that eventually killed it.

It seems almost inevitable, but our constant tendency towards centralism, which we can also see in the Bitcoin blockchain, property and shielding is widely present, driven by evolution and it is difficult to stop. It goes back to our ancient survival instinct. After all, you have collected something or found a fertile plot of land. This gives you freedom, welfare and a higher survival rate and you want to secure this. Everything that subsequently affects the status quo, you rather want to exclude. But it is like Leon Megginson⁷⁶ formulated it:

*"It is not the strongest of the species that survives, nor the most intelligent,
but rather the one most adaptable to change."*

You usually have gone to great lengths to acquire or experience what you currently possess. Therefore, we will defend the existing with all we have against 'robbers', competitors or people who want to share. After all, much energy and money were spent on collecting the existing. Of course, you will not give that up easily. The whole

⁷⁶ Leon Megginson, professor at the Louisiana State University tried to summarize Darwin's central idea in the book 'On the Origin of Species', during a speech for the Southwestern Social Science Association and came to this formulation (1963).

idea of ‘my property’ and of ‘wanting to control’ is the foundation for this. Because if you have more (goods, knowledge, food) than someone else, theoretically you have a higher survival rate. But apart from this pattern, we are also social animals and our survival rate and standard of living increases when we cooperate. This might surprise you, but our ancient instincts and dispositions still play a role, also in the development of blockchain. Maybe they play an even more important role than many might expect or hope for. At this moment in any case you’ll see that the technique does not immediately change the behavior of people (speculation, opportunistic behavior, etcetera) regarding for example Initial Coin Offerings (ICO) and the development of crypto futures.

Before we further discuss the two main flavors of blockchain (B4B and B4P) and the hybrid form (B4C), first an addition to the question: ‘What is actually the difference between central, decentral and distributed?’ We will explain the difference between these three forms in more detail in the following paragraph.

Central, decentral and distributed

As indicated before, blockchain technology can be used for maintaining a distributed ledger (divided across the participating nodes). However, when a ledger is technically distributed, it does not necessarily mean that power is also distributed. Everyone can buy bitcoins, but not everyone can open a bank account. A blockchain may be distributed in infrastructural terms, just like internet, but that does not mean that the power is also distributed. A widely used picture to display the difference between central, decentral and distributed networks is the depiction by Paul Baran⁷⁷. With his research, Paul Baran contributed to the design of internet. He suggested to make the communication network robust, (for example the failure of the network due to a nuclear attack), by working with packages that would be transported from sender to receiver via different nodes. At the receiver end, the different packages would ‘merge’ again. This principle is the basis for internet and therefore also for blockchain and is called ‘packet switching’, or package-switching (see also the TCP/IP protocol in paragraph 3.1).

⁷⁷ Paul Baran (1926-2011), originally a Polish engineer, but moved to the US later on. He studied packet-switched networks (Asynchronous Transfer Method) and worked for Rand Corporation, among others.

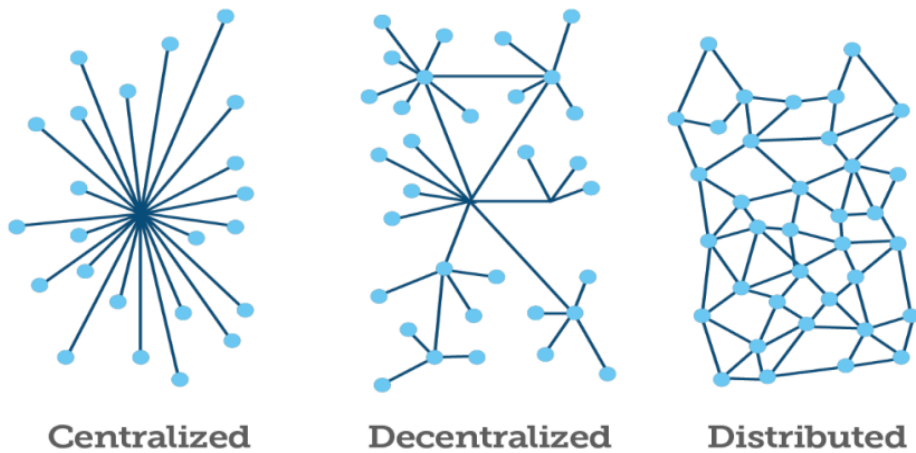


Figure 3.3: Paul Baran: difference between central, decentral and distributed network structure from infrastructural perspective

In figure 3.3 there are three different variants. But actually, there are four if you include the governance aspect, in addition to the technical aspect.

Four different forms

An extension of figure 3.3 does not only depict the technical aspect, but also the chosen governance model. As mentioned before, in addition to technology, especially the chosen organizational model and, within this the governance perspective, is important. What are old interests and legacies and how do these emerge in the new design? With a central protocol you can still have a lot of power in a peer-to-peer network. Examples are business platforms like Airbnb and Uber. Another and more comprehensive picture than figure 3.3, can be found in the blog of the company Maidsafe⁷⁸. This also includes the governance aspect, in addition to the technical aspect. We then have four variants:

1. (Fully) central
2. Distributed and within:
3. Decentral and within:
4. Peer-to-peer (fully decentral).

⁷⁸ See article by author Ioptio: 'Evolving Terminology with Evolved Technology: Decentralized versus Distributed' (4-12-2015).

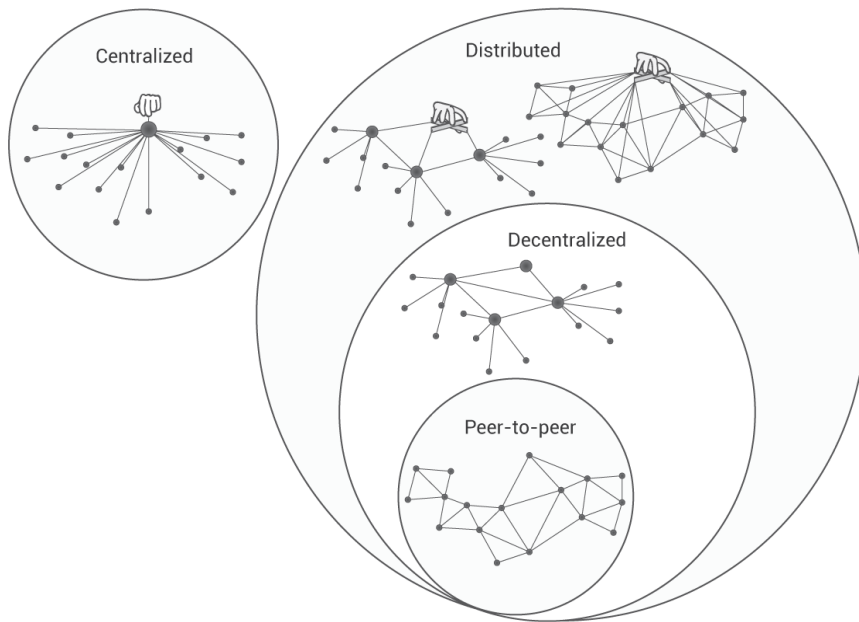


Figure 3.4: Four different networks from infrastructural and governance perspective

In Figure 3.4 you immediately recognize the difference between infrastructure and ‘being in control’. In other words: a distributed network (infrastructural) can indeed be controlled centrally (governance). An example of this is the SETI project⁷⁹, whereby thousand computers release a piece of computing power but are indeed controlled centrally. An important example of a decentralized blockchain, ‘without’ central stakeholder and with a peer-to-peer infrastructure, is the current Bitcoin network. Naturally you can call Nakamoto, who could mine the first bitcoin easily and therefore probably became ‘rich’, a stakeholder. But at this moment anyone who wants to join, can. Nobody will be excluded by a central authority. Whether you also have the opportunity and can invest enough to mine bitcoins is another story. But in the design, nobody and everyone has ‘the power’. Thousands of computers (nodes) assist to validate the transactions and to forward these to other nodes.

These are two important features of a technically distributed situation: validating and forwarding without exclusion. The so-called full nodes, guarantee the decentralized character of the network. For example, there must be ‘consensus’ in more than fifty percent of the entire network. Consensus about the validity of a transactions. There is not one node that can determine the sequence and what the others should do. The ‘democratic law’ of half + 1 applies in this system. If more than fifty percent of the

⁷⁹ See also website: seti.org

participating nodes agree with a transaction, this will also be valid and that will be the new shared reality. The new valid transaction is 'added' to the already existing shared reality (accounting system). The design is based on the fact that it would be impossible that one single node or party can solely achieve this majority. That this eventually could be different in the practice of mining pools, Nakamoto could probably not have imagined (or maybe he/she/they did?).

The Bitcoin network also tends towards a certain form of centralization because only a limited number of mining pools mine the most Bitcoins. This tendency towards centralization is a very natural process because it appeals to our natural tendency to scale up (becoming more and more efficient). But eventually it is about the balance between central (scaling up, mechanical organizing) and decentral (flexible, organic organizing). More about this central-decentral balance in the Weconomics theory book.

To be allowed to add a block to the blockchain in the Bitcoin protocol, it must be validated as well as 'mined'. Basically, validating means an answer to the question whether or not a transaction may be executed. So, the transaction is compared to the agreed rules that state that there must be an input reference before a value can be issued and if this is the case, that this value subsequently cannot be issued more than once. To get a better understanding why it is so important that there is a difference between decentral and distributed, the following is important. Mining has two purposes:

- guaranteeing the (technical) security of the network;
- an economic boost.

The latter means that there has to be a motivation or a temptation, so people *want* to perform work for others (checking the transactions, for example currently done by the accountant). This performance of work and the proving thereof, is called Proof of Work. Often it is said that a transaction within the Bitcoin network does not cost anything; that is not completely true. It costs energy of course and to grant a transaction priority it is wise to use a so-called 'miner's fee', otherwise you have to wait a long time before your transaction is 'approved'. A very small amount to pay the miner for the performed work. This amount, however, is still a lot lower than the transaction costs in the traditional monetary system. If you do not offer a miner's fee with a Bitcoin transfer, which is allowed, the crediting of the transaction will take longer. In addition to the miner's fee there is also another type of remuneration. As soon as a miner can prove that he has performed work (has correctly solved the puzzle), he receives a certain number of bitcoins.

From these three to four network variants, we can derive two different blockchains as extremes on a scale and a more hybrid form in between.

Public, permissioned and hybrid blockchains

As explained before there are basically two flavors of blockchains and a combination of both (hybrid) forms. The two flavors are centrally (dictatorial, authoritarian) controlled and decentral (consensus). These are the extremes on the scale, whereby many variants are possible between these extremes. An example of this is a democratic decision: half + 1. In terms of infrastructure there is elimination of intermediary entities or processes in all cases. This will certainly bring economic benefits from a procedural point of view. After all, 'agents' are removed and therefore the friction is reduced (unless the energy bill becomes too expensive). For example, clearing and settlement processes in the financial world, whereby it is possible that there are five parties between the provider and receiver, between the initial seller (cash machine) and end consumers (invoice shopkeeper).

We saw a similar layout, centralized/decentralized, when internet emerged. In the beginning, communication applications were built that (in terms of infrastructure) worked with the TCP/IP protocol and therefore infrastructural decentralized but still controlled centrally (intranet). This situation still exists: the communication department, for example, can determine who has access to the network and who has not and who can contribute, edit, delete, etcetera, and who cannot. Now we see a similar development again. Currently many blockchain developments are taking place within companies (central control), but with decentral blockchain technology. We call this permissioned blockchains as opposed to open, public or permissionless blockchains whereby there is 'no' central authority.

In principle, the infrastructural border is not relevant for Blockchain Organizing. We reason from the smallest building block and you as an organizational participant are part and total (holistic). A company, a city, a country: for Blockchain Organizing these are all just (collections of) connections between the smallest building blocks. Power and authority are all the more important to Blockchain Organizing. We want to get rid of all kinds of intermediaries to organize trust. We want to get rid of central powers of large companies and governments. We want to return the power to the people.

The following table gives a good overview of the difference between permissionless and permissioned blockchains:

Different blockchains		<-- Shades of grey -->	
		Permissioned Blockchain	Permissionless Blockchain
5 Key questions	1. Who maintains it? (edit mode)	1 or a small group of pre-selected and permissioned entities	Anyone that wants
	2. How are they incentivized to do it right?	Reputational risk	Crypto-economics' Direct economic incentive carrot & stick (e.g. token awards & fees versus power cost in Proof of Work)
	3. Who produces the underlying data? (can send their transactions)	Permissioned group of people (e.g. customers of a bank)	Anyone that wants
	4. Who has access to that database (read-only)	1 or a small group of pre-selected and permissioned entities (usually the ones that maintain it)	Anyone that wants
	5. Where is it stored?	Central servers	Massively distributed
Features	Need to trust central entity to secure it	+	-
	Transaction cost (direct function of costs of maintain the ledger)	-	+
	Speed	-	+
	Censorship resistance	No	Yes
	Need to use a token	No	Yes
	Examples	Hyper-ledger	Bitcoin, Ethereum

Table 3.2: Differences between permissioned and permissionless blockchains⁸⁰Public blockchains or permissionless blockchains

The public blockchain with Proof of Work (necessity for mining nodes), is currently the only 'fully' decentralized blockchain. After all, it is a public and transparent blockchain, with open source software, whereby every participant can and may send transactions to everyone. Nobody is owner. Consensus is proper consensus because each participating active node has the same 'vote'. There is no exclusion. The control of power lies with the majority of controlling miners. No external control can be exercised on, for example, the Bitcoin community or accounting system. There is a reason why governments, financial institutions and large companies are suspicious of this type of blockchain: they cannot exert any influence (at least not without adjusting

⁸⁰ Copied from the blog by Yacine Ghalim.

existing laws and legislation). The most important public blockchain, proven in practice, is currently the Bitcoin blockchain.

Private or permissioned blockchains

As soon as a blockchain is fully controlled by one party or organization, it is a private blockchain. This is an example of old thinking; after all it is based on the (legal) entity the firm, the bank or the government and not on organizing capacity. This situation does not require any or other types of ‘mining nodes’ then the current one. In any case, one does not have to be afraid of the so-called 51% attack, as the control is 100% within the own organization. The important question with private blockchains remains: why would you apply expensive database technology within your own company when there are cheaper alternatives such as a central database? We feel that some companies believe that they are a ship and by attaching wings they can turn it into an aircraft, while the ship is much too heavy to take off.

The consequence of private blockchains will be comparable to private communication systems such as Microsoft Outlook. Sharing information with the environment then becomes the sending and retyping of e-mails. Sharing transaction data will soon become the retyping of blockchain A to blockchain B. Analogue processes are digitized, but the way in which that demand and supply or trust is organized, changes little to nothing⁸¹. It is our opinion that private blockchains have little to do with the original idea behind Bitcoin. Private blockchain owners must first have the courage to give up some autonomy, let go of the existing and conditioned management model (the firm) and need to think more in terms of communities and shared transaction networks. Only then does the use of blockchain technology make sense. If we do not do this, the blockchain becomes the next internet which will provide little benefit to us, human beings. A typical statement to illustrate this is a statement from the Chamber of Commerce⁸²: *“The complex matter of blockchain can largely automate the administrative processes of the Chamber of Commerce, but the Chamber of Commerce will not be redundant for a long time.”*

A derived form of a permissioned blockchain is a consortium blockchain. We see this for example happening at banks and insurers. In the financial sector the idea has probably emerged to separate the term blockchain from the term Bitcoin. Because Bitcoin creates a serious threat while they can apply blockchain technology ‘in-house’. The consensus mechanism with consortia blockchains is vested in a number of privileged participants. The right of access and crediting is determined by the consortium and therefore by definition there is exclusion.

⁸¹ See for example also article Nikolai Hampton in Computerworld: ‘Understanding the blockchain hype: Why much of it is nothing more than snake oil and spin. Blockchains are an exciting new development, but the devil is in the detail’ (5-9-2016).

⁸² See also report (edited) by Marlous Plomp and Koen Hartog: ‘Results Blockchain pilots’ (June - November 2016), see also website: <https://www.blockchainpilots.nl/home-eng>

Incidentally, a partial right of access can be provided to the public as a result of, for example, the new privacy law and then you could speak of a hybrid situation or partial decentralization. But the control or power clearly lies with the consortium. This does not require any or other types of mining nodes. After all, the trust lies within the consortium itself. However you look at it, when you reason from the smallest building block, a company with CEO is also some kind of community whereby certain governance rules apply. A CEO cannot make all decisions and depends on statutes, management and a Supervisory Board. You cannot simply say what is central/decentral or permissioned/permissionless without also mentioning the unit of analysis and design. And the most important thing remains: with what purpose do I design organizing capacity?

Bitcoin versus blockchain

The idea that the only useable part of the Bitcoin ecosystem would be the blockchain technology and that you can just separate this from the entire ecosystem, is a bad one. This bad idea is subsequently positively framed by the existing establishment. Circumstances are created that make this bad idea come true. But only removing the blockchain technology (whatever that exactly might be) from a large total (Bitcoin ecosystem) is like trying to let a polar bear survive in the desert. It would also be a bad idea to apply Blockchain Organizing within the walls of existing companies. However, what could be useful is to prepare existing internal processes with the use of Blockchain Organizing, so that these can be transferred to a shared transaction network.

The disruptive nature of Blockchain Organizing mainly lies in the fact that with pure decentralization (infrastructure and governance are decentral) there is a serious knock on the door of the existing social-economic establishment. An establishment in which we still organize trust with incumbents, with companies, banks, governments and political parties. At this moment, the establishment seems mainly interested in the ‘underlying technology of the Bitcoin’. But if we had applied the assembly-line within the crafts workshop a hundred years ago, we would not have become so prosperous.

Apart from technology, leadership and the management model must also change at the same time and that will take more time, attention, money and resistance than only changing the technology. The question is, whether just changing the technology, or for example the medium, is actually progress? ‘And what is progress? When does technology provides progress and who does actually progress?’ Yacine Ghalim⁸³ says about this: *“As much as I don’t think one type of blockchain is inherently better than the other, I certainly believe that permissionless blockchains are far more disruptive and transformative. In the eyes of many the blockchain is seen as this disruptive piece of technology while Bitcoin is portrayed as a Napster-like experiment. I don’t think this is the right way to look at it.”*

⁸³ See also blog by Yacine Ghalim: ‘Why we should drop the whole “Bitcoin vs blockchain” discussion’ (7-10-2015).

Although blockchain technology can also provide efficiency improvement within companies (a company is also in fact a collection of smallest building blocks), the technology will mainly become disruptive when it is applied in shared transaction networks where not one company is or can be ‘the boss’. We would find it strange if each company would design and apply its own Windows or Linux operating system. However, you can develop different applications that work on an operating system. This means that we first have to organize supply and demand differently (management model), then we have to develop a joint infrastructure (data common, ontology, taxonomy), then a connection to operating systems (for example Ethereum based) and after that each company, but also each citizen, may develop his own blockchain or application (DAPP: Decentralize Application). This means that we can all build our own garden path but that the road network becomes a commonality. Because why should we do this for the road network and utilities and not for the data facility?

As a result of all this, there are only two flavors left for us that can make the difference in how we can support a completely new way of organizing capacity with organizational technology: the permissionless or hybrid blockchain. Only these setups ensure that we can organize trust without increasing bureaucracy, complexity and the power of large companies and central government. How this will develop will mainly depend on the setup and existing institutions (top-down) in relation to the success of start-ups (bottom-up), which are able to apply blockchain technology in shared transaction networks.

Now we will discuss a number of (technical) blockchain concepts and tools that we can use within Blockchain Organizing.

3.4 Blockchain concepts and tools

In this paragraph we will give a further explanation of a number of tools and concepts that are part of, and contribute to, Blockchain Organizing. Together, these tools offer the opportunity to bring supply and demand together in a decentral and democratic way without significant friction costs, while simultaneously reducing insecurity in an ever more digitizing society. We will limit ourselves to a number of technical concepts and tools for Blockchain Organizing:

- Data Logistics
- Data As A Service
- Personal Data Service
- Vendor Relationship Management
- Attribute Based Encryption
- Distributed Systems
- Smart contracts
- Decentralized Autonomous Organizations
- Ethereum

Data Logistics

People and tools generate more and more data: the house we live in, the car we drive, the sport we practice, the holiday destinations we visit, the goods we buy, the job sites we visit, the educational courses we book and our contacts with government. Even our body generates increasingly more data. The smarter the device and the more we do online as a person, the more data we generate. And all this data must be stored, preferably in an efficient way, and converted into useful information.

Opinions differ on usefulness and necessity. What remains is that only the most efficient tool that brings demand and supply of data together, will eventually survive. It is our belief and it is a law of nature. Water, air and electricity: they all seek the way of least resistance to go from A to B. An important tool to provide data to the buyer, at the right place, at the right time and in the right form, is Data Logistics⁸⁴. Data Logistics is also called information logistics, but information is derived from data.

As explained before, Blockchain Organizing, in its core, is: bringing supply and demand of added value together, with as little friction as possible. Bringing demand for data (information) together with supply of data (attributes). Logistical concepts can ensure that this is done in a quick, smart, affordable, safe, meaningful and reliable way. With the production, transport, storage and repacking of data (bits and bytes)⁸⁵ and the consumption of data (information), we can draw parallels with the physical world of production and distribution logistics. Logistics for bits and bytes is called data logistics in order to distinguish it from the 'normal' logistics related physical goods.

Logistics is mainly concerned with the production of goods from: raw material (producer) to end product (consumer). Logistics or broader supply chain management is the flow of goods from source to customer. Data logistics is about the transport, storage and repacking of data, bringing data to the next part of the chain or to get information. This can be between a single buyer and a supplier but there can also be logical preceding and follow-up processes in a chain. Basically, the Bitcoin protocol consists of a number of dependent processes such as finding other nodes, creating a private and public key, a digital signature, sending a transaction message in the network and subsequently the hashing, mining of transactions in order to realize a block and eventually a blockchain. In the figure below we show an example of data logistics regarding transport, storage and repacking of data between three suppliers and three buyers:

⁸⁴ More about data logistics in the Weconomics analysis book, paragraph 3.5: 'Weconomics communication'.

⁸⁵ Bits and bytes basically consist of two distinct voltage or current levels allowed by a circuit (or it may be the two stable states of a flip-flop, two positions of an electrical switch, two distinct levels of light intensity, two directions of magnetization or polarization) and are generally not seen as a physical product or material with a specific weight and size.

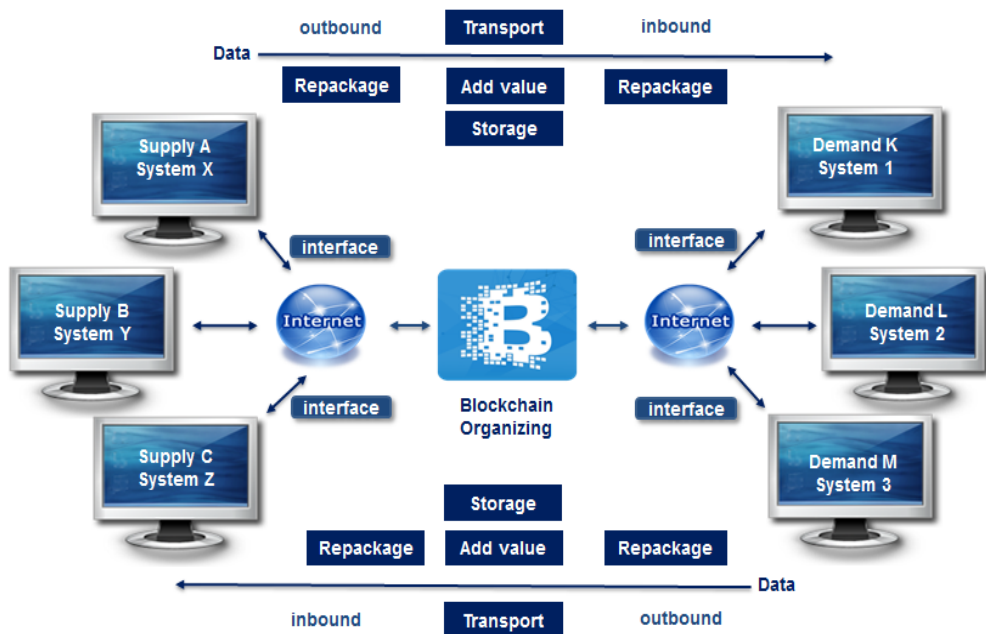


Figure 3.5: Data logistics, transport, storage and repackaging

Data As A Service

Data As A Service (DAAS) is a concept whereby data is offered from a service. Bring Your Own Data (see also Personal Data Service) is part of this. In this case, you as consumer/citizen/employee provide the necessary data that organizations require to deliver products/services to you. The organizations do no longer have to own data themselves, only ‘access to’ is required. DAAS is derived from an already existing concept called Software As A Service (SAAS). Hereby organizations do not have to install software themselves on for example their computers, they can use software in the cloud. DAAS assumes that data, needed to perform an organizational process, can be provided when this is required. The data can be provided by smart contract, blockchain technologies, data logistics and quick connections to one source file. This does not mean that physically there is one file or database, but in definition (in a logical sense).

There is one definition and with the use of a protocol attributes from different and decentralized databases are retrieved. This does not just improve the quality of the data. The hacking of data is also becoming more and more unattractive. Historically, organizations store data in their own database or repository⁸⁶. This was partly due to the physical condition of the first databases. Records went missing due to physical defects, power failures or software errors. That is why in the seventies and eighties,

⁸⁶ Literally: receptacle, reservoir, container or storage space.

most organizations started to manage a database with a backup service in order to be able to restore lost records. But the quality of databases has improved enormously and the storage and transport (internet connection) costs have been reduced rapidly. This means that we can move to another and more distributed concept of data storage: from database to DAAS with the cloud as interim solution.

A database, or more specifically a data structure, model or description, is an important part of the blockchain infrastructure as we already saw in paragraph 3.1. Traditionally (the sixties), a database is a physical archive (similar to a filing cabinet), usually at one location, to record, store, consult and delete digital data. The new invention (database) resembles the old functionality (filing cabinet). Attributes were usually connected to a unique key, such as the personnel number, national insurance number or bank account number. This way a record of attributes was created that were connected for example to an employee, citizen or tax payer.

Initially records were stored on tapes and later on disks and hard drives. But as indicated before, the physical storage is more and more separated from the description or data model. Another aspect, not related to the physical way of storing, is the relationship between data and the process for which the data is used. To appreciate this you need a bit of historical development. In the early days of automation (from the middle of the last century), mainly sub-processes were automated. By taking separate processes the automation became less difficult. But if subsequently these separate processes are not correctly aligned, a situation is created that becomes increasingly complex (combination of super specialization and bad alignment, see also trust paradox in paragraph 1.1). Due to a bad coordination between specialized processes the total does not become more productive.

The solution for this problem is sought in a joint database per company or function: for example an ERP, CRM and HR-systems⁸⁷. So, in this stage, data is separated from sub-processes. The ideal situation, storing all data in one database, is not possible, for example because of political or privacy reasons. This creates solutions such as data warehousing and the development of interfaces such as webservices and APIs. Since then more and more automation companies are working on the development of conversion and migration software to unlock various databases. In addition, the functional needs with sub-processes increase and subsequently the applications must be adjusted again per sub-process and therefore also the interfaces. This creates increasingly more complex IT systems that few people will understand, and therefore increases the risk of failure and data leaks. This is still happening at company level. We see this with banks that still run old mainframe systems or with executive organizations such as the Tax office and the Social Insurance bank.

From the nineties, the need increases to also perform processes together with the (business) environment. For example, the use of a procurement system by the

⁸⁷ ERP: Enterprise Resource Planning, CRM = Customer Relation Management, HRM = Human Relationship Management.

outsourcer, to which a supplier must connect. Fortunately, the solutions are also increasing, especially internet offered solutions. The ‘website-webserver architecture’ is actually a copy of the ‘client-server architecture’ (terminal-mainframe). The resources change but the fundamental way of organizing between data-supply and data-demand does not change substantially. The company remains the starting point for automation, data models and setup of a database.

Blockchain will also have little effect as medium, as tool, if we do not change the organizational design and take data and individuals as the starting points rather than companies and sub-processes. By continuing to choose for internal solutions, in a world with increasingly more outside connections, new (integration) processes and complexity are created. As long as the internal process is leading over available data, the data is still organized per company. The solution is sought in webservices and APIs, that subsequently also increase in numbers so they are more and more difficult to control. Here we see it again: by reducing insecurity, the insecurity increases. This is not because of the technology but because of applying new technology in an old management model. The firm is still the unit of thinking, analysis and design when automating. The question is if Blockchain Organizing, or for example the new European privacy legislation, can break this cycle, this paradigm.

Personal Data Service

Personal Data Service (PDS), Personal Data Management (PDM) and Bring Your Own Data (similar to Bring Your Own Device, BYOD) are part of a wider and aforementioned concept: Data As A Service.

PDS is the possibility to keep your own data and to give organizations and other natural persons only permission to use your data for the purpose that they must specify in advance: for example the organization of a project or the processing of an order. For instance, a webshop may only use your address details for sending a product, after that these details may no longer be present in their system for example to be resold, whether or not anonymous, to data brokers or advertising companies, unless you give permission yourself. You can also see PDS as a digital bank with safes. Each safe contains an attribute; a characteristic that belongs to you. For example, ‘street name’ or ‘date of birth’. In these safes, you can securely store your personal data. You can send other people a key to access a certain safe. The key (target group) comes with an access code (purpose): you give someone the key to your house but with a specific purpose: only to water the plants. You need another key to get to your jewelry.

PDS is made up of open and free-source software, an agreement system and protocols for autonomous implementation. Combined with internet and blockchain technology they make a secure, distributed and shared data network possible. A network of various actors such as users and associated organizations that conduct transactions with each other and agree to handle personal data in a responsible manner. For example, requesting personal data by a webshop can be seen as a transaction that can take place on a blockchain. Apart from using blockchain technology, PDS can be

connected to a general data utility facility. This is a kind of road network or power network, for personal data and (value) profiles. Examples of PDS applications (context- or domain-driven) are:

- Electronic⁸⁸ Portfolio (e-portfolio): structured overview of work and leaning experiences with connected values⁸⁹ such as: training courses, tasks, projects, employers, positions, assessments.
- Electronic Medical Record (EMR) and Personal Health Record (PHR): structured overview of values connected to a person in the role of patient. The PHR is seen more as a file that belongs to the person and the EMR as a file that belongs to the care provider and of which there are several per individual, in case a person uses several care providers. Actually, both files are also resources connected to a person (day hospital bed, medication X, treatment Y, doctor conversation Z etcetera).
- Electronic Financial Record (EFR): structured overview of connected financial products with connected values such as: a credit, an insurance policy, a pension, shares and bonds. At this moment, there is hardly any cooperation in this area between seller and buyers (consumers and companies)⁹⁰. The purpose of the EFR is to provide people and organization insight into various financial products, via one application, which may or may not belong to different sellers or buyers.
- Consumer record: structured overview of values connected to a person in the role of consumer. For example, an overview of all products and services purchased online. You could make the metadata from this overview available (sell) to advertisers or use it for benchmarking and reputation management in a community that works with blockchain technology.
- Citizen Record (CR): structured overview of values connected to a person in the role of citizen. For example, permits that you request, an allowance that you receive, an application for a new passport or the payment of taxes.

Via a PDS, various organizations such as insurers, training institutes, temporary employment agencies, governments, banks and webshops can offer their products and services via your own channel. Your own channel means that you have an own website or app, to find services and products and to conduct transactions. When this website is connected to a general data utility facility you can easily let the 'access to data' flow between seller and buyer whereby the buyer only needs to maintain data at one location and the buyer can search, book and conduct (blockchain) transactions with his own application. An organization may only use your data to perform an internal process, for example the processing of a permit request. Via PDS you are assured (with mathematical proof) that companies and institutions can only use your

⁸⁸ Most applications start with an 'e-' from electronic and mainly illustrate the transition from paper to digital. When electronic has become self-evident (and paper is seen as not normal) the 'e' before the name of a record could better be cancelled.

⁸⁹ With records it implies that connected values can also have a status such as: planned, approved, signed up, confirmed, evaluated, invoiced, paid, registered, archived.

⁹⁰ Most people store their (insurance) policies in paper form in their archives. In the Netherlands the Dutch Association for Insurers opened a website in 2018 where customers can access their policies digital, independent from the insurance company. This project already started in 2015.

data for the purpose for which you have granted access. It is expected that the market for PDS services will rapidly develop and grow. The Boston Consulting Group⁹¹ estimates the value of the European market for Personal Data in 2020 to become 1,000 billion euros. Personal data is a potential large market that is expected to grow by 20% every year.

PDS and General Data Protection Regulation (GDPR)

To prevent that organizations use your data without your permission PDS is an important addition to the stricter European legislation via for example GDPR. But only legislation is not enough and it provides a false security. Existing institutions like governments, banks and companies, will quickly go back to solutions within the same frame of mind, more institutions such as: rules, supervision, IT and more employment. These adjustments will not affect the right to exist of the institutions that offer these institutions but mainly strengthen it. But the question is not what institutions want, the question is what is best for the people and our welfare. There are better solutions from the human perspective.

PDS is an important new concept for this that can be built with blockchain technology. PDS is a cloud-based system to securely store your data, to make it accessible to or to share it with other persons who may use your attributes to get an answer to a question. It seems that GDPR will become the main guideline for the protection of personal data and thus the design of IT systems for the coming decades. GDPR will have a great influence with this, not just on IT companies and IT organizations but also on all organizations that record personal data in HRM, CRM or ERP systems, for example. We believe that it is important to know more about the new privacy law in relation to PDS. To prepare yourself now will prevent high fines and costs for adjustment later (for example the problems regarding the millennium bug). In addition, it also offers new opportunities for individuals and organizations that indeed organize correctly, from the design.

Not property but access is relevant

In order to comply with the new privacy law, to become more productive and to be less dependent on existing online platforms, there is a growing belief that not the property of data is relevant, but the access thereto. If an organization does not own any data it cannot leak this data. In a knowledge-intensive society much data is connected to knowledge workers who increasingly work for several clients. We see a change in the models used to organize work: from company, via cooperation in a business eco-system, to a network community with shared transaction networks. In this community, there is smart cooperation with customers, suppliers, employees, governments and other stakeholders. When we continue to cooperate with the firm as starting point and unity of analyses, IT systems become more and more complex. Plus, the number of (internet) connections with the company's environment

⁹¹ See report of the Boston Consulting Group: 'The Value of Our Digital Identity' (2012).

increases and the risk of data leaks continues to increase as well. To solve this problem, we often go back to more security or more use of IT and lawyers. This traditional way of thinking does not solve the problem but rather makes it worse. It is better to look at Privacy by Design. The use of Blockchain Organizing can help with this.

PDS on a blockchain

With PDS you cannot just assure your privacy and the value of your personal data better, but you can also become more productive. PDS is the possibility to maintain your data yourself at one location. PDS can be connected, for example, to Personal Data Search (non-manipulative search engine) and Personal Transaction Services. These create their own blockchain (for example e-portfolio) where there is consensus by the community (that a transaction has taken place) while you can still indicate yourself who can access your personal data. Only part of the data is decentralized on each node (distributed ledger, on-chain, for example: sender, receiver, timestamp, amount and key to off-chain). Using a unique key you can retrieve additional attributes from a PDS (off-chain). When you connect transactions (processes, workflows) that are automatically conducted, to your own records, your Personal Health Record (PHR) or your e-portfolio, using a smart contract for instance, updates become automatically reliable, secure and meaningful. Examples are: purchasing medication from a pharmacist or subscribing to a course with a training institute.

With PDS, you only allow e-commerce companies to use your data for the processing of an order. Or you allow a trainer to use your e-portfolio for the composition of a course group, but when the event has finished the data may no longer be available to the trainer. Eventually, PDS ensures that your personal data no longer roams the internet, but is stored in your own safe location and is only available when you allow this. Blockchain Organizing can help to design and realize this.

PDS uses the blockchain principle, whereby humans and resource are the two smallest building blocks. On the supply side (S) these two are connected to a specialization, which, whether or not with the use of coordinating capacity (C), are connected to a demand or finalization (D). This creates two transactions (resource to human as a specialist and a specialist to finalization). Transactions are created and can be executed and registered via Blockchain Organizing principles. A resource (value) is transferred from seller to buyer. The required data for his process can be retrieved from a PDS. A PDS can be connected to a peer-to-peer platform that works with blockchain principles. One of these principles is that the sender must prove ownership of this value and singular spending capacity. Subsequently the value flows from sender to receiver. This is not so much about the flow of a physical product (book) for example from author to reader), but about the associated product information (price, author, dimensions). This data also represents a certain value that flows from sender to receiver.

PDS with Blockchain Organizing starts with creating a profile (personal data and resources). Subsequently the buyer can find a product or service and make a purchase and payment. A peer-to-peer platform brings buyers and sellers directly into contact with each other, as much as possible, for example: consultancy services or training courses. It makes transactions possible without having to trust each other and without being able to spend a value twice. As mentioned before the most important processes involved are: profiling, selecting, purchasing and payment.

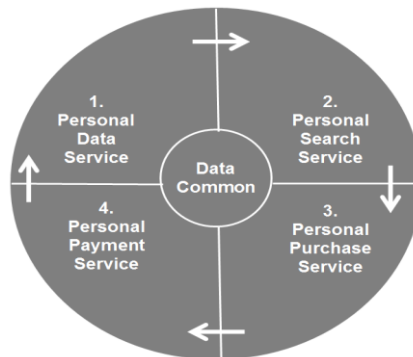


Figure 3.6: Processes related to the preparation, implementation and registration of transactions

In more detail, this means paying attention to the following concepts:

1. Personal Data Services (PDS) whereby users (sellers) can maintain their profile and can decide who is allowed to join.
2. Personal Search Services (PSS): this concept offers the possibility of Non-Manipulative Search⁹² whereby not the advertiser (seller) determines what you find, but the user (buyer).
3. Personal Purchase Service (POS, O=Order): this concept offers the possibility that the seller and buyer jointly set up the workflow consisting of: approval, confirmation, delivery and registration.
4. Personal Payment Services (PPS): this concept offers the possibility that the seller and buyer jointly select the payment system. For example, with bitcoins but also via an own community token.

Vendor Relationship Management (VRM)

The way to organize demand and supply is still largely determined by the producer, despite the development of internet. You must always enter the same data on various websites if you want to order something. Products and services are 'pushed up' the market through sales. The question is: can Blockchain Organizing can provide a reversal of the supply and demand model. Internet has not sufficiently managed this yet. Many 'Big Data' investments also do not contribute to the reversal of the model. They mainly focus on using data to promote sales. Can Blockchain Organizing

⁹² Many search engines are manipulative and use for example advertising banners at the top.

ensure that we switch from a push to a pull economy? One of the concepts that can help with this is Vendor Relationship Management (VRM).

VRM and CRM

You can consider Vendor Relationship Management as the counterpart of CRM (Customer Relationship Management). CRM is often supported by a CRM system. This is usually an IT system with data and characteristics of:

1. suspects (leads, Market Qualified Leads: MQL)
2. prospects (promising leads, Sales Qualified Leads: SQL)
3. customers

The purpose of a CRM system is to take parties from suspect to customer as quickly and cheaply as possible. The purpose of many organizations is to lock-in third parties (other organizations and consumers). Systems for communication, sharing information and making transactions are currently often still driven by the supplier and not by the buyer. Blockchain Organizing, and embedded therein VRM and PDS, are aimed on reversing this: the buyer has an own portal (pull instead of push) to find and book products and services and to make transactions.

VRM is aimed at Customer Driven Leads (CDL): giving a prospect the tools to commit to you and to become a customer when there is a need. A seller must enable a buyer to find his product (information) when the buyer needs this product. And when the buyer has found this product, a transaction must be enabled at minimum transaction costs and without lock-in, using buyer's own data to perform the process.

We expect that push tools, such as direct-marketing, will be used less due to VRM systems. Therefore, the advertising revenues could be structurally reduced in the coming years and decades to come, undermining the business model of many social media platforms. With the increasing power of large data companies, and for example the problem of fake news, we believe that more people will connect to a data utility. With this they can convert the present data into relevant information, at the moment they require it. VRM can be seen as a new dance, a tango whereby the seller no longer leads, but the buyer does. Despite empty cries like 'putting the customer first', most organizations still use their own CRM system and not a VRM system. Essentially, there does not have to be a technical difference between these two systems. There is mainly a difference in perspective on how we can organize trust, work and economy with the less friction and the highest standard of living per person. How do we organize supply and demand, using these techniques, enabling the buyer to be in control and not the seller? A situation where transactions can be made against much lower costs, with less complexity and with much lower waste as a consequence.

What is VRM and the ProjectVRM?

The term VRM was first used in an article of ComputerWorld magazine in May 2000, but with another meaning. The meaning of VRM we use, was first used in a

conversation⁹³ between Mike Vizard and Doc Searls⁹⁴. Vizard saw VRM as counterpart of CRM. The conversation was about an initiative, the ProjectVRM, which Doc Searls had just started. The project mainly focused on developing tools and services, but from a new perspective on CRM. CRM involves high transaction costs such as: branding, marketing, targeting, capturing, acquiring, lock-in and managing. With VRM, customers are more like partners who want to connect against much lower transaction costs and therefore a lower end price. According to ProjectVRM the main theory behind VRM is: ⁹⁵: *"Many market problems (including the widespread belief that customer lock-in is a 'best practice'), can only be solved from the customer side: by making the customer a fully empowered actor in the marketplace, rather than one whose power in many cases is dependent on exclusive relationships with vendors, by coerced agreement provided entirely by those vendors."*

VRM 'inventor', Doc Searls is, among others, the author of the book *'Intention Economy: When Customers Take Charge'*. According to Searls it is not about the seller but about the buyer. Since the internet was 'released' in 1995 we have seen that internet is a tool that can provide the buyer more 'power'. However, not a lot has actually changed since then in the fundamental way we organize trust and consumer interaction. The solution for lower transaction costs cannot only come from the technique. Several powers are at play here that we want to make meaningful within Blockchain Organizing in relation to finding a solution to improve productivity, privacy and balance of power.

In 2012 Customer Common⁹⁶ was created from the Project VRM with as purpose: education, research, support and the development of VRM tools for individuals when they get or have a relationship with organizations. VRM is also part of a large development that we usually call Life Management Platform. These are platforms that enable people to control their own data in relation to organizations that need this data to perform processes. This concerns data from all types of domains, for example data from: your body, home, car, work, education, purchases, finances and your relationship with the government.

PDS and Vendor Relationship Management (VRM) want to achieve the same thing: more direction and control by individuals in relation to organizations that use, process and sometimes even resell data. PDS and VRM reason more from humans or individuals while existing CRM and HRM systems reason from the tool humans creates such as companies, banks or government authorities. You could see VRM as

⁹³ Podcast from Doc Searls with Mike Vizard (1-9-2006) about a project that Searls started as fellow at the Berkman Centre for Internet & Society at Harvard University.

⁹⁴ Doc Searls (1947-), American journalist, open-source promotor, fellow Center for Information Technology & Society (CITS) at the University of California, Santa Barbara, and an alumnus fellow (2006–2010) of the Berkman Center for Internet & Society at Harvard University.

⁹⁵ See website: cyber.harvard.edu/projectvrm/Main_Page

⁹⁶ See website: customercommons.org

counterpart (but also addition) of the traditional CRM system that is more based on a push-market than a pull-market⁹⁷.

Attribute Based Encryption (ABE)

More and more companies discover the convenience of cloud services to store data. The use of cloud services offers many advantages such as: scalability, flexibility, accessibility, ease of use and low costs. But ‘the cloud’ also has concerns that cannot always be covered legally and therefore require technical measures. Because many cloud services (for example Dropbox and Google Drive) were initially primarily used by private individuals, relatively less attention was paid in the design to the security of data. The use of own resources, the so-called Bring Your Own Device⁹⁸ concept and the increasing number of cyber-attacks, require extra attention to security and encryption. Encryption is used to guarantee the reliability and integrity of data, but it can also be used for access control. When for example patient data or personal data is put in the cloud, you want to be able to offer proper access security that goes beyond the existing authorization systems with a login and password. You want to be able to connect certain rights to a role that has access to certain attributes of a person. These rights are based on a certain policy or on certain instructions or conditions. Only the role, with associated rights, that complies with the instructions or the protocol can access a certain range of attributes. Plus, it should not be necessary to give access to the value of an attribute (for example date of birth). It must be sufficient to get a correct answer to a clearly formulated question: may I send person X a ballot paper because this person has turned eighteen today or is older?

When you assume that you cannot trust a cloud server, you must add more security at data level. You can do this by encrypting attributes, in such a way that the owner of the data can determine via a key who (which role) gets access to which attribute or date (range) and for which purpose (rights). This is similar to placing data in a safe, whereby you send a key to the receiver. So, we see many similarities between the ABE concept and the Bitcoin protocol with key and digital signature. You could say the blockchain technology uses ABE technology. You could also say that ABE uses blockchain technology whereby you see the request for information (are you eighteen today or older) as value exchange, as a transaction, that can be run via a blockchain.

ABE is an important security concept that can be used with almost every role-based system that must offer trust and integrity. A role-based system means that you connect certain rights or policies to the role of ‘manager’. These policies indicate what the role can and cannot do. The advantage of this system is that it does not depend on the person who fulfils a role. Persons can change jobs, but roles usually last longer. A role-based system is also called Role Based Access Control (RBAC). Cloud services often work within a federative environment. This means that you for example can use your Facebook account to log on to your cloud service (Single Sign On). To

⁹⁷ See also the Weconomics practise book, paragraph 3.5.6: ‘Communication matrix’.

⁹⁸ See also the Weconomics practice book, paragraph 5.4.1: ‘Bring your own device (BYO)’

improve the security within federative environments, policies are often based on attributes. We call this Attribute Based Access Control (ABAC).

ABE goes one step further. Within ABE a user also has a set of attributes (such as age, or country of origin) in addition to an ID (for example Social Security Number). Parties that want to access the data of users must have a correct combination of ID and attributes. For example, you can program that certain IP addresses or GPS coordinates do not get access. Values that are outside the normal user pattern can also be excluded. When a user always requests access during the day and suddenly once at night, this could mean that someone from a different time zone is trying to log on. By logging the behavior of users, patterns are created. A login that differs from this pattern can for example be excluded. Concrete applications of ABE can be found in initiatives of Harvard and MIT: Sieve and Enigma⁹⁹.

Distributed Technology

Distributed Technology is a collective name for a number of concepts and technologies that are used by Blockchain Organizing. In figure 3.4 we saw four different types of networks: centralized, distributed and within this decentralized and within this peer-to-peer. With a centralized system, all users (clients) are connected to one central server. This server records the time and the server contains one database where one reality, one transaction sequence in the shape of a ledger is maintained. A distributed system is a hybrid form between central and (fully) decentral. It includes a central unit (protocol), a number of hubs (full nodes) and many users (wallets). With a distributed network, the nodes (users) on the one hand are connected sequentially, but there are several paths that lead to the final node, therefore several paths can be followed simultaneously (parallel) between the first and final node.

Internet actually has the same structure. A ‘message reproduces itself’ via several paths, from node to node, till the final node or the receiver has been reached. With internet, messages are divided in small parcels that eventually merge again at the receiver. Therefore, the transport is secured better. You can compare this to using a large money van to drive from A to Z. If this is robbed, you lose all your money. You can also use two small cars to drive from A to Z via two different routes. The notes have been cut through the middle so the serial numbers are no longer recognizable. At Z the notes are merged again to a unique number. If one car is robbed, it has little use to the robbers because the money cannot be used without the second part.

In principle, you can look at distributed systems in various ways. You can use a scale for this and ask to what degree a certain perspective or aspect is organized central or decentral:

Fully central  Fully decentral

⁹⁹ See paragraph 6.7 in book6.

Using this scale you can 'look' at the degree of (de)centralization. You can view it from the following perspectives:

- Physical, description of the physical architecture such as: which circuit of physical processors or nodes are required in the network?
- Logical, description of logical architecture such as: what must be included in the database, what is the physical position and what is the relationship between the logical parts?
- Governance, who has the actual power such as: who decides what, with how many votes?
- Logistics, how the total is structured such as: where is the Customer Order Decoupling Point, is there a general data utility facility to which applications can connect to and which application is used by the end user?

Uber and Airbnb for example are not connected to a general data utility facility. From a logical point of view they are distributed at application level (buyers and sellers run an application on their smartphone). In terms of governance they are managed centrally and the physical servers are probably installed decentralized. The way you could organize a distributed network can be seen from different aspects:

- From the physical aspect, you look into a distributed network to see how computers or nodes are installed compared to each other. You also wonder: how many computers must fail before the entire network fails? Can the network continue to function when ten percent fails? Is there a critical path?
- From a logical aspect, you look at how data can be stored from a logical point of view. The holistic character of your logical architecture plays a role in this: if you would separate a part of the network, would the new separate parts still function correctly? What is the smallest building block, what is a part and what is a total and is there a logical hierarchy in this?
- From the governance aspect, you look at how many individuals or companies have the actual power and how are decisions made.
- And finally, from the logistics aspect you will look at the structure of the network and which layers there are: application, data contact, network and general data utility facility.

In terms of governance, traditional companies are often organized centrally (there is one CEO), and also centrally from a logical point of view (when you 'cut off' some departments from the total, both parts can probably not immediately function independently). You could characterize a public blockchain like Bitcoin as decentral from the physical and governance aspect and as central from a logical point of view (one protocol and one shared reality). It seems that distributed systems only offer advantages, but there are also dangers:

- what if you cannot trust the designer or software developer?
- nodes use the same software, what if this contains an error?
- what if an update is suggested when everything is running correctly and the update contains an error, or the creators have bad intentions?

- what if the aforementioned graphic maps to mine are produced by one company and they can manipulate a blockchain via a backdoor?
- what if the number of miners reduces. You'll get a form of centralization and these miners also become nationalized or worse: they must be closed down.

A condition for peer-to-peer networks that use the Blockchain Organizing concept, is that there are many sellers and many buyers (n:n community) and that the infrastructure construction and maintenance are separated from the usage. That people can choose between different alternatives (applications, wallets, exchanges). With existing (social media) platforms these conditions are often not met. Initially internet was also a distributed peer-to-peer platform and is now largely centralized. This means that one should not only look at the infrastructure with Blockchain Organizing, but also at the power and independence aspect. In particular at the balance between scalability (centralization) and flexibility (decentralization). There must be a good balance between cooperating and competing and between individual and company/government. With Blockchain Organizing we must not become dependent on a few American tech companies with questionable intentions and non-transparent algorithms, just as with internet. In fact, there must be a better balance between the different stakeholders such as shareholders, maintainers, miners and users.

Smart contracts

A core attribute of capitalism is property. Capitalism needs a system with which we can trust each other regarding property rights and the possible transfer thereof. Why else would you invest your time in something that you are not sure of if it is, or will be yours? To realize this we need a variety of lawyers, contracts, banks and notaries, while this can also be done in a smarter way, for example, with a smart contract. A smart contract is an algorithm that performs a number of actions (for example workflow), as soon as one or more conditions are met. For example, each time that you log on to a website with the use of your Facebook or Google account, a smart contract is entered as it were. Behind the Facebook button that you use to log on there is a set of agreements. You must first (and often only once) agree with these, before you can continue to log on. Making an agreement about the use of your data, or for example providing access to your address book, is often part of this set of agreements.

You can see a smart contract as the digitizing of a paper contract, but it is better to see it as the digitizing of a workflow (collection of processes) between seller and buyer. A smart contract is a mechanism to formalize and automate the relationship between seller and buyer. The second characteristic is usually not possible without the first characteristic. Before you can program processes or tasks you must first formalize these, more or less linguistically (semantically) or mathematically, whereby you agree that $1+1=2$. So, with a smart contract, agreements are converted into programmable rules whereby the 'If This Then That' statement is one of the most commonly used rules.

Smart contracts reduce the transaction costs that are made with the preparation, execution, inspection and registration of transactions. Thereby users (often computers with algorithms) can continuously learn from the result of a transaction (for example to what extent does the outcome deviate from the expectation). By using evaluations, you can make a better choice next time. The result of a measurement can also feed another algorithm with a value (for example price or delivery time) so the values in a smart contract can be adjusted what can then lead to another result with a next transaction.

The execution of a smart contract is less sensitive to human errors and manipulation. What remains are errors, defects, incorrect estimates, manipulations, etcetera that can be made by the designer and programmer of a smart contract. Smart contracts are still sensitive to programming errors or attacks by hackers¹⁰⁰. But when a smart contract is running smoothly and you cannot hack a contract against acceptable costs, the risk of human errors at the execution will reduce. A computer does not easily make a calculation error or make a mistake in a date and it has no problem performing a large amount of complex calculation in a short period of time. Plus, a fault-tolerant distributed system will faster find a consensus on a value that is proposed but not yet determined.

A smart contract also has less trouble with remembering the conditions that lead to the performance of certain tasks or processes. The recording of the answer to a question whether the value of a condition has been reached or not is also much easier for a smart contract to execute than it is for a person. A smart contract does not sleep, is never tired or in a bad mood. This does not mean that we as users do not have to do anything anymore. We must mainly think about the meaning and consequences of technology for us, people. We have an important role in the design process and the transition from used institutions to programmable institutions.

When we go back to the example where you use Facebook and Google to log on to your favorite news site, you must trust that Facebook does its work correctly and sends the right data to your news site. In this example, we still need a central trusted third party, for the execution of a transaction. With Blockchain Organizing it is possible to skip the role of Facebook¹⁰¹. However, you must identify yourself once. This is, as an example, possible via your bank or the city council where you live. Identification, authentication and the secure use of data will then be smarter with Blockchain Organizing. With blockchain technology we try to address problems of fake work, privacy, power of large tech companies and fake news. We can manage or solve these problems if the various powers go into the right direction. The aforementioned consensus system will in particular play an important role in this. We will no longer depend on existing and often central institutions, to exchange data and to forward

¹⁰⁰ Besides programming errors and hacks, smart contracts are also subject to other powers such as legislation, human interaction, politics, economy and moderators.

¹⁰¹ However, we still need a trusted third party for linking the real and digital (virtual) world (also called Oracle).

transactions, but we'll have peer-to-peer systems whereby trust can be programmed with mathematical proof as foundation. When we bring a smart contract on a blockchain, participants can find consensus on the value that is used for the execution of a contract.

The final pieces of Nakamoto's puzzle also seem important for the further development of smart contracts. An important part of smart contracts is letting go of your own reality in relation to the reality of the counterparty, within the use of a shared system. A system of distributed databases and a protocol to find consensus regarding the shared reality within the system, will help with this. Therefore, two parties that execute a transaction have the same reality about whether a condition has been met and a transaction is or is not executed. They can operate within the same reality, they view the same facts on which there is consensus that these are true. For instance: about the price of a product or the agreed delivery time. With blockchain technology it is possible to realize a shared reality about possessing a certain value or not, or about including a transaction in a block or not. This creates a new shared reality as basis for trust and for subsequent transactions.

Smart contracts are also very useful in situations with many parties (n:n communities), non-transparent markets, high transaction costs and few laws and regulations. The blockchain that is created, after execution of the contract, becomes a shared single source or reality. Information from a blockchain (ledger) can be used for a number of processes such as suggested in figure 3.8 (smart contract between supply and demand). Examples of sub-processes with the execution of a smart contract are: profiling, selecting, approving, ordering, delivering, paying and registering.

We finish this part with naming a number of advantages of smart contracts with respects to the current institutions:

- Costs: lower (transaction costs) because fewer intermediaries and institutions are required.
- Objectivity: less human interpretation possible.
- Productivity: increases because people can do the same work in less time. For example, they have to retype or interpret less data. This also reduces the time for fixing errors.
- Speed: executing different processes and finding consensus is possible with the speed of internet and the speed of the connected processors.
- Risk: less risk because there is less interpretation and human interaction and therefore the risk of failure of parts is reduced.
- Reliability: a computer makes fewer calculation errors than man, forgets less and can perform routine processes much better and quicker in a relatively quiet environment.
- Security: less risk of human errors when transferring messages.
- Meaning: by using a joint ontology, taxonomy and synonyms list, the human language will become susceptible to fewer interpretations, will improve communication and thus reduce the chance of conflicts.

- **Maintenance:** as soon as there is for example 90% consensus on a new release of the protocol, everyone will then use the same protocol at the same (logical) time.

There are many applications for smart contracts. One example is an application in the shared economy where shared bikes, shared cars and shared rooms contain a chip that can be activated with your smartphone. As soon as you have accepted the conditions and complied with these, including the payment, you can open the shared room with your smartphone. There are also applications for offline shopping: you can just take clothes and other goods from the store. Only if you have an insufficient account balance you will not be able to pass the detection gates.

Decentralized Autonomous Organizations (DAO)

There are no (human) managers in a DAO, just as there are not drivers anymore in a self-driving car. A DAO protocol enables participants (these no longer have to be humans) to make transactions with each other without central authority such as management. Managers will not disappear, but fewer will be required and their roles will change. The core of management includes a mechanical part (planning, distributing resources, checking, registering) and an organizational part (motivating, developing). These are all actions to reduce insecurity whereby especially the mechanical part can be automated.

As we have often seen in this book, we want to reduce the insecurity between demand (finalization) and supply (specialists), but this can be done in various ways. Many coordinating actions that are done by managers involve bringing supply and demand together (connecting people and resources to a task, assignment, project or problem). The way that managers do this (this in the meaning of: ‘... if this is the situation I must do that...’), can be converted into a policy or recipe. So, it can be converted into a kind of recipe with which the organization can control itself. Mechanical management is then replaced by a protocol that enables participants to find consensus themselves. Participants in the decision making all have the same shared reality and can all make a proposal that others vote on. In some situations, for example the processing of transactions, DAOs are much more able to share information and to make decisions, than organizations that use a social script. A social script can be motivated by culture, interpretation or emotions for example and act differently in decisions whether a transaction took place or not.

A DAO has no traditional company boundaries, only smallest building blocks, connections, instructions and conditions, contained in a protocol. Of course, you can incorporate as a limitation that an actor of a DAO may not communicate outside the boundary of the DAO whereby you determine the boundary in advance by making connections. But this limitation is a simple ‘yes’ or ‘no’ in a protocol. A connection that you can switch on and off.

The separation of information provision between companies is much more complex and much more difficult to adjust with the current working method, than within and

between DAOs. With traditional companies, you have to deal with IT systems, collective employment agreements, buildings and fixed contracts. With DAOs you mainly have to deal with building blocks, connections, instructions, data and software code. Legislation may change but the principles with which you can execute a smart contract will not change often because they can be reduced to an IFTTT statement. A DAO itself often does not have an impact on legalization or government rules, for example: a postal company has to replace flex-work contracts for permanent contracts, but a DAO can easily change the value of a contract or connection.

For a DAO any contract is just a set of variables that can change. The values of variables can change but the principles will stay. The relationship between seller of work and buyer of work is a connection for a DAO. There are different types of connections for this whether or not made possible by politics. But the data and software to perform a certain type of connections, does not have to be influenced by politics, provided you have correctly addressed this from the design. This also applies to the performance of governmental transaction such as benefits, grants, allowances and taxes. DAOs are also little affected by mergers and takeovers. What is created after a merger is a new collection of connections that in the core (ontology) will not change through a merger. This offers cost benefits for IT. In addition to this advantage, within DAOs ideas will emerge and be used more effectively and efficiently. Each actor can propose an idea or software change for instance and the network votes on whether to use recourses for this or not. Working with other organizational forms such as the DAO, also requires another expertise of organizational advisors. You can compare this with the car industry: in the past, the mechanics mainly had a background in mechanical engineering, but now more and more in electronics and computer science.

Ethereum

An initiative that works on the practical implementation of DAOs is Ethereum. In 2013 Ethereum was started by a number of programmers, led by Vitalik Buterin who is also involved in the Bitcoin community. The project started with the publication of a white paper by Buterin called *"A Next-Generation Smart contract and Decentralized Application Platform."* The financing of the project was made possible by raising eighteen million dollars with crowd funding. Ethereum started as an internet project that wanted to realize the recording and exchange of information, decentral and without the intervention of traditional institutions. It is an extension of the Bitcoin protocol that mainly focuses on digital money. Ethereum wants to be able to facilitate more different values than just money. Gradually the project has grown into a platform with which you can develop applications that work with blockchain technology. You can also put applications on a blockchain yourself. The underlying mathematical principles of blockchain technology are not always immediately clear to many developers, but that is not necessary.

Ethereum ensures that the development of applications based on blockchain technology, becomes accessible to a larger group of developers. Ethereum could develop into some kind of Appstore for blockchain applications. The different decentral applications, also called DAPPs¹⁰² can for example be used for social media services (such as Steemit), cloud storage services (such as Dropbox) and even a replacement for supercomputers (such as Watson). An important motive for Buterin is the development of technology that will make the economy and community more inclusive, safe, free, efficient and reliable. According to Buterin, a decentralized economy and community is a logical consequence of the current system and manifests itself in movements such as the shared economy, the interactive Web and more recently: blockchain. Ethereum ensures that it becomes easier to development applications with the use of blockchain technology and on a blockchain. Bitcoin focuses more on digital money, Ethereum focuses more on messages via social media, storage services, contracts and other peer-to-peer transactions and the recording of values.

Currently Ethereum is mainly an open source, public platform based on blockchain technology for the development of smart contracts. The software code itself is put on a blockchain as value so you cannot mutate this. Once the code is in the blockchain, you cannot change this anymore. However, you can create a new value and put this in the blockchain as transaction that replaces old software, but you cannot undo transactions that have been made.

This concludes the description of the different technical concepts as part of blockchain technology and therefore Blockchain Organizing. We will continue with the application thereof within new organizations (models).

3.5 Blockchain for communities

We as authors are mainly interested in blockchains for communities. These are hybrid forms between fully central and decentral. As there are many types of communities this requires some explanation. A community is often seen as a company that forms a community with its customers, or a council with its citizens. We call this 1:n community. One organization maintains a connection with many of its members, citizens, customers or employees. Often communities are associated with an online community or group on LinkedIn or Facebook. Most of these communities were created at the beginning of the twenty-first century. They were created from a technology and business push. What we mean by communities are n:n communities: many sellers and many buyers who cooperate with smart contracts in a shared transaction network. Most existing n:n communities that are currently engaged in sharing information and executing transactions, are platforms or business communities¹⁰³ like Uber and Airbnb. There is hardly any decentralized authority or financing possibilities with this.

¹⁰² DAPP: Decentralized Application.

¹⁰³ Examples are Amazon or Bol.com, companies that organizes payments and logistics when you offer products via their platform.

Given the experiences with Facebook¹⁰⁴ and the Uber-drivers protests for example, the next generation of n:n communities will no longer belong to one company, but will be more decentralized, not just in terms of infrastructure but also in terms of governance. They not only enable information sharing but also making transactions: not depending on time, place and organization. Plus, more and more communities will be connected to a shared transaction network, which again is connected to a general data utility facility. To communicate and make transactions with each other, you use community software, similar to wallet software for Bitcoin. Community software will more and more replace company software. The thinking unit with organizing and automating demand and supply will change more and more from company to collaborations and eventually shared transaction networks. Transferring (valuable) information is also a transaction. The insecurity in transactions will increasingly be reduced by protocols that will replace existing institutions such as purchase and sales conditions. We are at the beginning of a network economy whereby companies will review their own role in solving management problems and organizing trust. Because companies and non-transparent markets have high transaction costs, they will be increasingly replaced by cheaper resources. Companies will probably become more and more a major node (hub) in shared transaction networks.

The firm and the market as ideal forms

Cooperating and partnerships play an important role within Blockchain Organizing. For example, when we look at the organization of work, we currently choose between doing the work ourselves, via the own company, or outsourcing the work via the market. A partnership is usually seen as an organizational form between these two extremes: company and market¹⁰⁵. Important differences between these two default forms are:

- The firm: a company performs processes itself. For example, Philips a Dutch Electronics concern, at the beginning of the twentieth century. They produced the glass for the light bulb and the paper for packaging themselves because the transaction costs would be too high when outsourced. This form of organizing is used when the need for cooperation is clear and long term. The extent to which joint decisions are taken and tasks, authorities and responsibilities are structured, is higher than for the market. A company is mainly scalable in its structure, but not so flexible.
- The market: when temporary employment agency provides a temporary worker to a company, it is a market transaction as the temporary worker does not gain full employment. This form of cooperation is short term (only the transaction) and the degree to which organization make agreements together is low (at the most a purchase contract or provision agreement). This form of organizing normally involves high transaction costs because you must search for a supplier, enter

¹⁰⁴ The unauthorized use of Facebook data via Cambridge Analytica, for example.

¹⁰⁵ See also Ben Gomes-Casseres in the Strategic Organization magazine (Vol 1(3):327-33): 'Competitive advantage in alliance constellations' (Augustus 2003).

separate contracts and you require institutions to reduce the insecurity. A market is mainly flexible, but not so scalable.

In principle, cooperating is inherent to organizing. Cooperating mainly involves the coordinating function between supply and demand. There is cooperation, but also competition, both within companies as well as between buyers and sellers in a market. Companies and markets are examples of partnerships. Cooperating constitutes a main part of organizing and can be seen as a counterpart to competing. When you distribute tasks to specialists (competitors), coordination or cooperation is necessary in order to achieve the desired end result. In recent years the western world has been shaped mainly by Darwin's evolution theory who assumed that animals must compete with each other to survive. But Darwin conducted a research on the Galapagos Islands where there was overpopulation. The Russian anarchist Peter Kropotkin conducted a research in Siberia at about the same time, where there was under-population. Here animals had to cooperate to survive.

So both can be helpful to survive: competition and cooperation. It depends on the situation and time which is best in any case. Both in philosophy and organizational theory the concept of 'contingency' plays an important role. In philosophy it means that every objectively perceivable object or subject has something coincidental that could be true but also be false, that could just as well have been different. It has properties that accidentally happen to it. In organizational theory it means that can be stated that characteristics of an object of research depend on various factors from the environment of the object. The contingency approach is based on the idea that every environment requires adapted behavior. See for example Lawrence and Lorsch, who developed an open systems theory of how organizations and organizational sub-units adapt to best meet the demands of their immediate environment.

The examples from nature (Darwin and Kropotkin) are also applicable to our social-economical order. Due to aging in European societies we are increasingly dealing with 'under-population', so to increase productivity it is better to cooperate than to compete. The main issue is that both (competing and cooperating), depending on context and finalization, must be able to take place in good balance. Blockchain Organizing with a suitable infrastructure and management model, plays an important role in this. To find a good balance between cooperating and competing there are five important organizational principles of Blockchain Organizing: 'specialize, coordinate & finalize'; 'programmable institutions'; 'structured flexibility'; 'apart together' and 'back to basic'. These are all addressed in paragraph 2.5.

A partnership or community can be positioned on the scale between the two extremes: the company and the market. Cooperating is often linked to the company and competing to the market. But we all know that there is also competition within companies. We also know that competitors sometimes cooperate in the market place. The idea that a company hardly has any transaction costs is also not correct. When we no longer take the company as unit of thinking, but the smallest building block,

the human being, a company also has transaction costs. People have to be recruited for a project, agreements must be made and there must be a safety net in case something goes wrong. In addition, companies have a turnover of staff what leads to higher transaction costs. People compete and cooperate, within and between organizations.

A community that works with the principles of Blockchain Organizing can both be scalable as well as flexible. Cooperating and competing are in balance and the transaction costs are minimal. A Blockchain organization is preferably a partnership, but not all partnerships are blockchain organized. For example, a joint venture (partnership) can also become a company that performs transactions via the market and operates internally based on its own business culture. Therefore, it is not necessarily a partnership that is blockchain organized. A partnership is often implemented in a new fiscal legal entity. The NAM (a Dutch oil & gas company), for example, is a joint venture of Shell (50%) and Exxon Mobil (50%). The NAM itself is also organized as a 'normal company'.

Cooperation options

To be able to properly place Blockchain Organizing it is useful to make a distinction between the different cooperation options that can apply to organizations (we do not mean the outsourcing of work here). Roughly we see a number of cooperation options between organizations:

1. Bilateral contracts and agreements

This cooperation mainly focuses on a 1:1 relationship, between two existing fiscal legal entities. Usually an agreement is concluded between two companies (but it can also be countries or councils for example). Examples are:

- Joint venture: usually a new entity formed by two organizations with two or more shareholders; has its own management and systems.
- Alliance, cooperation, franchise or distribution agreement.

2. Classic community

A classic partnership has a 1:n relationship. One (1) organization enters a relationship with its customers, citizens or for example its members (n). Microsoft for instance has an online community with its users. The communities are often created by the rise of Web2.0: the interactive web. With the use of these communities it is easy to share target group-oriented information and perform transactions. Companies can make transactions online and communicate with their customer or for example councils with their citizens. But classic partnerships essentially change little in the way production and consumption is organized. The producer, the government authority or the company still dictates the way consumers, citizens and employees must communicate and transact. Only the modality or medium, from offline to online, from physical to digital, changes.

3. Modern community

A modern community is characterized by a n:n relationship: many sellers and buyers cooperate smartly in a shared transaction network. The network is often provided by an independent network provider that executes and settles the transactions as a fiscal-legal entity under supervision of a foundation. This network provider is part of a trias politica system with a foundation that represents the interests of users and a supervisor. One can also opt for a (new) fiscal-legal entity such as a cooperative society (co-op). In any case a governance structure is chosen that usually also includes a supervisor and where the various stakeholders make policy and choices in a democratic way. In this form cooperation mainly means sharing information, communicating and making transactions, from the smallest building block via a shared network or shared service center above and beyond the boundaries of existing companies, whereby no party, or parties, can play a dominant role. In addition to the first two forms of cooperation, the modern community offers the possibility to transfer certain processes of the company to a new building and shared transaction network.

Cooperating with blockchain principles

As company, you usually have a vision: what is happening to the world, with my company, my market? As a company, you choose a certain mission (where do I stand?). Finally, you choose a target and target group and you develop a strategy. Strategy can be seen as the way you want to reach your goals. You usually require people and resources for the implementation of the strategy. With Blockchain Organizing, mainly the cohesion, the connection between the smallest elements is important. Because you very rarely will have infinite resources and capabilities, it is necessary to cooperate with your environment. You can do this by using your own systems and rules (we usually call this outsourcing), but you can also realize a cooperation within a joint venture or alliance (1:1) or classic community (1:n) or for example modern community (n:n). When you lack certain capabilities as an organization, you have a number of options, for example you can:

- train people or hire expertise
- outsource work
- take over a (part of a) company
- enter a cooperation

All these options have the purpose to add value, make connections and perform transactions. A certain choice usually has advantages and disadvantages. In economic theory, these can usually be traced back to the transaction costs and you could also call these 'connection costs'. It is generally assumed that the connection costs within a company are lower than the transaction costs between companies (via a market). In the past, the connection costs (transport, communication, coordination and risk) were relatively high with physical production and companies did most of the work themselves. But, with the switch from physical to digital production and consumption, the arrival of internet and now the rise of Blockchain Organizing, the

connection costs in a digital community are almost zero. Oliver Williamson¹⁰⁶ is seen as one of the founders of an economic theory that can be used for Blockchain Organizing.

According to Williamson, organizations exist because of the limited span or control of a human being (insecurity reduction in managing people) and because of the high connection costs associated with outsourcing or cooperating. Managers can only manage a limited number of people. Companies use a structure with a hierarchy to solve this, whereby a team is not greater than, for example, fifty people and a departmental staff does not exceed one hundred and fifty people. In doing so, a certain business culture is developed that causes employees, more or less automatically, to adhere to the desired behavior. It is thought that scalability is an advantage of a company. Due to structure and culture you can use a larger span or control and you probably have less connection costs. A disadvantage of a company, with high fixed costs, is in any case that it is less flexible. Basically, Williamson's theory means that you could choose between concluding separate contracts (connections) to realize a certain objective. Separate connections with individual people and resources. The alternative is to conclude one contract with one company, some kind of black box, or nexus of contracts that has already organized the separate connections in its structure and culture. Both alternatives have advantages and disadvantages. You can see them as the two extremes of a scale.

You can work with:

1. separate connections or transactions via the market;
2. a collection of fixed connections via a company.

An example of a combination of these two applications can be found in the movie industry. It would not be smart to employ actors because the demand is different every time. Actors are casted and usually work as freelancers. However, the equipment is used in every film. So, you can permanently employ lighting, sound and camera people and therefore save transaction costs to select and contract them every time. There are all kinds of partnerships between the two extremes of doing it yourself and outsourcing. The Blockchain Organizing concept is fairly new because it primarily focuses on a new balance between demand and supply, structure and flexibility, cooperating and competing and between separate and together. Blockchain Organizing has low transactions costs because the connection, the coordinating capacity between demand and supply is programmed in a protocol. This makes Blockchain Organizing scalable and flexible.

Classic organizing takes existing companies and markets (metaphor: Lego building kit), as principle. Blockchain Organizing takes organizing capacity (metaphor: Lego building blocks) as principle. Classic organizing often (implicitly) reasons from the

¹⁰⁶ See for example also paragraph 2.6 in this book and the Weconomics analysis book, paragraph 5.9: 'Making choices and considering interests'.

supply (what is already there). Blockchain Organizing (explicitly) reasons from the demand and starts to organize (connect) people and resources within a certain organizational model, transition program and certain infrastructure, at minimum connection costs. If you, as an organization, want to outsource a certain process, you can do this by entering cooperation with a community. For example: a group of self-employed people, united in a co-op¹⁰⁷. With Blockchain Organizing you can set up the process of searching and booking flex workers in a community. Many hirers (n) and many portfolio workers (n) cooperate in this n:n community. The community has the advantage that the relationship between scalability and flexibility is often better than within a company or via a market. Blockchain Organizing does not only provide a better balance between scalability and flexibility, but also between cooperating and competing, part and total.

Examples of blockchain communities

The main experience that we have and use as writers of this book, is our involvement (since the rise of internet), in the setup of n:n communities. We are convinced that Blockchain Organizing works especially well in n:n communities. In figure 3.7 you see an example of a n:n community for the Dutch training market. In this example, you see three sellers and buyers, but the principle also works with 300, 3000 or 3 million sellers and buyers.

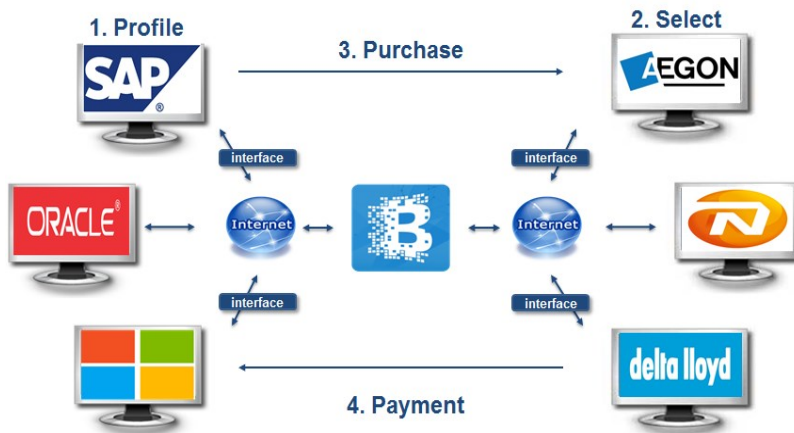


Figure 3.7: Example blockchain community for the Dutch training market

Between supply and demand you could develop and implement a smart contract (programmed workflow where for example a copy can put in a public library when the document is approved or where step 3 can start when 2 is finished). In figure:

¹⁰⁷ In December of 2012 a collective of sixty self-employed lawyers wins a European tender for the provision of legal services to the Ministry of Economic affairs in the Netherlands. These sixty lawyers are supported by the central organization: 'Network of Enterprising Lawyers'. They smartly cooperate via a shared (information) network.

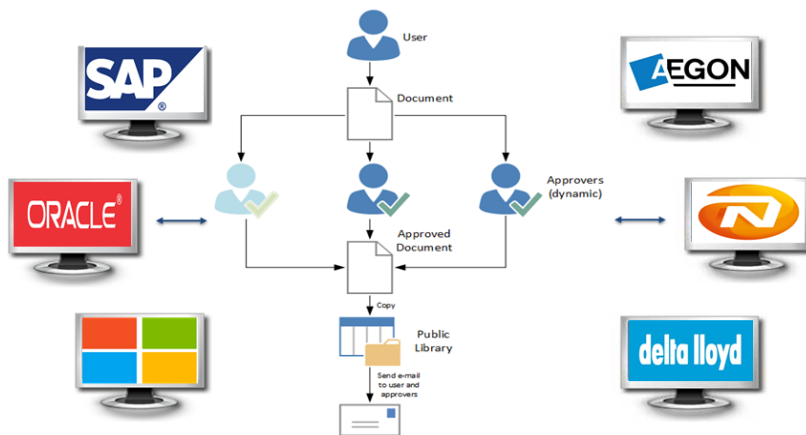


Figure 3.8: Smartcontract between supply and demand

Since the rise of internet, we have built up experience with organizing process step 1 through 3. Step 1 is creating a profile of the product/service (the value, for example a time management course) that you offer. Step 2 is searching and selecting a value and step 3 the purchase and delivery of the value. Step 4, the payment or the delivery of the counter-value will now be designed by using blockchain technology. Examples of n:n communities whereby we have used or use these principles of Blockchain Organizing are: Buro Staff Support, for the combination deployment of trainees and advisors in consultancy projects, Confluent Logistics to better organize supply chains, Prodis for the Dutch training market, School for Cooperating for a fundamentally new perspective on cooperating between positions and organization and Weconet Blockchain Technologies, a community for Blockchain Organizing. In addition, we currently cooperate in the setup of a HR, care and social coins community, and in the PLATO project for a smarter cooperation between different company positions to address the challenges of the new privacy legislation GDPR, which is further explained in paragraph 5.3.

3.6 Blockchain standardization

Many of the agreements that we make with each other, especially in political terms, are agreements between people and therefore also interpretable by people. According to George Packer¹⁰⁸, politics (also politics within companies) is about transactions, it is opaque and consists of hierarchical power relationships. They are sealed with handshakes and not recorded. When they are recorded, it is ensured that messages can be adjusted or interpreted afterwards. Technology however is empirical and often transparent, driven by data. This means that we can reduce insecurities in transactions

¹⁰⁸ Journalist at The New Yorker.

with the use of protocols, by removing the subjective interpretation of man. Whether 'politics' or 'man' also want this, is a different matter altogether. We can best explain the difference between politics/man and technology, and between the 'social script' and the 'technical script', with the Pythagorean Theorem:

In Politics: A^2+B^2 could be equal to C^2

In Technology: $A^2+B^2=C^2$

A protocol is a behavior or implementation agreement, usually in the form of a number of steps to be implemented sequentially, in parallel and / or 'under condition'. An example is the Internet Protocol (IP): an agreement whereby computers can communicate with each other in a network. When a protocol (by the international community) is structured, formalized and recognized, we usually call this a (ISO) standard. A standard usually is a document in which agreements are structural recorded and as such recognized, acknowledged and controllable. They could be related to a product, service, process or method. A protocol can lead to an ISO standard. So, you could see a standard as a protocol recognized by the international community.

As indicated before, you can regard blockchain as an additional layer to internet while internet can be seen as a network to share information peer-to-peer. Blockchain can be looked at as a network to execute transactions peer-to-peer without having to trust each other. Therefore, we also call blockchain the Trust Web. Internet can only exist because we have made agreements with each other about how we prepare (provide), send (transport) and receive (request) data with a (computer) device. Because we must be able to trust each other that this process is going well, we reduce the insecurity by developing a protocol and by agreeing, standardizing, recognizing and normalizing its use. So, to make reliable transactions we use agreements and programmable trust. Examples of these agreements or (uniform) protocols for internet are¹⁰⁹:

- TCP (Transmission Control Protocol)
- IP (Internet Protocol)
- URL (Uniform Resource Locator)
- HTML (HyperText Markup Language)
- XML (Extensible Markup Language)
- HTTP (HyperText Transfer Protocol)

Currently there is no (uniform) protocol for blockchain. However, there is a Bitcoin protocol that 'automatically' performs a number of processes at the transport level of Bitcoin from seller to buyer. You can also see Bitcoin as proof that blockchain technology works. But this does not mean that the Bitcoin protocol will automatically become the (yet to be developed, or agreed on) blockchain protocol. Parts of the Bitcoin protocol are for example the:

¹⁰⁹ For more explanation, see also the Weconomics practice book, paragraph 3.4: 'Weconomics and the World Wide Web'.

- downloading of wallet or mining software
- collecting and subsequently finding 'close' neighbors
- creating private and public keys and digital signatures
- implementing hash functions
- implementing mining
- managing and removing unsorted transactions
- including a block in the blockchain

So, the Bitcoin protocol is not the same as the blockchain protocol. As stated, there is no standard for a blockchain protocol yet. If you are now investing, or betting on the wrong 'standard', you could become the Video 2000 (Philips and Grundig) or Betamax (Sony) system of the video recording technology. Eventually, VHS (JVC) became the standard. Currently there are developments that can contribute to the creation of a blockchain protocol and standard. There are standards under development that indicate how we can determine, manage and share a certain identity quality in a similar way. There are also standards (being developed) to create a personal profile, to enter a training course or vacancy or to execute an economic transaction.

To be able to make a shared transaction network, in which existing organizations cooperate and function properly, it is important to make agreements about the exchange of data. This is often a sticky and lengthy process because existing organizations have their own interests and (legacy) systems. A newly established network organization or community will have fewer problems with this. Based on the last insights and technology, a new blockchain organization or DAO can be set up and subsequently the existing internal processes can be transferred to the new construction. However, this new construction will eventually have to deal with standards, which it probably contributed to, but often cannot determine. The advantage of blockchain technology is that it already contains a certain standardization: it is open source and set up by a community. This is different from a situation where software is developed within companies involving patent applications and a push to become 'the standard'. Nicolas Cary, cofounder and CEO of the company Blockchain says the following about this¹¹⁰: *"It's hard to regulate the blockchain technology in this sense: It's not centralized, there's no overall committees or the general authority. However, it's already regulated by its open-source nature and the fact that it's regulated by mathematics."*

Webservices and APIs are currently also widely used to exchange data, but this is often laborious and increases the risk of data leaks or unauthorized use of personal data. So, we are looking for more sustainable solutions to securely, meaningfully and reliably transport data from seller to buyer, to store it and to wrap it. Eventually we will be able to use a general data utility facility, similar to the electricity network.

¹¹⁰ See article by Alicia Naumoff in the Cointelegraph: 'Blockchain Co-Founder Nicolas Cary: You Can Be Your Own Bank' (11-4-2016).

In paragraph 6.3 of book6 we give a few examples¹¹¹ of initiatives in terms of standardization that can be useful for making sure shared transactions networks function properly. For example, it is important to make agreements about a Personal Data Service, when companies no longer have their own data to perform processes. Personal Data Service on a blockchain is an important application of Blockchain Organizing. Agreements in terms of identity management and authentication are important parts of Blockchain Organizing where agreements are necessary. The mentioned initiatives indicate the important and possible effect of standards and protocols. Later on, they can contribute to the development of a blockchain protocol and the standardization and normalization thereof. Most initiatives are private-public cooperation, illustrating that cooperating and open-innovation is important for Blockchain Organizing.

Open standards

In the Weconomics practice book¹¹² we already extensively discussed the development and the use of open data and standards. New for example is a recently adopted motion¹¹³ that makes it mandatory to use open standards by the Dutch government. This obligation is included in the GDI. The Generic Digital Infrastructure Act will then fall under administrative law. With the new law the House of Representatives can forcefully ensure that the government applies as many open standards as possible.

The House, having heard the deliberations, considering that the Open Government Action Plan 2016-2017 will focus on actively making public sector information available and stimulating an open attitude and behavior among governments; considering that the use of open standards is essential in actively making information available to citizens and in addition ensures for more freedom of choice in ICT suppliers, promoting legal certainty, reducing administrative burdens and being able to work efficiently and in chains as a single government; considering that open source software provides more competition in the software market, so the government can stipulate better conditions and prices for software; noting that open standards and open source software are not yet used everywhere within the national government and local authorities and that is partly due to the lack of knowledge and expertise with the contracting of software; requests the government to oblige the use of open standards by law; also requests the government to make information it has on open source software available via a knowledge center, and proceeds to the order of the day.

(Motion of House of Representative the Netherlands: Oosenburg, proposed 4 October 2016)

According to MP Astrid Oosenburg, the new legislation will soon lead to the choice of open source or free software in a tender, in case of equal suitability. If, for example, this is not yet technically possible, an exit strategy must be drawn up detailing how a government organization can terminate a contract with a non-open source supplier within a maximum of five years.

¹¹¹ No full overview, only to illustrate. We based our opinion regarding these examples on public information.

¹¹² See also the Weconomics practice book, paragraph 5.3.5: 'Free and open source'.

¹¹³ Motion of the Dutch Member of Parliament: Astrid Oosenburg (4-10-2016).

Top-down versus bottom-up

Most standards that are developed are based on the existing world. Many IT projects focus on the automation of analogue processes, without questioning these analogous processes or the institutions that perform them. They try to simplify the existing reality, with all its complexity, by standardizing it. This is a long term, time-consuming and expensive process that provides a lot of employment, but with little effect. It is better to start with a new organizational design that already contains a lot of simplicity. A design that reasons from agreements, building blocks and principles and reduces insecurity, without increasing the complexity. The only fruitful way to come to a shared and long-term standard is: working from a joint ontology.

A number of fundamental questions, on top of the used technical, legal and economic aspects, regarding the initiatives for standardization and regulations explained in this paragraph, are:

- To what extent is the organizational aspect taken into account?
- Are there thoughts about a new organizational model or design?
- What is the role of the subjective person in the performance of processes?
- Do these initiatives take the existing ideal form such as the firm, the government and non-transparent markets as their starting point and unit of analysis or not?

It almost seems impossible to take into account a large number of existing standards and hundreds, maybe thousands of companies with their own interests and their own IT systems. With many initiatives, it is unclear whether Greenfield organizing has been chosen. With Greenfield organizing you reason from the agreements (that transactions and standards have in it), and set up a new network and transfer existing processes from companies to networks. If we look at the way the internet standardization has taken place we must conclude that existing organizations played a big role in this¹¹⁴. This is probably one of the reasons why internet has fundamentally not changed a lot in the way we organize demand and supply. Pilot projects of the Dutch government seem to indicate that existing institutions and analogue processes are taken as unit of thinking and analysis in the design of blockchain processes.

New Organizing means that you must confront the status quo and your stakeholders, where it is not self-evident that firms, government institutions or banks continue to carry out transactions. It would be smart not to develop a blockchain standard on basis of the existing organizational model, but to immediately eliminate the boundaries of companies and to think and work from the smallest building blocks, connections and a shared transaction network. When existing organizations resist this, it can mean that you, in setting up blockchain projects, miss the connection of mainly large organizations in the beginning and mainly cooperate with start-ups, the SME and employee co-ops.

¹¹⁴ See also W3 consortium in Weconomics analysis book, paragraph 6.6: 'Internet of the people'.

Bitcoin protocol

Nobody, but everybody is ‘owner’ of the Bitcoin network and protocol. You can compare this with the fact that nobody is owner of the protocol that makes e-mail traffic possible. The Bitcoin protocol is controlled, updated and managed by a community of Bitcoin users and developers. Software developers are continuously improving the software but cannot enforce changes in the Bitcoin protocol without convincing others of the use, necessity and efficiency. Users are free to choose which software (version) they use. To stay compatible, nodes must use software that can cooperate with a lower level (stack). The Bitcoin protocol only works when a certain degree of agreement has been reached between the users. This can be compared with the required agreement in the Linux community¹¹⁵. To make the Bitcoin Protocol and the underlying principles work for, say, decentralization work, users and developers know that they must find consensus. But this principle is not much different from finding consensus on traffic rules for example. The difference is that we use a community in case of Bitcoin, while traffic rules are decided by representatives of the people. Although the latter is not completely correct, because city councils for example may also use certain regulations that affect traffic.

Blockchain protocol and standardization

In his book ‘The Fourth Industrial Revolution’ Klaus Schwab¹¹⁶ writes: *“The digital revolution is creating radically new approaches that revolutionize the way in which individuals and institutions engage and collaborate. For example, the blockchain, often described as a distributed ledger, is a secure protocol where a network of computers collectively verifies a transaction before it can be recorded and approved”*¹¹⁷. In Schwab’s vision, blockchain essentially is a shared, programmable, cryptographic, secure and reliable ledger, without central control, that can be checked by nodes in the network on input reference and double spending.

Different blockchain protocols will be created. After the Bitcoin protocol, mainly the development of the Ethereum platform is currently important. You could see this as a blockchain of software program code in decentral applications. Currently we see, mainly in the financial sector, a multitude of different blockchains and blockchain companies. Different technologies and protocols are used for different applications, whereby there is little agreement yet on a standard to be used such as TCP/IP with internet. Large companies preferably shield their innovations and ensure that customers are faced with a vendor lock-in. There is little consideration for interoperability, because this is not in the interest of an existing company. Even a number of important blockchain principles are abandoned. An example is the ability to adjust a transaction after this has been agreed between the different nodes. New technology

¹¹⁵ There is not just one Linux community, but several. However, there is one Linux Foundation to promote, protect and improve ‘collaborative development’ of Linux. Linux is an open-source control system.

¹¹⁶ Klaus Swab (1938-), founder and chairman of the World Economic Forum.

¹¹⁷ See paragraph 2.1: ‘Megatrends’ in that book.

competes with existing organizational forms and interests. Therefore, technical possibilities lose their strength and the risk occurs that companies with the most financial means, pull the standard towards them and shape this in their own interest. Because these are usually large, wealthy companies, there is a high risk that blockchain, just as internet, will change little in our productivity, privacy and powerlessness of people and small entrepreneurs.

But there is also a growing awareness and also maybe belief that we (and we mainly mean us people, small entrepreneurs and in a wider context the European Union or maybe the United Nations), are able to do things differently this time. We know the interoperability of data and standardization is necessary to use the complete potential and capacity of blockchain technology. But in the bottom layer data consists of zeros and ones, that is also a form of standardization. The obstacle is not in the standardization but the fact that almost every company wants to have its own database. Even when we look at the new European privacy law there is a need for data transport from one system to the other, if the user requests this. It is better not to connect a blockchain to a company or government but directly to the smallest building block from the start. Starting points for blockchain standardization should be the individual who, for example, has his or her own portfolio, patient file, consumer file and citizen file. This principle is far away from the interests of the most used (social media) platforms and business models of large companies. We know that Facebook is designed so you are tempted as much as possible to click on adverts and to spend as much time as possible on it. It is not designed to easily take your own data, to make transactions with parties that do not pay Facebook.

Besides the development of diverging and eventually possibly converging blockchain protocols, the need for blockchain standards will also increase. There are already some private, public and private/public initiatives, but there is still a lot of work to be done. Apart from the technical work, we believe that it is especially important to work on awareness that blockchain will change little if we do not change the model of organizing demand and supply. To illustrate this better, we list a number of initiatives in terms of cooperation and standardization below:

- The Hyperledger Project¹¹⁸ of the Linux Foundation: is a common effort to develop blockchain technology by developing domain crossing open standards for distributed ledger. Many large organizations are a member.
- Ripple Lab¹¹⁹: introduces the InterLedger Protocol (ILP), a free open source protocol for secure and efficient payments in a network.
- Digital Asset Holding: uses Distributed Ledger Technology for 'tracing and tracking' financial assets.
- R3 CEV consortium: worldwide consortium of large banks managed by the company R3. With 42 banks a test period of sixty days is used to use five different blockchains to spend, trade and redeem financial products.

¹¹⁸ See also website: hyperledger.org

¹¹⁹ See also website: ripple.com/technology

- **B3i consortium:** consortium of insurers who start a payment platform based on blockchain technology. The project initially focuses on the reduction of paper work and the cheaper setup and execution of insurance contracts. B3i is a first step to a standard for blockchain within the insurance world.
- **Muskoka Group:** On the initiative of Don and Alex Tapscott¹²⁰, a group of blockchain experts meets in 2016. The group primarily does not focus (yet) on standardization, but actually on the stage before awareness. To properly use the potential of blockchain we must first agree on a number of principles, before we are able to standardize.
- **The International Organization for Standardization (ISO):** To be able to improve data exchange between users and applications, the standardization institute of Australia suggests to develop international standards for blockchain technology. ISO has asked all 161 members to submit the proposal to the national stakeholders.

So in 2016 ISO also started a consultation round in response to the proposal from Australia to investigate the desirability for standards in terms of blockchain technology. Opinions differ on the extent to which and the speed with which we can come to standardization. People rightly fear a similar development as with the standardization of internet, whereby via the World Wide Web Consortium (W3C) mainly large (American) tech companies had a lot of influence¹²¹. The central influence is at odds with the decentral principle of internet/blockchain, that should give people (citizen, employees and consumer) more power and freedom. Apart from this aspect, it is also a question if blockchain developments are ready 'yet' for standardization? Therefore, the management model and the infrastructure must probably be slightly further developed and (successfully) used.

The development of internet, as we know it today, did also not happen within a few years. The same will apply for the development of standards for blockchain technology, because the interests are too big and we do not have a worldwide knowledge network yet to quickly and professionally publish organizational problems, to exchange solutions and to reach a consensus. Compared to internet, we are now, in 2018, in a similar stage as the beginning of a standard for e-mail traffic, one of the many parts of internet. One of the many initiatives that eventually led to the TCP/IP protocol was Open System Interconnection (OSI). OSI started in 1977 and was an attempt by ISO to realize a standard for a network of computers to exchange data. But many saw the standard as too complex and inefficient to be used widely. There was also resistance from existing suppliers who had already invested a lot in other systems. The TCP/IP protocol is much more pragmatic and cheaper to realize. Eventually, after about fifteen years, it has become the standard for data exchange via internet.

¹²⁰ Authors of the book: 'Blockchain Revolution: How the Technology Behind Bitcoin is Changing Money, Business, and the World' (2016).

¹²¹ More about this in the Weconomics analysis book, paragraph 6.6: 'Internet of the people'.

3.7 Blockchain is industry agnostic

Agnostic, from an organizational model perspective, refers to a concept where data is generalized in a way that it is operable among different information systems. The term can also refer to IT-systems or more specific blockchain technology, blockchain practices, processes and businesses. In our opinion Blockchain Organizing is industry agnostic if you design blockchain organizations with a common ontology, taxonomy and organizing fundamentals. Industry agnostic means that is basically doesn't matter in which (traditional) industry you want to apply Blockchain Organizing. Blockchain is about transporting a value, translated in bits and bytes, from supply to demand, from A to B with a consensus 'Yes' to these four and earlier mentioned important and fundamental questions:

1. Are sender and receiver the persons who they claim they are?
2. Has the transport between supply and demand been executed safely?
3. Was the sender the owner of the value?
4. Did the sender spend the value only once?

Once there is consensus on the 'Yes' of these questions the transaction is valid and processed into a block and a block into to a blockchain. If we reason from the situation that every industry has transactions, a transaction is a value transport and every value can be translated in zero's and ones, all transaction can be designed in the same way at the lowest levels of shared information and transaction networks. If you digitalize values, for instance the claim that you are owner of a house, is a digital paper stored at the land register. If you sell a house you will have a digital transport from supply to demand. So selling a house could be seen as digital transport or transport of a closed box. And it doesn't matter what is in 'the box'. A blockchain doesn't care. Transporting a value from supply to demand can be seen as transporting a closed box from A to B. You could open the box and see a bitcoin, ten bitcoins or the ownership of a house. Still a closed box will be transported from A to B. Money, goods and services they all can be translated into bits and bytes and as a message be transported from A to B. So Blockchain Organizing is about transporting bits and bytes and these can represent all kinds of products and services from all kinds of industry.



Figure 3.9: Blockchain is industry agnostic: it doesn't care what's in the box

Most of the companies or governments we as authors are asked to work with, use their own business model, context, processes and their own industry as a starting point to design and develop a new blockchain organization. This is understandable but not necessary and 'dangerous'. Dangerous in the sense that technology alone won't change enough to address the problems we are facing as a society and as mentioned in paragraph 4.1. We all witness an 'explosion' of possible use cases. Every organization wants to be the first with a blockchain application and think they are unique. Every industry and almost every organization wants to start a blockchain project that starts from their own situation and context. And very soon they find themselves in a situation whereby a disruptive technology has been integrated in an old organization model: a new soul has been fitted in old body. When substantial investments in a new technology have been made, over a five to ten year period you are not going to change your strategy that easily. New technology will, again, be applied in outdated organization models with the sole intention to protect existing revenue models. And as a consequence, in ten years we all will conclude: blockchain is a nice decentral technology, but it is still business as usual.

If we, together with a disruptive technology, don't change the way of organizing supply and demand, nothing will really change fundamentally. Just automate analogue processes won't help us solve the trust and productivity paradox or the power and privacy problem for example. When we analyze Blockchain Organizing with a deductive approach you could say that:

1. Different industries process transactions (major premise)
2. All transactions are basically the same from a digital perspective (minor premise)
3. All industries are the same (conclusion from a transaction perspective)

An agnostic way of doing transactions within the concept of Blockchain Organizing means that you execute transactions via different channels (see also figure 2.1 and 3.5), but with a common basic infrastructure. The supplier can choose an application to profile and offer products and services and, connected to the same data common, the customer can choose a channel to select, purchase and pay. Together with a smart contract they can execute the transaction workflow because they both want to execute a valid transaction. Agnostic means: make your own content available through every contribution and distribution method possible. The opposite of agnostic would be a situation whereby every electronic device in your house or office needs a different type of power socket.

Blockchain is basically a simple model to transport a digital value from A to B with only four conditions the network has to agree upon: authentication of sender and receiver, safe transport without interference, was there an input reference and was the value send only once. When all existing organizations and industries start to apply blockchain from their context or perspective, the power of blockchain will soon

disappear and business as usual will take over (again). We hope that blockchain will not enter history books as the simplest but at the same time the ‘most-complex-to-implement-innovation’ in decades.

Conclusion

It is still too early to come to a blockchain standard as part of Blockchain Organizing. Developing standards before there is a common ground, or a joint understanding on what blockchain exactly entails, what it could mean for us people and which problems it addresses, is not very smart. On the other hand there is already a lot of commonality and standardization in the blockchain protocol itself.

There is a large difference of opinion between, academia, organization designers and software developers on the one hand and business leaders and commercial parties who decide on investments and the development of consumer and business applications on the other hand. First we will have to learn and develop more use cases before we take them into production. But before that we have to agree on the social problems we want to solve and the organizational model we are going to use for this. This ‘exercise’ does not only deal with the technical part, but also the:

- Economic aspect: how does blockchain smartly combine supply and demand and what will this mean for the existing economic actors and business models?
- Political aspect: which powers, regulation and legislation will come into force if blockchain is used in such a way that existing powers will lose their power basis?
- Legal aspect: what are the legal consequences of another management model to organize supply and demand? What if there is a better balance between buyer and seller due to better (more objective synchronous real-time) information exchange? What is the need for contract in a new digital context?

And more widely, for example in the Weconomics philosophy, the following aspects should also play a role:

- Ecological: does blockchain ensure that we will consume in a more sustainable way and waste less?
- Psychological aspect, mainly identity: man shapes tools and then tools shape us. Would we behave differently as a human being in our interactions, economic traffic, home situation if we have different tools to organize trust?
- Social aspect: will inequality reduce when more people have access to a general data utility facility to which a blockchain infrastructure will be connected later on?

In addition to the importance of blockchain as technology, we mainly see blockchain as ‘a way of organizing’. To understand this new way of organizing, we must first find consensus on the problem we want to solve and the direction we want to go. What is progress and how do we get there? Therefore, we will have to redesign the existing systems with requirements, wishes and processes. So, before we develop technology (standards) from the existing mind-set and system, we believe that we must first contemplate the system itself. We will have to pay attention to the principles that we will use in further developments and use of the technology. Otherwise

blockchain will change as much or as little as internet (depending on your views and interests).

Primarily, a tool like Blockchain Organizing should contribute to the solution (and definitely not the creation) of social organizational problems. We will not benefit at all if a blockchain standard must include exceptions for a specific domain or certain sectors. It is not necessary at all to make a distinction between different domains when you reason from similarities, foundations and smallest building blocks. Inherent to the existing software industry, is that they benefit from differences and complex systems. This has little to do with standardization and technology, more with perspective and politics. It would be great if we could get 'the industry' interested in a new target for organizational technologies that we develop: new organizational technology and standards within, must contribute to the improvement of the meaning of human work, productivity, privacy and power for the citizen. As authors of this book, we are happy to contribute to this endeavor.

4. What does it mean for your organization?

When we see Blockchain Organizing as a mechanism, ability, enabler or a tool to manage transactions within an organization and especially between organizations, there are many valorization or application options. Here are some examples of Blockchain Organizing:

- selecting, hiring and connecting humans and resources to tasks, projects and finalizations;
- organizing Employee Self Service whereby employees can profile, select, order, pay and register HR instruments on a blockchain themselves;
- organizing trust within your organization, for example through identity management and document control;
- smart contracts for workflows, employment contracts, framework agreements;
- expense payments, payroll and personnel administration, personnel information systems.

In general, you can apply Blockchain Organizing well in those domains with many buyers and sellers (fat markets), little regulations, relatively expensive intermediaries (such as temporary employment agencies, payroll companies and interim-management agencies) and enough innovative capacity. In this chapter, we start with the introduction of some domains (industries) which are relevant for managers and their consultants. We will mainly talk about organizing:

- money, trade and economy (trust and economics)
- HR and management domain (work)
- supply chains, cooperation between organizations (efficiency, transparency)
- government tasks (commonalities)

You might also expect an explanation of applying Blockchain Organizing for the financial sector, also called Fintech, but there are already several specific books about this. In addition, we currently have the feeling that in this sector, although it sounds paradoxically because most investments are made here, the least will change fundamentally in the short run. This is mainly due to the fact that blockchain is an important mechanism to render many parts of the financial sector irrelevant. The reason that a lot of money and effort is being invested now has, in our opinion, mainly to do with internal efficiency improvements and self-preservation. At the beginning of 2018 we have the feeling that more people are involved in buying and selling cryptocurrencies than actually designing and developing useful ideas and applications to solve social problems.

Before we go into these four domains, we start with the question: which problems can you solve with Blockchain Organizing?

4.1 Which problems can you solve with Blockchain Organizing?

The rise of most populist movements in Europe, at the beginning of the twenty-first century, can largely be contributed to the fact that our welfare state has become too expensive. The expectations cannot be met through sufficiently high productivity growth. If we do not want to take on more debt, privatize care duties, pay tax, work longer and still keep our welfare intact, we must become more productive¹²². Since the 1980s the productivity growth has lagged behind the rapidly growing costs of care, education, security and climate change. This is starting to become a worldwide and fundamental problem, which unfortunately is acknowledged by few politicians, policy maker and business leaders¹²³.

One of the main challenges that we address with Blockchain Organizing, is the improvement of the productivity of information workers. We can easily switch to a 6-hour working day. The technology, to realize this, is available. All we need now is the (political) insight and the will to change. When you improve productivity, without producing more or firing people, you save time. Blockchain Organizing can ensure that you can do the same amount of work in less time. The surplus time is then available for tasks that the government can no longer afford to pay for at the contemporary level, such as healthcare, education, safety and the consequences of climate change. Therefore, we can maintain the same welfare without additional costs.

Blockchain Organizing is a form of organizational technology. Organizational technology can help us become more productive provided we find a meaningful interpretation of the surplus time that will become available. Now we invent fakenwork at the office to avoid real work where it is needed. Apart from productivity, there are other problems that we can solve with organizational technology. In this book we will mainly focus on the following challenges and drivers:

1. Purpose: why innovate (computer) machines and still want to work?
2. Productivity: improving the productivity of information workers
3. Privacy: stricter European laws for the protection of personal data
4. Power: becoming less dependent on capital, governments and tech companies
5. Propaganda: more towards a shared reality
6. Process: efficient performance of processes within existing organizations

Purpose

We require a new perspective related to the question: where are we going and how do we get there? What is progress? When will our children be better off? More and more people lose the direction, the purpose, the reason of living. It was always clear: you worked to provide for your family and to give your children a better life. That was progress: making sure that your children would be better off and for example could buy a house or car or complete a study. This form of progress is 'bluntly'

¹²² Other alternatives are 'making' more children or allowing more immigrants.

¹²³ See for example article in the Financial Times: 'Weak productivity turns into a problem of global proportions' (25-05-2015).

translated into: economic growth. But (continuous) economic growth is not the same as progress. In fact: it endangers progress. The sadness is that our own invention, you could call it capitalism and within that neo-liberalism, will eventually kill us.

We see that the purpose in our lives is moving from more and more welfare to more well-being. With this movement, the meaning of work and how we will organize 'new work' also changes. For example, if you as government, the unemployment and benefit office, trade union or employers' association are continuously engaged in the employment market and want to give people the prospect of a permanent job, you probably use the idea of having permanent jobs as the basis. It then becomes self-evident that creating employment and having a permanent job is desirable and good. The idea that job creation is a good idea is framed: policymakers create the circumstances that their idea becomes true. The question is who benefits from this? The people, the government or the institutions and companies that are engaged in employment services?

A permanent job is only a regulation principal to reduce insecurity in supply and demand of work, something that provides 'security' of income or adds value. But we know that is a 'fake' security to increase supply if there is no demand. Arrangements such as permanent jobs, seems fine during the Industrial Revolution. However, if we cannot meet the expectation of our welfare state with a higher productivity, hierarchies, permanent jobs and full employment can have a negative effect if used as regulation principle. More and more people are looking for the meaning of work and employment organizations and can no longer find this in the answers of the establishment. The 'basic income' theme also continues to return when we talk for example about robotization and the disappearance of jobs.

When filling or organizing permanent jobs is your task, you probably no longer wonder about the question what a job actually is. Why there are jobs and why we actually work (in an economic sense). The theme employment also works well during elections: nobody is in favor of more unemployment and nobody is against more jobs. Right? We have become a supply economy. We have offices, we employ people and to give them security, we subsequently start looking for finalizations because our fixed costs still have to be paid.

That is the opposite world of how it should be. The effectiveness question is apparently no longer important. Why do we, people, innovate (computer) machines (we suppose with the idea to take over our work) and at the same time we want to continue to work? Why do we strive for full employment? Why do computers not ensure that we become more productive in offices so we have to work less and can spend more time for example on education, care and leisure? We know that we do not have enough time for the things that we find really important in our life. This means that we must transfer time from non-productive working in offices, to a meaningful contribution to our direct environment and society.

Increasingly more people are critical regarding the perspective of full employment and are looking for a new meaning and intention of work, employment organizations and unemployment. Blockchain Organizing, in combination with a new perspective on work and employment organizations, can ensure that we have more time available. In relation to the organization of work, Blockchain Organizing will lead to us having to work less. We will not spend the surplus time on tasks that are irrelevant as Cyril Parkinson¹²⁴ noted (also called bullshit jobs¹²⁵), or on producing things we don't need, but on sustainability.

If we move to programmable institutions and Decentralized Autonomous Organizations (DAO)¹²⁶ with the use of Blockchain Organizing, we, as human beings, must mainly and firstly think about what we will do with the spare time. Or as formulated by Karl Marx: *"The production of too many useful things results in too many useless people."* People who no longer have to go to the office unnecessarily because smart machines, computers and protocols have taken over the work, will have more spare time. They can use this time to perform tasks that the government is handing back to the citizen¹²⁷. This way we can slow down the crumbling welfare state and bring it back to the old level.

Productivity

Since the 1970s (rise of offices) we see a drop in productivity growth. You could say that the productivity growth, on average in Western countries in agriculture and industry is above the level of cost increases in welfare. The problem is in the productivity growth of information and knowledge worker (see also figure 4.1). When we look at the figures¹²⁸ of the development of productivity growth during the period 1800-2010, we see the following picture:

¹²⁴ See also book: 'Parkinson's Law: The Pursuit of Progress' (1955).

¹²⁵ As these are called by David Graeber. David Graeber (1961-) is an American anthropologist, anarchist and professor of Social Anthropology at Goldsmith University of London. Graeber is activist (Occupy Wall Street). His contract at Yale University would not have been renewed for questionable reasons.

¹²⁶ Decentral Self-Operating Organizations: similar to autonomous driven cars.

¹²⁷ Consider the concrete interpretation of the participatory society where the central government transfers tasks to councils and councils to the citizen. Enforces, because we, as citizens, do not want to governments to borrow more money, to privatize more public facilities or to charge more taxes.

¹²⁸ England is the country where Industrial Revolution is first visible in the improvement of productivity. In figure 4.1 we see the productivity growth since 1800. The figures are from the Bank of England. See also website: bankofengland.co.uk/research/Pages/onebank/threecenturies.aspx

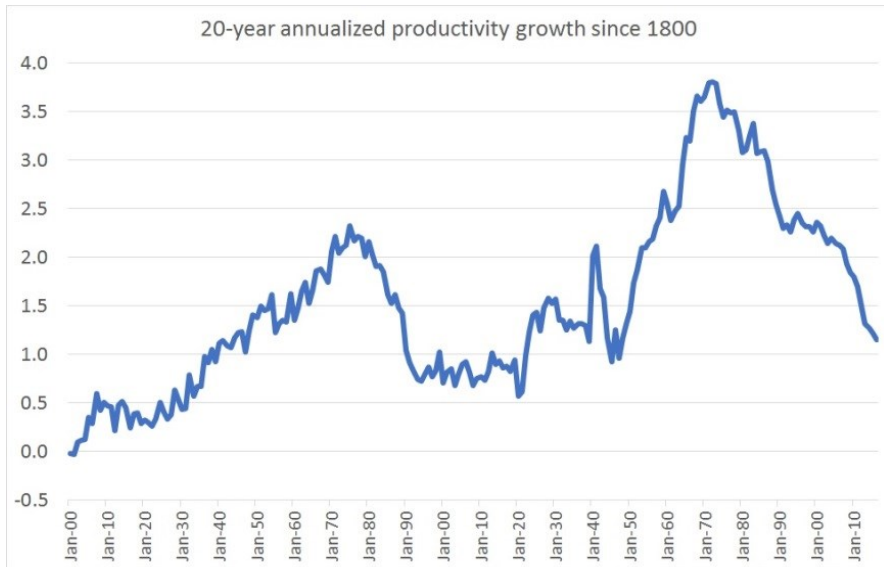


Figure 4.1: Productivity growth all industries since 1800 (Jan-00 from left to right is: January 1800, January 1900 etcetera)

A drop in growth does not have to be a problem. It will only become a problem when the welfare costs increase faster than productivity. After all you must be able to justify more consumption with more production. Because our welfare costs increase faster than our productivity, a gap occurs that has mainly been filled with privatization, borrowed money and more taxes, since the eighties. As a result, the national debt in the Netherlands for example has increased from 86 billion euros in 1980 to more than 450 billion in 2016. After the crisis of 2008 we believed that budget cuts would solve the problem, but that failed. The national debt has almost doubled since the crisis.

The main and most fundamental way to balance production and consumption is the improvement of productivity: as a person, you may spend more on welfare if you produce more. But our productivity does not grow as much as our welfare costs. Especially information workers become insufficiently more productive because they use expensive industrial institutions to reduce insecurity, they always must retype the same data and they find it difficult to share information because there are walls between departments, companies and systems. Partly because we still organize demand and data in a traditional way, the productivity in offices does not grow sufficiently.

It is time that we start to use an 'internet assembly-line' in our digital society, whereby values are automatically transferred from workstation to workstation. Blockchain technology can provide this. Work and employment organizations must fit together. They have not done so since the end of the twentieth century.

To meet the increasing demand for productivity growth, we must first switch to a fundamentally other organizational design for our digital society. This new design also has the advantage that it aligns better with the new and much stricter European privacy laws.

Privacy

Everybody has to face it, only a few are prepared for it¹²⁹. Since 2016 there is new European legislation regarding the protection of personal data. The new privacy law has an impact on both individuals as well as organizations, mainly within IT organizations. Each organization has a HRM, CRM or ERP system with personal data. Different articles¹³⁰, make it clear how naïvely many people handle their personal data. As a result their insurance policy could be cancelled or a mortgage application is refused inexplicably. They could be stopped at an airport because someone has stolen their identity. When it concerns the use of personal data, the naivety of many people can be compared with entering into more debts during the 2008 crisis.

Initially people often react dismissive when we invite them to provide input regarding the consequences of the new privacy legislation. For them it is still too far away. Or they do make it a priority, or, as with entering debts, that the government will keep ‘an eye on things’. Unfortunately, governments themselves are guilty of not handling personal data carefully. At the same time the phenomenon emerges whereby tools start influencing our behavior. That is actually the intention of tools or institutions: they are there to reduce the insecurity in our behavior. Our behavior is not just formed by laws and rules, but also by IT systems. We like ‘free’ social media, we use it every day, we get used to it, but eventually we cannot do without. We must take into account that man forms the tool and then the tool forms us. This means that if we are not aware, tools will become more and more powerful and therefore reduce the power of people.

In addition to citizens, organizations are not (yet) really impressed by the *General Data Protection Regulation* (GDPR). Directors often haven't even heard of it. However, a data leak that is reported too late can easily cost your organization up to four percent of the worldwide annual turnover! And the risk of data leaks is ever increasing with the complexity of IT systems that are continuously added with new functions. Plus, more and more data is stored in ‘the cloud’ and more and more systems are connected to each other via internet. Lately a customer told us: “*We have connected so many IT systems that we did not even notice that there was a back.*”

Blockchain Organizing ensures, with new concepts like Personal Data Service (PDS), Bring Your Own Data (BYOD) and Data Logistics, that companies and governments have to ‘own’ less data to perform their processes. Not ownership, but access

¹²⁹ In 2016 a total of 5,500 data leaks were reported to the Data Protection Authority (DPA) in The Netherlands. Most reports are from (healthcare) insurers, hospitals, banks and governments. See also: <https://autoriteitpersoonsgegevens.nl/en>

¹³⁰ See for example article from Jérôme Groetenbriel: “Tinder must be joking with their “download your data” option’ (October 31, 2017).

becomes important. So, you cannot leak data as an organization if you don't have the data in your own systems. For example, if a webshop requires your address to send a product, they 'ping' a request to your PDS: What is your address? Subsequently they receive a reply and as soon as the transaction is completed your address will no longer be present in their system (unless you give explicit permission for this, but why would you do that?).

By fundamentally changing the organization of (personal) data with blockchain technology, we also address the problem of the increasing power of large, often American, data companies.

Power

In Europe, there is a growing need to become less dependent on powerful technology companies¹³¹. Currently you as consumer do not have a real alternative for free of charge platforms, where your personal data is the product that generates revenue. The business model of most free (social media) platforms is simple: *the platform is free but you pay by giving your data*. You pay through exchange: your data as raw material for the production of adverts or datasets by Facebook, in exchange for free use of the infrastructure.

It seems that we are not aware what it can mean to be dependent on a limited number of companies from Silicon Valley. You can compare it with the dependence on cheap oil supplied by a limited number of countries from the Middle East. If Europe wants to become less dependent on American tech companies, it is important that there will be a better cooperation to organize the supply and demand of data. Just as when the borders were opened in 1992¹³², logistics, in this case data logistics, also plays an important role again.

For many technical innovations, it is necessary that these are realized based on a strong cooperation (co-creation) between knowledge institutions, enterprises, governments, interest organizations and users. Of course, we can wait till large tech companies come with a standardization of the blockchain protocol, but it is better to contribute to this ourselves from a joint European cooperation, or maybe even stipulate it.

By further developing and applying innovations such as PDS, we can respond to the undesired power growth, not just of American tech companies, but also of central governments, e-commerce companies, banks and insurers. Related to the power of large companies is the power of financial capital. The improvement of the productivity provides more advantage for money lenders than for human capital. Apart from the lagging productivity growth, the available growth is disproportionately divided.

¹³¹ Also called the 'big five': Alphabet (Google), Amazon, Apple, Facebook and Microsoft.

¹³² See also: Bessems, P., R. Schenk & L. Verspui (red.): "International distribution logistics after 1992", final report Industria study trip 3-24 April 1989. TU-Eindhoven: Study association of the Faculty of Business Administration (Dutch only).

Real incomes are lagging behind productivity growth. Much employment is replaced by technology and mainly money lenders make money on this.

Propaganda

2016 was not just the year of Brexit, increasing populism and the victory of Donald Trump, but also of the renewed attention to fake news and manipulated news. Fake news is news of which the creator is consciously aware that the news is incorrect and distributes this with a certain intention. Manipulated news does not necessarily have to be true. For example, facts can be omitted or certain messages have a higher level in search results or on timelines. Propaganda, that covers a part of the fake news and manipulated news, has been around for some time, but it is becoming an increasingly bigger problem through the intensive use of internet and social media.

For example, it cannot just influence elections, which it has always done, but maybe even determine. By using propaganda in combination with social media, people with more money can easier influence decisions than people with fewer resources. An investor in Facebook will not directly influence the timeline algorithm, but definitely indirectly. This process can undermine our democracy. Even one of Facebook early investors and former ‘coach’ of Mark Zuckerberg worries about Facebook's algorithm and power¹³³. Blockchain technology can help to slow this process down.

Internet critic Evgeny Morozov¹³⁴ is right to indicate that technological progress does not automatically lead to social progress. The modern world has perhaps become too complex to understand. This has everything to do with the aforementioned trust paradox. With Blockchain Organizing a new way is created to address this. The number of (technical) tools with which we can form, realize and control a shared reality will improve. But whether we use technology properly is up to people. Because despite all technological improvements, human contact and interaction still remains the most important way to try to understand the complex world together in a dialogue, and to form a picture or opinion. Technology cannot replace all aspects of human interaction.

Process

In the short term blockchain technology will help to ensure that processes within organizations can be carried out more efficiently¹³⁵. Especially within financial institutions, there will be more and more investments in blockchain applications, for example to make the payment traffic more efficient and quicker¹³⁶. Banks also invest in Fintech start-ups that can perform certain processes cheaper. But in the short term

¹³³ See for example article from by Roger McNamee: ‘How to Fix Facebook - Before It Fixes Us’ (January 2018).

¹³⁴ Evgeny Morozov wrote the book ‘The Net Delusion’, in which he shows that authoritarian regimes use internet to suppress resistance.

¹³⁵ Currently, 70 per cent of all financial transactions are executed via algorithms.

¹³⁶ See for example report from Morten Bech and Rodney Garratt, employed at The Bank of International Settlements (BIS): ‘Central Bank cryptocurrencies’ (September 2017).

these Fintech companies often require a bank again because of the banking license. That means an extra intermediary in the chain rather than fewer intermediaries. This is at odds with the intention of Blockchain Organizing: reduction of the number of intermediaries in order to reduce insecurity and to organize trust without increasing bureaucracy and complexity. The improvement of existing processes is mainly useful when the activities are aimed at 'preparing' processes, so they can later be transferred to a new blockchain organization. You can read more about the transition of existing organizations to new blockchain organizations in chapter 5.

So far, we have seen six challenges Blockchain Organizing can contribute to. Now we continue with a number of applications of Blockchain Organizing subdivided into four themes: economy, work, supply chains and government.

4.2 Blockchain for the organization of money, trade and economy

The basis for a smoothly running economy is trust. When trust is not, or insufficiently, present, we, as human beings, will invent all kinds of tools to reduce the insecurity in transactions and to organize trust. We innovate the purchase order, the invoice, the double bookkeeping and audit for companies. To be able to trade internationally, we have export credits, exchange rates, we conclude trade agreements and we develop import charges if we are not confident that the other country does business in an honest way. To organize our welfare we must pay taxes and we receive surcharges and grants, and everything is checked several times. When we have a financial crisis, like in 2008, this is not solved by thinking differently, but by installing more rules for banks and financial institutions.

According to Nancy McKinstry, CEO of Wolters Kluwer, every year thousands of new rules and procedures are added to the financial sector. And if the government ratifies a new privacy law, it will soon generate tens of thousands of new jobs for Data Protection Officers, all institutions that should reduce insecurity, but at the same time increase the number of parties and therefore the complexity between supply and demand. We solve coordination problems with even more super specialists and institutions that we subsequently badly coordinate. These are all examples of not considering what is the most effective way to handle these issues and shows little knowledge of the most basic organizational principles. Policy makers expect that more rules and procedures (and therefore staff and management) will improve the performance of transactions, while the opposite is the case.

The previously mentioned research by Hamel and Zanini shows that the costs of an excessive bureaucracy in the United States constitute seventeen percent of the GDP. The average span or control is 1:4.7, one manager/staff for almost five direct employees. In 'lean' organizations such as Morning Star and Svenska Handelsbanken the ratio is 1:10. The main reason for a larger span or control is decentralization. The figures in the US can be compared with other OECD countries. By using new coordination mechanisms, such as Blockchain Organizing, the productivity can grow

additional by 1.3% per year¹³⁷ because we require less bureaucracy and institutions. This growth would already suffice to keep up with the increasing welfare costs.

So, we must not contemplate how to regulate banks, but we must ask ourselves how we can make banks less relevant by becoming less dependent on money as ‘medium of exchange’ for transactions. By continuing to think in the same manner we did before, we’ll try to solve new problems with the same frame of mind that created them in the first place. Instead, we must think in a new way and towards a new system that is less dependent on social institutions. We must switch to a system whereby two parties (preferably two people), can do business with each other with as little friction as possible. Blockchain Organizing can help with this by ‘programming’ institutions into smart contracts. These will execute transactions based on a technical script, no longer a social script.

Various steps with transactions

When you buy something in a store, using your credit card, it often requires five to seven parties to make sure the payment is executed correctly¹³⁸. There are institutions between the cash machine and bank account of the shopkeeper, that must reduce the insecurity. After all, you do not want your money to go to the wrong party and the shopkeeper wants to make sure that there are sufficient funds available. Trust is fundamental for trade and economy. Without trust our community would not function. The only question is: What is the best, most efficient way to reduce insecurity in transactions. The core of organizing trust in trade includes a number of parts:

1. The seller must proof: ownership of the value that is to be rented or sold, and the right to rent out the property.
2. The seller can only spend a certain value or right once.
3. A party must be able to be authenticated, sometimes without revealing their identity.
4. The value must arrive at the receiver for whom it is intended.
5. A non-mutable registration of the transaction must take place.
6. It must be possible to securely execute the transaction.

To be able to perform these steps, costs have to be incurred. These costs are part of the direct transaction costs. Apart from these direct transaction costs, there are also costs to maintain the reputation of institutes or to maintain the trust in the money system or currency. Money is still one of the main institutions to organize trust. Money is a medium of exchange. It is an important tool, also called the lubrication for our (open) economy¹³⁹. Most money that we use is fiduciary money. This means: money that you can trust as a citizen because the government monitors that it is and remains legal tender. In the past, coins had an intrinsic value. The intrinsic value of

¹³⁷ See also article by Gary Hamel and Michele Zanini: 'Excess Management Is Costing the U.S. \$3 Trillion Per Year', in Harvard Business Review (5-9-2016)

¹³⁸ Also called clearance & settlement.

¹³⁹ More about the role of money as currency, as medium of exchange in the Weconomics analysis book, paragraph 5.4: 'Economy and the role of money'.

tangible money is the value of the material from which the coin is made. Currency money consists of coins and notes held by the public. The nominal value of money is the amount that is printed on the coin or note.

In addition to currency money, we have started to use more and more giro or digital money. Here money actually only consists of zeros and ones in the accounting system of banks and trust is even more important. When we use money as medium of exchange, and that definitely applies for digital money, you must be able to trust that you can buy a comparable value with the received money. This means, for example, that the government must assure that you can use money as general tender, that the expected value doesn't change too much, that the inflation does not get too high or too low and that your savings are guaranteed up to a certain amount.

These are all tools to organize trust, to reduce insecurity and to enable transactions so the economy is running smoothly. The trust in a good payment system with the settlement of transactions is essential. The use of fiduciary money helps with this, but it also has disadvantages such as speculation and the additional printing of money without a real production or economy in return or without a democratic decision made.

Your own medium of exchange for your community

A community can decide to use an own coin or token within the community as calculation unit to be able to exchange values more easily. In order to trust digital money, the various parties of a community (some kind of mini-economy) must co-operate smartly¹⁴⁰, harmonize processes and work from a joint ontology. They must have a connection (see also figure 2.6: ten core connections of a community) and be able to trust each other. The use of an own coin can help with this. In addition, it should be possible to exchange the own coin against other currencies, like euros or dollars. It is also important that a community ensures that they adhere to existing standards¹⁴¹, so that transactions with other communities are also possible. When this is met, Blockchain Organizing can ensure that the number of intermediaries and human activities can be reduced.

The exchange of values, without having to use money as medium of exchange, look like an important way to bypass the existing powers and regulations. You only use a calculation unit. The disadvantage of this solution is that it requires more energy in clarifying values at the supply and demand site of the (possible) transaction. This is necessary to quickly, easily and reliably find values to trade with. In the past, it was easy on physical markets to exchange, for example, a bag of potatoes for a kilo of cotton. Markets were clearly and physically visible. But now we have to deal with worldwide trade on the one hand, what makes organizing trust more difficult, and on the other hand we have internet to better match supply and demand. Now there

¹⁴⁰ See for example: Dutch Payment Association that organizes the collective tasks in the national payment system for its members, <https://www.betalvereniging.nl/en/>

¹⁴¹ For example SEPA: Single Euro Payments Area.

is also Blockchain Organizing, so we can make transactions without having to trust each other. Therefore, we must switch to another organizational model.

Blockchain Organizing looks at organizing demand and supply of money, from several perspectives, other than only making the process more efficient, which is done by many banks and fintech start-ups for example. Blockchain Organizing also looks at the effectiveness question: do we really need money as medium of exchange? Blockchain ensures that there are also other ways of exchanging values. You do not always need money as medium of exchange. This means that we become less dependent on existing and mostly centralized institutions in the payment system.

Money has many advantages, but also disadvantages. So, we have developed all kinds of financial products, not with the purpose to reduce insecurity but to get a higher yield. Making money with money is of course a really stupid idea in the long run. A relationship must be retained between the money supply and real value. Financial products have become so complex that nobody can properly assess the risks anymore, which increase the insecurity. Blockchain Organizing, in relation to money and economy, is about designing a new ‘welfare machine’, that is less dependent on money and institutions, to organize demand and supply.

Apart from organizing trust and economy, Blockchain Organizing can make an important contribution to organizing work, also an important pillar of our welfare.

4.3 Organizing work

Blockchain Organizing can make an important contribution to the productivity of information workers, provided it is used with a new management model. When organizing work (the HR and management domain), it is important to bring supply and demand together with as little friction as possible. For example: bringing the demand of a job applicant together with the supply of a vacancy. The tool that does this at the lowest costs, will eventually survive. Blockchain Organizing enables us to connect people and resources to an objective of an organization, at minimum costs. Let’s look at one example. Suppose you are a consultancy firm. You can define an innovation project to develop blockchain advice services. Because you have permanent employees, you want to quickly employ these people in the development of the new service provision. Therefore, you ignore the question whether these people are the most suitable? Perhaps you could better engage external experts, but this involves high transaction costs to find and to engage these.

Blockchain Organizing is a form of organizational technology, also called Orgtech. For example, you could see Orgtech as ‘mother’ of Fintech (financial sector), HRtech (Human Resource) and GovTech (government). Blockchain Organizing is a combination of a new organizational model (secondary), with a technique (tertiary) that focuses on the improvement of productivity (primary). Organizational technology is a concept that is primarily based on an organizational or management problem, that

develops a suitable organizational design with this and only then looks at a suitable implementation technique or tool. Contrary to many hyped tools such as Telecommuting, Agile Working and Cloud Computing, organizational technology is not based on the tool, but on the problem to solve and the design with which you coordinate supply and demand.

Organizational technology answers the question how you can organize as efficiently as possible with the use of technology. It is an applied science, or field of expertise, that is part of (or rather should be) Organizational Science¹⁴². Organizational technology should be a field of expertise that provides principles, foundations and recipes to managers, consultants, trainers and blockchain designers and developers. Organizational technology should be a field of expertise that mainly promotes the cooperation between education and research on the one hand, and practice on the other. This practice mainly consists of knowledge-intensive organizations, where relatively many information or knowledge workers are used, where there are relatively many investments in digitization, but where the productivity increase does not keep pace with these costs.

We saw earlier that the transaction costs in a digital economy almost go to zero, so it becomes more attractive not to perform activities yourself but to outsource these to suppliers. It is even better to use smart networks, with smart contracts or agreements between the various stakeholders in the ecosystem. If activities in the ecosystem are a routine, occur frequently and fundamentally consist of IFTTT actions, the network of nodes could develop into a Decentralized Autonomous Organization. In the Netherlands, most people working in organizations currently mainly process data into information and information into knowledge. The costs for finding, selecting, storing, processing and sending data is almost zero. This creates partnerships that in the past would mainly take place within the walls of a company, but can now be more and more organized within blockchain communities. Using organizational technology makes the firm and traditional markets less and less attractive as ideal form to prepare, execute and register transactions.

Until recently, when you googled 'HR', 'HRM' or 'management' in combination with 'blockchain', you would only get little to no relevant results. This is changing quickly. It is expected that Blockchain will bring a similar change in our life as internet, but also for the organization of work. After the mainframe, the PC, internet and social/mobile, we now see the next disruptive innovation: blockchain. The question is now: 'When will HR and management use blockchain technology?' Large ICT companies, banks and insurance companies already invest billions in blockchain technology and the legal domain is also becoming more and more active. So, when will HR and management follow?

¹⁴² See also the Weconomics theory book, paragraph 3.6: 'Different disciplines of organizational science'.

In our view and experience as blockchain consultants, HR and management are mainly engaged in finding and connection people/resources to tasks, projects and targets. At this moment, there is technology that is so powerful that it will not only make the existing way of finding and connecting redundant, but even has the potential to make a large part of the activities of the default positions and organizations, that are currently engaged in this, irrelevant! Imagine that you ignore this, because it seems too difficult, too technical or too far away yet. Where will you stand after ten years when blockchain technology has indeed undergone the same development as internet? You will not directly notice it when you are too late.

Changes happen incremental, but when you compare the way we organized music or film distribution now, with ten years ago, the change is revolutionary. From the examples of organizations like Blockbuster or Freerecordshop we see that this only leads to problems many years later. But then it is often too late to adjust. If your organization uses Blockchain Organizing too late, it will for example be more difficult to find and hire the best talents, because they do increasingly work with modern organizational technology and are less and less willing to work in traditional companies.

For the HR and management domain it is also important to delve into Blockchain Organizing, so they can assess what they can and cannot do with it. Blockchain Organizing is not technical or difficult. It is a concept that can be used by most professionals if they want this. However, a condition is that you dare to let go of your old self-evidence and that you are open to a new perspective on organizing, so you will continue to provide added value in a world of a new way of organizing and new organizational technology.

What does blockchain mean for the organization of work?

Blockchain literally means a chain of blocks, whereby a block is created from a number of transactions. A transaction can also be the purchase of a management book or training course and the performance of a project as portfolio worker. When you 'connect' all transactions of a person in time together, a chain is created of connected tools to a person and of supply to demand. We also call this an e-portfolio. It is a sequence of controlled and non-mutable transactions that the community has consensus on. As indicated before, you can see blockchain as an additional layer on top of internet. This layer ensures that we cannot only exchange information between sender and receiver in a secure and reliable way, but also make transactions.

Now many HR professionals and managers will say: But we have been making transactions via the internet for quite some time? That is correct. You probably have registered yourself online for a course or ordered a management book at Amazon. But at this moment you order via the website of the seller, while you as buyer, as employee or as citizen would like to order via an own application, so you do not continuously have to enter the same data and your data is secure. Thereby it would be

handy if the creation of a purchase order and the payment would more or less go automatically via a protocol.

The large difference between the current way of working and Blockchain Organizing is that we in principle no longer require training institutes, publishers, payroll companies and temporary employment agency for the preparation and execution of transactions. We also no longer require an administration to create purchase orders, to place orders, to check invoices and to make payments. We do not even require a bank for the payments. Also, employment contracts, whether or not for a permanent job, fixed or flex organizing, assessments, etcetera, they all get a different meaning or they are no longer required when you use a smart contract which is based on the concept of Blockchain Organizing. In its core, organizing becomes switching connections on and off so transactions can be executed almost fully automatically.

This entails quite a bit. Blockchain Organization therefore has the potential to make many (administrative) activities within the HR and management domain redundant. Even the firm as default organizational form is largely no longer required. Because the HR department, the manager and the company belong together, mainly traditional companies with their own HR departments and traditional managers will change or disappear. Many departments still lean heavily on existing analogue processes that are digitized. But we will more and more organize the finding and binding of resources with Decentralized Autonomous Organizations. Self-operating processes in a shared transaction network, similar to self-driving cars. So, it is important for you as HR professional and manager to delve into Blockchain Organizing.

Blockchain Organizing integrates concepts and developments like Personal Data Service for employees, peer-to-peer platform such as vacancy databases, sharing knowledge via communities, self-organizing capacity, digitization, crowdsourcing, cloud computing, social media, open source and open innovation. Blockchain Organizing provides a new perspective on concepts like more autonomy, self-organizing capacity, more inspiration, decentralization, accountability, trust, value, trade, leadership and entrepreneurship. It ensures that you can cooperate smartly with both your colleagues as well as your environment (portfolio workers, customers, suppliers, talents, etcetera.) with the same organization. It provides an effective policy related to the organization of a fixed-flex shell.

Blockchain Organizing is mainly suitable for digital production and consumption and will make information and knowledge workers much more productive, flexible and innovative. This way we can strengthen our competitive position in the future and organize a sustainable welfare. For example, the whole discussion about replacing permanent jobs for flex contracts would not be any problem within Blockchain Organizing, because it uses smart contracts¹⁴³, prior consensus and reputation management. A fixed or flex contract within Blockchain Organizing is nothing more and

¹⁴³ See also for example Ethereum: A decentralized software platform (for smart contracts).

nothing less than a type of connection. This way you connect people and resources to each other to realize an objective that you cannot realize as an individual entrepreneur.

Why should HR and management pay attention?

Organizations in the service and knowledge domain have a growing need to cooperate in a smarter way, not just internally, but mainly with their environment and with professionals such as portfolio workers. This environment does not only include customers, competitors and suppliers, but also talents, trainers, temporary employment agencies, portfolio workers, governments and interest organizations. The necessity to cooperate in a smart way is only growing because we must become more productive due to aging and increasing welfare costs.

The environment is becoming increasingly bigger, more dynamic and more complex. Old rigid forms no longer work. More and more organizations are therefore looking for new models and organizational forms. HR professionals and managers, who were already working in the nineties, can still remember the question that many organizations were facing that time: ‘Must we join internet or not?’ Must we put our vacancies on line? Must we put our manual and training offers on intranet? And when will we switch to e-HRM and Employee Self Service? It is hard to imagine these questions now. But still, in the coming years, HR professionals and managers will have to answer a similar question: Should we switch to a blockchain or not?

To be able to answer this question correctly, it is useful that you first delve into Blockchain Organizing. Because designing, setting up and managing blockchain organizations is not a daily job for most people, and requires a lot of expertise, more and more often blockchain professionals are also deployed within the HR and management domain. An expert in Blockchain Organizing, for example, is competent in devising, designing, setting up, managing and implementing coordinating capacity that uses blockchain technology. A good design is also important for this (especially) within the HR and management domain to prevent problems in the implementation later on. The better the design, the less management and HR is needed in the implementation and exploitation phase. Blockchain Organizing for example means that you already consider during the design, that you will cooperate more and more with your environment and less and less with your ‘regular’ colleagues. It does not only consider the smallest building blocks and the connections between them: Blockchain Organizing takes this as a fundamental principle.

Blockchain application for HR and management

For the HR and management domain many applications of Blockchain Organizing can be imagined. Everything that in essence can be reduced to the smallest building blocks and the connections between them and the connection of these capacities to a finalization can be organized with blockchain principles. When discussing a number of examples, it is important to mention that there is not just one blockchain, not even for the HR and management domain.

Just like internet, an organization can for example opt for a public internet website, an internet website with login and an intranet website. As organization you can choose to join a public blockchain, but you can also decide on having your own private or permissioned blockchain. Hybrid forms are also possible and more likely. What is wise and practical, will differ per situation (see also paragraph 3.3). A few examples of blockchain applications for the HR and management domain are:

- Identity, assigning and assuring; within HR and management it is important to be able to establish the identity of an employee or flex worker and to monitor this. With Blockchain Organizing you only have to do this once and you can use existing identity systems.
- Purchase, searching and booking products and services: for example, searching and booking training courses whereby the seller only has to keep track of training information at one location and the buyer can search and book a training course with an own portal and workflow.
- Expat, requesting fiscal advice, residence permit and relocation: expats must organize a lot when they go abroad. This can be done in an easy, secure and efficient way via Blockchain Organizing.
- E-portfolio, for work and lifelong education: when you have followed a training course, finished a task or project, you receive feedback in your portfolio in a secure, reliable and meaningful way. Once approved and recorded in your portfolio, you can no longer change this yourself without creating a new transaction.
- (Employment) contracts, drawing up, distributing and storing: HR and management are often engaged in drawing up, distributing and monitoring contracts. This can be easily realized with smart contracts. An employment contract actually includes a number of conditional actions, so it can be programmed.
- Payrolling: with payrolling, transactions are continuously executed via a certain script between sender (payroll company) and receiver (employee). Within Blockchain Organizing there are no restrictions to the form or conditions of a contract. Everything that can be included in an algorithm, can be organized via a blockchain. The payroll company becomes redundant for the execution of transactions.
- Taxes, surcharges and grants: as employer you are responsible for paying premiums to the Tax office, education fund and pension fund for example. This can easily be done via Blockchain Organizing.
- Communication: HR and management communicate often with employees via all kinds of media. A document or a message is also a tool that exists, has value and with which you can make transactions. With a blockchain you can easily organize this whereby you, regardless of the medium (WhatsApp, website, Twitter, e-mail, etcetera), only have to keep track of your source at one location and the receiver can read your message with an own application, in a reliable, secure and meaningful way.

Which tasks will change or disappear?

Blockchain Organizing provides transparency and offers reliable, meaningful and well secured data. Many routine and mechanical tasks and processes will be

programmed in a smart contract and performed by a protocol. Whether party A will make a transaction with party B and how this will be organized with a blockchain, will also depend on the reputation of both parties. The reputation can be traced back to previous transactions and is confined in a blockchain. When a transaction is made, the chain will be further complemented and therefore also the reputation of party A and B and therefore also the reliability of the entire blockchain.

An e-portfolio, for example, is a collection of training courses, tasks and projects that are connected to a person. When you book a course with a training institute, it will be approved a third party and your supervisor and trainer will provide you with feedback in your own blockchain (e-portfolio). When adding content to your e-portfolio, you could see this as a (data) transaction with a sender and receiver. Both parties¹⁴⁴ must agree on the question: is this data:

- always public
- never public
- public on request

In the last case both parties must agree on this question: To access the data you must have:

- the key of the sender
- the key of the receiver
- both keys (also called a pair)

So, values are added to your portfolio that the community has reached consensus on. In this way, others can make a better decision when executing a next transaction (for example starting a new project). We can better connect specializations (supply) to finalizations (demand) without increasing the complexity. The e-portfolio as an expression of Blockchain Organizing can be seen as coordinating mechanism to bring demand and supply together in a smarter way. Smarter than your LinkedIn profile, where you can basically post anything, without anybody checking this. Plus, a blockchain will run the entire ledger, from the moment that you started to bring your e-portfolio on a blockchain. A date and time stamp are set per transaction, and the public address of buyer, seller and product are known. We must take into account that transactions will not only be executed with money as medium of exchange but also with other forms with value, such as hours, credits, rights, energy, waste, other products, etcetera. This offers, for example, options to convert a part of your salary into additional free hours or parental leave without any problems. For example, you can program that if an irregularity surcharge exceeds €1,000 the surplus must be converted into free hours at a valuation of €50 gross per hour.

Finally

¹⁴⁴ This system could also be applied to health records for example where general practitioner and patient must agree on data access.

HR, management and technique do not always have a frictionless relationship. But you don't have to be familiar with how the TCP/IP protocol works to be able to put the staff manual on intranet. There is at least the same need, if not more, for people who know how to design future organizations or use Blockchain Organizing meaningfully for the organization of work, than for people who can program a blockchain application¹⁴⁵. Because using new techniques for the solution of non-problems, doesn't make any sense.

4.4 Organizing supply chains

From the eighties onwards, more and more work has been outsourced, mainly due to Japanese companies. This leads to so-called supply chains. When a company no longer performs a certain process itself, this process is outsourced to a supplier. This supplier is also an organization (factory) that probably performs the process in a similar way, but usually much cheaper, because it produces a larger volume and it is probably specialized in a certain process. When outsourcing work to a supplier, an additional relationship, an additional connection is added between outsourcer and supplier and therefore also additional transactions.

It is especially important to investigate this new relationship. Because if the supplier must work precisely under the (transaction) conditions and must connect with the systems of the outsourcer, the transactions costs across the entire chain will increase. We saw before that a bilateral relationship (1:1) or an outsourcer with several suppliers (1:n) is not the same as a business ecosystem (n:n). Supply chains usually consist of several 1:1 or 1:n intermediaries between the first seller and final buyer. This is not the same as a business ecosystem. The latter is more based on equal relationships between actors that use a smart contract and shared transaction network connected to a data utility facility.

Marco Iansiti¹⁴⁶, among others, professor at Harvard Business School, has done research on business ecosystems. Iansiti has contributed to the development of insights related to the development of ecosystems and the relationship between organizations and their ecosystem. This field of expertise is also called Ecosystem Strategy. According to him, the success of companies likes Walmart (and according to us at ASML in the Netherlands) mainly leans on developing and managing a complex and extensive ecosystem. He formulates this as follows¹⁴⁷: *"Many industries today behave like a massively interconnected network of organizations, technologies, consumers and products. Perhaps the most dramatic and widely known example is the computing industry. In contrast with the vertically integrated environment of the 1960s and 1970s, today's industry is divided into a large number*

¹⁴⁵ A lot of blockchain code is open source and can for example be found on websites like GitHub.

¹⁴⁶ Marco Iansiti is mainly engaged in research on the interface between technology, operation and innovation management.

¹⁴⁷ See also his paper, together with Roy Levien (NERA): 'Keystones and Dominators: Framing Operating and Technology Strategy in a Business Ecosystem' (24-2-2004).

of segments producing specialized products, technologies and services. The degree of interaction between firms in the industry is truly astounding, with hundreds of organizations frequently involved in the design, production, distribution, or implementation of even a single product. And because of this increasingly distributed industry structure, the focus of competition is shifting away from the management of internal resources, to the management and influence of assets that are outside the direct ownership and control of the firm.”

Because the bureaucracy in chains with 1:1 and 1:n relationships can quickly increase (see also Hamel and Zanini research) we see that supply chains are growing towards business ecosystems based on n:n relationships. Blockchain Organizing will play an important role in this. Currently, especially the financial sector invests a lot in blockchain technology. There are probably even more opportunities in the Supply Chain Management and logistical domain, also called SCM. Not in the least because SCM is actually already a chain of transactions. In principle processes take place within an institute (intra-firm), between institutes (extra-firm) and in a chain, supply chain or shared transaction network (more firms).

Supply chain in the food chain

More than a hundred years separate a more agriculture-based society and our present one. More than hundred years ago a large majority of the people was still working in agriculture while production and consumption took place locally without intermediaries. Now a chicken breast ‘flies’ across the entire world, before it arrives on your plate at home. The main reason for this is making processes more efficient through specialization and scaling. Each actor in the chain adds a little value, but indeed with a large volume. Blockchain Organizing has other principles, than just improving the efficiency in existing organizations. Blockchain Organizing also focuses on the productivity demand at macro level. To realize sufficient productivity growth, it is important that final demand and supply are correctly aligned and to keep the aggregated transactions costs low.

In SCM we explicitly see the rise of institutions to reduce insecurity, with as a result that the ‘chain’ becomes more complex. This is definitely the case in the food chain. It is no surprise that Blockchain Organizing will find more and more applications in this domain. Walmart, for instance, the largest retailer in the world, uses blockchain technology for tracing & tracking pork meat from China¹⁴⁸.

Another project is the chain of packed fruit and vegetables that is produced in the US. With the use of blockchain technology and an online accounting system, the identity and processes of a large number of parties, in these often complex supply chains, can be better monitored. The main purpose is to make the food supply safer. There are producers, slaughterhouses, transporters, freight forwarders, inspectors, traders and consumers in the food chain. In addition, standardization, normalization,

¹⁴⁸ See for example article in The Wall Street Journal: ‘Walmart Turns To Blockchain For Tracking Pork In China’ (19-10-2016).

software, research, education, legislation, etcetera is used. A large number of actors that must cooperate smartly but that do not use the right tools for this.

Together with IBM and the Tsinghua University from Beijing, Walmart is building a system to verify the identity of these parties and subsequently connect these. Transactions are made and logged with this system, without being able to adjust these afterwards. Because supply chains in the food industry are long and complex, the risk of fraud increases. Worldwide¹⁴⁹, but also in the Netherlands, several fraud cases¹⁵⁰ are known.

Besides fraud, food safety is also an important aspect. A study by Robert Scharff, associated professor at the Ohio State University, estimates the annual costs for food related consequences at 55 billion dollars for the economy in the US¹⁵¹. For example, when food poisoning¹⁵² occurs, it must be possible to quickly identify the source. Blockchain Organizing also offers advantages for the 'speed' aspect. Because 'input reference' is a basis of blockchain technology, it is possible to quickly identify the previous party.

Trust and insecurity reduction are an important aspect within the food chain. As we saw before, the core of Blockchain Organizing is the reduction of insecurity, not by adding even more intermediaries in the supply chain, but by programming insecurity reduction in a (blockchain) protocol. This does not increase the complexity, unless we also strive for full employment besides (blockchain) innovation.

Other chains

Apart from the food chain, it is also important to know that clothes are not made with the use of child labor, and factories are safe. Remember the reports of fires in unsafe clothing factories in Bangladesh for instance. For the organization of a sustainable welfare, it is important to invest in sustainably organizing chains. Consumers, governments as well as interest organizations have an increasing need for information about the way products are made and to what extent this is done, for example, socially and climate- and energy-neutral. Generally, actors in a chain have no interest in making this information transparent. But due to social pressure and technological progress, we can make supply chains much more transparent and gain trust without additional intermediaries.

Using Blockchain Organizing within the SCM domain provides the option to identify and monitor entities in the chain from source to consumption. Tracing & tracking is

¹⁴⁹ See for example: https://en.wikipedia.org/wiki/2008_Chinese_milk_scandal

¹⁵⁰ See for example the fipronil fraud in the egg sector in The Netherlands (2017).

¹⁵¹ See for example article in Fortune.com by Beth Kowitz: 'America's food industry has a \$55.5 billion safety problem' (6-5-2016).

¹⁵² The Centre for Disease Control & Prevention (CDC) in the US, estimates that 48 million people become ill every year due to a food-related cause (also called food-borne disease), 128,000 people therefore end up in hospitals and 3,000 people die.

a relatively old concept within SCM, but now gets new impetus through improved technology. It offers the opportunity to execute transactions at minimum costs and there is a shared reality about the status of a process or the question where a product originates from.

We previously indicated the importance of Data Logistics. SCM, logistics and Data Logistics are very much connected to each other. They are not only an important application area, but also a basis for Blockchain Organizing.

Examples

In addition to Blockchain Organizing for food and clothing chains, there are other examples of applications for the SCM domain:

- *Toyota Motor Corporation*: wants to organize the tracking of car parts with blockchain technology.
- *Everledger*: is an initiative to trace the origin of diamonds, especially to counteract the trade in blood diamonds.
- *Provenance*¹⁵³: organizes supply chain certification (digital passport) on a blockchain. The passport guarantees authenticity (is the owner the person he claims to be) and proven origin (input reference, where does the product come from?). A digital product passport is a kind of Bill of Materials (BOM) that indicates from which parts make up a completed product. A part is a value (asset) and the origin of this value is transparent.
- *Port of Rotterdam*: a partnership of, among others, the Port of Rotterdam, ABN Amro and the Technical University of Delft. They conduct research into the possible applications for, and later maybe the development of, blockchain technology. The research mainly focuses on sharing logistical information and contract information¹⁵⁴.

Advantages

Using Blockchain Organizing within SCM provides advantages such as:

- Real time smart data: quicker and more targeted data from a shared reality. Identifying the source (traceability) can for example be reduced from days to minutes.
- Fraud: becomes more difficult because the accounting system is shared, cannot be changed and can be checked by the participating actors.
- Tracing & tracking: where are goods in the supply chain, where do they come from, how many intermediaries are used etcetera.
- Costs: the transaction costs are reduced because more and more intermediaries (institutions) are programmed in a blockchain protocol.
- Structured flexibility: using blockchain technology in a supply chain with many actors makes it possible to organize tools (such as software and shared databases)

¹⁵³ See also website: provenance.org

¹⁵⁴ See for example article ABN-Amro: 'Logistics sector gets to grips with blockchain technology' (16-11-2016).

in a scalable way, while through scripts the differences between actors can be respected and people remain deployable in a flexible way.

So far, using Blockchain Organizing in supply chains. Now we continue with the use of Blockchain Organizing for the organization and performance of government tasks (regardless the question, what we, as democracy, believe what government tasks should be).

4.5 Organizing government tasks

Governments can play an active role in the further development of blockchain technology and Blockchain Organizing. Not just as early adaptors for their own use, but also a director and facilitator for other blockchain projects¹⁵⁵, and mainly as initiators of a general data utility facility. A number of practical effects of the role could be:

- stimulating and facilitating more research and education;
- taking the lead in the construction of a general data utility facility¹⁵⁶;
- stimulating community working with shared information and transaction networks;
- active role with standardization of an ontology, taxonomy, and technology based on new thinking and a new organizational model;
- using Blockchain Organizing with internal and external transactions.

For the government, we see a large number of applications. According to the aforementioned Agora model we live in four atmospheres: personal, private, public (government) and politics. Both citizens and companies can influence the public atmosphere via the political atmosphere. A government authority itself has both relationships with citizens and companies, as well as with own employees and other institutions. This provides a large number of transaction options that can be organized smarter with Blockchain Organizing,

A few examples:

- Identity: creating an identity by linking an identity tool (for example a passport or fingerprint), to a citizen. There is consensus about this link, that can be included in a blockchain and no longer be changed or connected to another person afterwards. After all, you can only spend (read connect) a value once.
- Voting: a ballot paper is sent via a blockchain infrastructure and can only be used once to cast a vote (value) by the person with the right key of the (virtual) safe that contains the ballot paper. In exchange for a ballot paper, maximum one vote can be cast (counter-value). Ballot paper and vote are logged in a blockchain, can easily and reliably be counted and cannot be changed

¹⁵⁵ See for example Amendment 34550-XIII-18, about stimulating blockchain technology by the Dutch government by D66 MP Kees Verhoeven and VVD MP Erik Ziegls (27-10-2016).

¹⁵⁶ See for example the 'Discipl' project from ICTU.

- Permit: requesting a permit includes a number of processes. This workflow must be monitored so certain handling times are not exceeded. In addition, it must be possible to make the different steps, considerations and decisions (all values) transparent. The permit can be paid with digital currency for example, and the permit can be connected to grants via a smart contract. This means that a company only receives a certain grant if it also has the right permit.
- Implementing bodies: many administrative and executing bodies of the government such as land registry and the tax office make numerous transactions every day. Requests, decisions and payments must be executed very efficiently. Subsequently the agreement must be performed via an IFTTT statement, whereby it is important that the payments are made to the right person. In addition, it can be programmed in a study allowance agreement, that the allowance can only be used for payment to educational institutions.
- Document control: the government develops and controls many documents. These must be controlled in a systematic, secure and meaningful way. This can be done by considering a document as a value and by including this in a blockchain.
- Employment organization: the government is also an employment organization where, for example, the e-portfolio of civil servants can be organized on a blockchain. Each time that a civil servant follows a course, does a project or changes position, this is recorded in an e-portfolio. A group of connected portfolios is the new personnel information system

Giving the many applications, it is clear that Blockchain Organizing is also relevant for the government. But also for the government it applies that blockchain technology is not decisive, but the combination of technology with a different way of thinking and a fundamentally different organizational model for the coordination of demand and supply, is.

Advantages Blockchain Organizing for the government

Blockchain Organizing provides many advantages for the government and therefore also for citizens. The large 'disadvantage' for the government is that Blockchain Organizing requires a lot less government. The question here is again: what will we do with all these people who currently work for the government? Will we dismiss half of them or will everyone who works 8 hours per day go back to a 4-hours working day if technology doubles our productivity? In any case, Blockchain Organizing also offers for the government an answer to the challenges of the 6 Ps: : purpose, productivity, privacy, power, propaganda and processes. The advantages of Blockchain Organizing for the government are:

- A permit request can be processed quicker without requiring personal data of the requestor.
- The improvement of lead time, price and quality of identity products like passports.
- The improvement of transparency without additional costs. Policy, memos, minutes, e-mail, and decisions: everything with value can be included in a blockchain, cannot be changed and can be made transparent.

- Participation-community gets content: returning more tasks to the private atmosphere by offering citizens and organizations a blockchain structure so that older people for example can find care providers themselves and make transactions.
- Only having to maintain personal data at one location and to make this available, for example, via a Personal Data Service, to buyers who require this data to perform processes.
- Designing, developing, implementing and checking new laws, permits and regulations can be done quicker and better by putting these in a smart contract. For example, a farmer can automatically get a permit for a maximum expansion of X, when a similar farmer discontinues his company within a radius of Y kilometers.
- The improvement of the trust in the government of citizens and companies by making data meaningful, reliable and more secure with the use of Blockchain Organizing.
- The easier, quicker, less complex and cheaper development of applications that connect to a general data utility facility.

So far, a more detailed explanation of a number of application options of Blockchain Organizing and the problems that can be solved with this. In the next chapter, we continue with the question how you can start with Blockchain Organizing within your organization/network and what you require for this.

5. How to start with Blockchain Organizing?

Because designing, setting up and managing blockchain organizations is not a daily job for most people, and requires a lot of expertise, more and more often, blockchain professionals are deployed. An expert in Blockchain Organizing is competent in devising, designing, setting up a blockchain organization. A good design is important to prevent problems in the implementation later on. Blockchain Organizing means that you already consider during the design, that you will cooperate more and more with your environment and less and less with your ‘regular’ colleagues. It takes into account that you can mechanize more processes. Plus, Blockchain Organizing does not only consider the smallest building blocks and the connections between them: Blockchain Organizing takes this as its fundamental design principle.

When solving management problems within or between organizations you often hear: ‘we must cooperate better or smarter’. But often this is not possible in a satisfied way. The experience is that cooperating is difficult and usually leads to many meetings and unfinished action lists. Budgets and lead times are exceeded and cooperation rarely leads to the desired result. How could this happen? The problem with cooperation is that we take our own company as the center of attention and principle. People base their thinking on their own company and our own interest. Companies make products or develop services with the intention to sell them. Organizations reason from their own supply and then look for customers to go with this. When we take the company as the unit of analysis and design, we can distinguish the following partnerships:

1. Within existing institutions¹⁵⁷: teams, departments, business units, concern.
2. Between existing institutions: joint venture, alliance, franchising, etcetera.
3. Within (newly set up) partnerships: cooperatives, communities, shared networks, shared service centers and blockchains.

Organization principles play an important role in all these forms. The main principle of organizing and managing is that there is always specialization, coordination and finalization. The still prevailing thought is that the coordinating function should lie with firms with classic management, traditional markets or governments, but it is better to reason from the demand and the smallest building blocks and to look for coordinating capacity that best suits this. This can be a hierarchy or a market, but also a hybrid in the shape of a shared transaction network for example that uses blockchain principles. Blockchain Organizing is often a much more efficient way to coordinate and to reduce insecurity and therefore a better way to cooperate. Especially the latter form (newly set up partnerships) can be organized well with Blockchain Organizing. It is a form whereby the smallest building blocks and

¹⁵⁷ An institute is a collective name for companies, institutions, independent administrative and governmental bodies. To make this easier to read, we sometimes replaced the term ‘institute’ by ‘company’ because many government institutions, for example, are also organized like a company.

organizing cooperation between humans and resources is already considered at the design phase, whereby people not necessarily need to have a permanent ‘connection’ (for example a permanent employment contract) with one or a few fiscal-legal entities (for example a company). It is possible but not necessary. Permanent connections, such as: positions, employment contracts, collective employment agreements and management development programs of companies, have more and more disadvantages. Blockchain Organizing does not have this problem. It does not take the existing as starting point (there is a company with a building and permanent employees and objectives are sought for this...) but the demand, the need, the problem to solve. The existing organizations or organizational forms are not important with Blockchain Organizing, but the coordinating capacity between demand (finalizing: target and task setting) and supply (specialists: man and resource) is.

In this chapter we discuss some aspects that you must be aware of when you start with Blockchain Organizing in your organization or network, such as:

- What is a blockchain lead and what does he or she do?
- The design and development of a business ecosystem
- How do you start with Blockchain Organizing?
- Project-based approach
- Dealing with resistances

5.1 Blockchain lead

We advise larger company’s en network organizations to start with appointing a blockchain lead. This person will keep track of blockchain developments and assist directors with questions like: What does Blockchain Organizing mean for our business and what should we do with it and what not? Of course, you can decide to do nothing, to sit back and see how the future develops. You can catch a ride with other people, not knowing when you will arrive at your destination. But maybe you also believe that it is wise to keep your eyes on the road and your hands on the steering wheel and to determine yourself which decisions to make. To get people interested in your blockchain innovation you must be able to provide a clear picture of what the future will bring for your business and how you want to get there.

A blockchain lead¹⁵⁸ is the first designated person within an organization or network to keep track of blockchain developments, to advice the board of directors and management about (dis)investments and to set up and guide projects. An internal/external blockchain lead prepares an organization to use Blockchain Organizing and is therefore some sort of quartermaster. Some examples of tasks:

- keeping track of developments, visiting conferences, building a network;
- start a Community of Practice;
- designing and setting up Proof of Concept projects and experiments;

¹⁵⁸ See for example also Weconet Blockchain Academy on website weconet.org

- creating conditions for a successful implementation;
- identifying and developing common values, principles, perspectives, targets, vision and ambitions;
- coordinating different internal roles like: purchasing, sales, administration, legal affairs, ICT and HRM;
- alignment with external roles such as: consultants, architects, developers;
- project implementation, including: planning, documentation, business model development, prototype delivery.

Many managers want to use blockchain technology within their organization to make the company stronger than the competition. They want to continue to exist as a company, while a company is only a tool to produce welfare. Management is an important part of a company and management is mainly a tool to coordinate (human) recourses to a target. And a large part of traditional management tasks can be transferred from a social to a technical 'script', so traditional management becomes less relevant. And when traditional management becomes less relevant we must stop using it. The competition between Real Madrid and Barcelona becomes irrelevant when people stop watching soccer.

Traditional companies become less relevant when mechanical management tasks can be put in a protocol. A blockchain lead first looks at the intention of the tool he wants to use. Blockchain leadership is not about leadership whereby an IT manager or CIO knows everything about digitalization or successfully leads an IT company. Not the technology should be primary, but the question how you can prevent and solve management problems with technology. Blockchain leadership currently mainly means that you dare to be a pioneer, think differently, 'be a cross-thinker' and dare to renew. A pioneer is not someone who asks people to follow him or her, but who challenges them to join the expedition.

Despite our structured approach to realize shared transaction networks, for many organizations the use of Blockchain Organizing is currently still mainly pioneering. Blockchain Organizing is so new that you cannot expect that you can use a navigation system. There are no road signs yet, sometimes not even a road or path. You could see blockchain technology as a Tesla, but the way how we manage organizations is still a dirt road. As we already concluded in the introduction of this book, we need to reinvent management as we know it today. Therefore, we require pioneers who are not afraid to deviate from the beaten paths and take a new path. A path is created when you walk on it. The more people subsequently walk on this path, the bigger the chance that it becomes a road and eventually is included in the route planner.

A blockchain lead is currently mainly a person who learns and shares the accrued knowledge and experience. Blockchain still requires a lot of failures in order to become successful. But if you are not prepared to fail, you probably also not come up with original ideas and applications. We must also share failure knowledge. And the better we share our experiences, the easier it gets to put Blockchain Organizing on

the map together, and therefore fundamentally innovate management. Blockchain Organizing can still make the promise of internet come true: a tool to become more productive so we can sustain our welfare.

In this paragraph we will further discuss the type of leadership that we believe is required for Blockchain Organizing. We mainly need leadership with the insight that digitization is a tool to solve management problems. In addition, we mainly need authentic leadership. That is leadership whereby followers have enough trust that what their leader says and writes, can also be implemented reliably. People should know what they can expect and which character, which knowledge and experience contributes to the identification and solving of management problems. An authentic leader is also original and pristine and averse of self-evidence, status and linear thinking.

Organic versus mechanic organizing

An important insight into the organizational theory comes from sociologist Tom Burns and psychologist G. M. Stalker: mechanical versus organic organization. According to Burns and Stalker, the connection between people in mechanical organizations lies mainly in the (decision) structure, hierarchy and formal agreements. This can increasingly be organized with blockchain as a coordinating mechanism and is industry agnostic. The connection between people in organic organizations lies mainly in shared values, perspectives, principles, objectives, governance and culture. For this we need specific leadership: leaders who know how things work in a certain industry or organization, in a certain context. It is not one or the other, but depending on the situations you choose the most appropriate coordinating capacity. According to the contingency theory, the best organizational form depends on the environment in which the organization operates. Digital leaders synchronize the way they organize trust and work with the possibilities that technology offers, in order to solve real social problems.

Digital leadership requires vision

According to a research by Bart van Ark¹⁵⁹, the rapid growth of ICT investments and the 'New Digital Economy' (including mobile, cloud computing and internet), has not yet given us a visible improvement in the productivity growth. According to van Ark, this is because ICT is still in the installation stage and will only improve productivity when ICT is in the deployment or implementation stage. According to us, we are still hunter and collector when it concerns information. ICT will first have to be organized industrially (for example with an internet assembly line for data), to be able to sufficiently contribute to the productivity growth that we need. Blockchain Organizing will ensure that we move towards a conveyor belt of nodes that cooperate smartly via a protocol and automatically process transactions and perform certain management tasks. To make this process from traditional to industrial go smoothly

¹⁵⁹ See article by Bart van Ark, The Conference Board and University of Groningen: 'The Productivity Paradox of the New Digital Economy' (autumn 2016).

we have less need for opportunistic IT suppliers and more need for authentic blockchain leadership. We mainly need leaders who ensure that we use technology smarter to solve management problems. Therefore, we do not just need new technology, but first and foremost also another organizational model and paradigm shift. For this paradigm shift we need binding leadership at all levels.

We think, to quote Otto Scharmer¹⁶⁰ that most people know indeed what they want and what the 'emerging future' is. We move from an industrial community to a welfare community, from more to better. Especially care for purpose, people and planet, whereby profit is a tool to be able to invest. Many people assume that technology will solve the problems that we have caused ourselves as human beings. For some people technology has become the new belief system; they sacredly believe in robotics and that technique will save our planet. However, technology itself can do little about it, it is man who uses technology. For example, when we invented the internal combustion engine, we had no idea what burning fossil fuels would do to earth and that it would cause a number of disruptive problems (global warming, climate refugees, etcetera).

It is much wiser not to just use technology to solve problems, but mainly think in advance which problems can be caused by technology. You can only do this based on a certain value system. This also means attention to the moral and ethical side of technology. This means that there are three other organizational perspectives, in addition to technology, economy and politics, that are important: psychology, ecology and sociology. Blockchain technology could solve but also cause problems. If we only use Blockchain Organizing to make us more productive, but do not think about what we do with the surplus time, this can, for example, result in many people sitting at home without a future, unable to find a meaningful way to spend the available spare time. Currently it is more important, as we believe John Maynard Keynes meant it in his article '*Economics possibilities for our grandchildren*', to think about the question what we will do with the surplus time, than to think about whether we have sufficient technology to become more productive.

Trust in leadership at the lowest point

According to the Edelman Trust Barometer, 53% of the worldwide population no longer trusts existing leaders and institutions. Another 32% is not sure. Only 15% trusts existing institutions including politics, government, media and companies. Two thirds of the countries they are investigating are now 'distrusters'. This means that the trust in the mainstream institutions has dropped below 50%. People no longer trust that these institutions act in ways that benefit the population. They no longer believe that the current system still works for him or her. A vast majority of the people have not enough trust in the institutions we invented to organize trust! And further globalization and technological changes only reinforces this image. According

¹⁶⁰ Otto Scharmer (1961-) American professor and creator of the Theory-U.

to the distrusters, the system fails in leadership and in more protection of the population against for example globalization and technologic renewal.

There is growing doubt about the future and the question whether our children will have a better life. Many people have lost their faith in systems and organizations. The only way to regain this trust is through people. People we know, people who are authentic, competent, people we can trust. Authentic leadership is more important than ever. Authentic means that statements and deeds must correspond. We need new leadership and more specifically a new and authentic digital leadership. New means that we must dare to say goodbye to the existing management models that no longer work in favor of us, but against us.

Being good and competent in your field of work is therefore not sufficient for a blockchain lead. We also require leaders who have, for example, a vision on the purpose of work and our identity with economized work. We need leaders who, together with others, dare to face questions like:

- how do we relate to technology as humans?
- what does digitization mean and how can we best organize it?
- what are the consequences of digitization on organizing trust?
- how will the new digital network communities and society look like?

A blockchain leads recognizes the trend that the organization of trust, work and economy will be digitized more and more and will be organized in both local and global communities. He or she sees the consequences of this, not just for IT systems, but mainly for the existing IT architectures, organizational forms, management models, economic models and our social-economic order. As indicated before in this book, we develop more and more towards a digital society. Digitization starts from the foundations: how do I organize demand and supply of data at minimum transaction costs? Digitization does not just mean the replacement of analogue processes, but mainly organizing, for example, privacy, power, democracy and news provision, differently. This requires a digital strategy and a wide and thorough understanding of new management models, technical concepts and transition programs.

Together on expedition

Digitization happens in almost all aspects of management and organizations. Almost all positions are faced with digitization. The question here is not so much about the differences between digitization for management, sales or HR for example, but mainly concerns what the similarities are? What are the deeper underlying principles and how can you digitize from the smallest building blocks and basic processes, something almost any coordination of supply and demand entails. A blockchain lead thinks and works from the core and from organizational principles.

With this way of thinking he/she determines a direction, a strategy how to get there and persuades others to join the expedition. After an organization has gone on an expedition together with its stakeholders and has created a successful Proof of

Concept application, the blockchain lead will start to further develop Blockchain Organizing and will eventually implement this. Not only within the own organization, but mainly through transferring existing processes to a new blockchain organization, that is part of a shared transaction network. A blockchain lead thinks from a new frame of mind and knows which management problems must be solved. A blockchain lead has sufficient knowledge and experience and knows the digital concepts that are required, or knows to find these. New digital leadership knows how to find specialists and how to manage these in a modern way, with a peer-to-peer platform and how to use these at minimum transactions costs to achieve better results.

Blockchain leadership

You can vest blockchain leadership with a project leader, community manager or blockchain lead. A blockchain lead knows how to assess the organization on questions like: What is our function in a network economy? Or: Which processes can I best do myself, which ones should I outsource and which ones fit better in a shared network? According to our research, a blockchain lead investigates what the organization must do and which competences must be developed to use Blockchain Organizing. A digital leader new style is aware of the fact that employees, customers and citizens want to communicate and coordinate differently from how they do it now and want to be less dependent on existing companies with their own websites, applications and databases. They no longer want to reason from the company as center of doing business, but from the individual.

But what does this mean for the connections between organizations and their stakeholders? Which fundamental adjustment is required from the management model, governance, infrastructure and culture? A blockchain lead wonders how New Organizing can continue to provide effective added value at minimum transaction costs and what the role of digitization, and more specific Blockchain Organizing, can be in this. A blockchain lead thinks in hybrid forms: not just doing things yourself or outsourcing, but also considering a crossbreed of company and market in a shared network.

More and more blockchain communities will be established, all with their specific needs in terms of management, coordination and communication and therefore with a different leadership and management. A blockchain lead must prepare the management for participation in network organizations. And finally, a blockchain lead must of course be aware of opportunities and threats. Currently, we mainly find threats in terms of cybersecurity with wiretapping, extortion and data leaks. A blockchain lead will not only respond to this with practical solutions, but mainly with the installation of fundamental safeguards to prevent problems. A suitable leader knows for example that owning data is not important, but access to data is. Or more specific: organizations don't need ownership of data, even not access to data: they need answers to question to processes requests. A digital leader is interested in answering questions to be able to execute own processes. A true digital leader is not interested in owning data that you are not allowed to use.

Important features of a blockchain lead

A blockchain lead must be able to deal with often complex projects that lead to the development and setup of a shared transaction network. These are projects with often ten to twenty different parties and stakeholders involved, all with different interests and looking at the problem from different perspectives. Plus, the focus of the project is not always clear enough in the beginning. But to achieve a focus the leader cannot suffice by only stating what the focus is. Via a process of steering and letting go, of control and trust, the group must become focused. This requires authentic digital leadership.

A leader must take initiative, make participants think from the total, rather than from their own interest, not give up easily and keep the deployed transition program in place. The leader must dare to let go of control, to give space to self-organizing capacity of project members. Control is not the same as discipline. Discipline is required from both the leader as well as the participants. This means: honor your agreements (effort obligation). Especially the leader should know, and express this, that discipline can keep the group together on the often, difficult expeditions.

Authentic leadership requires a leader who can make connections, not just between the group members, but also externally and in terms of the ten core connections of a community. A leader is someone who knows that work is being done on a structural solution of management problems and that it will require a long road to go, with small steps forwards and sometimes a few steps backwards. Blockchain Organizing mainly requires a leader who knows that there is often not even a road yet and that the group will enter territory where nobody has gone before. This can cause confusion, anxiety and sometimes raise fear among the project members. The leader is a pioneer who is creative and knows ways to achieve the necessary clarity. Someone who, together with his group, can find the right way to the final destination,.

Within your own organization, you, as blockchain lead, develop an integral vision and strategy for a digital transformation of your organization, to a shared transaction network. You are able to make colleagues aware, to develop new frame of minds and to share new insights with others. You also have the capacity to convince others of your vision and your organizational plan and how to realize this. Your organizational plan, your knowledge and experience make your digital strategy concrete. Furthermore, it is especially important to get the right people interested from the start to realize the desired digital transformation.

In summary, we can formulate some important properties for a(n) (authentic) blockchain lead:

- **Authenticity:** trust your own common sense and do not succumb to the investor with a large wallet or a director with a budget that has other interests. You are first and foremost a leader of yourself. According to Mahatma Gandhi, our greatness as a human being lies not so much in the capacity to re-create the world but in the

capacity to recreate ourselves. Your working method must be a reflection of your own personality. Digital transformations and transferring processes to blockchain organizations will not always be easy. When it matters you show character and persistence and you trust your values, vision, principles and your personal ambition.

- **Integrity**: digital leadership is also moral leadership and observing codes of conduct and general agreements that have been agreed in your community. You are transparent and you allow an independent third party as supervisor into your community.
- **Consistency**: you must show consistent behavior. Your behavior must correspond with your statements and principles. This requires courage. Can others always rely on you, also in difficult times? Because courage is not the absence of fear but the judgement that something else is more important than fear.
- **Authority**: be regarded as a recognized expert in a specific area such as Blockchain Organizing for a certain industry. Ensure to accrue sufficient knowledge and experience that corresponds with your 'personal brand'. Develop a reputation management system within your community in which you also participate.
- **Distinctive**: distinguish yourself based on your 'leadership brand'. Your qualities, knowledge and experience must be expressed in a distinctive way. A way that is not just different from that of your competitors, but that also provides added value. When you show your profile clearly and honestly, your target group will also be able to quickly understand what you stand for.
- **Relevant and original**: what you stand for must be connected to what is important for you and for others. Do not be a parrot, but come up with original and innovative solutions. Provide added value and relate this value to solving organizational issues, not just a higher yield for shareholders. Do not adjust yourself to the world, but challenge others to follow you in your vision and organizational plan
- **Visible**: Your leadership must be carried out in a continuous, consistent and repeated way, until it is embedded in the heads of the target group. This involves repetition and long-term exposure.
- **Persistence**: your leadership needs time to grow. It must be organically developed. You must persist, not give up, believe in yourself and be patient: *"A river cuts through rock, not because of its power, but because of its persistence."* (James Watkins).

Repairing the roof when the sun is shining

The time has come for radical changes in the way we organize trust, work and economy. Time for change in thinking, time for changing instruments that we use to reduce uncertainty and time for other leadership. A blockchain lead will play an important role in this within Blockchain Organizing. Now is the time to show what you are worth as blockchain lead for your organization, your network and the community you are part of. In 2017 and the beginning of 2018 the economy is doing well again and people will think: why would we change anything? But when the economy is doing badly, there are other priorities and you do not have time to repair your system. As the saying goes: *"You must repair the roof when the sun is shining"*. Now it is definitely

time to repair your roof, but it is even better to build a new house if your roof is leaking because the foundation is rotten. Now is the moment to invest in new organizational forms and in blockchain technology.

Blockchain Organizing starts with an idea. Subsequently you want to test how this idea, or your innovation, is received in the ecosystem you are part of. You could start this testing phase for example by organizing a kick-off meeting whereby you invite stakeholders to present them your idea. This kick-off might be the start of a project to realize a new business ecosystem in which you can usefully apply blockchain technology. To illustrate a concrete approach of a blockchain project, we'll discuss the PLATO case: a real project to prepare your organization integrally and 'by design' for the new European Privacy law (GDPR) in 2018.

PLATO case

In this chapter we take a concrete Weconet project that we support as consultants. It is an example to clarify Blockchain Organizing and the way you can (project-related) approach this. The project is called 'PLATO' and is named after the insight of the Greek philosopher Plato: *"The cure of a part should not be attempted without treatment of the whole"*. The project is about the smart cooperation to realize a better design of our data sovereignty and our privacy protection. As indicated before, there is a new European privacy law (General Data Protection Regulation, GDPR) in place. To comply with this, various positions and organizations must cooperate smartly to realize Privacy by Design. Through the project, it is examined how Blockchain Organizing can play a role in solving privacy problems, or better in preventing them. The final goal of PLATO is a blockchain community to bring supply and demand of different Privacy by Design instruments (courses, advise services, impact analyses instrument's etcetera) together. A schematic representation of this network is:

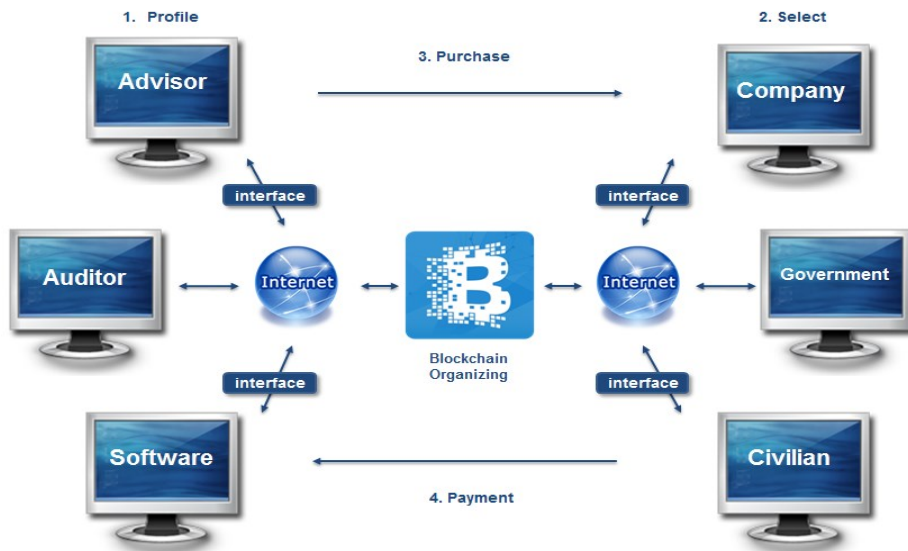


Figure 5.1: Blockchain community for new privacy law (GDPR)

Projects start with an idea. Subsequently you want to test whether Blockchain Organizing can bring something meaningful to the implementation of your idea, your innovation. We have developed a quick scan and an extensive questionnaire for this which are part of our Blockchain Organizing transition program¹⁶¹.

5.2 How do you start and how do you move

Blockchain Organizing is mainly suitable for situations where there are many buyers and sellers, each with a relatively small market share in a non-transparent market, with relatively new instruments but without many rules, access thresholds, with intermediaries, high transaction costs and sufficient financing options. In the example of PLATO there are a number of positions, both with companies as governments, on the demand side (on the right in figure 5.1) that have to deal with GDPR, for example: purchasing, sales, HRM, communication, marketing and management. On the supply side, there are service providers such as advisors, trainers, architects, lawyers, developers and auditors. When we bring all these positions to an individual level (the smallest building block) there are often dozens of people involved in designing, developing and implementing solutions to comply with GDPR. All these people must be managed at minimum transaction costs. You can solve this with the use of organizational technology and more specific with Blockchain Organizing.

¹⁶¹ See also 'solutions' tab on website weconet.org

Say you have the idea to set up PLATO and you want to test if this idea can be implemented with the Blockchain Organizing concept. To be able to make a well-founded choice on what the best way is to start your blockchain project, it is useful to first answer a few questions and to set criteria. A possible research question is: Can I use Blockchain Organizing to bring supply and demand of services together at minimum transaction costs, so an answer can be given to the challenges of GDPR? Possible sub-questions are:

- Describe your project in your own words: what does it entail and what is the role of your (project) idea in the total (domain: market for 'Privacy by Design instruments)?
- Which problem will be solved by the final result of the project?
- What are the main developments in the domain, especially regarding the development and application of organizational technology?
- Which instruments (values) are required and offered?
- How many buyers and sellers are there in the domain?
- Is the domain transparent: do buyers and sellers have similar information at the same time?
- Are you dealing with a local or global market?
- Which resistances do you expect? Who are the opponents of your idea and are there barricades such as laws and access thresholds?
- Who are the ambassadors of your idea?
- Are there competitors with a similar idea and what is their investment capacity?
- Which blockchain concepts do you think you need?
- With how many stakeholders do you want to start a project? Which of their interests correspond with yours and which differ?
- Who are the intended customer, project leader and blockchain expert?
- With which target do you want to start the first project? How do you think that the dot on the horizon looks like (Z) and how does the first step look like (from A to B)?
- Who wants to participate in your project?
- How will you finance the first process (from A to B)?

Development model

A workable model, to create a focus and development plan, after asking and answering these questions and to determine what you will start with, is the model from Iansiti and Lakhani that appeared in the Harvard Business Review¹⁶². Using this model, you can draw an analogy between 'e-mail', as first application of internet technology and 'Bitcoin', as first application of blockchain technology. Based on their experiences, Iansiti and Lakhani have developed a framework for blockchain adaptation. We have used this model and adjusted it for Blockchain Organizing that is wider than just blockchain technology. The framework consists of two axes:

¹⁶² See also article in Harvard Business Review by Marco Iansiti and Karim R. Lakhani, both Harvard professors: 'The Truth About Blockchain' (January-February 2017).

1. Degree of complexity and coordinating capacity that you require. The more parties are involved in the innovation, the more difficult it becomes to manage this. It takes longer to reach standardization and consensus, but once there is consensus, you have a greater chance and a greater domain to apply your innovation. The success of a network organization (also called network effect) often depends on the number of participating nodes.
2. Degree of novelty of the required technology and organizational model and therefore adaptability for the user. The newer the application, the more effort must be spent on the adaption (adaptation/adoption?) by a wide audience.

This creates four quadrants that can also represent four development stages. In each quadrant, there is an example of Blockchain Organizing. The example from the internet development is printed in *italics*. Developments with a low degree of novelty and complexity will be embraced first.

		Degree on Novelty	
		Low	High
Amount of Complexity and Coordination	High	3. Substitution Loyalties based on Bitcoin <i>Online bookstore</i>	4. Transformation Self-executing smart contracts <i>Skype</i>
	Low	1. Single Use Bitcoin Payments <i>E-mail on ARPAnet</i>	2. Localization Private online ledger <i>Intranet</i>

Table 5.1: Framework for planning blockchain developments/investments¹⁶³

We explain the separate quadrants:

1. Single Use

Like e-mail was the killer application with the development of internet, Bitcoin is the killer application for blockchain technology. While e-mail was the replacement for faxes and letters, with all its well-known advantages, Bitcoin is a replacement for traditional money with immediate, clearly visible, advantages for the user. At the beginning of a technical development there is often only one application for the technology. Think of pumping water from mines with the steam engine. The steam engine only became a success when several applications were developed such as the steam locomotive and drives in the factory.

2. Localization

With this development, you do apply a new technology, but you do this in your own environment. Therefore, you are only dealing with a limited number of stakeholder

¹⁶³ According to the model of Marco Iansiti and Karim R. Lakhani © HBR.ORG.

(parties/departments). At this moment, we especially notice the development at banks and insurance companies to create an own, private or permissioned blockchain. This is similar to the development of an own communication platform within companies, based on the TCP/IP protocol. Another comparison is the difference in the development of internet and intranet applications. Companies have much further progressed in the transfer of (analogue) processes to their intranet environment, than they have to internet communities. Examples of these intranet applications are Employee Self Service applications, employees can use to request holidays, choose employee benefits and exchange overtime for salary. This quadrant is mainly related to the sixth driver for Blockchain Organizing that we described in paragraph 4.1 (Driver 6: process improvement).

3. Substitution

In the third quadrant, technological innovation is not that big, but there are several parties involved that require coordination. With this transformation we see the transfer of processes to a partnership or platform. There is replacement, in case of Blockchain Organizing, when a limited number of internal processes are transferred to an external platform while the technology to perform these processes does not have to change a lot. An example of this is the transfer of processes to a shared service center that still communicates via e-mail with the different actors on the network.

4. Transformation

There is transformation when processes are transferred to a shared transaction network and these processes are performed using a new technology and a new management model. This means, for example, that checking the execution of an agreement between a buyer and supplier is no longer done based on random checks by an employee of the quality department. The execution of a transaction and monitoring thereof is done fully automatically and conditionally via a blockchain protocol.

The risk of failure increases from quadrant 1 up to 4. For example, if you can invest little and you want to/may be at little risk, you can best start with quadrant 1. Once this innovation is running smoothly, you continue with quadrant 2, etcetera. A relatively simple and safe way to start with blockchain technology, is adding an additional process to existing peer-to-peer platforms or online communities, where you can buy products or services. An example is the market for business courses in the Netherlands, where a number of platforms are used that could be expanded with the provision and registration of the service (purchase) via blockchain technology. After that the payment could also be made via for example payment in Bitcoin or an own currency. But this relatively simple expansion also requires a careful planning and study because processes must be performed securely and faultlessly, especially in the payment system. Plus, large companies have definitely already invested a lot in internal systems for payments, a connected bank book and accounting system. This could mean that you could probably best start with the medium-sized and small enterprises and not with the large companies. Anyway, it seems that we are still awaiting for real

transformational blockchain applications. This will change when we become aware of our social problems and how organization technology can help to solve these.

How do you move?

According to the aforementioned John Hagel, large scale interventions are usually initiated by two alternative sources. A senior officer where you work, could read for example this book and subsequently wants to apply, top-down, Blockchain Organizing in your organization. We often know from our own experience and from management literature, that this usually does not work properly. The second option is the formation of a team within the company, that develops an initial idea further so it can be implemented later in the existing operation and structure. The success ratio of this approach is often also low because the existing organization is involved in the innovation too late. So, we usually opt for a hybrid form of the following movements:

1. Top-down and bottom-up, whereby we design and develop a new organization and transfer existing processes to this new organization
2. Short and long term, we also call this a double decision: you solve short term problems (for example making payment transactions cheaper) and in parallel, with at least the same budget and invested at the same level, a long-term project (for example making money redundant as medium of exchange).
3. Exploitation and Exploration, also known as ambidexterity, whereby you combine walking over a bridge with building the bridge as to speak.

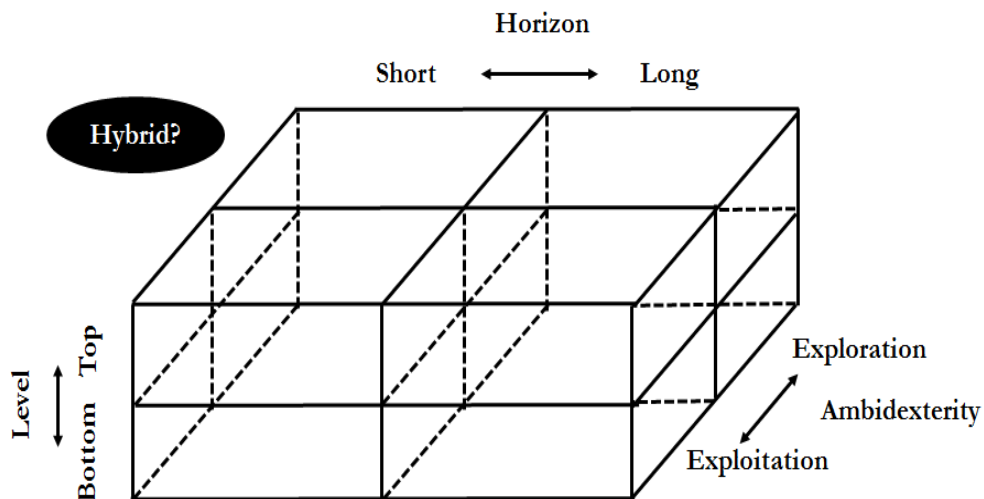


Table 5.2: Framework for how you can move

We will further explain the axes (Level, Horizon and Ambidexterity):

Level: top-down and bottom-up

Blockchain Organizing is building from a new perspective, on new foundations and with a new strategy. Blockchain Organizing is Greenfield organizing (building a new

house), without the ballast (legacy) of Brownfield organizing (repairing old house). This means that especially old institutions like non-transparent markets, industrial companies, government authority and traditional management will lose part of their power, their autonomy and interest because they become less relevant. This will happen gradually because more and more processes are transferred from classical institutions to modern network organizations.

You can distinguish two movements with this (extremes on a scale): on the one hand there is the top-down approach, whereby existing institutions decide to transfer a part of their processes to a new organization. On the other, there is the bottom-up movement, whereby portfolio workers will organize trust and work fundamentally different. Like in the case where self-employed professionals will organize themselves via communities in the shape of cooperatives. With the use of Blockchain Organizing individuals are able to develop organizing capacity similar in size to large heavy institutions at much lower management costs.

By combining top-down and bottom-up approaches, a strong foundation is created for Blockchain Organizing. An example of this hybrid form is a shared transaction network where portfolio workers can unite themselves to be able to perform large assignments. It is a network that is joined by both portfolio workers as well as large organizations. This shared network creates a capacity similar to a large engineering firm or law firm.

Both movements, top-down and bottom-up, require a different approach, but must eventually come together and be used from the same strategy. Many existing companies still think from the competition perspective and less from the cooperation perspective. But this image becomes less relevant with Blockchain Organizing. Think of the aforementioned competition between major soccer teams, it becomes irrelevant when people stop watching soccer.

With big assignments, portfolio workers think of cooperation with other freelancers and with small assignments of competition. Whichever way you turn it, we need a management innovation that better balances cooperation and competition. You cannot renew the existing, without respecting the old. We see, for example, that new initiatives like Uber and Airbnb are faced by existing legislation and interest organizations. You can fight the old world in court, or you can also take into account in your design that existing processes must be transferred to your new building and that you can better realize this step by step by involving the existing world that is open-minded in this respect.

Top-down

With the top-down movement, existing institutions are the unit of thinking, analysis and design (remember the metaphor: a Lego set). From this position, you can

reorganize processes with the help of partnerships, shared service centers, supply chains and new concepts like ‘circular economy’. A company will then become the fixed core or hub in a network of professionals and transfers certain processes to a flexible shell or partnership. An example of top-down movement is an existing company, that wants to switch to a 6-hour work day with the use of a CSR program¹⁶⁴.

The reason behind is that employees can go home at three o’clock because they have caring tasks at home. The advantage for the company is that employees come to work the next day well rested and much more productive during the six hours they work. But to realize a six-hour working day, the company must cooperate with other organizations in the region including the government. When a company stands alone in improving productivity, the company will probably think from a competitive perspective: if we become more productive, I am able to produce more with the same resources or if I continue to produce the same and I can dismiss people. By cooperating, for example under the direction of a council, from a same interest, by keeping our welfare state affordable, this will succeed sooner.

Bottom-up

The unit of analysis and design with the bottom-up movement is organizing demand and supply with the individual as unit of thinking, analysis and design (remember the metaphor: smallest Lego blocks). A group of self-employed accountants can offer a similar organizing capacity, via a partnership (for example a cooperative), as a large accounting firm. In theory, this movement means: reorganizing and coordinating transactions and making these more transparent and better scalable with the use of partnerships. The transaction costs can be reduced by using organizational technology.

To join this movement, existing markets must become more transparent and cheaper and existing companies will have to change their way of thinking, their strategy and structure. In case of tenders they should give cooperatives the same chance as large service providers. There is especially an important role here for governments, when they outsource services. Because the government in particular wants to avoid risks and wants to reduce insecurities as much as possible, there is an opportunity here for innovation. When cooperatives show that they can do the same work, with the same quality or even better, but at much lower costs, there is an opportunity. New organizations, for example, will be created because portfolio workers, with the help of a shared transaction network, can achieve a similar organizing capacity as large companies, but at much lower transaction costs and without increasing the complexity.

Hybrid

An example of a hybrid form is demonstrated by Constantijn van Oranje-Nassau, the youngest son of princes Beatrix of the Netherlands and brother of King Willem-Alexander. After his law study in Leiden, he followed an internship at the cabinet of

¹⁶⁴ Corporate Social Responsibility Program.

the Commissioner of the European Commission. Then he followed a MBA course at INSEAD, worked for RAND cooperation and became Head of Cabinet of European Commissioner (Digital Agenda) till the end of 2014. According to an interview in Elsevier Magazine¹⁶⁵ he could have continued working at the European Commission, but he was spending eighty percent of his time with people who did not want to change and who benefitted from a status quo. Member states do not want to move and neither do established organizations with vested interests.

From the Digital Agenda, he tried to get companies and governments interested, but according to him this was: ‘moaning and dragging’. This top-down way of working, did not work for him. He prefers to work with people who like to get things done. Therefore, he started his own company after his time in Brussels. Now he is, among other things, Special Envoy for StartupDelta¹⁶⁶. He also works on the implementation of the Digital Agenda, but now bottom-up, in combination with his business and government contacts.

A hybrid movement is a movement whereby you bring a top-down movement and a bottom-up movement together. This means that the stakeholders in your project are large companies, entrepreneurs and portfolio workers who want to unite themselves. Large companies see that they can develop a function as hub in a network organization and portfolio workers need assignments from companies. Traditionally, a large company could opt for an innovation by either use their own R&D department or purchase the innovation by acquiring a start-up. In both cases little will change fundamentally because the management model is not renewed. Innovation is still done with the existing unit of thinking, analysis and design. A fundamental innovation is created when large companies start fundamentally reviewing their management invention of hundred years ago and start replacing their bureaucracy with organizational technology.

A second movement that you can make in your blockchain project, but then via another axis (see table 5.2) is the combination of a short and long-term project. When we look at the PLATO example, large companies should rely more on Privacy by Design in the long term, instead of using bandages in the short term with more IT and more procedures. On the supply side professionals should unite to be able to deliver a customized product to clients through cooperating smarter at lower costs. Cooperation is required between the different specializations such as an organizational expert, an IT architect and developer, lawyer and auditor. Specialization should not lead to more complexity and bureaucracy. Therefore, coordination receives at least the same attention within the PLATO project as specialization and finalization.

Horizon: short and long term

¹⁶⁵ Article in Elsevier magazine by Carla Joosten and Jean Dohmen (31-12-2016, Dutch only).

¹⁶⁶ Accelerator for Dutch start-ups.

Many organizations that we advise, ask for solutions for short term problems or how to apply blockchain technology in order to make internal processes more efficient. They hear about the disruptive character of the technology and want to do something with this new technology without understanding what is really is. They do not want to miss out¹⁶⁷, but they are not thinking about a long-term vision or strategy. Sometimes they hear that blockchain is aimed at making existing institutions redundant, but that will not apply for the own company, they think: My position and my company will continue to exist, as long as I embrace new technology on time.

But this, often linear adjustment, is not enough. It is no longer just about technology. It is also about a fundamental management innovation. It is about the contribution of your organization in solving organizational and societal problems. When you only look at Blockchain Organizing from a short-term self-interest, your investment will probably fail. The paradox is that you must first dare to let go of your own position and organization in order to continue to exist. This requires courage and a short-term and long-term approach, preferably combined in one program or strategy. Therefore, we use the double track decision (hybrid form) more and more at the organizations that we advise.

An example of a double decision was The NATO Double-Track Decision. It was a decision of NATO (1979), to offer the Warsaw Pact a mutual limitation of medium-range ballistic missiles and intermediate-range ballistic missiles combined with the threat that in case of disagreement NATO would deploy more middle-range nuclear weapons in Western Europe. This reciprocity turned out to be successful in the end. Short-term changes will often be organized from a top-down movement and long-term changes (actual innovation), will be organized more bottom-up and with external pressure, including technology.

A double track decision combines both into a hybrid form. You could see a double track decision as building a bridge (long-term) while you walk across (short-term). A double track decision is a strategy to combine practice (short-term) and principle (long-term) into something new. This strategy ensures that short-term problems are solved within existing companies and markets, under the condition that a long-term project is also started at the same time with a similar budget and invested at the same level. The long-term project is regularly tested and no resources may be transferred from the long-term to the short-term project.

Ambidexterity: exploitation and exploration

The word ambidexterity means that you can use both right and left hand equally. In management science it means that you can combine exploitation (management) with exploration (innovation). Organizational ambidexterity needs leadership that sees that both exploitation and innovation are important. They should be in balance and need diversity in the people and resources. The one cannot do without the other.

¹⁶⁷ Also called Fear of Missing out (FOMO).

You cannot innovate without revenues, unless you collect the necessary resources from outside the organization but that gives you other challenges such as keeping shareholders satisfied the next quarter. And probably, without innovation revenues will stop in the future.

So ambidexterity has also to do with short and long term, but innovation is not always important only for the long term. Innovation could help your organization to stay restless, young and challenged at the short term. Mostly innovation gives people new energy, but it could also frighten them to lose their job for instance. You should give attention to both sides of the effects of innovation.

The question is not whether or not you should improve existing business or disrupt, the question is how you combine these two and whether or not you do execute it within the same fiscal legal entity for example. You can imagine that you do business as usual in your existing company and parallel you innovate via a new organization (form). You cannot stay in business without changing and adapting, but that could be changing within the system: Brownfield organizing, operational excellence, improving the candle or sustainable innovation. Most managers are optimizing the existing system with a mindset of structuring, Lean Six Sigma, Agile or Self Organizing Teams for instance. But what if the existing system is the problem?

For changing the system (from candle to light bulb) you need disruptive innovation, you need Greenfield organizing and Blockchain Organizing at the moment is an example of the latter. But Blockchain Organizing goes further than what a lot of people mean with exploration or even further than disruptive innovation. Most businesses keep their own organization in the middle, between purchase and sales or collecting and distributing and they think of innovating new products, services, and processes but still organized with or from their existing organization. But as we learn from this book: blockchain start with giving up some autonomy. Your organizing is not the center of the business universe, but just a part of a whole. Without changing your worldview, blockchain will not work the way it was designed for.

5.3 Project-related approach

As we stated before, you can come up with many application that can work with blockchain technology: from maintaining your e-portfolio or patient file, to selling/buying your house. Everything with value, the ownership of which can be transferred (digitally) from seller to buyer, can in principle be registered on a blockchain. Plus, according to the Blockchain Organizing concept the technology is not primarily important, but the organizational problem that you want to solve and the organization model you use for that. This means that Blockchain Organizing does not primarily focus on the improvement of traditional management, but on reinventing the management itself. It is based on Innovation by Design. This concept assumes that you can better start a new organization and transfer existing processes to this new organization. You'll have less problems with legacies, traditional management and

old IT systems. When we, apart from improving processes, look at the other drivers for Blockchain Organizing (purpose, productivity, privacy, propaganda and power), they can only be solved in a structural way by going back to the drawing board.

Innovation by Design means that you bring a number of elements together, as described in this book. Some examples are:

- what is your perspective, from which one do you look at the future?
- what is your wide and deep analysis of the current management concepts
- which management problems do you want to solve?
- which (technical) concept will you use for this?
- which organizational and management principles will you use?
- how are you going to fill in the ten core connections?
- what does your innovation program looks like?

The last question deals with the process that must lead to your management innovation and the use of Blockchain Organizing. A program includes various (sub) projects that you plan to get from A to Z. Therein you must always indicate how your project and eventually your innovation will distinguish itself (focus, branding) from the rest. What is your target and target group (the stakeholders), who will be the client, project leader and the organizational expert? When you have completed this, you can define (in a 'SMART' way) the first step (from A to B).

When you want to start with Blockchain Organizing within your organization, supply chain or ecosystem, it is important (as blockchain lead) to develop a vision. Subsequently, you can share this vision with other stakeholders, for example in a workshop. A workshop 'Blockchain by Design' can be held based on a vision document. This vision document contains a perspective, problem and solution direction. This document can be elaborated on by the client, blockchain lead (project leader) or for example a blockchain consultant. A vision includes a number of insights such as: 'Technology solves little if we do not first look at the organizational design'. During the workshop, the functional needs, available concepts and pre-conditions can also be sketched for the first time.

Participants in the meeting are asked to respond to the vision document. Preferably, the participants in the workshop should be stakeholders in your project. The developed vision document will be discussed during this workshop based on the (pre-introduced) perspectives of the participants. Subsequently, vision, needs, concepts and pre-conditions will be included in your project plan for the implementation of Blockchain by Design. If desired by the client, participants in the workshop can subsequently be invited to participate in your blockchain project. To reduce resistances and to increase the success rate, the workshop should have a high interactive content and mainly focus on the widest possible supported commitment for the start of your project. The basis of interaction should not be the discussion, but the guided dialogue. By asking open and non-suggestive questions you could together determine what the old conditionings of the traditional management are and discover new

perspectives. During the workshop, the various stakeholders are separately and jointly challenged to arrive at an open innovation, to take opportunities and to solve problems.

Example workshop program

An example program of your workshop could be:

1. An introduction to the problems or the challenges of the project that you want to start, seen from the perspective of the relevant domain and from the project initiator/client. Questions that will be discussed could include:
 - a. What can new management concepts, like Blockchain Organizing, mean for solving the problem?
 - b. How can we cooperate smarter between business, government, knowledge institutions and communities?
 - c. What are the principles and foundations of New Organizing, management innovation and Blockchain by Design?
 - d. How can we better solve our problem based on these principles?
2. A series of presentations in which the various participants present their own perspective on one or more aspects from your vision, idea or problem. Plus, they share their ideas regarding some directions to arrive at an imagined destination. These insights are sent to the project leader in advance.
3. A number of round table discussions in small groups. Ideas about problem definition, functional needs, technologies, management models, pre-conditions and a first project assignment are exchanged and determined here.
4. Finally, the results of the discussions are summarized and bundled and will serve as input for the next step. This creates a chain of progress and the actual start-up of a Blockchain by Design project.

Blockchain by Design is not change, but renewal

As mentioned before, Blockchain by Design is: Greenfield organizing instead of Brownfield organizing. It is innovation instead of change. Despite the fact that a billion dollar industry has developed around change management and change projects, very little research has been done into the effectiveness of all these projects. The first important question is already a difficult one: how do you measure the success of a change project? When exactly is a project successful? Many ideas circulate around the success of change projects, but these myths are not really (scientifically) verifiable. One figure that is being tossed around is that seventy percent of change projects fail. It has also been said that you need charismatic leaders for changes that self-steering teams would perform better and that early participation of stakeholders would lead to less resistance. What is not a myth is that every failure should be better in the process. Your blockchain project will probably follow the next steps:

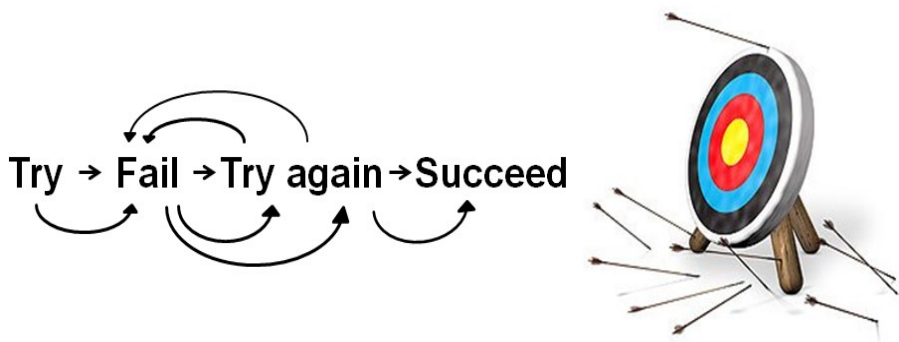


Figure 5.2 Next time: fail better

The success of a project could be measured by the percentage of transactions that are no longer purely done by yourself or via outsourcing, but within new entities such as a blockchain community. This number is currently still very low, but when measured in a verifiable way it could better assess the development of Blockchain Organizing. In the long-term, this number could indicate whether these projects contribute to the improvement of the productivity and the creation of more surplus time.

So far, the beginning of your Blockchain by Design project. Now we continue with a more detailed look at the project approach that we have used often with a number of our blockchain projects.

Starting-up a project yourself

You can start a project yourself. It does not matter whether you have your own business or for example work at a large organization. With our working method, we focus on enterprising people who want to solve organizational problems with Blockchain Organizing such as the low productivity growth and the new privacy law. The projects that we guide as consultants mainly focus on using organizational technology, to become more productive or to better assure privacy and security. Projects take place in knowledge-intensive domains with many buyers and sellers. A project must eventually lead to the exploitation of a shared transaction network, with which you can offer, order, administratively settle and digitally pay products/services/data requests. The aforementioned PLATO project must lead eventually to a situation whereby a company can quickly assemble the right building blocks and connect these at low transaction costs to enable Privacy by Design.

To quickly test whether your ideas can be organized with blockchain, we have developed a quick scan method with ten questions. In addition, there is also an extensive questionnaire once you are further progressed into your project. After the quick scan the next 'test of the quality of your idea' could be a kick-off. Try to get ten to twenty stakeholders interested in your idea. Try to 'tempt' them to come to your meeting and provide an active contribution without them knowing exactly what they will

receive in return. During the first meeting, you want to test your initial idea with the people present as much as possible. This means that you are open to additions and criticism from the participants. On the one hand, it is advisable to first explain your idea in general terms, so there is room for the participants to make modifications. On the other hand, your story must not be too abstract and vague because people will then drop out because it is not concrete enough.

When participants wonder after the kick-off whether they should spend their valuable time on your project, you must make a choice: continue to try to make people enthusiastic or to draw the conclusion that you do not fit together. It is also possible that your idea is good, but that the timing is not right. The trick is to find a good balance between open and structured. A nice dialogue form for this is for example Openspace, Roundtable or ThoughtExpedition.

Anyway, you must already have a well thought-out idea of what will be your target and target group of the first (sub) project. These details can be made more precise by points brought up by the participants. When you continue with your own idea, without using the feedback, it is possible that participants drop out. It is also possible that you encourage other participants with your clear and focused idea, to join. Between the first and second meeting you have individual contacts with the participants to evaluate and to ask whether they want to continue or not, and if so in which role. Based on this evaluation you tighten the target and target group. After the second meeting, you must ensure that each participant has a similar understanding level of your idea, your domain and theme. Anyway, the target of your project must be clear at the second meeting.

Blockchain project goal

A project can have different goals. These can also be different stages of your transition program, for example:

1. Start: exploring each other's values, perspective, target, visions and ambitions. Choose project leader, expert, fill in quick scan, prepare kick-off etcetera.
2. Awareness: why do we need Blockchain Organizing for our organization or our domain? What problem do we solve?
3. Proof of concept: setting up a test/trial organization and showing that blockchain works with a simple, but workable demo.
4. Setting up and conducting a feasibility study that must give a definite answer within a certain time whether it is feasible to use Blockchain Organizing (model, infrastructure and program) in the researched domain. You can research the technical feasibility but also the market opportunities.
5. Setting up and implementing an open-innovation project, that must define the functional needs, wishes and pre-conditions within a certain time frame, for an organizational model, infrastructure and transition program for domain.
6. Setting up and implementing a development process that develops an organizational model, knowledge structure and transition program for the domain, based on functional wishes, needs and pre-conditions.

7. Setting up and implementing a (new) organizational entity (for example foundation, limited company or cooperative), which can exploit a blockchain community within a certain time.

The main target of setting up projects is the development and eventual exploitation of a shared transaction network for your organization within a certain domain. A shared transaction network is a network with many sellers and many buyers (n:n community). With this network you can, almost fully digitally and at minimum transaction costs, profile, select, purchase and pay products/services. Using organizational technology helps with this. Plus, technology is the coordinating capacity between supply and demand (see also figure 1.3). The underlying infrastructure (general data utility facility) for blockchain communities is in principle the same for any community. The differences are mainly in the focus (target -and target group-oriented communication) and could be programmed with script language.

In addition to a clear target for your project you must also define a clear target group for your project. On the one hand, there is a target group with which you want to implement your projects (also called project stakeholders), and on the other hand you have the target group (your customers) that benefit from the result of your project.

Target group project

Once you have a project plan ready and you want to start with your blockchain project, the choice of your project members, your project stakeholders, is important. These stakeholders can differ from the participants in your first kick-off. It can be possible, for example, that there are mainly experts present during the first kick-off and then each expert will approach a number of market parties to participate in the project. In the beginning, you must especially trust experts and communicate clearly which market parties you are looking for. Subsequently you must trust the experts that they will find the right parties. Usually we advise a project group of ten to twenty stakeholders and divide these equally into for example:

- Market parties: this group can usually be sub-divided into:
 - Primary market parties, those who will later on be seller or buyer in a transaction in the community;
 - Secondary market parties, those who offer products and services with the exploitation of the community to enable or facilitate transactions: hosting company, consultancy or training agencies, legal advice, software developer, digital currency platform and intermediaries.
- Project suppliers: only present in project: project leader, organizational expert, blockchain consultant, research agency, media partner, IT architect, software developer, tester, hosting company for Proof of Concept.

You will notice that as soon as your idea gets traction, there will also be resistance from existing parties that feel threatened. That is why it makes sense to take into

account from the beginning that there will be resistance. A good principle for this is: 'hope for the best, prepare for the worst'.

5.4 Dealing with resistances

How can you prevent that resistances occur that slow down your blockchain project? In this book we have discussed the rise of destructive populism that is partly caused by fear and helplessness of people. When we continuously change things but eventually have the feeling that we gain little from it, we will become apathetic in the next project and perhaps oppose it. In professional literature, this is called learned helplessness. Learned helplessness is a characteristic of people which can occur when the desired result is not achieved after a number of attempts to avoid an unfavorable situation. When attempts to change continue to fail, you eventually become apathetic and you not even try it the next time, although there is a (new) chance of succeeding.

We have become powerless to escape an unfavorable situation. So, learned helplessness describes a situation whereby people become apathetic as a result of futile attempts to escape the situation. A good example is traffic jams. When you want to avoid a traffic jam and take another route, but time and time again this route is not faster, you will eventually not even try to take another route. You undergo the situation although there is another chance that the other route is indeed quicker.

This behavior is substantiated with scientific research by the American psychologist Martin Seligman¹⁶⁸, among others. He conducted extensive research into the phenomenon of learned helplessness. According to Seligman, learned helplessness is learned during the exposure to uncontrollable events. These are events whereby it is impossible to respond to a negative stimulus. Seligman performs experiments with dogs where he puts the floor of the dog house under a weak electric current. Certain dogs seem to be able to jump over the fence to avoid the expected electric shock. Other dogs 'passively' undergo the pain of the electric shock because they cannot escape. When these dogs do get the opportunity to escape they do not do this. Apparently, they have developed learned helplessness. They have become apathetic. With man we see learned helplessness with change processes within companies. They are tired of change and expect that the project will not change their situation. So, when a project is started with new opportunities with which they can indeed change their situation, they probably will not join. They undergo it, but do not participate actively.

Blockchain Organizing will only succeed when we manage to reinvent the latest management innovation of Taylor and Ford and adjust our perspective on the organization of work and welfare. We must convince people that we can easily switch to a 6-hour working day for those who are interested. There will be resistance from the

¹⁶⁸ Martin E. P. Seligman (1942-), is an American psychologist, author of self-help books and professor at the University of Pennsylvania. He conducts research into learned helplessness in humans (and animals).

‘old-world’ that is still convinced that the firm, the bank, the government are the right tools to coordinate supply and demand. Most managers will say: If organizational technology ensures that we finish the work sooner, I will look for additional tasks or I cut the budget of that department. We must dare to let go of this expansion and old budget thinking. We must also let go of the thought that working hard is good. When working hard means that we buy products that we do not require, then working hard is not right¹⁶⁹.

From resistance to awareness

Blockchain Organizing is still mainly in the awareness stage and this will be followed by the design and development stage. After that, Blockchain Organizing will be brought into reality and applied more and more. How do we deal with this innovation, which resistances will arise and how can we cooperate smartly with the existing and conditioned world? How can we make sure that the improvement of the productivity indeed leads to a six-hour working day for example. Then it is not useful to immediately ‘import’ a new technology into an old organizational structure, without thinking about the meaning of work and unemployment.

As to speak with Plato again: with the body you should also look after the soul. It is better to first work on ‘awareness’, to look at more fundamental questions and for example learn from the internet development and the consequences thereof on the organization of work. In the first stage, it is useful to become aware together with the stakeholders and to find answers to questions like:

- What is progress? When will our children have a better life?
- Can we first think about the questions what we will do with the surplus time when technology makes us more productive?
- What will be the role of companies in shared transaction networks? Will Blockchain Organizing disrupt existing organizational forms and organizations or can they adapt quickly enough to, for example, a hub in a network?
- Are directors, politicians and policymakers aware that internet has fundamentally changed little in the way we organize supply and demand? What will we learn from this regarding the development and application of blockchain?
- At what rate will the innovation (have to) take place? Which drivers are there and what is the urgency of the new privacy law for example?
- To what extent will blockchains become interoperable, mutable or replaceable? Will we mainly start with private blockchains or will blockchain be used more in communities?
- Will existing companies make disinvestments in existing systems such as CRM, HRM, ERP, purchase and payment systems to be able to transfer processes to blockchain communities?

¹⁶⁹ See for example article by Mark Whitehouse in 'The Wall Street Journal: 'Number of the Week: Americans Buy More Stuff They Don't Need', Americans spend 1,200 billion dollars per year on things they do not need. (23-4-2011).

- How quickly will education, consultancy firms, business administration and managements training courses and training agencies switch to new insights related to organizing trust, work and economy?
- Which sectors or countries will be the first to take on Blockchain Organizing? Where is the most low-hanging fruit?
- How quickly will the legislator follow?
- How quickly will software developers take on new technologies and train themselves for this? What will be the supply of blockchain developers later on?

When you get insight into the answers to these questions you will be able to better estimate resistances within and outside your organization. When you complete the project group, you can for example start with research. This can be a feasibility study, but also a design assignment. In all cases it is wise to perform research, design, innovation and development process in a structured way. Of course, the design itself does not have to be done in a structured way, iteration and creativity, play an important role in this.

We do not know exactly how long it will take before blockchain has sufficient applications and volume, so people will consider it as ‘the new normal’. When we take into account how for example the TCP/IP protocol has developed itself in three decades, we expect that a major breakthrough of blockchain technology and therefore Blockchain Organizing will take one to two decades. Our society is some kind of super tanker that does not suddenly change direction. Still it is useful to already start today with the development and application of ideas and innovations surrounding Blockchain Organizing.

The main lesson of ‘large scale interventions’ is to start as early as possible with controllable learning. According to the insights of John Hagel, you need to track down opponents of innovation (resistors of change), weaken them and strengthen supporters (ambassadors of change), as early as possible. For example, you can strengthen supporters with a good workable and useful Proof of Concept (POC) application. This means: taking acceptable risks to prepare people and organization step by step for what is to come.

According to Hagel, we must be careful that we not just see large scale interventions as a rational process with which you move towards the final target, step by step. According to Hagel, it is mainly a political process¹⁷⁰. We agree with this. Since the beginning of the nineties we have experience with the use of internet technology, for example with the smart organization of supply and demand in the SCM, HR and government domain. It is our experience that fundamental innovation cannot just be realized by using a new technology, not even in combination with a new management

¹⁷⁰ See also video of Singularity University: ‘How to Become a Champignon of Change’ (14-11-2014).

model. There must also be leadership and problems to be solved. According to us, that is now even more relevant than in the 1990s.

Furthermore, there are also other forces that play a role, such as dis-rationality¹⁷¹. Often you are not able yet to reason or rationalize a choice, whether to embrace a system or not. The innovation is still too abstract. It differs too much from what you are used to. Abstract is not the same as complex. Abstract means omitting what is not relevant. *We 'reject' a simple world because we do not know yet what we must do with the surplus time.* Plus, many choices are made emotionally, instead of rationally and are often politically driven. We assess, often implicitly, an innovation on the risk of losing our power base.

In addition, with the rise of a technology many people do not take the time to understand it. For most people, and rightly so, technology must be easy to use. We want to experience technology as something self-evident, something that works for us and that is how it should be. Many people do not have the time to delve into it or think that they will not understand it anyway. Maybe they are also afraid to discover something that can jeopardize their current reality. However, when you delve into it too late, in this case Blockchain Organizing, the result may be that you are too late with learning, with innovating and adjusting yourself to a new order.

John Hagel sees three important parts of large change processes, or as we call them: innovation processes:

1. Identify and neutralize the enemies of innovation. This sounds easier than it is, because during a project meeting or kick-off, for example, nobody will be against renewal or innovation. But if this innovation undermines their own position, people in the comfort zone will start to oppose from their own position or own organization without this being noticed by the project group.
2. Identify the supporters of innovation and give them as much as possible support with the process of: developing awareness, feasibility studies, open-innovation, development and implementation.
3. Before you start implementing an innovation you must make sure that the opponents are sufficiently weakened, and the supporters sufficiently strengthened, so it is evident to implement the innovation.

If you, after reading this book, have become interested in using Blockchain Organizing within your own organization or ecosystem, it is important to know that there are solutions for problems that you will probably face. However, these are often not practical, short term solutions that usually make the problem worse anyhow, but fundamental solutions that will last longer. From own experience we know that the further development and application of disruptive innovations can encounter a lot of resistance. You have to imagine that most positions and organizations that

¹⁷¹ Characteristic, that we are often unable to make rational decisions despite our intelligence. This is because we are conditioned.

currently exist, have been created to reduce insecurity. We have started to globalize demand and supply more and more, which increases the need for insecurity reduction. We find it normal that there are purchase and sales conditions, while we forget to coordinate these.

Due to efficiency pressure, we have started to use more and more specialists. When these are not properly coordinated, the complexity increases and therefore also insecurity and inefficiency in supply-chains. Subsequently we are conditioned in the solutions we come up with. We solve new problems with the same frame of mind from which they originated. We find it normal that we use money to trade while this was not the case in most of our human existence. We find it normal that we organize work with companies, while the company, in the form as we know it currently, is only a few hundred years old.

In Western Europe we find it normal that there is a government that is responsible for our life, from when we are born till we die, while we forget that more than hundred years ago almost everything was organized privately. And some people believe that we are entitled to welfare while they forget that you must become more productive if you want more welfare. All conditionings that we find difficult to get lose.

New starts by letting go of the old

When you ask someone: What is hard to forget, the answer usually is: cycling, skiing or driving a car. When you learn how to cycle for example, you build up reflexes. These are some kind of conditionings. They have become a kind of permanent paths in your brain. The advantage is that you do not have to think when you cycle in a busy town. The disadvantage is that you find it difficult to get rid of these conditionings. Well-known is the YouTube video *'The Backwards Brain Bicycle - Smarter Every Day'* where Destin Sandlin, an engineer from the US, has a bike made with two gears between the bicycle handlebar and the wheel. When you steer the bike to the left, the bike goes to the right. It seems difficult to learn how to cycle on this new bike. Not because cycling is difficult to learn, but we find it difficult to reduce existing reflexes and to let go off conditionings. It is no surprise that it is easier for children to learn how to cycle on the reversed bike. Innovation therefore starts with letting go of the old. This is difficult, but indeed necessary. So, if you want a disruptive management invention, like Blockchain Organizing, to be a success, you must actively start with letting go of old conditionings. Luckily there are all kinds of tools for this.

So, there is a way of thinking, a way of organizing with a new technology that enables us to organize trust, without increasing the complexity. That alone is revolutionary, because in the past centuries the complexity increased more and more when organizing trust. But even this revolutionary 'invention' will only succeed if we first and foremost address the question how we can usefully spend surplus time that arises when we smartly use organizational technology to reduce the number of intermediaries.

If we use management technology without considering what we will do with the surplus time, we, as human beings, will not benefit much from the use of technology. When we reassure people with the idea that robotics is also creating new jobs, this is a form of self-imposed immaturity. When we do not think about a meaningful interpretation for the intermediaries that are made redundant, these intermediaries will continue to oppose innovation and find other ways to preserve work. When we do not learn that internet has not made us more productive or democratic, blockchain will also change little. It is up to our generation to proof that we are smart and have learned from the past.

Innovations do not work with full employment

The main problem, according to us, is the contradictory policy of both innovation and full employment. The meaning that we give to (un)employment and the way we organize trust plays a crucial part in this whole story. Managers (and directors and supervisors in general) also play an important role in this. Therefore, our attention to the theme: Blockchain Organizing for managers. The most resistance to innovations occurs because we find it difficult to let go of the old conditioning.

The idea that working hard is good, that entrepreneurs must grow and that we need full employment is framed. Innovations will only work if we identify ourselves less with the existing world including economized work. We innovate more and more (computer) machines to do the work for us. It is time that we, as human beings, contemplate what we will do with the time that is given to us otherwise the system that brought us so much prosperity and helped us to pursuit happiness, will, eventually make us very unhappy.

Conclusion

As mentioned in the preface, many people do not see management as an invention. They probably also have a different view on management innovation. They cannot form a clear picture for themselves of a disruptive innovation such as replacing traditional management with a protocol, similar to replacing the candle with the light bulb. The light bulb was not created by improving the candle step by step, but through pioneering, through the presence of certain materials (like tungsten) and through other inventions like electricity. The invention of the light bulb, same as the invention of a blockchain, cannot be assigned to one person, but it is a process of trial and error. Plus, not just one aspect, like technology, is important, but also for example successfully convincing people to use the product. One of the light bulb pioneers was Royal Philips, a Dutch technology company headquartered in Amsterdam, probably became successful with the light bulb through the cooperation between Gerard Philips (engineer) and his brother Anton (the entrepreneur) and their father Frederik who acted as financier.

Much later, the invention of the light bulb led to other applications, such as the electron tube that is used in televisions and later computer monitors. With the development of Blockchain Organizing there will be more aspects that will be decisive, and

the invention will lead to a large number of applications and new inventions. However, the time must be right for this. Technology must solve social problems and various positions will have to cooperate smartly to achieve the target. But the main thing is that you dare to let go of the old establishment. If you still have candles in stock for the next ten years, you will not quickly switch to the light bulb, even though it has been proven that it is better and cheaper.

Asking the right questions is essential in this. Philosophy, curiosity, research and education play an important role in the transition phase we are in. In the past we saw that *Sophia* (the Greek meaning for insight and wisdom) is more important in fundamental transitions, than *Techne* (art and science). But many politicians, economists, managers and education institutions still focus on the latter with the current transition. When it concerns blockchain, they are very busy preparing managers and also tax specialists, lawyers, accountants, HR professionals and administrative staff and with training them for other work within the existing organizational form and the associated compartmentalization. They say: Robotization is not bad, because it creates new jobs. However, these advocates of employment ignore the intention of innovation and the use of tools for the benefit of people.

They mainly think of the contribution of innovation to the economic growth. They think linearly and are insufficiently curious about a different meaning of work and unemployment. It is time to let go of dogma for a new vision, or as formulated by Abraham Lincoln: *“The dogmas of the quiet past are inadequate to the stormy present. The occasion is piled high with difficulty, and we must rise with the occasion. As our case is new, so we must think anew and act anew.”*

So far, some insights regarding (project-related) innovating and developing Blockchain Organizing for your organization of network. We finish this book with a look at the future of Blockchain Organizing.

6. The future of Blockchain Organizing

For those of you who are not yet aware of it: our society is squeaking and creaking and has started to move on the surface with a strong undercurrent. There is anger, insecurity and a sense of injustice that affects many people more than the lack of economic growth. The old establishment has become too unstable and will have to be replaced. What could be the underlying causes (and similarities) of Trump, Brexit, fake news, populism, propaganda, climate change, immigration, increasing inequality, debts, problems in healthcare, education and security? And what could Blockchain Organizing do about it, you might wonder? Not everyone will immediately see a connection between a management innovation and solving all kinds of social problems. However, in this book we show, from a new perspective, that digital transformation and blockchain technology, in combination with a new organizational model, can actually help solve organizational problems and therefore social problems. Because we will become more productive in offices, for example, we have more time available to at least think about social problems. Because who has got time lately to read background articles for elections for example? A democracy presupposes that we have time to inform us before we take a decision.

The core of our argument is that we are in a transition of an establishment that was aimed at ‘creating’ *more and more* welfare, to a system that is aimed at controlling what we have. Organizing a sustainable welfare mainly costs time and attention. In order to create time and attention we must especially become more productive in offices. Blockchain Organizing can achieve this, provided it is applied correctly. Blockchain, and tackling the unrest and insecurity that we experience in our community, has one main similarity: organizing and managing trust. The unrest that we experience, especially since the economic crisis in 2008, is all about trust and uncertainty and the way we organize these. Obviously, there are people, when given the opportunity and a motive, cannot be trusted. So, we need tools to organize trust. Trust is the basis of work, trade and economy and therefore for our society. But trust arrives by foot and leaves by horse. It is important to take a thorough look at this and to ask ourselves: what is the best way to organize trust, also for the future. The future is where our children will live in. In this final chapter, we take a look at the future of Blockchain Organizing. The near and far future, but we start with distinguishing a number of stages in the development of blockchain that we can already observe.

6.1 Blockchain power

If we take a look at the distribution of power in the Bitcoin blockchain¹⁷², we can conclude that the old law of sociologist Robert Michels¹⁷³ also applies to this situation. The ‘iron’ law of the oligarchy states that all organizational forms, irrespective

¹⁷² For example: the centralization of mining pools and distribution of bitcoins ownership.

¹⁷³ See also the Weconomics theory book, paragraph 2.3: ‘Organization basic forms’.

of the democratic or decentralized governance structure in the beginning, will inevitably become oligarchic. The cause of this law is the trend among the rulers to organize around more or less natural behavior of our species:

1. Defend own interests
2. Gratitude of the guided group
3. Passivity of the masses

Michels observes that directors, in the bureaucratic model of his mentor Max Weber, gain more and more power and can also maintain this, because they have more resources and information as a result of their position. They also have power over other people's careers, so they will always choose directors who will strengthen their own position and will not overthrow them.

Another consequence of this concentration of power is the unwillingness to change. The status quo is the foundation of power. As a result, those who have the power to change have every interest in changing as less as possible. The present situation is the basis for their power. Michels came to this conclusion from his practical experience that every group of rulers such as factory directors, politicians, church-leaders but also labor unions, has a tendency to evolve into an oligarchy.

Democratically elected rulers will often find themselves in a small group of their own kind, 'yes-man' and co-rulers and lose contact with the people that have voted for them and put them in charge. By studying political parties Michels found out that the problem lies in the nature of humans and the way they organize trust and work via entities with a collective goal. In a modern democracy organizations emerge, and with the traditional (and natural) growth DNA, they will become more complex, become paradoxically less democratic with a higher need for central governance, unless we find a way to organize trust without increasing complexity.

There comes a time when every growing organization is confronted with coordination and ordering issues that until now could only be solved with more bureaucracy. The question in our new context of globalization, digitalization and sustainability is: will this coordinating mechanism be a social bureaucracy (social script) or technical bureaucracy (technical script such as a blockchain protocol). To make a bureaucracy efficient in traditional organizations, with little distinction between mechanic and organic organizing, a hierarchy was needed. Daily decisions cannot be taken by large groups of people. This means that the power in the organization comes into the hands of a small group or sometimes one person: the CEO with the danger of a single point of failure. In addition, the CEO, or small group of rulers, will do everything in their power to maintain their power and expand it to protect it.

Conversely, the most autocratic ruler also needs other people in and outside the organization. Power is made possible by the (initial) gratitude of the group that is led and the passivity of the environment in which 'the power' can develop. This makes it possible for the ruler to defend his or her own interest. In order to prevent this,

transparency is required and a policy, strategy and tools to prevent concentration of power. According to the iron law of Michels, the final result of an organization is that eventually every organization develops in an oligarchic direction. Maybe it is time that Blockchain Organizing is going to challenge this law with a good consensus mechanism, transparency, low cost decision-making and democracy on a decentralized network for and from the people. This sound unrealistic, but is it as unrealistic as having the idea in 1969 to operate Spotify one day on the internet?

6.2 Blockchain stages

In an article in the Harvard Business Review¹⁷⁴, Vinay Gupta¹⁷⁵ discusses a number of important stages that blockchain has undergone. According to him, we are in the middle of a 'silent' revolution that has taken place in the last ten years. We have expanded his overview in certain areas:

1. The first important innovation is Bitcoin¹⁷⁶, currently used by millions of people. Bitcoin started with the publication of white papers by Nakamoto, but continues to build on previous technologies, such as cryptography, database technology, the TCP/IP protocol and distributed systems.
2. The second important innovation is called blockchain, an underlying technology of Bitcoin. By separating the technology (blockchain) from the application (Bitcoin), an enormous potential of new innovations is created, such as smart contracts, identity management, document control and other applications outside digital currencies. Currently, almost every large financial or government institution is investigating the possibilities of blockchain technology. In 2017, about fifteen percent of the banks, expects to actually use this technology with the execution of their transactions¹⁷⁷.
3. According to Gupta, the third innovation is: smart contracts. These smart contracts are made in a next generation technology of which Ethereum is the first and most important development platform at the moment. Ethereum is a kind of (development) platform with hundreds of different projects that are being worked on.
4. The fourth innovation is the development of technology that is no longer based on (being secured with) Proof of Work, as within Bitcoin, but based on 'Proof of Stake'. The Proof of Work concept within Bitcoin becomes more and more dependent on a limited number of miner farms, especially in China. As a result, the danger of centralizing power is lying in wait. The miners provide security and control on input references and double spending and are therefore the heart of the

¹⁷⁴ See article by Vinay Gupta in the Harvard Business Review: 'A Brief History of Blockchain' (28-2-2017).

¹⁷⁵ English thinker, blogger and writer on the interface of (the history of) technology and philosophy. He is involved in, among other things, the Ethereum platform.

¹⁷⁶ Bitcoin Core (BTC).

¹⁷⁷ See also article in Fortune by Lucinda Shen: 'Blockchain Will Be Used By 15% of Big Banks By 2017' (28-09-2016).

Bitcoin protocol to prevent fraud. Proof of Stake is more based on the use of complex financial tools that can realize the same or higher level of security.

5. According to Gupta, the fifth innovation on the horizon is: 'blockchain scaling'. Currently, each node must process transactions. This makes the total slow, non-efficient and therefore less scalable. According to him, a scalable blockchain must improve the efficiency, without reducing the level of security. The protocol also investigates how many nodes are required to validate each transaction and distributes the work across the nodes. The scalable blockchain is required, among other things, to be able to settle transactions on the Internet of Things.

To pick up on these latest developments: we expect that blockchain organizations (including Decentralized Autonomous Organizations) and miners will also work via the ancient principles of specializing, coordinating and finalizing. This means that there will be specialized nodes within a shared transaction network, which can check and process certain types of transactions. Other nodes will then process other types of transactions. For example, a transaction of hundred million euros can be settled differently than a transaction of 1 eurocent. The settlement of the transaction can also depend on who the sellers and buyers are and their reputation. Machine Learning and Artificial Intelligence will also play an important role within blockchain protocols. The coordinating mechanism continues to focus on organizing sufficient (mathematically substantiated) trust at minimum costs.

We expect that Vinay Gupta will agree with use, that there is an important role for the government in the development of a blockchain infrastructure. It is our view, and that of for example Mariana Mazzucato¹⁷⁸, that it is not companies that will provide long term large social breakthroughs, but governments, whether or not in private-public cooperatives. In her book Mazzucato teaches us the following: *"The cutting edge of innovation, particularly in infrastructure, is often in the hands of the state, and that seems destined to be true in the blockchain space."*

Especially when it concerns the underlying basis infrastructure, data utility facility, we believe that the government¹⁷⁹, and in Europe we think of the 'European Commission' in particular, will be of decisive importance. The thought behind this is that a 'road network' is required to process transactions (transport from supply to demand). Hereby it is important that there is standardization, network neutrality and that there are traffic rules and an infrastructure no longer owned by a limited number of commercial companies with other intentions. Companies only have to build a garden path (own application) from the front door to the road. Local roads can be built by councils and national roads by the national government. Therefore, we have to use a layered infrastructure within Blockchain Organizing. We believe that in the US there is less cooperation between private and public and the big five (Amazon, Alphabet, Apple, Facebook en Microsoft) will try to preserve their position in the

¹⁷⁸ See book by Mariana Mazzucato: 'The Entrepreneurial State' (2011).

¹⁷⁹ More general: a commonality with democratic basis.

organization of the data infrastructure. In countries like China and Russia, we expect that the state will have an important stake in the development of a blockchain basis infrastructure.

From our experiences in digital transformation and developing communities, we can imagine several stages in the digital transformation and more specific in Blockchain Organizing when we look at the aspect of central versus decentral organizing:

1. Central governance /database, without interface applications. For example: central database of a bank, no sequence problem for transactions and the bank can choose who gets a bank account and who not.
2. Central database with interface. For example: banks in the PSD2¹⁸⁰ era. In this stage other entities can develop applications for consumers for example to access financial information, independent from the bank or other financial service companies. Think of the aforementioned Electronic Financial Record, as part of your Personal Data Service.
3. Company platform that facilitates 1:1 or 1:n transactions without interface application. For example Airbnb, Ubers or Booking.com. Still one party or central authority is leading in the supply chain and can impose discounts or higher cost for other actors without a democratic decision system or consensus mechanism.
4. Company platform with application ecosystem and (API) interfaces, for example Google with Yet Another Mail Merge and other application that work on Gmail or Google calendar or maps.
5. Decentralized platform, for example Ethereum¹⁸¹ is managed by the non-profit Ethereum Foundation and enables developers to develop decentral applications (so called DAPPS) on this platform. In this case it is possible to design decision-systems that require a 90% consensus before releasing a new version of the software, or forking in the transaction future.
6. Decentralized information and transaction network with no central authority or single service provider. Governance is Trias Politica and in Europe, most likely, investments are done in a public/private cooperation
7. Agnostic ecosystem: a shared information and transaction center independent from the existing industries for transport, storage and repackaging of data in order to give answers to digital questions. Working with a certain consensus on the ontology, taxonomy, synonym list and semantic procedures.

At this moment most of the organizations are still in the first phase where they work with local, not well coordinated systems for communication and processing transaction with tools such as Microsoft Office, local (paper) forms, phone talks and face to face meetings. The use of cloud service is not yet mainstream, for example. That has almost nothing to do with the technology, but with the way we are used to organize work. Most people share information by sending emails with attachment and

¹⁸⁰ Payment Service Directive 2(PSD2): directive for banks to give third parties access to their data.

¹⁸¹ Or for example Synereo that wants to build a peer-to-peer social platform without central authority such as Facebook.

typing data in own Excel sheets or Word documents. When it comes to organizing information processes, most of the organizations are still in the hunter-gathering period, some of them are in the crafts periods and only a few are working with an assembly line for data processes. And that's only information sharing. It is probably a good idea that when you are going to design and develop a shared transaction system, is to skip some phase and bring information sharing immediately in a shared transaction center and see sending and receiving data as a transaction (which it is!).

6.3 Where blockchain should not lead to

From the white paper by Nakamoto¹⁸², it appears that Satoshi Nakamoto has carefully considered the gap between the status quo of laws and regulations and the potential of the Bitcoin protocol. An example of this is the gradual and controlled growth of the Bitcoin network. It should not go too quickly as it could endanger the robustness of the network. If the network would become too big too quickly, sleeping dogs could be woken up. As far as robustness is concerned, we like to point out an interesting technical characteristic of blockchain: a 'young' blockchain is quite vulnerable to external danger (hacking). Regarding the sleeping dogs, a technology that has not yet reached its full potential and is exposed to (often commercial) short term strategy, is at risk of being dismissed as unimportant or dangerous. This was also tried with Bitcoin, by connecting it to criminal activities and the Dark web.

Premature exposure can also lead to 'quick money earners' getting in on the act and applying the technique within their current thinking and management model, with the only intention to turn this into cash as soon as possible. In that case, Bitcoin mainly becomes a change of medium, just like internet. At this moment (beginning of 2018) it seems that more traders, investors and financial professionals are busy buying and selling cryptocurrencies then designers and developers who build blockchains. Instead of creating jobs in trading or worse speculating on something that has to be developed: work on the design and development itself is our believe. It seems that greed (again) is becoming more important than understanding. The following figures illustrates this situation which also occurs with blockchain start-ups, although most entrepreneurs and in particular investors will deny this:

¹⁸² As well as the many conversations that he had afterwards with early interested parties such as Hal Finney and Martti Malmi.

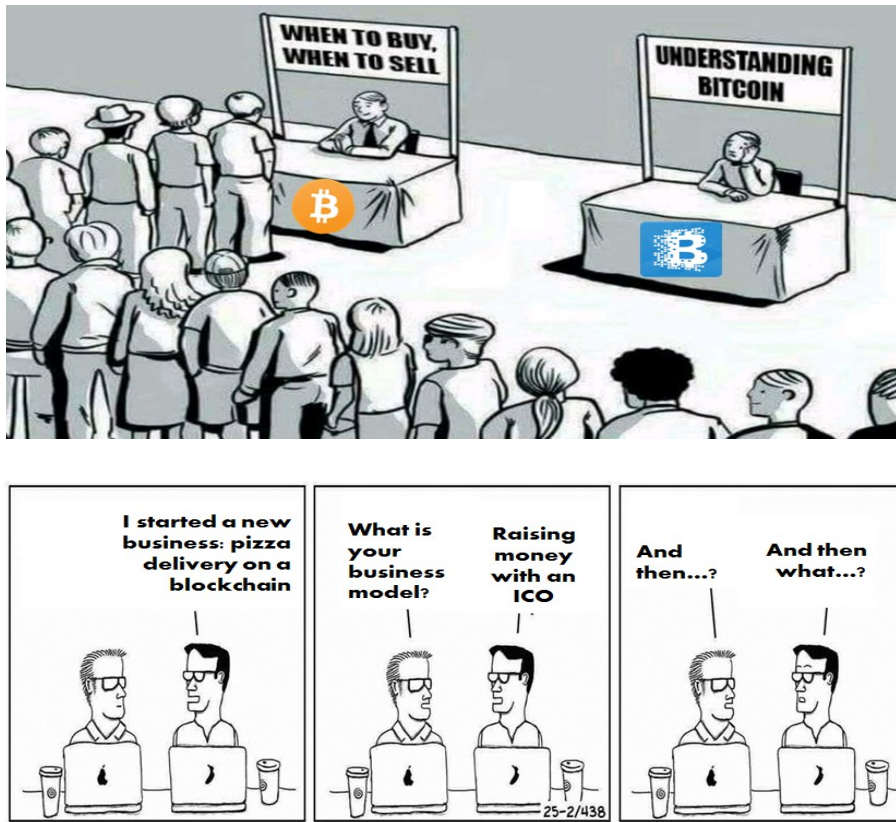


Figure 6.1: 'Greed versus understanding' and new business models¹⁸³

In addition to start-ups and investors, we also see questionable developments taking place in existing companies. An almost epic example of this is the proposal of Accenture¹⁸⁴ to make blockchains mutable. It is a proposal to make it possible to enable changing written transactions. Admittedly, under an extremely strict regime, but still the ultimate potential of blockchain will *definitely* be lost if this occurs. After all, one of the main characteristics of blockchain is its immutability. But mistakes are human, and we do not live in a perfect world, do we? Absolutely. But we believe that you should not undo a transaction once it has been executed, by changing that same transaction afterwards. What you should do is creating a new transaction that 'repairs' the 'incorrect' or 'wrong' transaction. That you 'cannot undo' the past in a non-mutable blockchain, or actually create transparency, is (still) dead scary to many people. But transparency is an important driver for a better and more democratic world.

¹⁸³ Image from: NRC.Next and DeMorgen.

¹⁸⁴ See also the PDF of Accenture: 'Editing the un-editable blockchain: Why distributed ledger technology must adapt to an imperfect world' (2016).

Once you know that you cannot change anything afterwards, you will carefully consider the consequences of an action or behavior in advance.

You might think that all of this is not so significant, a blockchain that may or may not be mutable. But precisely this immutability is, in our view, a hard condition for calling something a blockchain. Otherwise it would just be a central organized database in a world, where the reality for the seller and buyer does not have to be the same. The development of including a newly designed technology, beneficial to all people, in an existing social economic order, is of course not new. It is the old 'fight' for power. But evolution teaches us that eventually not the existing organizing regime or system will win, but the system that can best adapt to the changing circumstances and the system with the lowest fiction costs.

Another development that (sometimes justifiable) is criticized by the existing establishment such as financial regulators, are the ICOs (Initial Coin Offerings). These are somewhat similar to the IPO (Initial Public Offering), whereby a company registers on the stock market and collects money by issuing shares. But an own coin or token can, in addition to the function of share, also have the function of calculation unit or currency within a community. Exchange markets subsequently ensure that you can also exchange a cryptocurrency into dollars, for example.¹⁸⁵ The criticism on ICOs by regulators is that it is impossible to check, for example, the identity of the person who is offering the ICO. Plus, there is a lot of speculation with coins. Comparisons are drawn with Ponzi schemes and there is often a weak relationship between the financial and real value of a company or project. These developments are also familiar. We saw the same thing occurring at the creation of internet. Technology alone will not change people's behavior.

Apart from these developments, we currently also see the occurrence of a fight 'for power' between public blockchains (new establishment) and private blockchains (old establishment). We saw the same fight when internet was introduced. The existing establishment, the existing way of organizing supply and demand clearly won at that time. Whether this will also happen with blockchain, still has to be proven. Hopefully, we have learned, and we will not make the same mistakes. Or as the famous Dutch soccer player Johan Crujff would have said: *"Before I make a mistake, I know that I did not make this mistake."* In other words: do not make the same mistake twice and learn from the past. The early internet gives us a realistic picture of a decentralized democratic world in which information exchange is possible without the intervention of third parties and at minimum transaction costs. But the internet has drifted far from its original intention. We all know the problems of the internet: fake work, loss of privacy, power rests with a limited number of companies, fast spreading of fake news, and it makes us insufficiently more productive. Companies, non-transparent financial markets and governments have become more powerful than even

¹⁸⁵ At this moment (beginning 2018) there are already more than 1300 coins (also called Altcoins in relation to the first: Bitcoin), see also: https://en.wikipedia.org/wiki/List_of_cryptocurrencies

before due to internet. Internet has damaged our democracy rather than strengthened it. Let's not make this 'mistake' again with blockchain.

What is blockchain not and what are the disadvantages?

Regularly, you see dialogues on online discussion groups debating the question: what is a successful blockchain application? But this question is similar to the question: name a successful student, for example, of the Harvard University. To judge whether the student is successful, he or she must have completed the university with a certificate, but then he or she is no longer a student. Even when the student has a certificate and starts to work, when will he or she be successful? Only when the learned is implemented step by step or also when he or she interprets changing circumstances and responds to learned principles? So, what is successful, how can you measure it? Will the Bitcoin blockchain become less successful because many greedy speculators use it to generate money with money?

The question is also who can claim the name 'blockchain' and what does it mean? The name probably first appeared in the Cipher Block Chaining concept (see also paragraph 3.1). Nakamoto does not use the term blockchain in the white paper. The text is about 'ongoing chain', 'honest chain', 'longest chain', 'proof-of-work chain', 'chain of blocks', etcetera, but not directly about blockchain. However, it is clear that this is meant but it is not called as such. So, when may/can a person claim a name and meaning and what is the relevance of it at this moment? For us, blockchain and mainly Blockchain Organizing is not something that already exists. It is not a fully developed concept or a way of thinking yet. It is especially something that could encourage you to think differently and anew.

We know situations whereby the CIO concludes that blockchain technology is not a smart choice to solve a certain problem, while the CEO still wants to go for it. The reason could be: Fear Of Missing Out. At this moment blockchain, and more specific cryptocurrency, is definitely a hype and a growing number of directors wants to participate and join the discussions, without knowing the advantages and disadvantages. We cannot prevent this behavior by publishing this book. It will probably lead to the implementation of a stretched concept of 'blockchain' in such a way that it still solves the problem and the CEO is eventually right. There are many discussions about the question whether, when and how an organization will start with a blockchain pilot. There is little thought and discussion about the why question. This turns blockchain into a hammer looking for nails. An important part of the blockchain development will also be the development of a conceptual framework so in the long term we will talk about the same things.

In this book, we have written a lot about what blockchain and Blockchain Organizing is, and less about what it is not. What Blockchain Organizing, and definitely blockchain technology, is not at all, is a panacea for all organizational and management problems. Blockchain technology also has some disadvantages, such as:

- You require more storage capacity because all transactions are stored on all nodes.

- You must pay connection costs between all nodes.
- To verify a transaction and to include this in a block with the help of miners, costs, relatively, a lot of time (10 minutes within Bitcoin) and energy (power).
- It is relatively expensive to verify each transaction in the same way. A smart verification learns from previous verifications and you could probably also verify batch-wise or let the type of verification depend on the reliability of the sender and receiver.
- The protocol must be maintained and finding consensus is in some cases probably more expensive than a hierarchy or dictatorship.
- Finally, you will have to replace existing systems, that have an almost totally opposite architecture. This cannot be done with maintenance or by continuously adding new functionality, but is only possible with a new building.

Some companies start with their own blockchain (permissioned) without miners. Verification can be done locally, but the governance of the protocol is central. Apart from the fact, that we benefit little from local blockchains, demand and supply is still not organized in a fundamentally different way. As a consumer you still have a lock-in with a bank and for example as citizen a lock-in with a council and it is not always a secure solution. After all, you can hack a so-called decentral network, that is centrally controlled, or the system can fail. Therefore, third parties might get full control over the 'so-called' decentral nodes. When you, as hacker, take over the control, you can also commit fraud. With the failure of a permissioned blockchain, political decisions can also be influenced. Some participants will be treated differently from others. For example, after the crisis in 2008 system, banks (shareholders) were rescued at the expense of others (the tax payer).

If there are no (democratic or decentral) incentives within permissioned blockchains to check transactions, why would you opt for blockchain technology? In that case, a blockchain will be no more than a cumbersome and expensive database technology. This means that Blockchain Organizing is much more about power and organizing than about technology.

6.4 Innovation Strategy: towards a decentralized cooperation perspective

In traditional management programs at universities, strategy is often taught as a company's coordinated effort to realize a chosen finalization. A strategy is a path to a chosen destination so to speak. Strategies are often aimed at developing core competencies in order to gain a competitive advantage over competitors or startups. The basic idea that a company or a tool in general, is a means to an end is forgotten. Traditional innovation strategies are often designed and processed from a protective perspective instead of a cooperative perspective (see also difference between Darwin's and Kropotkin's perspective in paragraph 3.5). A basis function of a hierarchy is to coordinate supply and demand, to connect humans and resources to activities in a way that productivity, and with that our standard of living, increases. The

fundamental goal of hierarchies was not to make rich people richer or stimulate inequality.

A traditional company is considered to have succeeded when it has implemented a strategy that its competitors found too hard or too costly to imitate. So basically the successful company will have more food than others and therefore more chances to survive. But we humans are not an individual species, but a social one. And social animals have a higher chance to survive. So at the end of the day, cooperation has probably a higher chance to survive than competition. Most of the strategies that are taught at business schools, ignore the fact that companies are just a means to an end, invented by humans to improve our standard of living. If some other coordinating mechanism between supply and demand is more efficient than the old one, the traditional one becomes irrelevant. And with this irrelevancy all kinds of MBA programs and business schools with old 'firm- DNA' will become irrelevant, unless they adapt.

If we take a closer look at the relevance of 'competition' nowadays we see that it becomes less relevant. Rosenberg Equities¹⁸⁶ made an overview of the frequency of use of the words 'competition', 'competitors' and 'pressure' in annual reports. The frequency of the word 'competition' for example decreased from 8 times per 10,000 words in 1995, to less than 2 times per 10,000 words in 2017. The Economist concluded that two-thirds of American industries were more concentrated and in the hands of a few firms, probably as a result of executed strategies they learned at business school (survival of the fittest).

Even if firms are competitors in a market, the equity is, more and more, held by a few parties. This is, from the perspective of decentralized organizing, democracy and self-sovereignty, not a desired development. But still a lot of traditional firms design and act from a centralized and competitive perspective, instead of a more decentralized cooperative perspective. The organization of work and economics should not be a matter of competition or cooperation, central or decentral, it should be about the balance between them. What matters at the end of the day is to achieve a better standard of living for more people. It's about the question: *How to connect people and resources to finalization in a way that has the highest aggregated productivity for society?*

In traditional companies managers have to invest enormous resources to protect a traditional way of thinking as if the competitive advantage is the only strategy for happiness, a higher standard of living, or a better life. This old way of thinking comes with a disadvantage when a new situation emerges and when other strategies (decentralized cooperation: learn, innovate, be open, crowdsourcing and work together) becomes more relevant. This traditional strategy partly explains, for example, Kodak's decision to put the digital camera 'on the shelf' in order to protect its core competence and cash cow. That didn't work out well. Using a strategy of protecting

¹⁸⁶ See also article: 'What annual reports say, or do not, about competition', The Economist (16-11-2017).

an existing competitive advantage, works in environments that have long term stability. However, such environments no longer exist.

The rapid pace of globalization, digital transformation and sustainability has changed the dynamics in most industries. In addition to these developments, more and more industries become ‘the same’ as we look at the way transactions can be designed and performed in shared information and transaction networks. It doesn’t matter what you order at Amazon, the underlying processes and logistics are basically the same and as a consumer you don’t see whether or not it is a complex or simple process. Most companies not only need to develop the ability to respond to change quickly, but also to give up some autonomy and ‘learn’ to cooperate with former competitors. Learn to give up some autonomy in order to gain more advantages per actor is very difficult in a world that in the last hundred years was educated with old organizing paradigms of protecting own revenue models and maximization of parts instead of the whole. Introducing a new paradigm is difficult because traditional management practices, that were needed to protect competitive advantages, are the opposite way of increasing cooperative advantages in a decentralized autonomous environment.

Unless you want to be trapped in continuous reorganizations where your employees are not interested in the next step anymore, leaders in, especially large organizations, need to change their approach to strategy. Strategy can no longer be defined as the singular exploitation of core competencies to maximize gains and protect existing business models. We need authentic digital leaders, and organization designers to ‘see’ this and act on it. Traditional management and innovation strategies are at the end of the S-curve: it doesn’t get any better anymore and things will get worse.

Modern organizations will have the ability to compete in mature markets with new technologies and organization models. Blockchain Organizing and its design principles can help with this. Blockchain Organizing doesn’t start with competition and protection but with cooperation and renewal. Stakeholders look for hygiene factors: processes in which they can better work together instead of competing with each other. And processes like data logistics, purchasing, invoicing and paying are not processes you can compete with. Customers just expect that you do these processes correctly with minimal cost. You can’t compete on doing your booking keeping correctly, you can only fail.

6.5 Blockchain in the near future

Since the Renaissance our social-economic establishment is more and more formed by institutions that are aimed at reducing insecurity and organizing trust. When Ann wants to buy a book from Ben, there are several parties involved in this transaction, such as: a bank, an insurance company, a bookseller and an accountant to process the transactions in the accounting system. In principle a trusted third party, such as a notary when you sell or buy a house performs three processes: checking whether Ann is the owner of the house she wants to transfer to Ben, whether she only spends

this value once and the entire process must go securely. Basically, we can automate these three processes and we no longer need the notary (and other parties such as bank and accountant). See figure 6.2:

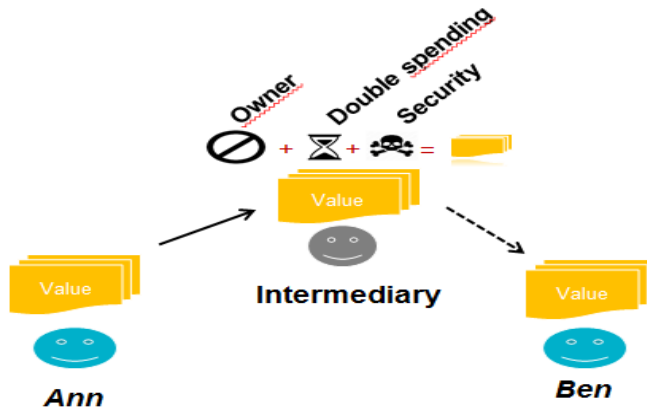


Figure 6.2: Programmable trust replaces Trusted Third Parties

From the moment we are born till the moment we die, we have innovated all kinds of organizations, with the intention to make our life more pleasant. Our value system is more and more formed and is more and more dependent on tools like: money, debts, insurances, pensions, economic and monetary policies, continuous economic growth, trade agreements and institutions such as the IMF, the EU and the G8. These institutions and tools have increasingly distanced themselves from most people and lack a human dimension.

Now there is a tool that can organize trust, with the human dimension as principle. We can make connections again, share information and do business, from person to person, with less intermediaries. We can reverse the chain, from power with the companies, to power with the people. Figure 6.3 shows an example of what is possible with Blockchain Organizing. Companies are leading in the old model. The same data must be entered at different locations and a bunch of keys are needed to log in differently everywhere. In the new model, a company 'pings' a question to a Personal Data Service. With the answer, this company can start its process, such as shipping a book by Amazon. Amazon asks a question to your PDS: what is your 'name', 'street name', 'postal code' and 'place of residence'? In a millisecond, the Amazon system receives an answer and it can start a business process. After this, these four attributes no longer have to be present in the database of Amazon. So, in the new model people bring their own data when they place an order with a company. The company can simply process this data because it is connected to the same data utility. You can compare this with the power from your solar panels that is used by a company for the lighting of an office.

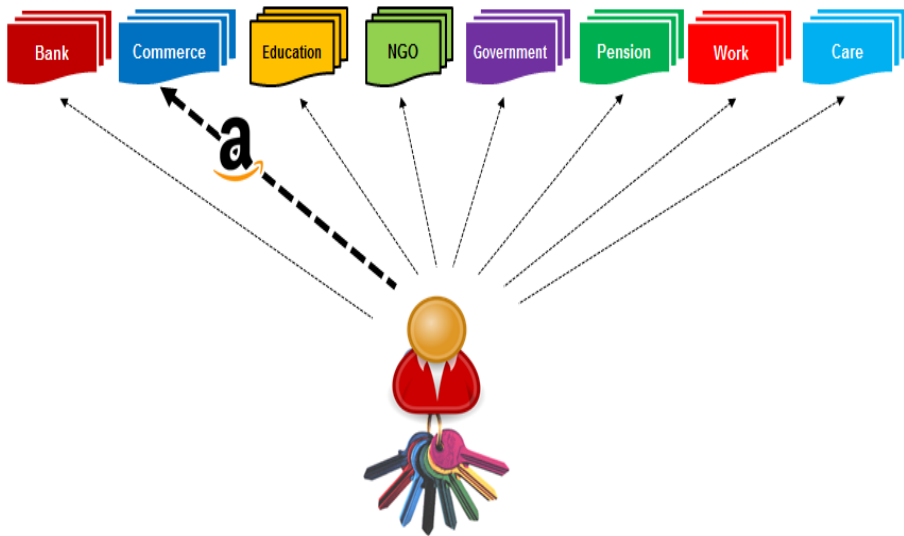


Figure 6.3: Old Organizing: organization in the lead

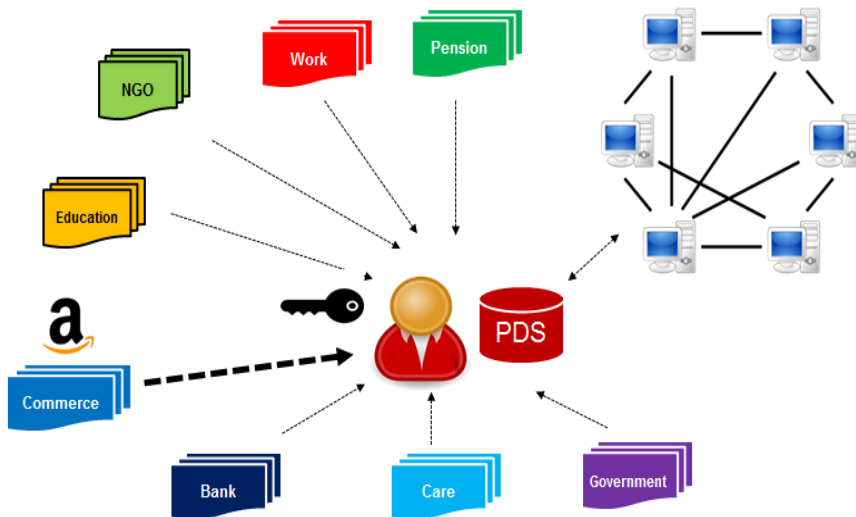


Figure 6.4: New Organizing: humans in the lead.

In the new situation organizations ‘asks’ questions to Personal Data Services (PDS) and attributes are scattered over the network so there is less motivation to ‘hack’ a server because it contains attributes of millions of different users without any relation between the attributes and users. When we look at cybersecurity most people think about tooling to protect data. But when a new tool is invented by the protector, the

aggressor will find a new tool that will be stronger. This action-reaction 'game' will never end. So we need a more fundamental approach.

When an inspector tries to solve a crime, he or she will always look at three aspects: tool, circumstances (change to get away with it) and motivation. In cybersecurity not much people look at the motivation to steal data. When a thief steals a truck load he or she will have a customer for the load, for example thousands of laptops. But when attributes are scattered over the network and a server has attributes of millions different users, hacking a server would be like steeling a truck with thousands of different products, a truck full of laptops, cloths, food, construction materials etcetera. There will be no customer for the truckload, so there will be no motivation to steal despite circumstances and tooling. The new privacy law GDPR should not be about more tooling and more institutions, but about another mind-set and IT-architecture.

We require less and less existing institutions. We can organize trust again without increasing complexity and therefore distrust. Whether the tools, that are currently available to us, will be used (properly) in the coming decades, will not so much depend on the technology, but on pioneers, authentic digital leadership and the political and human willingness to fundamentally innovate the existing establishment for their own benefit. Now with blockchain there is an advantage compared to the rise of internet: time is starting to run out to solve a number of potentially disruptive, social problems.

Of course, we do not have a crystal ball. We cannot look into the future and predicting how Blockchain Organizing will further develop itself, remains difficult. But we do know that we determine the world in which we live, with the choices that we make. We know that you can better organize in the future, when the existing world is less complex. The world becomes less complex when we base ourselves on organizational principles and not just try to solve practical problems in a way that will cause other problems. But simplicity does not provide sufficient employment, complexity does. It seems that we refuse to accept a simpler world, because otherwise we have nothing more to do. The world becomes less complex when we give another meaning to growth, to work and therefore also to unemployment. We have the technology to work less, but we still want to keep everyone employed, so we create fake work which makes the world unnecessarily complex.

Every intermediary in the system might become more productive with IT investments, but we continue to add new intermediaries, new parts in the system, because we strive for full employment. These new intermediaries are subsequently badly coordinated which makes the world unnecessarily more complex and can no longer be understood by a growing number of people. The only way to understand a complex world is to review our identity with economized work and to go back to the organizational principles. This might be the most important feature of Blockchain Organizing; going back to the origin. Also, back to the origin of organizing and managing. Many people have forgotten why we work and why we organize work. We only think

in permanent jobs or more or less flex jobs or how we regulate them. We are so busy with economics that we forget to ask ourselves some more philosophical questions. Or as the ACM Turing Award winner Alan Kay puts it: *“Our (real) job is to remind us that there are more contexts than the one that we’re in, the one that we think is reality.”*

Reinvention of management

Till now we had to use a management system that was invented at the beginning of the twentieth century. We think that management innovates, but these are often small adjustments within the existing invention. The system itself, with the firm as superior organizational form, has not changed. Dominant management theories also do not contribute to the change thereof. Management science is still mainly seen as a form of applied social science, while a large part of the tasks that we perform as a person, can be put in a technical protocol, which, with mathematical proof, can perform tasks must more accurate, more secure and quicker. With organizational technology, the field of expertise of management science and organization theory will also change and can become more of a hard science.

Management and therefore all kinds of management resources, such as books, courses, consultancy and assessments, will also fundamentally change. We, as a person, have become more productive because in the last hundred years we have started to use technology, including in particular management technology. We’ll become less productive when we use the wrong tools. The latter is clearly visible in the productivity figures since the 1970s, when a lot of people started to work in offices. At this moment, most management tools make our world more complex which costs us more time and attention for something that can also be automated.

We can more easily imagine what the negative consequences of robotics are than the positive effects for our busy lives. We can get rid of institutions that we use to organize trust and create fake work or non-meaningful work. Trust is the lubricant for the organization of work and economy. But when this lubrication turns into sand, things will go wrong with the ‘machine’ that we use to produce welfare. With this book, we have tried to offer a basis for a new social-economic establishment with organizational technology, and more specific with Blockchain Organizing as a tool that can contribute to this. That is not yet easy for everyone to imagine. This is not because Blockchain Organizing is abstract, but because it differs from what we are used to.

What does success depend on?

The success of blockchain technology will depend a lot on the question whether we can achieve standardization in the short term. The TCP/IP protocol ensures that internet could get off the ground quickly. Due to standardization, more and more people will invest in this and therefore more and more people will start to use it and the number of applications will increase. There is already some sort of standard for the organizational model of Blockchain Organizing that reasons from the smallest

building blocks and organizational principles that have been around for a long time. With regards to the use of the developed ontology and taxonomy, additions will definitely follow when more communities will start to use this as a basis in their own context. In addition, the development will also depend on the acceptance by people. People like to be in control and with Blockchain Organizing you might quickly get the impression that you, as a person, are no longer in control and that the protocol decides. It will scare many people that they cannot implement mutations in transactions that have taken place. Algorithm aversion could also cause a delay. We, as persons, sooner rely on people than on algorithms, even when it has been proven that the latter are better.

A main conclusion of this book is that Blockchain Organizing, like internet, will not fundamentally change a lot if we do not simultaneously change the model that we use to organize supply and demand. Blockchain Organizing is based on a technology whereby the ability is strongly reduced the more it is used within the walls of companies. Internet is a great tool to bring supply and demand together, without almost any friction. But this technology, without the additional help of a new organizational model, is insufficiently capable of solving problems in terms of fake work, privacy, propaganda and low production growth for example. Science, democracy and law are more and more undermined, and we 'make' large companies and central government more and more powerful. This is not due to the technology, but due to our awareness, lack of trying to understand, behavior and due to the form in which the technology is used. Technology is just a lever. You also need a foundation or common ground to stand on.

Blockchain Organizing offers foundations for apparently different organizational and management problems, which often have the same origin. Problems in healthcare, home care or education are exposed in different ways, while they often still have the same origin. Usually the undercurrent of the problems that we experience, is not mentioned in the media. That is too abstract. Apart from these organizational problems in the public sphere, there are also problems in the private sphere that are solved with old thinking and old organizing. These problems often have the same cause and undercurrent: insufficient coordination of supply and demand and solving problems with the same level of thinking that created them in the first place. Often driven by political motives, opportunistic behavior and wanting to maintain the existing power base. When you want to solve an organizational problem and can let go of the following assumptions, you will make some progress:

- the firm and government as ideal form for organizing work and commonalities;
- working hard and continuous economic growth are good;
- competing has the exclusive rights to innovation and progress;
- innovating in combination with full employment is a good idea.

Blockchain Organizing offers principles to solve these problems in a different and structural way. The advantage of principles is that these do not change very often. The disadvantage is that many people find them abstract. Stephen Covey once said:

“There are three constants in life: change, choices and principles.” The world changes, therefore we must choose and when you choose you can best do this from a number of principles and foundations. Because you do not need so many foundations to solve organizational problems, an organizational expert can be of great significance for the problems that will keep us more and more busy. When you understand these foundations and apply them, you can handle almost every management problem.

Technically, there are still (especially with public blockchains) some problems to be solved such as: centralization of mining capacity, speed of transaction processing, synchronization and scalability. Security also remains an action-reaction game because whatever security is developed, eventually this can probably be cracked again. The question is at what costs? But currently, not many fundamental technical problems are expected for Blockchain Organizing. The speed of development will be determined more by the adaptability of people: (still not enough people take the time for a fundamental deepening of their understanding¹⁸⁷), and the adaptability of existing:

- organizations that must disinvest and switch to another standard;
- powers that benefit from changing as little as possible;
- intermediaries that live from a complex world;
- standards and legislation.

In the chain between first seller and last buyer, Blockchain Organizing cannot perform all tasks autonomously. We need people, not just for inventing and designing processes, but also for the input of values in a blockchain (so called ‘Oracles’), solving errors or for example mediation in case of disputes. According to us, the following is important for Blockchain Organizing in the short term: being aware of the problems, trust that we can do it, change of behavior, applying new organizational forms, standardization (in identity management and blockchain protocol), algorithm aversion, regulation and legislation.

6.6 The future of Blockchain Organizing

When we look a little further into the future of Blockchain Organizing, psychology (identity), anthropology (our connection with other people) and philosophy (asking the right questions) will become increasingly more important. An important question is for example: what is progress?

What is progress?

An important driver for people to make an effort is the expectation that we make progress. But maybe for the first time in our existence we ask ourselves the question: what is progress? When do we grow? When will our children be better off? Is it more or better? Is the sustainability of what we have also growth? We believe it is. And as we have shown in many places in this book, Blockchain Organizing contributes to growth. If we become more productive and continue to produce the same, we have

¹⁸⁷ Except, of course, the readers of this book.

more surplus time and attention for the things that really matters to us and for sustainability and well-being. We think, that we will move more and more from the offices with fake work to our homes, neighborhoods and community's for real 'work'. Do not take neighborhood literally. It can also be an online community.

For us as authors, an important condition to putting energy into something is an answer to the following question: 'Does the innovation contribute to a better future for our children?' However, better is a subjective concept. But we are convinced that more and more people in the West will find that better mainly means sustainability of what we have. The welfare crisis, we believe we are still experiencing, will actually only stop if we can image a new reality and have a plan to achieve this. The crisis is only over when we have accepted a new establishment that no longer strives for more welfare, but a sustainable welfare or well-being. When this would be possible within a few decades, it would be a success for the current generation of policy makers, CEO's, politicians and mainly pioneers. When this is not possible, we are probably part of the first generation in modern times that ensures that the next generation is not better off.

Beyond the hype

At the moment (beginning of 2018) you can definitely call the blockchain development, and more particularly: cryptocurrencies a hype. But a hype with much potential. A hype is usually characterized by 'blown up' expectations and no real results. Not in the least by investors, crypto traders, dealers and consultants who believe they can quickly make some money. But there are also 'world improvers', who want it differently this time. People that experienced the internet development, also at the core a decentralized and democratic system, and after decades concluded that we drifted away from the basic idea. Or as Cypherpunk Jameson Lopp puts it: *"The older you are, the less excuse you have to be skeptical of the positive benefits of technological innovation because you've witnessed the revolution firsthand over the previous decades."*

The question is whether anarchist of today becomes the traditional CEO's of tomorrow? Do they 'succumb' to big money? Blockchain will probably also follow the 'Hype Cycle for Emerging Technologies' by Gartner¹⁸⁸. The process of this cycle usually looks as follows: at first hardly anyone knows the technology, except of course the 'inventor' and a surrounding small group of 'tech nerds'. Then someone manages to persuade the media to write about it, the expectations increase, investors and traders do not want to miss out and consultants will give lectures and provide advice. But the technology remains vague for many people and the useable applications fail to turn up. So, the expectations drop. Only a few years after the peak of the hype cycle, the technology gets sufficient traction because a standard is created, the strongest ideas survive and the number of application increases.

¹⁸⁸ See for example: 'Gartner's 2016 Hype Cycle for Emerging Technologies Identifies Three Key Trends That Organizations Must Track to Gain Competitive Advantage' (16-08-2016).

The development of Blockchain Organizing is a different story and not the same as the development of blockchain technology. One of the pillars of Blockchain Organizing (the organizational model: Weconomics) has already had its dip and since 2010 it has been on the rise with real projects and an increasing number of participants. However, Blockchain Organizing, like ICT and internet, is also still in its infancy. We are still hunting for information and we still do not have an internet assembly line. There will certainly still be projects that fail. We learn from this. The development of the assembly line for example, did also not immediately go well. Henry Ford went bankrupt several times and especially the Second World War and the need for mass production ensured a real breakthrough.

So also the circumstances can make or break an innovation. Internet did also not immediately go well. For example think of: World Online, Myspace, Pets.com, and Netscape Navigator. Blockchain Organizing will develop itself in the same way: by trial and error. The technology will find its way in this, it has always done so. The question is what we, as a person, will do with this technology. As a person, we have always struggled socially to put new technologies into place. That will not be different now. Not the technology, but man is the most important part in Blockchain Organizing. The examples of using Blockchain Organizing, which you have been able to read in this book, can best be seen as examples to make you think and change your perspective on management and organizations. Most examples in this book are not tested in terms of answers to the question whether an idea or plan is actually implemented, still exists or is sustainable. For us, Blockchain Organizing is not something that already exists, or a way of thinking, but much more a concept that makes you think. This also applies to this book.

Organizing the future?

Based on recognizing patterns from the past and extending these patterns to our future, we can make a reasonable assumption of the type of work that we will do, how we will organize this and which management concepts we will require for this. As a result of this we can make a reasonable assumption of the type of welfare that we moving to, how we will organize this and what the contribution of Blockchain Organizing can be. We can translate this into a long term vision. We plead that political parties¹⁸⁹, in addition to a short-term plan (one to four years ahead) also make a 'next generation plan' (three decades ahead). Every four years the short-term plans should be tested to the original long-term plan. This double track decision must ensure that we do not cause long-term problems by solving short-term problems.

Now we can hear you say: how can we look forward one generation if we cannot even look forward one year? This is thinking from an old paradigm that the world is complex and dynamic and that there would not be any fundamental and recognizable patterns. But these are there and have been described in detail in chapter 8 of the

¹⁸⁹ Apart from political parties, NGOs and companies, for example, should also do this.

Weconomics analysis book: 'Patterns that predict the future.' It is not always clear how quickly patterns will develop themselves, but the direction usually is. So, our development more and more focuses on wanting to control our environment, in order to increase our survival rate. After controlling fire, controlling food production and muscle strength, it is now the turn of information, and later on our welfare, to be controlled. We do not longer want to hunt for information, but be able to acquire this in a controlled manner. Because we cannot imagine that our (working) life ends as shown in the figure below.

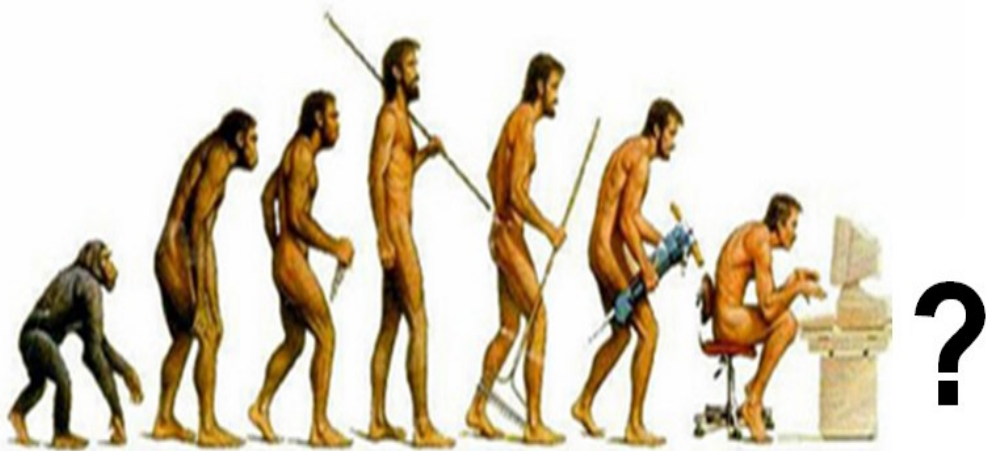


Figure 6.5: Evolution of working man

The next stage of the 'working' person

We will not be in front of the computer for decades. The question is what comes after the information worker? We believe that we will initially mainly focus on the smarter organization of the work of the information worker (sphere 3 in figure 6.6). More and more data will soon be processed into information by a 'blockchain assembly line'.

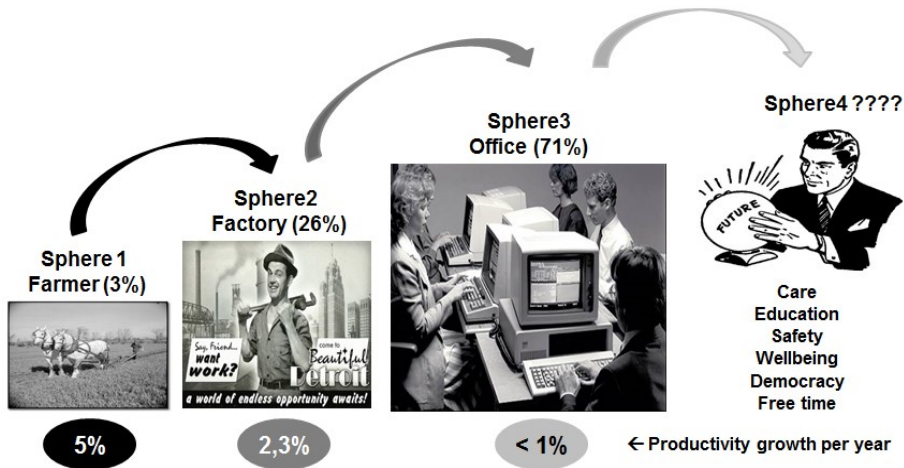


Figure 6.6: Development to a fourth 'working' sphere (The productivity growth numbers, 5% - 2.3% and <1%, come from the Netherlands but are comparable in other OECD countries)

The farmer (sphere 1; 3% now works here), was able to become more productive, not just because the horse and carriage were replaced by the tractor, but also because 'the surplus' was able to work in factories: *you mainly become more productive when there is a replacement for your surplus time when technology makes you more productive*. The same happened in the factory (sphere 2, 26% now works here). The assembly line made us more productive and the surplus went to offices (sphere 3, 71% now works here). More than 70% of the work force in The Netherlands now works in offices, but we will not become more productive if the surplus cannot go somewhere (sphere 4). We see that the productivity growth in sphere 3 (less than 1%) is far too little to keep up with rising welfare costs (often more than 2%).

Generally, you could say that sustainability (sphere 4) mainly requires time and attention and Blockchain Organizing creates this. Therefore, man will also become, in addition to (professional) specialist, a (private) generalist, so a hybrid day or week format. But that is great, we like change and want to continue to develop ourselves. Information workers will become more productive due to Blockchain Organizing, which creates surplus time for sustainability. ICT becomes more and more mature and the processing of data becomes more and more sustainable, this needs less attention. The privacy can be assured better, we become less dependent on a limited number of data companies from Silicon Valley and preventing fake news becomes easier. As a person, we are chrono-centric. We live in this age and think that Twitter, Facebook and Google will be there forever. But before e-mail and social media existed, there were the: petroglyph, smoke signals, printing, letter, telegraph, telex and fax. And now we have e-mail and social media to communicate. So, there is something before Facebook and Twitter... and there will be something afterwards.

The next generation of (technical) innovations will continue to build on the existing internet technology, but hopefully with more social relevance and a higher democratic level. Inventions that do not solve a social problem, will not survive long. So, there will be new communication tools whereby we probably do not have to submit our personal data. Tools that are less driven by adverts and consumption. The advert machines, that now make ‘free’ communication and sharing possible (but also fake news), will lose meaning and be replaced. Anyway, a shared network is about and turns the model around: in exchange for a few bucks per month:

- you remain the owner of your personal data that is worth a lot more;
- you receive discounts on orders because adverts are no longer required;
- you can select news, information, products and services more objectively.

But... first there is a lot to learn

As long as there is so much free money available in the market, investments in non-useful applications will remain possible. A current IPO that is symbolic for this is that of Snap Inc. This company is listed on the stock exchange at a value of 28 billion dollars. It is not clear yet to us, what the social relevance is of the main application of this company: Snapchat. At this moment, it is mainly used to take selfies and to do sexting.

Investors are wild about a company that suffers losses¹⁹⁰, whereby only the founders have voting right and whereby it is also not clear how they will make a profit. Apparently, investors have learned little from similar IPOs like Twitter. Apparently, there is so much cheap money in the economy¹⁹¹, that money flows to this type of companies. This is in stark contrast to the somewhat less sexy companies in, for example, healthcare, that do solve real social problems, but that find it more difficult to find funds because they are probably less speculative and ‘sexy’. We can no longer imagine that, partly due to pension funds, money is still being invested in companies that are so far away from the real problems that we face as humans and society. Or would our problems be so great that it would be better to go down while ‘snapchatting’?

With Blockchain Organizing we prefer to work on solving real organization problems. With the use of Blockchain Organizing we will organize the work of information workers less and less within the walls of companies and more and more within shared information and transaction networks. Not companies, but humans and protocols are important for the organization of our work. Money and contracts will be less and less required and therefore also all kinds of institutions, like banks. Man is increasingly becoming a prosumer and can no longer be put in a ‘box’ as citizen, consumer or employee. The artificial distinction between the personal, private and public spheres will disappear more and more in the organization of work, welfare and wellbeing.

¹⁹⁰ In 2016 more than half a billion dollars.

¹⁹¹ Partly thanks to the QE programs and opening the money tap by the FED and ECB.

Siloization will only lead to more complexity and distrust. Man is above all human and will organize trust and work, more and more via shared transaction networks. Because institutionalization has gone too far, and the welfare state has become too expensive, more and more people will (have to) perform more tasks themselves, instead of outsourcing these to a private or public sector. This is not just for economic reasons it is also embedded in our being. We are social animals and time and attention for our environment is still worth the most. We can only conclude that the current social-economic order imprisons us into something that we are not. Man is therefore the only animal species that refuses to be what it is as the French philosopher Albert Camus formulated. Man is the only animal species that requires intelligence to be in balance with his environment. Linear rationalization has brought us to a point of irrationality and dis-rationality that borders on a new form of ridiculousness. Our innovations have brought us to a crossroad, where choosing the right way is more important than ever.

What is success? The identity of the working person

During the House of Representatives elections in 2017 in the Netherlands, 'the Dutch identity' was an important theme¹⁹². We understand this, but will we find out, in the near future, that this type of identity issues contrasts sharply with a completely different identity issue. For a long time, it was clear what progress entails. That threatens to change for the first time in our modern history. Chances are that our children will no longer be better off in the sense of more welfare. But the question is then: what is better? This is also a form of identity. What is more successful and will this make us happy? When Albert Einstein was traveling to Japan in 1922 to give a series of lectures, he heard that he was awarded for the Nobel Prize. He was overwhelmed with all the attention of people who wanted to meet him. Back in the safe environment of his hotel room he wanted to give a messenger a tip. The messenger either refused to accept the tip, in line with local practice, or Einstein had no small change available to thank the messenger. Instead of giving money he wrote a note to the messenger: *"A calm and modest life brings more happiness than the pursuit of success combined with constant restlessness"*.

For how long can we still identify ourselves with more success, more welfare, income, paid work and consume more and more? Will it make us happy or can we also develop another identity? Anyway, our western way of life, our human identity is at stake. This goes much further than our own sovereign state identity. Sovereignty is a relative concept when you think from the starting point: human self-sovereignty. As we saw earlier in this book, Ubuntu means I am because we are. The question what 'we is', what we want and what we see as progress are very difficult questions to answer, especially because these questions were never asked. We have always been busy with gathering food, hunting, production of food and products and processing data into information.

¹⁹² Same 'identity' issues we saw at the French and German election and the aftermath of Brexit referendum in 2017.

More and more people do not know who they are, what they want and where they are going. We will form a new identity because economically and materially you cannot grow infinitely on a finite planet. Claiming or implementing the opposite should be made a criminal offence, as far as we are concerned. We start looking for answers to questions like: Who am I and what do I want? With which welfare do I want to live? What is my relationship to myself, to others, to my job, to technology and especially to the future of our children? People become increasingly aware of the consequences of unnecessary use of scarce raw materials and tools.

We hope that tools will become more functional again and that there is less money available for brand and lifestyle products. These will probably also be taxed heavier. Sharing will become more important because in the short term this provides the best contribution to the reduction of the use of raw materials and energy. In a circular and sharing economy, Blockchain Organizing will play an important role. Our lives and survival as a specie is especially threatened by our current system based on continuous economic growth. We must grow two percent per year to repay debts that we have created to make grow possible. Economics growth is inherent to the system. But if you continue this, our children must consume eighty percent more and our grandchildren three hundred percent. That is impossible on a finite planet.

Time for blockchain pioneers

Blockchain Organizing mainly needs pioneers. A pioneer is an entrepreneur who is one of the first to enter a certain domain, and has to find his way in there. Plus, a pioneer cannot use the experience of another person who has ever been there. Along the way there are no tools to be found, this must be considered before the pioneer starts the expedition. A pioneer has vision, whereby an effective vision is a sustainable imagination that has the power to become reality. Characteristic of pioneering are the setbacks one experiences, the hardships one undergoes, the delayed desire, the success that one must deny oneself and the large effort that one has to make because all kinds of facilities are still missing. The work of a pioneer is often accompanied by many failures, but the pioneer learns from that. A pioneer does not have methods that have been proved in practice. A pioneer must have courage and be prepared to make a journey nobody else has made. Courage is not the absence of fear, but the notion that something is more important in your life than you. Or as the American writer Meg Cabot writes¹⁹³: *“Courage is not the absence of fear, but rather the judgment that something else is more important than fear. The brave may not live forever, but the cautious do not live at all. From now on you'll be traveling the road between who you think you are and who you can be. The key is to allow yourself to make the journey.”*

A pioneer knows that a lot of people are afraid, not knowing where society will go. But this vacuum gives space for hope. It gives space for letting go the old and a time

¹⁹³ Originally by: Ambrose Hollingworth Redmoon, pseudonym for the writer James Hollingworth (1933-1996).

for new thinking and doing. The work of a pioneer is very important, so other can build on it. A pioneer looks at a problem in such a way that the solution for the problem is evident. In this dynamic and hectic time, pioneering has the character of admitting color: show who you are and what you stand for. A pioneer shows and experiences inspiration in the work performed and how it is done. A pioneer is able to give substance to developments that are necessary and important in the current world. A pioneer dares to let go of old securities in order to create space to look from a new perspective. In doing so, rely on abundance, in spite of those who regard scarcity as a fact. A pioneer dares to rely on imagination, in spite of those who feel captured by existing conditioning. A pioneer is able to let go of comfortable ideas because these do not challenge the way of thinking. Being able to offer genuine value by serving the best to a next development stage in humanity is characteristic for a pioneer. The road to more maturity is cleared. A road more and more people are ready and longing for. A pioneer is deeply convinced that we humans are capable of much more than what we now deliver. A pioneer knows that not experiencing the results of his effort is no reason not to make the effort.

Is it time for a new Enlightenment?

The question is: will man, will democracy survive, will or prosperity sustain in a rapidly globalizing, digitizing and centralizing society when we continue to use old forms to organize supply and demand? We believe it is time for a new form of Enlightenment with which we can separate ourselves, as human beings, from existing conditions, institutions and management paradigms. For example, in the past we have separated ourselves from dogmas of 'the church', the State and other self-evident things like exclusion of women and minorities. Now it is time that we separate ourselves from organization paradigms and tools which are more and more of a burden to us than a pleasure. We mean tools that we use, but that are not transparent, that we cannot control and for which there are actually no alternatives. Tools that, step by step, without being aware of it and perhaps not so intended, take away our human freedom and keep us more trapped. One of the best-known Enlighteners is Immanuel Kant, whose offers the following insight: *"Enlightenment is man's emergence from his self-imposed immaturity. Immaturity is the inability to use one's understanding without guidance from another."*¹⁹⁴

His insight is still relevant. Immanuel Kant bases himself on the autonomous person, who will not be led by others. 'Dare to think' he says. With this he responds to people who state: 'I cannot help it because I was born this way'. According to Kant, this means that you do not even accept your own dignity. Many people cover themselves in advance, also with regard to using traditional management concepts like: 'It is never my problem, but always the problem of the other person'. By saying or seeing this, people make themselves incapable of action and thus inferior, not compared to other people, but more and more often compared to the tools they use. Besides 'the other person', the circumstances are usually blamed. Never myself. But

¹⁹⁴ For 'Guidance from another', you can also read 'help from a certain tool'.

circumstances and tools cannot choose, only people can choose. You cannot blame the actual circumstances or tools, of the world you want to live in. You cannot blame the rain that you have forgotten your umbrella. The right to individual self-development can increasingly only be exercised by people who also have access to the data that is known about them. We mean 'data self-determination' and 'data self-sovereignty'. 'Be a person', says Kant and carry your responsibility. The situation that we can think, be aware and relate to, what affects us, makes us human and distinguishes us from other animal species and tools. We will desperately need this feature with all technological violence that comes our way (unless we choose differently) such as¹⁹⁵: Artificial Intelligence, Machine Learning, Robotics and Internet of Things. A democracy can only survive, if the decisions that we take as a person, are based on data that people can also acquire in freedom and is not pre-sorted by a tool. When we cannot make a choice in freedom, it undermines our constitution, our community and our fundamental human rights. We are clearly at a crossroad in our existence. If we take the wrong decision now, it can have serious consequences for the future of our children. Anyway, Blockchain Organizing offers a way for people to 'regain control' of their own data and the tools that they use to transfer data from seller to buyer.

Homo prospectus

People will try to keep what they have got. That is not a status quo, but progress. Less growth and more control does not mean that we take a step back in our evolution, but rather a step forward. From uncontrolled welfare growth, to sustainable welfare, wellbeing and more surplus time is also growth. It means that we replace a system that is threatening to become unstable. Evolution is not always quicker, bigger or more. Evolution is mainly adapting when the circumstances require this, so a better chance of survival is created. More and more people will enter the highest development stage (holistically), known to us and be aware of our relationship with the total. This will create a less complex society. A society that is much more based on man and less on what man has created. A society with a human dimension where common sense and virtue beat the rules, where values beat procedures and where people are again more important than tools.

That may sound contradictory coming from two blockchain evangelists, but we hope that this book has made it clear what we stand for: we not only see blockchain technology as a tool to solve management problems, but also social problems. Organizing trust plays an important role in this. And Blockchain Organizing main characteristic is probably that it can organize trust, without increasing complexity and therefore distrust. If that is possible, we may have made an important 'discovery' and reinvented management.

When we discover this, we will start solving problems that really matter. The way we currently look at problems, is the problem. We, as humans, we are a kind of problem-

¹⁹⁵ Cybernetics is the bases of many of our new technologies. Cybernetics is derived from the Greek 'helmsman'. It is the science that deals with the management of natural organization and mechanical systems that learn with the help of feedback (loops).

solving super organism¹⁹⁶. Not so much due to our individual intelligence but due to the fact that we as human beings are able to cooperate and image something that is not there yet. The Homo Sapiens, is able to image the solution of a problem. Therefore, the name Homo Sapiens is probably less suitable. We are defined as wise people, but we are mainly the Homo Prospectus: the foresighted man. To understand this, the insights from the book 'Homo Prospectus'¹⁹⁷ will help. This book is about positive psychology. Evolution has developed some sort of brain in man that mainly focuses on 'being able to think ahead'. As man we first 'started' a social-cultural and later a technical evolution line. Our brain is able to anticipate and to visualize. We are able to come up with ideas, these ideas are our future and we can transfer these to others. Unfortunately, we are less able to stop bad ideas on time. Certain forms of capitalism, such as the ideas from Ayn Rand¹⁹⁸, come close to this last phenomenon. Former director of the New York University, John Sawhill, formulated it as follows: *"A society is defined not only by what it creates, but by what it refuses to destroy."*

Our brains try to understand the past and the present, to be able to choose actions that lead to progress and a better future for our children. This 'thinking ahead' provides us with an enormous advantage compared to other animal species. The possibility to be able to 'see' and evaluate future actions gives us an enormous advantage. Returning to the past, a past that we have left for good reasons, will therefore not help destructive populists in the long term. We 'do not look back', we 'look ahead'. Much existing literature looks at the past as predictor of the future. According to us, and the authors of the book Homo Prospectus, this is only partially true. Looking ahead is using learned experiences from the past, but also from our capacity to image solutions. In this book we discussed the negative influence of old conditionings when forming solutions for social problems. We must isolate and weaken old ideas and concepts that no longer work and strengthen new ideas, which are expected to work (according to imagination). As said before: Blockchain Organizing refuses to be what it is. It is mainly a concept what can make you think and help you with the reinvention of management. It would already be a great start when Blockchain Organizing is an idea for you with which you can experiment without immediately having to accept the idea. We like to hear your views on Blockchain Organizing and what you can and cannot do with it. We like to finish with a nice insight of Aristoteles:

"It is the mark of an educated mind to be able to entertain a thought without accepting it."

¹⁹⁶ Unfortunately too often a problem-creating super organism.

¹⁹⁷ Book by Martin E. P.Seligman, Peter Railton, Roy F. Baumeister and Chandra Sripada: 'Homo Prospectus', Oxford University Press USA (2016).

¹⁹⁸ See also the Weconomics analysis book, paragraph 1.2: 'Where did it fundamentally go wrong?'

About the authors

Paul Bessems is an international speaker, blockchain consultant and (co) author of more than ten management books. He has more than 25 years experience in designing and developing new organizations that align better with our new context (data economy) and new organization technologies such as blockchain. He is an expert in the field of Blockchain Organizing. He helps organizations with the transition *'From Business to Networkorganization'* and answering questions like: *'What is Blockchain Organizing, why is it important, how can I use it (and how not) and how should we start with it?'*

Besides his work as a consultant, he is also chairman and fellow of the Weconomics Foundation. Weconomics is a network that facilitates the transition to a more sustainable prosperity. The core in this development is the deployment of organization technology such as blockchain to create surplus for a new 'work sphere'.

Paul is also founder and CEO of Weconet Blockchain Technologies, located in Eindhoven, The Netherlands, one of the smartest regions in the world. Weconet helps organization to use new technologies such as blockchain and align these new technologies in new organization models such as shared transaction networks. Paul is also founder and fellow of The Institute for New Organizational Thinking.

Paul studied industrial engineering and management science at the Technical University in Eindhoven. In 1992 he introduced the Harvard Case Method in the Netherlands, developed several master programs for universities and developed two PHD's research proposals. In 1992 he started Buro Staff Support, his first professional community. Since then he helped dozens of professional communities. In recent years, these communities more and more work with the concept of Blockchain Organizing.

Walter Bril is a process knowledge management expert. For more than 25 years now he has been involved in analyzing, (re) designing business processes, with his focus mainly in the field of practical knowledge retention management. Currently this shows concretely in the form of contributing to the, San Francisco based, SaaS start-up Elements.cloud. As one of the original co-creators of the Universal Process Notation (UPN) and member of the core team of Elements.cloud, he adds value to mainly Salesforce customers.

In addition, somewhere in 2014 he started to discover and study the interesting developments around blockchain and joined one of the first blockchain registered certification MOOCS. Particularly because his concerns related to the sustainability of traditional business models and the potential socio-economic opportunities that blockchain has to offer, he has turned into a fervent blockchain evangelist. The combination technologist, process knowledge 'nerd' and blockchain evangelist turned out to be particularly valuable!

Walter studied business informatics at the Rijkshogeschool IJsselland in Deventer the Netherlands, has a semi-technical background, is author of several blogs, management books and international trainer and speaker. Besides that he actually does have a social life and loves to perform music, read, enjoy running and traveling.

Weconet Technologies

Both authors work for Weconet Technologies BV, a spin-off from the Weconomics Foundation. Weconet brings information technology and organizational science together to solve social problems. Weconet Technologies was founded in one of the world smartest regions: Brainport Eindhoven in the Netherlands. We see blockchain technology not just as technology, but as a form of organization technology (#or-gtech) that enables us to manage transactions and organizations decentralized, autonomous and almost without any friction between supply and demand.

The right deployment of Blockchain Organizing means that your organization will be more productive and flexible, that you comply with the new privacy law (GDPR), will be less depending on big data- and software companies and give a new meaning to human work. For more information about our solutions go to www.weconet.org/solutions

For more background information with this book, including a literature list and blockchain knowledge center, see also the website: weconomics.org (Dutch) and weconet.org (English).

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